Pathway Analysis Report

This report contains the pathway analysis results for the submitted sample ''. Analysis was performed against Reactome version 95 on 19/12/2025. The web link to these results is:

https://reactome.org/PathwayBrowser/#/ANALYSIS=MjAyNTEyMTkwODU2MzdfNTM0NA%3D%3D

Please keep in mind that analysis results are temporarily stored on our server. The storage period depends on usage of the service but is at least 7 days. As a result, please note that this URL is only valid for a limited time period and it might have expired.

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1. Introduction

Reactome is a curated database of pathways and reactions in human biology. Reactions can be considered as pathway 'steps'. Reactome defines a 'reaction' as any event in biology that changes the state of a biological molecule. Binding, activation, translocation, degradation and classical biochemical events involving a catalyst are all reactions. Information in the database is authored by expert biologists, entered and maintained by Reactome’s team of curators and editorial staff. Reactome content frequently cross-references other resources e.g. NCBI, Ensembl, UniProt, KEGG (Gene and Compound), ChEBI, PubMed and GO. Orthologous reactions inferred from annotation for Homo sapiens are available for 14 non-human species including mouse, rat, chicken, puffer fish, worm, fly and yeast. Pathways are represented by simple diagrams following an SBGN-like format.

Reactome's annotated data describe reactions possible if all annotated proteins and small molecules were present and active simultaneously in a cell. By overlaying an experimental dataset on these annotations, a user can perform a pathway over-representation analysis. By overlaying quantitative expression data or time series, a user can visualize the extent of change in affected pathways and its progression. A binomial test is used to calculate the probability shown for each result, and the p-values are corrected for the multiple testing (Benjamini–Hochberg procedure) that arises from evaluating the submitted list of identifiers against every pathway.

To learn more about our Pathway Analysis, please have a look at our relevant publications:

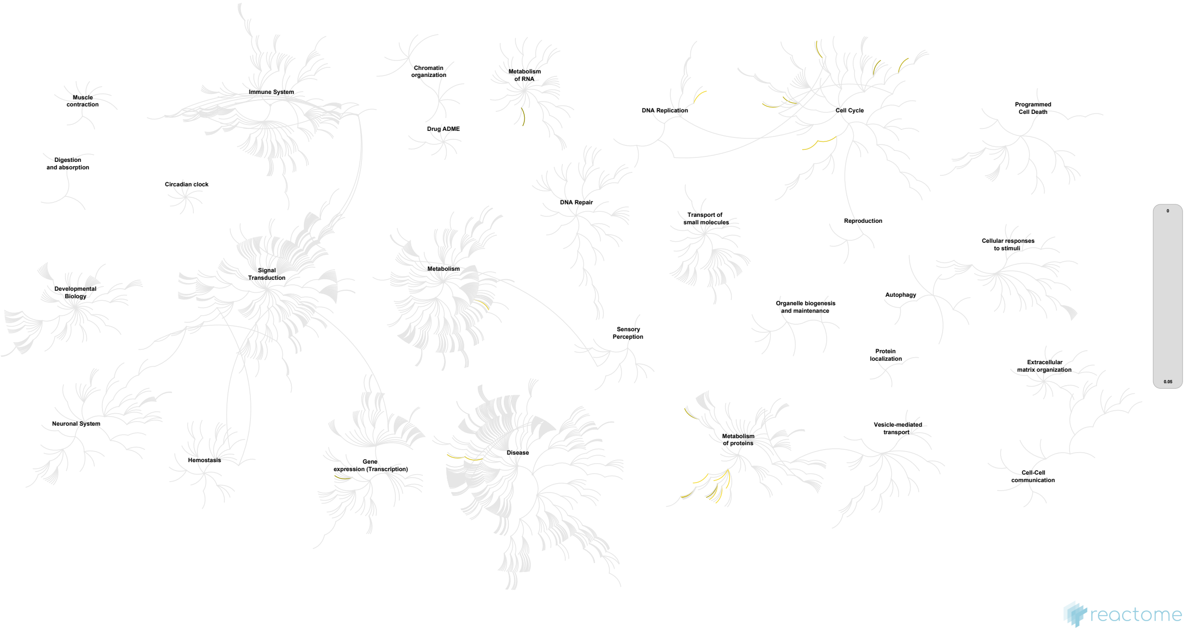
Fabregat A, Sidiropoulos K, Garapati P, Gillespie M, Hausmann K, Haw R, … D’Eustachio P (2016). The reactome pathway knowledgebase. Nucleic Acids Research, 44(D1), D481–D487. https://doi.org/10.1093/nar/gkv1351.

Fabregat A, Sidiropoulos K, Viteri G, Forner O, Marin-Garcia P, Arnau V, … Hermjakob H (2017). Reactome pathway analysis: a high-performance in-memory approach. BMC Bioinformatics, 18.

2. Properties

- This is an **overrepresentation** analysis: A statistical (hypergeometric distribution) test that determines whether certain Reactome pathways are over-represented (enriched) in the submitted data. It answers the question 'Does my list contain more proteins for pathway X than would be expected by chance?' This test produces a probability score, which is corrected for false discovery rate using the Benjamani-Hochberg method.
- 4474 out of 6773 identifiers in the sample were found in Reactome, where 2610 pathways were hit by at least one of them.
- All non-human identifiers have been converted to their human equivalent.
- IntAct interactors were included to increase the analysis background. This greatly increases the size of Reactome pathways, which maximises the chances of matching your submitted identifiers to the expanded pathway, but will include interactors that have not undergone manual curation by Reactome and may include interactors that have no biological significance, or unexplained relevance.
- This report is filtered to show only results for species 'Homo sapiens' and resource 'all resources'.
- The unique ID for this analysis (token) is MjAyNTEyMTkwODU2MzdfNTM0NA%3D%3D. This ID is valid for at least 7 days in Reactome's server. Use it to access Reactome services with your data.

3. Genome-wide overview



This figure shows a genome-wide overview of the results of your pathway analysis. Reactome pathways are arranged in a hierarchy. The center of each of the circular "bursts" is the root of one top-level pathway, for example "DNA Repair". Each step away from the center represents the next level lower in the pathway hierarchy. The color code denotes over-representation of that pathway in your input dataset. Light grey signifies pathways which are not significantly over-represented.

4. Most significant pathways

The following table shows the 25 most relevant pathways sorted by p-value.

Pathway name	Entities				Reactions	
	found	ratio	p-value	FDR*	found	ratio
SRP-dependent cotranslational protein targeting to membrane	86 / 119	0.005	4.08e-05	0.114	5 / 5	3.15e-04
Eukaryotic Translation Termination	76 / 109	0.004	7.41e-04	0.725	5 / 5	3.15e-04
Formation of a pool of free 40S subunits	81 / 113	0.005	8.16e-04	0.725	2 / 2	1.26e-04
ZNF598 and the Ribosome-associated Quality Trigger (RQT) complex dissociate a ribosome stalled on a no-go mRNA	76 / 109	0.004	0.001	0.725	4 / 4	2.52e-04
Peptide chain elongation	74 / 106	0.004	0.001	0.725	5 / 5	3.15e-04
Orc1 removal from chromatin	48 / 64	0.003	0.002	0.95	4 / 4	2.52e-04
Selenocysteine synthesis	77 / 118	0.005	0.005	1	5 / 7	4.41e-04
Mitochondrial translation termination	89 / 134	0.005	0.006	1	7 / 7	4.41e-04
Deposition of new CENPA-containing nucleosomes at the centromere	37 / 53	0.002	0.009	1	4 / 4	2.52e-04
Nucleosome assembly	37 / 53	0.002	0.009	1	4 / 4	2.52e-04
Aberrant regulation of mitotic G1/S transition in cancer due to RB1 defects	15 / 17	6.85e-04	0.013	1	2 / 2	1.26e-04
Defective binding of RB1 mutants to E2F1,(E2F2, E2F3)	15 / 17	6.85e-04	0.013	1	1 / 1	6.30e-05
SCF-beta-TrCP mediated degradation of Emi1	29 / 42	0.002	0.021	1	3 / 3	1.89e-04
E2F-enabled inhibition of pre-replication complex formation	9 / 9	3.62e-04	0.025	1	1 / 1	6.30e-05
Separation of Sister Chromatids	122 / 205	0.008	0.027	1	8 / 8	5.04e-04
Formation of the ternary complex, and subsequently, the 43S complex	35 / 54	0.002	0.028	1	3 / 3	1.89e-04
Autodegradation of Cdh1 by Cdh1:APC/C	35 / 54	0.002	0.028	1	2 / 2	1.26e-04
Prefoldin mediated transfer of substrate to CCT/TriC	21 / 29	0.001	0.029	1	2 / 2	1.26e-04
Mitochondrial translation elongation	75 / 122	0.005	0.035	1	7 / 8	5.04e-04
Recruitment of NuMA to mitotic centrosomes	57 / 97	0.004	0.036	1	2 / 2	1.26e-04
Post-transcriptional silencing by small RNAs	7 / 7	2.82e-04	0.044	1	3 / 3	1.89e-04

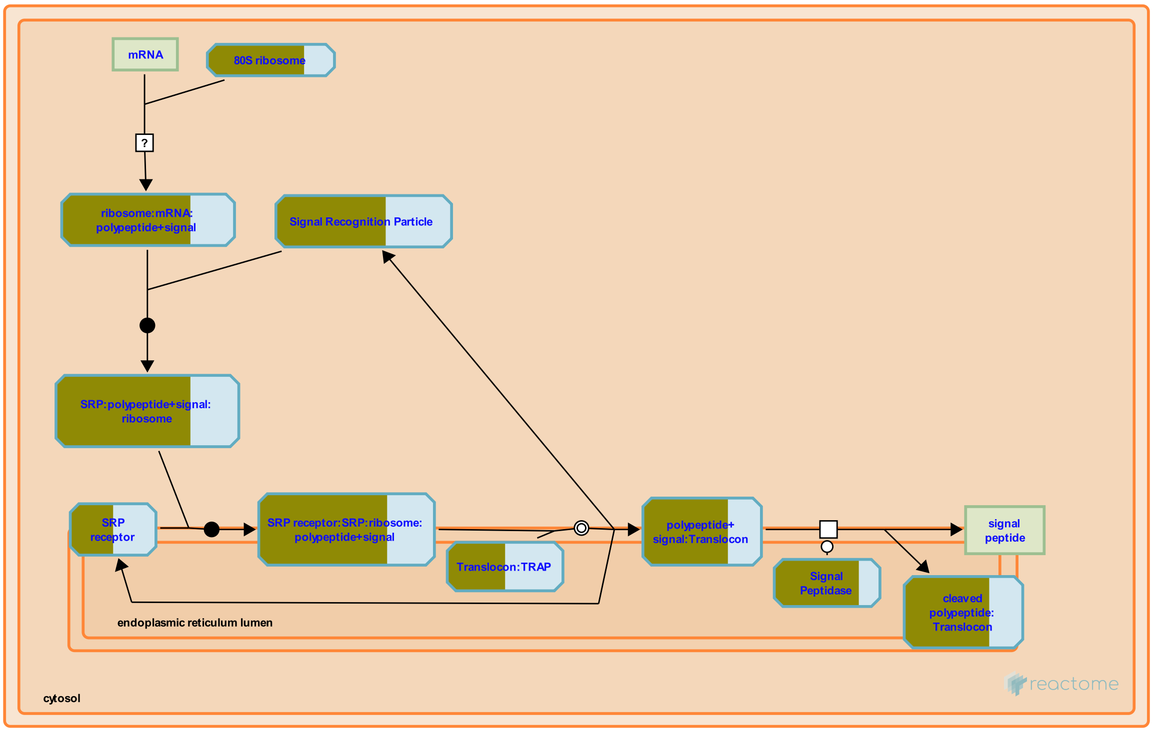
Pathway name	Entities				Reactions	
	found	ratio	p-value	FDR*	found	ratio
Nonsense Mediated Decay (NMD) independent of the Exon Junction Complex (EJC)	85 / 134	0.005	0.046	1	1 / 1	6.30e-05
PELO:HBS1L and ABCE1 dissociate a ribosome on a non-stop mRNA	84 / 129	0.005	0.052	1	3 / 3	1.89e-04
APC/C:Cdc20 mediated degradation of Securin	39 / 59	0.002	0.054	1	3 / 3	1.89e-04
Vif-mediated degradation of APOBEC3G	28 / 44	0.002	0.054	1	3 / 4	2.52e-04

* False Discovery Rate

5. Pathways details

For every pathway of the most significant pathways, we present its diagram, as well as a short summary, its bibliography and the list of inputs found in it.

1. SRP-dependent cotranslational protein targeting to membrane (R-HSA-1799339)



Cellular compartments: cytosol, endoplasmic reticulum lumen, endoplasmic reticulum membrane.

The process for translation of a protein destined for the endoplasmic reticulum (ER) branches from the canonical cytoslic translation process at the point when a nascent polypeptide containing a hydrophobic signal sequence is exposed on the surface of the cytosolic ribosome:mRNA:peptide complex. The signal sequence mediates the interaction of this complex with a cytosolic signal recognition particle (SRP) to form a complex which in turn docks with an SRP receptor complex on the ER membrane. There the ribosome complex is transferred from the SRP complex to a translocon complex embedded in the ER membrane and reoriented so that the nascent polypeptide protrudes through a pore in the translocon into the ER lumen. Translation, which had been halted by SRP binding, now resumes, the signal peptide is cleaved from the polypeptide, and elongation proceeds, with the growing polypeptide oriented into the ER lumen.

References

Edit history

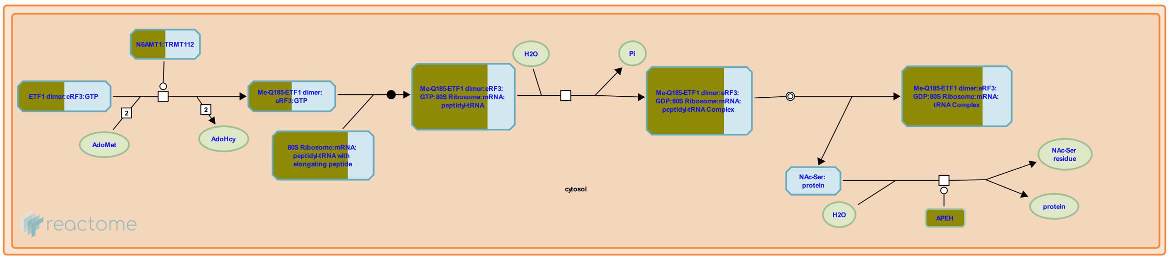
Date	Action	Author
2008-11-20	Authored	May B, Gopinathrao G

Date	Action	Author
2008-12-02	Reviewed	Matthews L, D'Eustachio P, Gillespie ME
2011-10-22	Revised	D'Eustachio P
2011-10-23	Edited	D'Eustachio P
2011-10-23	Created	D'Eustachio P

83 submitted entities found in this pathway, mapping to 102 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CRY2	P62263	FAU	P62861	RPL10	P27635, Q96L21
RPL10A	P62906	RPL11	P62913	RPL12	P30050
RPL13A	P40429	RPL15	P61313	RPL17	P18621
RPL18	Q07020	RPL18A	Q02543	RPL19	P84098
RPL21	P46778	RPL22	P35268	RPL23A	P62750
RPL24	P83731	RPL26	P61254, Q9UNX3	RPL26L1	Q9UNX3
RPL27	P61353	RPL27A	P46776, P61353	RPL29	P47914
RPL3	P39023, Q92901	RPL30	P62888	RPL32	P62888, P62910
RPL34	P49207, P62899	RPL35	P42766	RPL35A	P18077, P42766
RPL36A	P83881, Q969Q0	RPL37A	P18077, P61513	RPL38	P63173
RPL39L	Q96EH5	RPL41	P62945	RPL5	P46777
RPL6	Q02878	RPL7	P18124	RPL7A	P18124, P62424
RPL8	P62424, P62917	RPL9	P32969	RPLP2	P05387
RPN1	P04843	RPS10	P46783	RPS11	P62280
RPS13	P62277	RPS14	P62263	RPS15	P62841
RPS16	P62249	RPS17	P08708	RPS18	P62269
RPS23	P62266	RPS24	P62244, P62847	RPS25	P62851
RPS26	P62854	RPS27	P42677, Q71UM5	RPS27A	P62979, P62987, Q71UM5
RPS27L	Q71UM5	RPS28	P62857	RPS29	P62273
RPS3	P23396	RPS3A	P61247	RPS4X	P62701, Q8TD47
RPS4Y1	P22090, Q8TD47	RPS6	P62753	RPS7	P62081
RPS9	P46781	RRP1	P05386	SEC11A	P67812
SEC11C	Q9BY50	SEC61B	P60468	SEC61G	P60059
SOS1	P42766	SPC25	Q15005	SPCS1	Q9Y6A9
SRP14	P37108	SRP19	P09132	SRP54	P61011
SRP68	Q9UHB9	SRP72	O76094	SRPRA	P08240
SSR3	Q9UNL2	SSR4	P51571	UBA52	P62979, P62987
UBB	P62979, P62987	WBP1	P39656		

2. Eukaryotic Translation Termination (R-HSA-72764)



Cellular compartments: cytosol.

The arrival of any of the three stop codons (UAA, UAG and UGA) into the ribosomal A-site triggers the binding of a release factor (RF) to the ribosome and subsequent polypeptide chain release. In eukaryotes, the RF is composed of two proteins, eRF1 and eRF3. eRF1 is responsible for the hydrolysis of the peptidyl-tRNA, while eRF3 provides a GTP-dependent function. The ribosome releases the mRNA and dissociates into its two complex subunits, which can reassemble on another molecule to begin a new round of protein synthesis. It should be noted that at present, there is no factor identified in eukaryotes that would be the functional equivalent of the bacterial ribosome release (or recycling) factor, RRF, that catalyzes dissociation of the ribosome from the mRNA following release of the polypeptide

References

Salas-Marco J & Bedwell DM (2004). GTP hydrolysis by eRF3 facilitates stop codon decoding during eukaryotic translation termination. Mol Cell Biol, 24, 7769-78.

Ichikawa S & Kaji A (1989). Molecular cloning and expression of ribosome releasing factor. J Biol Chem, 264, 20054-9.

Edit history

Date	Action	Author
2004-11-09	Authored	Bedwell DM
2005-01-03	Created	Merrick WC, Bedwell DM
2025-11-12	Edited	Gillespie ME

71 submitted entities found in this pathway, mapping to 91 Reactome entities

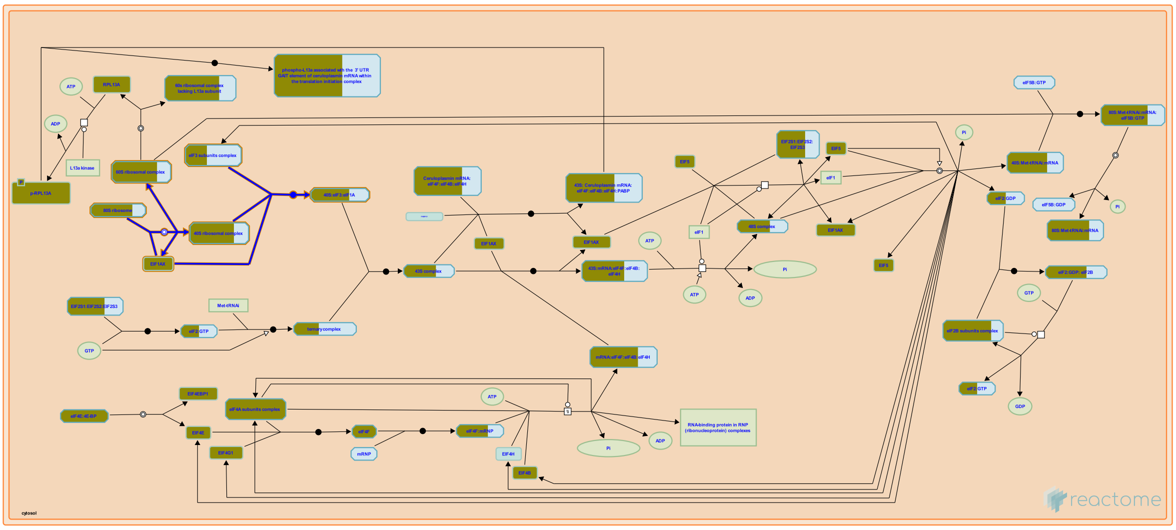
Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
APEH	P13798	CRY2	P62263	FAU	P62861
GSPT1	P15170, Q8IYD1	GSPT2	Q8IYD1	N6AMT1	Q9Y5N5
RPL10	P27635, Q96L21	RPL10A	P62906	RPL11	P62913
RPL12	P30050	RPL13A	P40429	RPL15	P61313
RPL17	P18621	RPL18	Q07020	RPL18A	Q02543
RPL19	P84098	RPL21	P46778	RPL22	P35268
RPL23A	P62750	RPL24	P83731	RPL26	P61254, Q9UNX3
RPL26L1	Q9UNX3	RPL27	P61353	RPL27A	P46776, P61353
RPL29	P47914	RPL3	P39023, Q92901	RPL30	P62888
RPL32	P62888, P62910	RPL34	P49207, P62899	RPL35	P42766
RPL35A	P18077, P42766	RPL36A	P83881, Q969Q0	RPL37A	P18077, P61513

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
RPL38	P63173	RPL39L	Q96EH5	RPL41	P62945
RPL5	P46777	RPL6	Q02878	RPL7	P18124
RPL7A	P18124, P62424	RPL8	P62424, P62917	RPL9	P32969
RPLP2	P05387	RPS10	P46783	RPS11	P62280
RPS13	P62277	RPS14	P62263	RPS15	P62841
RPS16	P62249	RPS17	P08708	RPS18	P62269
RPS23	P62266	RPS24	P62244, P62847	RPS25	P62851
RPS26	P62854	RPS27	P42677, Q71UM5	RPS27A	P62979, P62987, Q71UM5
RPS27L	Q71UM5	RPS28	P62857	RPS29	P62273
RPS3	P23396	RPS3A	P61247	RPS4X	P62701, Q8TD47
RPS4Y1	P22090, Q8TD47	RPS6	P62753	RPS7	P62081
RPS9	P46781	RRP1	P05386	SOS1	P42766
UBA52	P62979, P62987	UBB	P62979, P62987		

Interactors found in this pathway (2)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
APEH	P13798	P13798	IST1	P53990, P53990-3	P13798

3. Formation of a pool of free 40S subunits (R-HSA-72689)



Cellular compartments: cytosol.

The 80S ribosome dissociates into free 40S (small) and 60S (large) ribosomal subunits. Each ribosomal subunit is constituted by several individual ribosomal proteins and rRNA.

References

Edit history

Date	Action	Author

73 submitted entities found in this pathway, mapping to 92 Reactome entities

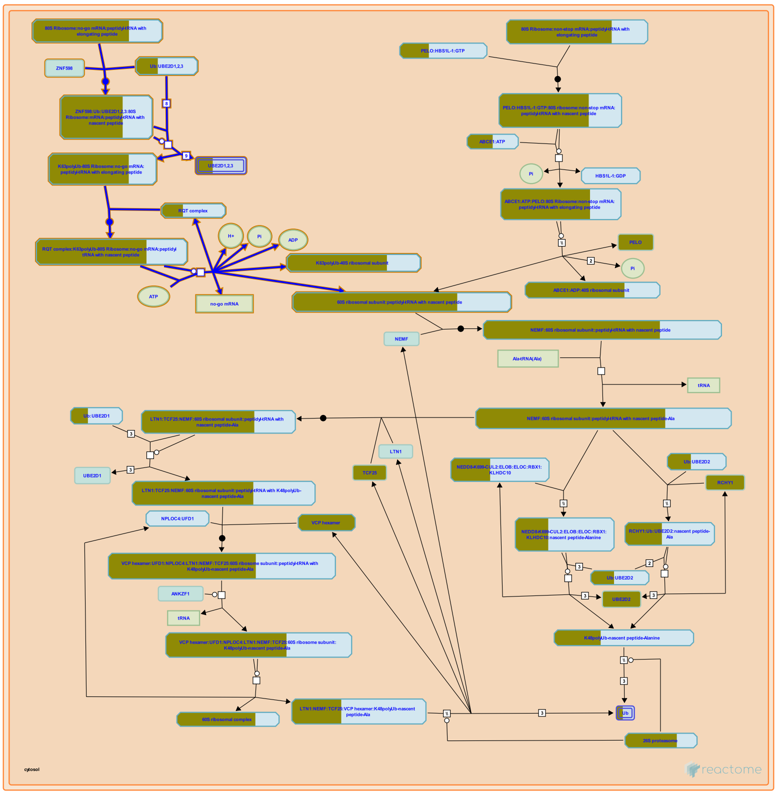
Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CRY2	P62263	EIF1AY	P47813	EIF3B	P55884
EIF3D	O15371	EIF3G	O75821	EIF3I	Q13347
EIF3J	O75822	FAU	P62861	RPL10	P27635, Q96L21
RPL10A	P62906	RPL11	P62913	RPL12	P30050
RPL13A	P40429	RPL15	P61313	RPL17	P18621
RPL18	Q07020	RPL18A	Q02543	RPL19	P84098
RPL21	P46778	RPL22	P35268	RPL23A	P62750
RPL24	P83731	RPL26	P61254, Q9UNX3	RPL26L1	Q9UNX3
RPL27	P61353	RPL27A	P46776, P61353	RPL29	P47914
RPL3	P39023, Q92901	RPL30	P62888	RPL32	P62888, P62910
RPL34	P49207, P62899	RPL35	P42766	RPL35A	P18077, P42766
RPL36A	P83881, Q969Q0	RPL37A	P18077, P61513	RPL38	P63173
RPL39L	Q96EH5	RPL41	P62945	RPL5	P46777
RPL6	Q02878	RPL7	P18124	RPL7A	P18124, P62424
RPL8	P62424, P62917	RPL9	P32969	RPLP2	P05387
RPS10	P46783	RPS11	P62280	RPS13	P62277
RPS14	P62263	RPS15	P62841	RPS16	P62249
RPS17	P08708	RPS18	P62269	RPS23	P62266
RPS24	P62244, P62847	RPS25	P62851	RPS26	P62854
RPS27	P42677, Q71UM5	RPS27A	P62979, P62987, Q71UM5	RPS27L	Q71UM5
RPS28	P62857	RPS29	P62273	RPS3	P23396

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
RPS3A	P61247	RPS4X	P62701, Q8TD47	RPS4Y1	P22090, Q8TD47
RPS6	P62753	RPS7	P62081	RPS9	P46781
RRP1	P05386	SOS1	P42766	UBA52	P62979, P62987
UBB	P62979, P62987				

Interactors found in this pathway (5)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
EIF1AY	O14602	P47813	EIF5	P55010	P47813
IPO13	O94829	P47813	NCR3LG1	Q68D85	P47813
RAN	P62826	P47813			

4. ZNF598 and the Ribosome-associated Quality Trigger (RQT) complex dissociate a ribosome stalled on a no-go mRNA (R-HSA-9954716)



Due to features such as damaged nucleotides, strong secondary structure, presence of stretches of more than four uninterrupted AAA codons in a mRNA coding region (designated no-go mRNAs) a ribosome translating an mRNA can stall and cause collisions with the ribosomes trailing it on the mRNA (reviewed in Joazeiro 2017, Eisenack and Trentini 2022, Filbeck et al. 2022, McGirr et al. 2025). In cases in which the ribosome stalls internally on the mRNA and a 3' region of the mRNA protrudes from the ribosome, ZNF598, a ubiquitin E3 ligase that is the homolog of yeast HEL2, binds the ribosome (Garzia et al. 2017, Juszkiewicz and Hegde 2017, Sundaramoorthy et al. 2017) and ubiquitinates the 40S subunit ribosomal proteins eS10 (RPS10) at residues K138 and K139 and uS10 (RPS20) at residues K4 and K8 to initiate splitting of the 40S and 60S ribosomal subunits (Juszkiewicz and Hegde 2017, Sundaramoorthy et al. 2017, Garzia et al. 2017, Juszkiewicz et al. 2018, Narita et al. 2022, Miścicka et al. 2024, also inferred from yeast homologs in Matsuo et al. 2017, Sitron et al. 2017, Ikeuchi et al. 2019, reviewed in Ford et al. 2024). ZAK⍰ also interacts with the collided ribosome at the 18S helix through its c-terminal domain, which triggers ZAK⍰ autophosphorylation and phosphorylation of p38/JNK in a MAPK cascade known as the ribotoxic stress response (Wu et al. 2020, Vind et al. 2020, Stoneley et al. 2022).

The ASCC2 subunit (Narita et al. 2022) of the ribosome quality control trigger complex (RQT complex, ASCC2:ASCC3:TRIP4) binds K63-linked polyubiquitin conjugated to the 40S protein uS10 (Hashimoto et al. 2020, Juszkiewicz et al. 2020, Narita et al. 2022, and inferred from yeast homologs in Matsuo et al. 2023). The ASCC3 subunit of the RQT complex splits stalled 80S ribosomes with K63-polyubiquitinated uS10 into 60S and 40S subunits (Hashimoto et al. 2020, Juszkiewicz et al. 2020, Narita et al. 2022, Miścicka et al. 2024) apparently by exerting a pulling force on the mRNA (inferred from the yeast homolog Slh1 in Best et al. 2023). The peptidyl-tRNA remains bound in the 60S subunit, with the tRNA positioned in the P site. The problematic mRNA dissociates after splitting and is thought to be degraded at this time. In the case of collided yeast ribosomes, the mRNA is first endonucleolytically cleaved and the cleavage products are exonucleolytically degraded by XRN1 and the exosome (Ikeuchi et al. 2019).

References

Yan LL & Zaher HS (2021). Ribosome quality control antagonizes the activation of the integrated stress response on colliding ribosomes. Mol Cell, 81, 614-628.e4.

Joazeiro CAP (2017). Ribosomal Stalling During Translation: Providing Substrates for Ribosome-Associated Protein Quality Control. Annu Rev Cell Dev Biol, 33, 343-368.

Eisenack TJ & Trentini DB (2022). Ending a bad start: Triggers and mechanisms of co-translational protein degradation. Front Mol Biosci, 9, 1089825.

McGirr T, Onar O & Jafarnejad SM (2025). Dysregulated ribosome quality control in human diseases . FEBS J, 292, 936-959.

Garzia A, Jafarnejad SM, Meyer C, Chapat C, Gogakos T, Morozov P, ... Sonenberg N (2017). The E3 ubiquitin ligase and RNA-binding protein ZNF598 orchestrates ribosome quality control of premature polyadenylated mRNAs. Nat Commun, 8, 16056.

Edit history

Date	Action	Author
2025-06-06	Edited	May B
2025-06-06	Authored	May B
2025-06-08	Created	May B
2025-08-09	Reviewed	Bennett EJ, Ford PW
2025-11-07	Reviewed	McGirr T, Jafarnejad SM

69 submitted entities found in this pathway, mapping to 89 Reactome entities

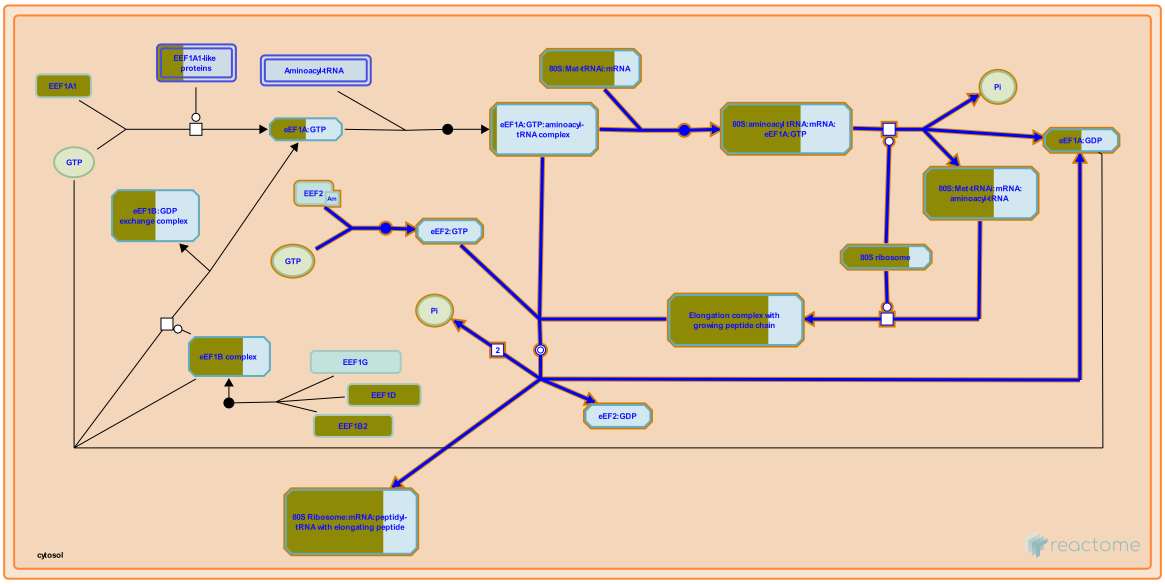
Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ASCC3	Q8N3C0	CRY2	P62263	FAU	P62861
RPL10	P27635, Q96L21	RPL10A	P62906	RPL11	P62913
RPL12	P30050	RPL13A	P40429	RPL15	P61313
RPL17	P18621	RPL18	Q07020	RPL18A	Q02543
RPL19	P84098	RPL21	P46778	RPL22	P35268
RPL23A	P62750	RPL24	P83731	RPL26	P61254, Q9UNX3
RPL26L1	Q9UNX3	RPL27	P61353	RPL27A	P46776, P61353
RPL29	P47914	RPL3	P39023, Q92901	RPL30	P62888
RPL32	P62888, P62910	RPL34	P49207, P62899	RPL35	P42766

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
RPL35A	P18077, P42766	RPL36A	P83881, Q969Q0	RPL37A	P18077, P61513
RPL38	P63173	RPL39L	Q96EH5	RPL41	P62945
RPL5	P46777	RPL6	Q02878	RPL7	P18124
RPL7A	P18124, P62424	RPL8	P62424, P62917	RPL9	P32969
RPLP2	P05387	RPS10	P46783	RPS11	P62280
RPS13	P62277	RPS14	P62263	RPS15	P62841
RPS16	P62249	RPS17	P08708	RPS18	P62269
RPS23	P62266	RPS24	P62244, P62847	RPS25	P62851
RPS26	P62854	RPS27	P42677, Q71UM5	RPS27A	P62979, P62987, Q71UM5
RPS27L	Q71UM5	RPS28	P62857	RPS29	P62273
RPS3	P23396	RPS3A	P61247	RPS4X	P62701, Q8TD47
RPS4Y1	P22090, Q8TD47	RPS6	P62753	RPS7	P62081
RPS9	P46781	RRP1	P05386	SOS1	P42766
UBA52	P62979, P62987	UBB	P0CG47, P62979, P62987	UBE2D2	P62837

Interactors found in this pathway (3)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
GIGYF1	O75420	Q86UK7	KBTBD7	Q8WVZ9	Q86UK7
YWHAE	P62258	Q86UK7			

5. Peptide chain elongation (R-HSA-156902)



Cellular compartments: cytosol.

The mechanism of a peptide bond requires the movement of three protons. First the deprotonation of the ammonium ion generates a reactive amine, allowing a nucleophilic attack on the carbonyl group. This is followed by the loss of a proton from the reaction intermediate, only to be taken up by the oxygen on the leaving group (from the end of the amino acid chain bound to the tRNA in the P-site). The peptide bond formation results in the net loss of one water molecule, leaving a deacylated-tRNA in the P-site, and a nascent polypeptide chain one amino acid larger in the A-site.

For the purpose of illustration, the figures used in the section show one amino acid being added to a peptidyl-tRNA with a growing peptide chain.

References

Green R & Lorsch JR (2002). The path to perdition is paved with protons. Cell, 110, 665-8.

Edit history

Date	Action	Author
2005-03-13	Authored	Gopinathrao G
2005-03-17	Created	Balar B, Ulloque R, Kinzy TG

68 submitted entities found in this pathway, mapping to 87 Reactome entities

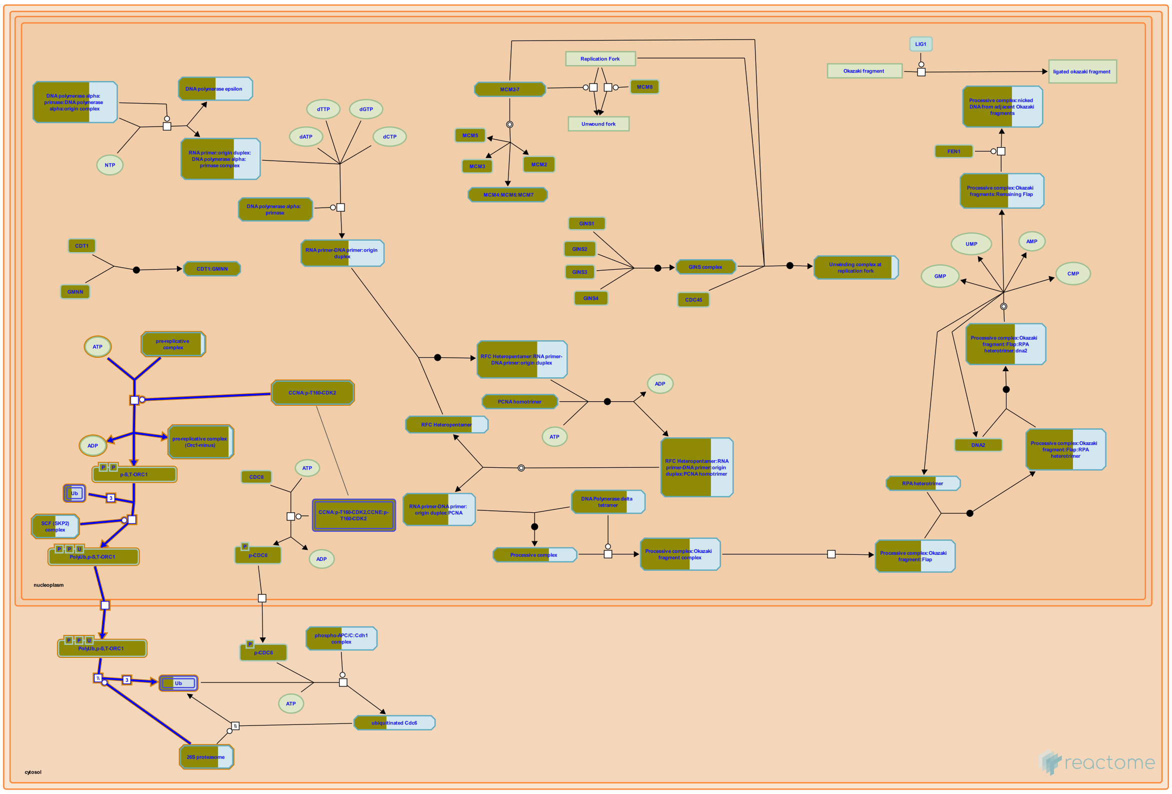
Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CRY2	P62263	EEF1A1	P68104	FAU	P62861
RPL10	P27635, Q96L21	RPL10A	P62906	RPL11	P62913
RPL12	P30050	RPL13A	P40429	RPL15	P61313
RPL17	P18621	RPL18	Q07020	RPL18A	Q02543
RPL19	P84098	RPL21	P46778	RPL22	P35268
RPL23A	P62750	RPL24	P83731	RPL26	P61254, Q9UNX3
RPL26L1	Q9UNX3	RPL27	P61353	RPL27A	P46776, P61353
RPL29	P47914	RPL3	P39023, Q92901	RPL30	P62888

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
RPL32	P62888, P62910	RPL34	P49207, P62899	RPL35	P42766
RPL35A	P18077, P42766	RPL36A	P83881, Q969Q0	RPL37A	P18077, P61513
RPL38	P63173	RPL39L	Q96EH5	RPL41	P62945
RPL5	P46777	RPL6	Q02878	RPL7	P18124
RPL7A	P18124, P62424	RPL8	P62424, P62917	RPL9	P32969
RPLP2	P05387	RPS10	P46783	RPS11	P62280
RPS13	P62277	RPS14	P62263	RPS15	P62841
RPS16	P62249	RPS17	P08708	RPS18	P62269
RPS23	P62266	RPS24	P62244, P62847	RPS25	P62851
RPS26	P62854	RPS27	P42677, Q71UM5	RPS27A	P62979, P62987, Q71UM5
RPS27L	Q71UM5	RPS28	P62857	RPS29	P62273
RPS3	P23396	RPS3A	P61247	RPS4X	P62701, Q8TD47
RPS4Y1	P22090, Q8TD47	RPS6	P62753	RPS7	P62081
RPS9	P46781	RRP1	P05386	SOS1	P42766
UBA52	P62979, P62987	UBB	P62979, P62987		

Interactors found in this pathway (3)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
DISC1	Q9NRI5	P13639	HSD17B10	Q99714	P13639
MAPT	P10636-8	P13639			

6. Orc1 removal from chromatin (R-HSA-68949)



Cellular compartments: nucleoplasm, cytosol.

Mammalian Orc1 protein is phosphorylated and selectively released from chromatin and ubiquitinated during the S-to-M transition in the cell division cycle.

References

Mendez J, Zou-Yang XH, Kim SY, Hidaka M, Tansey WP & Stillman B (2002). Human origin recognition complex large subunit is degraded by ubiquitin-mediated proteolysis after initiation of DNA replication. Mol Cell, 9, 481-91.

Li CJ & DePamphilis ML (2001). Mammalian Orc1 protein is selectively released from chromatin and ubiquitinated during the S-to-M transition in the cell division cycle. Mol Cell Biol, 22, 105-16.

Wenger CB, Roberts MF, Stolwijk JA & Nadel ER (1975). Forearm blood flow during body temperature transients produced by leg exercise. J Appl Physiol, 38, 58-63.

Ladenburger EM, Keller C & Knippers R (2002). Identification of a binding region for human origin recognition complex proteins 1 and 2 that coincides with an origin of DNA replication. Mol Cell Biol, 22, 1036-48.

Edit history

Date	Action	Author
2025-11-15	Modified	Weiser JD

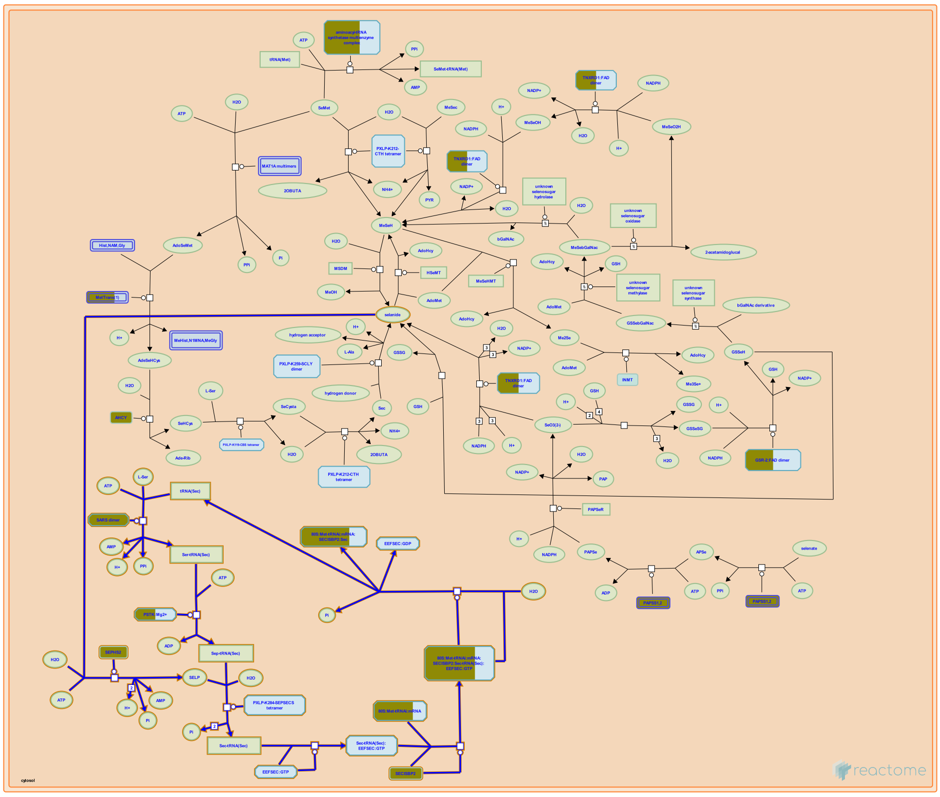
48 submitted entities found in this pathway, mapping to 53 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CCNA1	P78396	CCNA2	P20248	CCNL1	P49736
CDC6	Q99741	CDK1	P24941	CDK2	P24941
CDT1	Q9H211	MCM2	P33993, P49736	MCM3	P25205
MCM4	P33991	MCM5	P33992	MCM6	Q14566
MCM7	P33993	MCM8	Q9UJA3	ORC1	Q13415
ORC2	Q13416	ORC3	Q9UBD5	ORC4	O43929
ORC5	O43913	ORC6	Q9Y5N6	PSMA3	P25788
PSMA4	P25789	PSMA5	P28066	PSMA7	O14818
PSMB2	P49721	PSMB3	P49720	PSMB5	P28074
PSMB6	P28072	PSMB7	Q99436	PSMC1	P62191
PSMC2	P35998	PSMC3	P17980	PSMC4	P43686
PSMC6	P62333	PSMD1	Q99460	PSMD11	O00231
PSMD12	O00232	PSMD14	O00487	PSMD3	O43242
PSMD6	Q15008	PSMD8	P48556	RBX1	P62877
RPN1	Q13200	RPS27A	P62979, P62987	SEM1	P60896
SKP2	Q13309	UBA52	P62979, P62987	UBB	P0CG47, P62979, P62987

Interactors found in this pathway (8)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
CCNA2	P20248	Q13415	GOLGA6L9	A6NEM1	Q13415
ORC2	Q13416	Q13415	ORC3	Q9UBD5	Q13415
ORC4	O43929	Q13415	ORC5	O43913	Q13415
SKP2	Q13309	Q13415	TERF2	Q15554	Q13415

7. Selenocysteine synthesis (R-HSA-2408557)



Selenocysteine, the 21st genetically encoded amino acid, is the major form of the antioxidant trace element selenium in the human body. In eukaryotes and archaea its synthesis proceeds through a phosphorylated intermediate in a tRNA-dependent fashion. The final step of selenocysteine formation is catalyzed by O-phosphoseryl-tRNA:selenocysteinyl-tRNA synthase (SEPSECS) that converts phosphoseryl-tRNA(Sec) to selenocysteinyl-tRNA(Sec).

References

Palioura S, Herkel J, Simonovic M, Lohse AW & Söll D (2010). Human SepSecS or SLA/LP: selenocysteine formation and autoimmune hepatitis. Biol. Chem., 391, 771-6.

Sheppard K, Yuan J, Hohn MJ, Jester B, Devine KM & Söll D (2008). From one amino acid to another: tRNA-dependent amino acid biosynthesis. Nucleic Acids Res., 36, 1813-25.

Donovan J & Copeland PR (2010). Threading the needle: getting selenocysteine into proteins. Antioxid. Redox Signal., 12, 881-92.

Edit history

Date	Action	Author
2012-07-17	Created	Williams MG
2014-05-06	Authored	Williams MG
2015-08-29	Edited	D'Eustachio P
2015-08-30	Reviewed	Rush MG
2025-11-15	Modified	Weiser JD

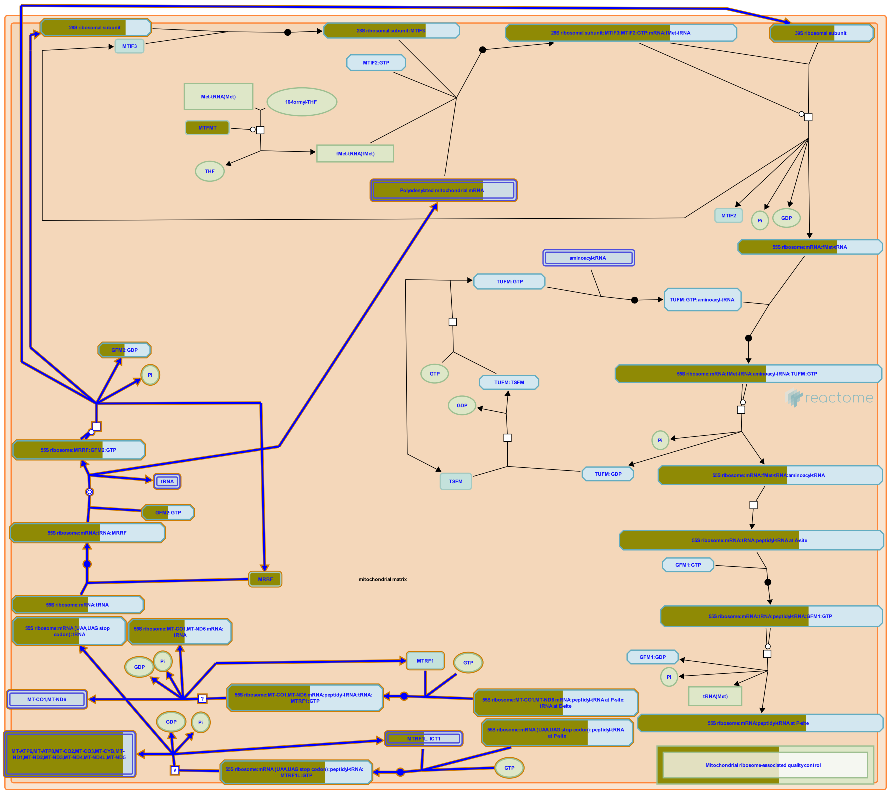
71 submitted entities found in this pathway, mapping to 90 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CRY2	P62263	FAU	P62861	PSTK	Q8IV42
RPL10	P27635, Q96L21	RPL10A	P62906	RPL11	P62913
RPL12	P30050	RPL13A	P40429	RPL15	P61313
RPL17	P18621	RPL18	Q07020	RPL18A	Q02543
RPL19	P84098	RPL21	P46778	RPL22	P35268
RPL23A	P62750	RPL24	P83731	RPL26	P61254, Q9UNX3
RPL26L1	Q9UNX3	RPL27	P61353	RPL27A	P46776, P61353
RPL29	P47914	RPL3	P39023, Q92901	RPL30	P62888
RPL32	P62888, P62910	RPL34	P49207, P62899	RPL35	P42766
RPL35A	P18077, P42766	RPL36A	P83881, Q969Q0	RPL37A	P18077, P61513
RPL38	P63173	RPL39L	Q96EH5	RPL41	P62945
RPL5	P46777	RPL6	Q02878	RPL7	P18124
RPL7A	P18124, P62424	RPL8	P62424, P62917	RPL9	P32969
RPLP2	P05387	RPS10	P46783	RPS11	P62280
RPS13	P62277	RPS14	P62263	RPS15	P62841
RPS16	P62249	RPS17	P08708	RPS18	P62269
RPS23	P62266	RPS24	P62244, P62847	RPS25	P62851
RPS26	P62854	RPS27	P42677, Q71UM5	RPS27A	P62979, P62987, Q71UM5
RPS27L	Q71UM5	RPS28	P62857	RPS29	P62273
RPS3	P23396	RPS3A	P61247	RPS4X	P62701, Q8TD47
RPS4Y1	P22090, Q8TD47	RPS6	P62753	RPS7	P62081
RPS9	P46781	RRP1	P05386	SARS	P49591
SECISBP2	Q96T21	SEPHS2	Q99611	SOS1	P42766
UBA52	P62979, P62987	UBB	P62979, P62987		

Interactors found in this pathway (3)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
CCL22	O00626	Q99611	SEPHS1	P49903	Q99611
ZNF408	Q9H9D4	Q99611			

8. Mitochondrial translation termination (R-HSA-5419276)



Cellular compartments: mitochondrial matrix, mitochondrial inner membrane.

Translation is terminated for 11 of the 13 mitochondrially encoded proteins when MTRF1L (MTRF1a) recognizes a UAA or UAG termination codon in the mRNA at the A site of the ribosome (Soleimanpour-Lichaei et al. 2007, reviewed in Richter et al. 2010, Chrzanowska-Lightowlers et al. 2011. Christian and Spremulli 2012). Translation of the MT-CO1 and MT-ND6 proteins is terminated at an AGG codon and an AGA codon, respectively. These two noncanonical stop codons are recognized by MTRF1. GTP is hydrolyzed, and the aminoacyl bond between the translated polypeptide and the final tRNA at the P site is hydrolyzed by the 39S ribosomal subunit, releasing the translated polypeptide. MRRF (RRF) and GFM2:GTP (EF-G2mt:GTP) then act to release the remaining tRNA and mRNA from the ribosome and dissociate the 55S ribosome into 28S and 39S subunits.

References

Christian BE & Spremulli LL (2012). Mechanism of protein biosynthesis in mammalian mitochondria. Biochim. Biophys. Acta, 1819, 1035-54.

Richter R, Pajak A, Dennerlein S, Rozanska A, Lightowlers RN & Chrzanowska-Lightowlers ZM (2010). Translation termination in human mitochondrial ribosomes. Biochem. Soc. Trans., 38, 1523-6.

Chrzanowska-Lightowlers ZM, Pajak A & Lightowlers RN (2011). Termination of protein synthesis in mammalian mitochondria. J. Biol. Chem., 286, 34479-85.

Soleimanpour-Lichaei HR, Kühl I, Gaisne M, Passos JF, Wydro M, Rorbach J, ... Chrzanowska-Lightowlers ZM (2007). mtRF1a is a human mitochondrial translation release factor decoding the major termination codons UAA and UAG. Mol. Cell, 27, 745-57.

Edit history

Date	Action	Author
2014-04-30	Edited	May B
2014-04-30	Authored	May B
2014-05-03	Created	May B
2014-08-29	Reviewed	Chrzanowska-Lightowlers ZM
2014-09-20	Reviewed	Spremulli LL
2025-03-04	Reviewed	Matthews L
2025-11-15	Modified	Weiser JD

73 submitted entities found in this pathway, mapping to 89 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
AIP	Q9NWT8	BRIP1	Q9P0J6	CHCHD1	Q96BP2
ERAL1	O75616	GADD45GIP1	Q8TAE8	GFM2	Q969S9
MRPL10	Q7Z7H8	MRPL11	Q9Y3B7	MRPL12	P52815
MRPL13	Q9BYD1	MRPL14	Q6P1L8	MRPL15	P49406, Q9P015
MRPL16	Q9NX20	MRPL17	Q9NRX2	MRPL18	Q9H0U6
MRPL19	P49406	MRPL22	Q9NWU5	MRPL27	Q8IXM3, Q9P0M9
MRPL28	Q13084, Q8TCC3	MRPL3	P09001	MRPL30	Q8TCC3
MRPL32	Q6P1L8, Q9BYC8	MRPL34	Q9BQ48	MRPL35	Q9NZE8
MRPL36	Q9P0J6	MRPL37	Q9BZE1	MRPL38	Q96DV4
MRPL39	Q9NYK5	MRPL41	Q8IXM3	MRPL44	Q9H9J2
MRPL45	Q9BRJ2	MRPL46	Q9H2W6	MRPL50	Q8N5N7
MRPL51	Q4U2R6	MRPL52	Q86TS9	MRPL55	Q7Z7F7
MRPL57	Q9BQC6	MRPL58	Q14197	MRPS10	P82664
MRPS11	P82912	MRPS12	O15235	MRPS14	O60783
MRPS15	P82914	MRPS16	Q9Y3D3	MRPS18A	Q9NVS2
MRPS18B	Q9Y676	MRPS18C	Q9Y3D5	MRPS22	P82650
MRPS23	Q9Y3D9	MRPS25	P82663	MRPS28	P82673, Q9Y2Q9
MRPS30	Q9NP92	MRPS31	Q92665	MRPS33	Q9Y291
MRPS34	P82930	MRPS35	P82673, Q9Y2Q9	MRPS36	P82909
MRPS5	P82675	MRPS7	Q9Y2R9	MRPS9	P82933
MRRF	Q96E11	MT-ATP8	P03928	MT-CO2	P00403
MT-CO3	P00414	MT-CYB	P00156	MT-ND1	P03886
MT-ND2	P03891	MT-ND3	P03897	MT-ND4	P03905
MT-ND4L	P03901	MT-ND5	P03915	PTCD3	Q96EY7

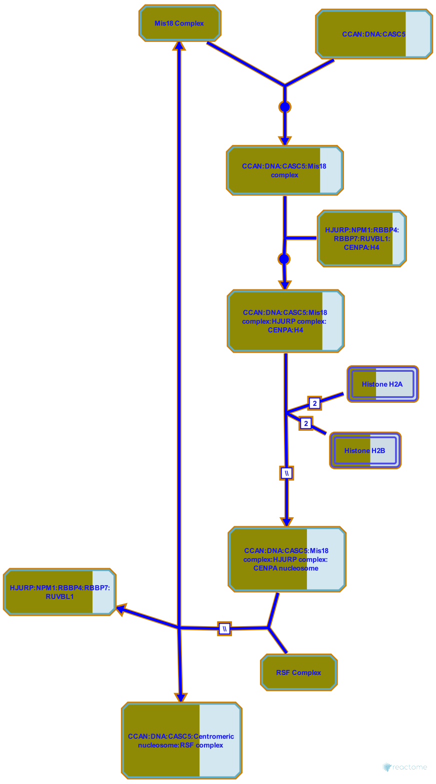
Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
MT-ATP8	ENST00000361851	MT-CO2	ENST00000361739	MT-CO3	ENST00000362079
MT-CYB	ENST00000361789	MT-ND1	ENST00000361390	MT-ND2	ENST00000361453
MT-ND3	ENST00000361227	MT-ND4	ENST00000361381	MT-ND4L	ENST00000361335
MT-ND5	ENST00000361567	MT-RNR1	ENST00000389680		

Interactors found in this pathway (7)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
BANP	Q8N9N5-2	Q96E11	DBT	P11182	Q96E11

Input	UniProt Id	Interacts with
FAM9B	Q8IZU0	Q96E11
HCCS	P53701	Q96E11
ZNF408	Q9H9D4	Q96E11
FHL3	Q13643	Q96E11
TRIM27	P14373	Q96E11

9. Deposition of new CENPA-containing nucleosomes at the centromere (R-HSA-606279)



Cellular compartments: nucleoplasm.

Eukaryotic centromeres are marked by a unique form of histone H3, designated CENPA in humans. In human cells newly synthesized CENPA is deposited in nucleosomes at the centromere during late telophase/early G1 phase of the cell cycle. Once deposited, nucleosomes containing CENPA remain stably associated with the centromere and are partitioned equally to daughter centromeres during S phase. A current model proposes that pre-existing CENPA at the centromere drives recruitment of new CENPA, however this has not been proved.

The deposition process requires at least 3 complexes: the Mis18 complex, HJURP complex, and the RSF complex. HJURP binds newly synthesized CENPA-H4 tetramers before deposition and brings them to the centromere for deposition in new CENPA-containing nucleosomes. The exact mechanism of deposition remains unknown.

References

Jansen LE, Black BE, Foltz DR & Cleveland DW (2007). Propagation of centromeric chromatin requires exit from mitosis. J Cell Biol, 176, 795-805.

Foltz DR, Jansen LE, Bailey AO, Yates JR 3rd, Bassett EA, Wood S, ... Cleveland DW (2009). Centromere-specific assembly of CENP-a nucleosomes is mediated by HJURP. Cell, 137, 472-84.

Dunleavy EM, Roche D, Tagami H, Lacoste N, Ray-Gallet D, Nakamura Y, ... Almouzni-Pettinotti G (2009). HJURP is a cell-cycle-dependent maintenance and deposition factor of CENP-A at centromeres. Cell, 137, 485-97.

Shuaib M, Ouararhni K, Dimitrov S & Hamiche A (2010). HJURP binds CENP-A via a highly conserved N-terminal domain and mediates its deposition at centromeres. Proc Natl Acad Sci U S A, 107, 1349-54.

Mellone BG, Zhang W & Karpen GH (2009). Frodos found: Behold the CENP-a "Ring" bearers. Cell, 137, 409-12.

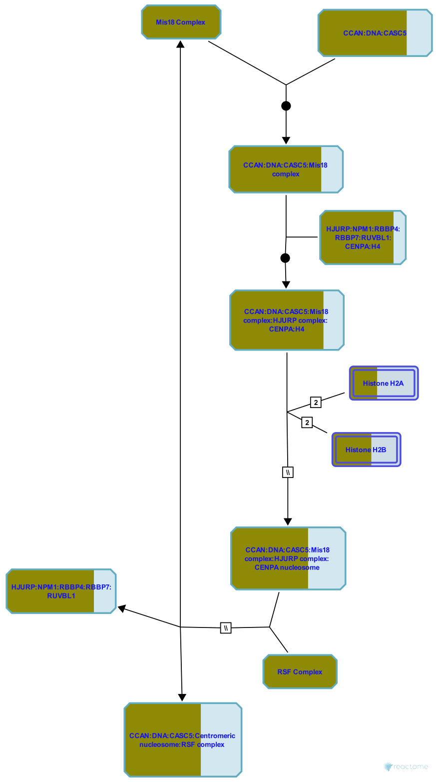
Edit history

Date	Action	Author
2010-04-15	Edited	May B
2010-04-15	Authored	May B
2010-04-17	Created	May B
2010-05-30	Reviewed	Dunleavy EM, Almouzni-Pettinotti G
2010-06-15	Reviewed	Foltz DR
2025-11-15	Modified	Weiser JD

40 submitted entities found in this pathway, mapping to 43 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CENPA	P49450	CENPH	Q9H3R5	CENPI	Q92674
CENPK	Q9BS16	CENPL	Q8N0S6	CENPM	Q9NSP4
CENPN	Q96H22	CENPO	Q9BU64	CENPQ	Q7L2Z9
CENPT	Q96BT3	CENPU	Q71F23	CENPW	Q5EE01
CENPX	A8MT69	H2AFJ	Q9BTM1	H2AFX	P16104
HIST1H2AC	Q93077	HIST1H2AI	P20671	HIST1H2BD	P58876
HIST1H2BG	P62807, Q93079	HIST1H2BH	P58876, Q93079	HIST1H2BJ	P06899
HIST1H2BK	O60814, Q99880	HIST1H2BL	Q99880	HIST1H2BM	Q99879
HIST1H4A	P62805	HIST1H4D	P62805	HIST1H4I	P62805
HIST1H4L	P62805	HIST3H2BB	Q8N257	HJURP	Q8NCD3
ITGB3BP	Q13352	KNL1	Q8NG31	MIS18A	Q9NYP9
MIS18BP1	Q6P0N0	NPM1	P06748	OIP5	O43482
RBBP4	Q09028	RSF1	Q96T23	RUVBL1	Q9Y265
SMARCA5	O60264				

10. Nucleosome assembly (R-HSA-774815)



The formation of centromeric chromatin assembly outside the context of DNA replication involves the assembly of nucleosomes containing the histone H3 variant CenH3 (also called CENP-A).

References

Edit history

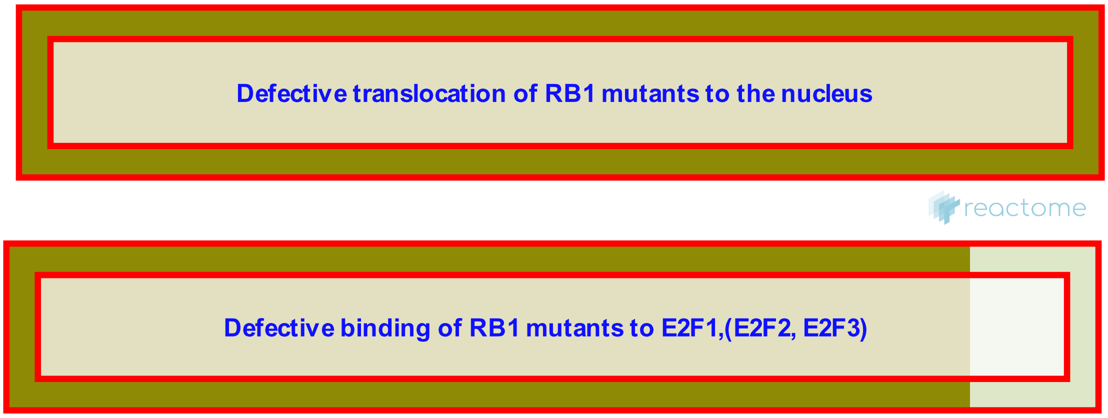
Date	Action	Author
2010-05-21	Authored	Matthews L
2010-05-21	Created	Matthews L
2010-05-24	Edited	Matthews L
2025-11-15	Modified	Weiser JD

40 submitted entities found in this pathway, mapping to 43 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CENPA	P49450	CENPH	Q9H3R5	CENPI	Q92674
CENPK	Q9BS16	CENPL	Q8N0S6	CENPM	Q9NSP4
CENPN	Q96H22	CENPO	Q9BU64	CENPQ	Q7L2Z9
CENPT	Q96BT3	CENPU	Q71F23	CENPW	Q5EE01
CENPX	A8MT69	H2AFJ	Q9BTM1	H2AFX	P16104
HIST1H2AC	Q93077	HIST1H2AI	P20671	HIST1H2BD	P58876
HIST1H2BG	P62807, Q93079	HIST1H2BH	P58876, Q93079	HIST1H2BJ	P06899
HIST1H2BK	O60814, Q99880	HIST1H2BL	Q99880	HIST1H2BM	Q99879
HIST1H4A	P62805	HIST1H4D	P62805	HIST1H4I	P62805
HIST1H4L	P62805	HIST3H2BB	Q8N257	HJURP	Q8NCD3

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ITGB3BP	Q13352	KNL1	Q8NG31	MIS18A	Q9NYP9
MIS18BP1	Q6P0N0	NPM1	P06748	OIP5	O43482
RBBP4	Q09028	RSF1	Q96T23	RUVBL1	Q9Y265
SMARCA5	O60264				

11. Aberrant regulation of mitotic G1/S transition in cancer due to RB1 defects (R-HSA-9659787)



Diseases: cancer.

RB1 protein, also known as pRB or retinoblastoma protein, is a nuclear protein that plays a major role in the regulation of the G1/S transition during mitotic cell cycle in multicellular eukaryotes. RB1 performs this function by binding to activating E2Fs (E2F1, E2F2 and E2F3), and preventing transcriptional activation of E2F1/2/3 target genes, which include a number of genes involved in DNA synthesis. RB1 binds E2F1/2/3 through the so-called pocket region, which is formed by two parts, pocket domain A (amino acid residues 373-579) and pocket domain B (amino acid residues 640-771). Besides intact pocket domains, RB1 requires an intact nuclear localization signal (NLS) at its C-terminus (amino acid residues 860-876) to be fully functional (reviewed by Classon and Harlow 2002, Dick 2007). Functionally characterized RB1 mutations mostly affect pocket domains A and B and the NLS. RB1 mutations reported in cancer are, however, scattered over the entire RB1 coding sequence and the molecular consequences of the vast majority of these mutations have not been studied (reviewed by Dick 2007).

Many viral oncoproteins inactivate RB1 by competing with E2F1/2/3 for binding to the pocket region of RB1. RB1 protein is targeted by the large T antigen of the Simian virus 40 (SV40), the adenoviral E1A protein, and the E7 protein of oncogenic human papilloma viruses (HPVs) (reviewed by Classon and Harlow 2002).

References

Dick FA (2007). Structure-function analysis of the retinoblastoma tumor suppressor protein - is the whole a sum of its parts?. Cell Div, 2, 26.

Classon M & Harlow E (2002). The retinoblastoma tumour suppressor in development and cancer. Nat. Rev. Cancer, 2, 910-7.

Edit history

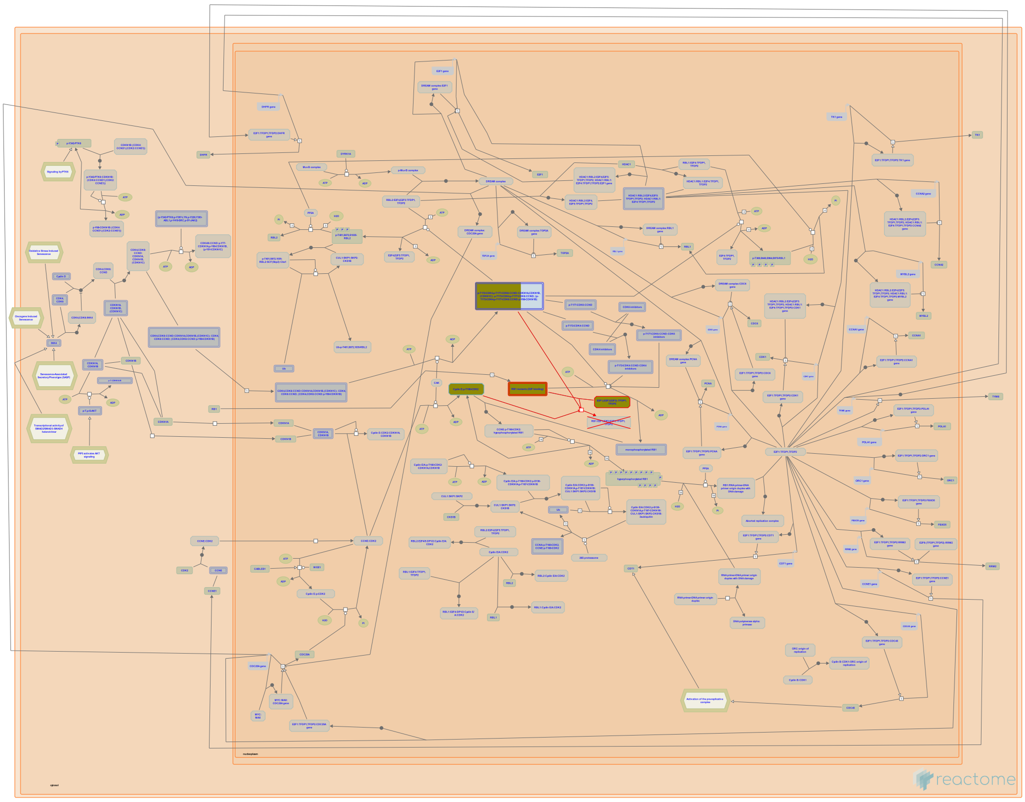
Date	Action	Author
2019-09-05	Created	Orlic-Milacic M
2020-05-07	Authored	Orlic-Milacic M
2020-05-17	Reviewed	Dick FA

Date	Action	Author
2020-05-18	Edited	Orlic-Milacic M
2023-03-08	Modified	Matthews L

16 submitted entities found in this pathway, mapping to 17 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CCND1	P24385	CCND2	P30279	CCNE1	P24864
CCNE2	O96020	CDK1	P24941	CDK2	P24941
CDK4	P11802	CDK6	Q00534	CDKN1A	P38936
CDKN1C	P49918	E2F1	O00716, Q01094	E2F2	Q14209
E2F3	O00716	RB1	P06400	TFDP1	Q14186
TFDP2	Q14188				

12. Defective binding of RB1 mutants to E2F1,(E2F2, E2F3) (R-HSA-9661069)



Cellular compartments: nucleoplasm.

Diseases: cancer.

This pathway describes impaired binding of RB1 pocket domain mutants to activating E2Fs, E2F1, E2F2 and E2F3 (Templeton et al. 1991, Helin et al. 1993, Otterson et al. 1997, Ji et al. 2004).

References

Otterson GA, Chen Wd, Coxon AB, Khleif SN & Kaye FJ (1997). Incomplete penetrance of familial retinoblastoma linked to germ-line mutations that result in partial loss of RB function. Proc. Natl. Acad. Sci. U.S.A., 94, 12036-40.

Templeton DJ, Park SH, Lanier L & Weinberg RA (1991). Nonfunctional mutants of the retinoblastoma protein are characterized by defects in phosphorylation, viral oncoprotein association, and nuclear tethering. Proc. Natl. Acad. Sci. U.S.A., 88, 3033-7.

Helin K, Harlow E & Fattaey A (1993). Inhibition of E2F-1 transactivation by direct binding of the retinoblastoma protein. Mol. Cell. Biol., 13, 6501-8.

Ji P, Jiang H, Rekhtman K, Bloom J, Ichetovkin M, Pagano M & Zhu L (2004). An Rb-Skp2-p27 pathway mediates acute cell cycle inhibition by Rb and is retained in a partial-penetrance Rb mutant . Mol. Cell, 16, 47-58.

Edit history

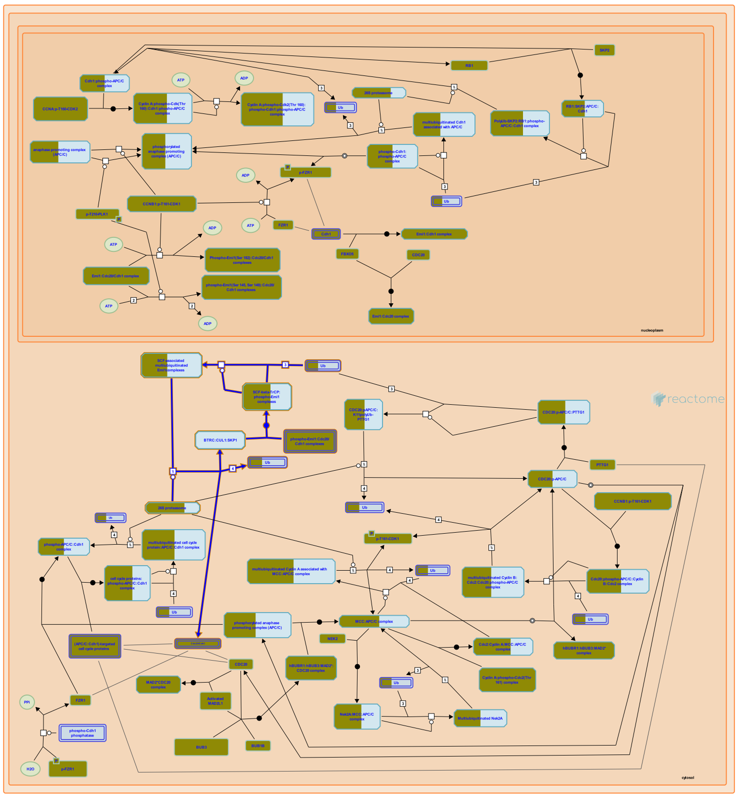
Date	Action	Author
2019-09-13	Created	Orlic-Milacic M
2020-05-07	Authored	Orlic-Milacic M

Date	Action	Author
2020-05-17	Reviewed	Dick FA
2020-05-18	Edited	Orlic-Milacic M

16 submitted entities found in this pathway, mapping to 17 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CCND1	P24385	CCND2	P30279	CCNE1	P24864
CCNE2	O96020	CDK1	P24941	CDK2	P24941
CDK4	P11802	CDK6	Q00534	CDKN1A	P38936
CDKN1C	P49918	E2F1	O00716, Q01094	E2F2	Q14209
E2F3	O00716	RB1	P06400	TFDP1	Q14186
TFDP2	Q14188				

13. SCF-beta-TrCP mediated degradation of Emi1 (R-HSA-174113)



Cellular compartments: cytosol.

Emi1 destruction in early mitosis requires the SCF beta-TrCP ubiquitin ligase complex. Binding of beta-TrCP to Emi1 occurs in late prophase and requires phosphorylation at the DSGxxS consensus motif as well as Cdk mediated phosphorylation. A two-step mechanism has been proposed in which the phosphorylation of Emi1 by Cdc2 occurs after the G2-M transition followed soon after by binding of beta-TrCP to the DSGxxS phosphorylation sites. Emi1 is then poly-ubiquitinated and degraded by the 26S proteasome.

References

Margottin-Goguet F, Hsu JY, Loktev A, Hsieh HM, Reimann JD & Jackson PK (2003). Prophase destruction of Emi1 by the SCF(betaTrCP/Slimb) ubiquitin ligase activates the anaphase promoting complex to allow progression beyond prometaphase. Dev Cell, 4, 813-26.

Edit history

Date	Action	Author
2006-01-26	Authored	Castro A, Lorca T
2006-01-30	Edited	Matthews L
2006-02-17	Created	Matthews L
2006-03-28	Reviewed	Peters JM

29 submitted entities found in this pathway, mapping to 33 Reactome entities

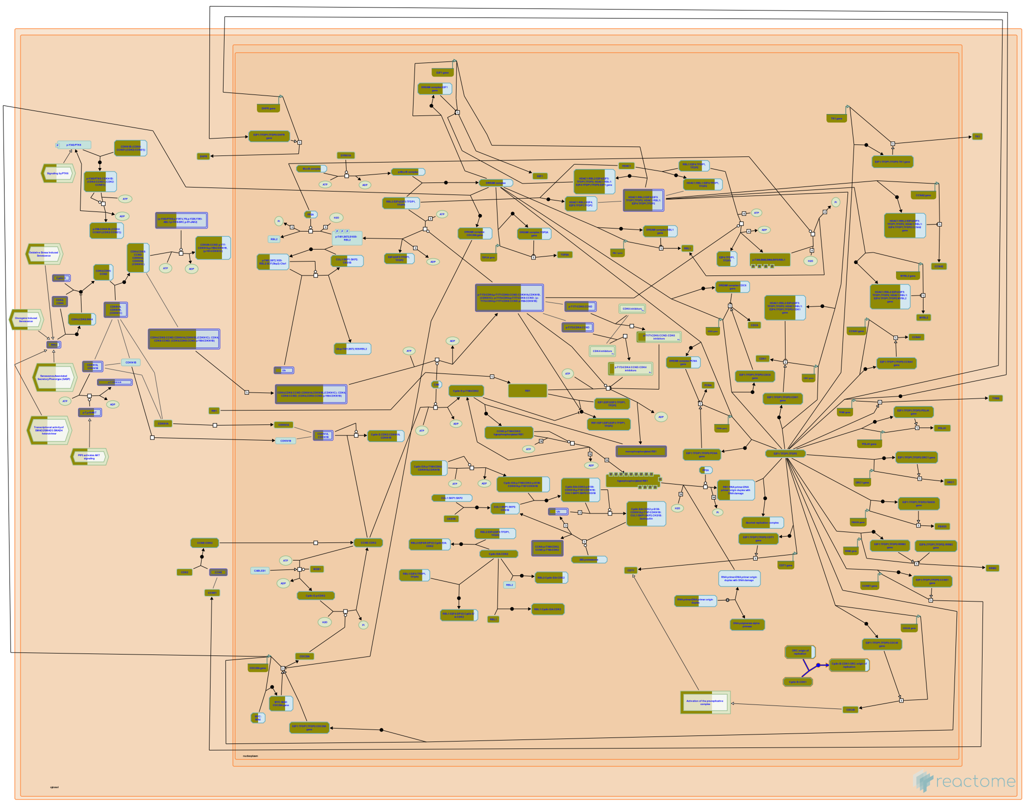
Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CDC20	Q12834	CDH1	Q9UM11	FBXO5	Q9UKT4

Input	UniProt Id
PSMA3	P25788
PSMA7	O14818
PSMB5	P28074
PSMC1	P62191
PSMC4	P43686
PSMD11	O00231
PSMD3	O43242
RPN1	Q13200
UBA52	P62979, P62987

Input	UniProt Id
PSMA4	P25789
PSMB2	P49721
PSMB6	P28072
PSMC2	P35998
PSMC6	P62333
PSMD12	O00232
PSMD6	Q15008
RPS27A	P62979, P62987
UBB	P0CG47, P62979, P62987

Input	UniProt Id
PSMA5	P28066
PSMB3	P49720
PSMB7	Q99436
PSMC3	P17980
PSMD1	Q99460
PSMD14	O00487
PSMD8	P48556
SEM1	P60896

14. E2F-enabled inhibition of pre-replication complex formation (R-HSA-113507)



Cellular compartments: nucleoplasm.

Under specific conditions, Cyclin B, a mitotic cyclin, can inhibit the functions of pre-replicative complex. E2F1 activates Cdc25A protein which regulates Cyclin B in a positive manner. Cyclin B/Cdk1 function is restored which leads to the disruption of pre-replicative complex. This phenomenon has been demonstrated by Bosco et al (2001) in Drosophila.

References

Kennedy BK, Barbie DA, Classon M, Harlow E & Dyson NJ (2000). Nuclear organization of DNA replication in primary mammalian cells. Genes Dev, 14, 2855-68.

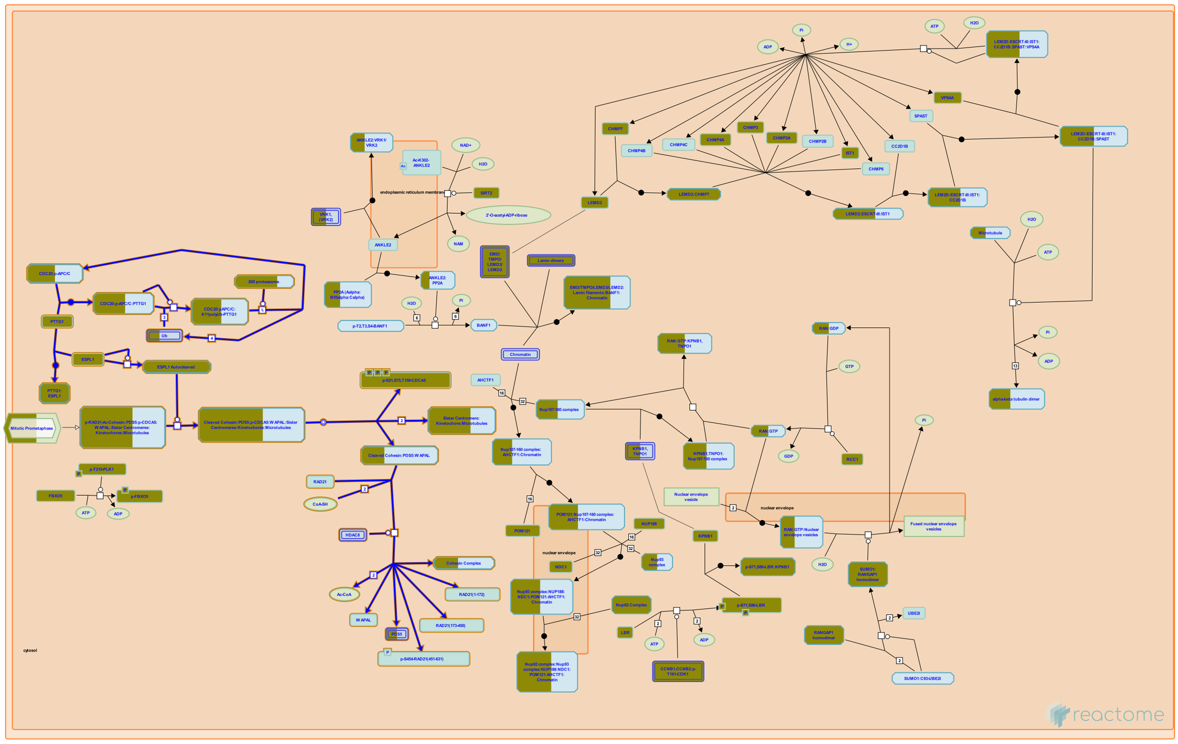
Edit history

Date	Action	Author
2004-05-26	Authored	Gopinathrao G
2004-06-16	Created	Bosco G

9 submitted entities found in this pathway, mapping to 9 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CCNB1	P14635	CDK1	P06493	MCM8	Q9UJA3
ORC1	Q13415	ORC2	Q13416	ORC3	Q9UBD5
ORC4	O43929	ORC5	O43913	ORC6	Q9Y5N6

15. Separation of Sister Chromatids (R-HSA-2467813)



Cellular compartments: cytosol.

While sister chromatids resolve in prometaphase, separating along chromosomal arms, the cohesion of sister centromeres persists until anaphase. At the anaphase onset, the anaphase promoting complex/cyclosome (APC/C) ubiquitinates PTTG1 (securin), targeting it for degradation (Hagting et al. 2002). PTTG1 acts as an inhibitor of ESPL1 (known as separin i.e. separase). Hence, PTTG1 removal initiated by APC/C, enables ESPL1 to become catalytically active (Zou et al. 1999, Waizenegger et al. 2002). ESPL1 undergoes autoleavage (Waizenegger et al. 2002) and also cleaves RAD21 subunit of centromeric cohesin (Hauf et al. 2001). RAD21 cleavage promotes dissociation of cohesin complexes from sister centromeres, leading to separation of sister chromatids. Subsequent movement of sister chromatids to opposite poles of the mitotic spindle segregates replicated chromosomes to two daughter cells (Waizenegger et al. 2000, Hauf et al. 2001, Waizenegger et al. 2002).

References

Hauf S, Waizenegger IC & Peters JM (2001). Cohesin cleavage by separase required for anaphase and cytokinesis in human cells. Science, 293, 1320-3.

Hagting A, Den Elzen N, Vodermaier HC, Waizenegger IC, Peters JM & Pines J (2002). Human securin proteolysis is controlled by the spindle checkpoint and reveals when the APC/C switches from activation by Cdc20 to Cdh1. J Cell Biol, 157, 1125-37.

Waizenegger I, Giménez-Abián JF, Wernic D & Peters JM (2002). Regulation of human separase by securin binding and autocleavage. Curr. Biol., 12, 1368-78.

Waizenegger IC, Hauf S, Meinke A & Peters JM (2000). Two distinct pathways remove mammalian cohesin from chromosome arms in prophase and from centromeres in anaphase. Cell, 103, 399-410.

Zou H, McGarry TJ, Bernal T & Kirschner MW (1999). Identification of a vertebrate sister-chromatid separation inhibitor involved in transformation and tumorigenesis. Science, 285, 418-22.

Edit history

Date	Action	Author
2012-09-13	Created	Orlic-Milacic M
2012-10-02	Authored	Orlic-Milacic M
2012-10-05	Edited	Matthews L, Gillespie ME
2012-10-22	Reviewed	Zhang N
2012-11-20	Reviewed	Tanno Y, Watanabe Y

115 submitted entities found in this pathway, mapping to 123 Reactome entities

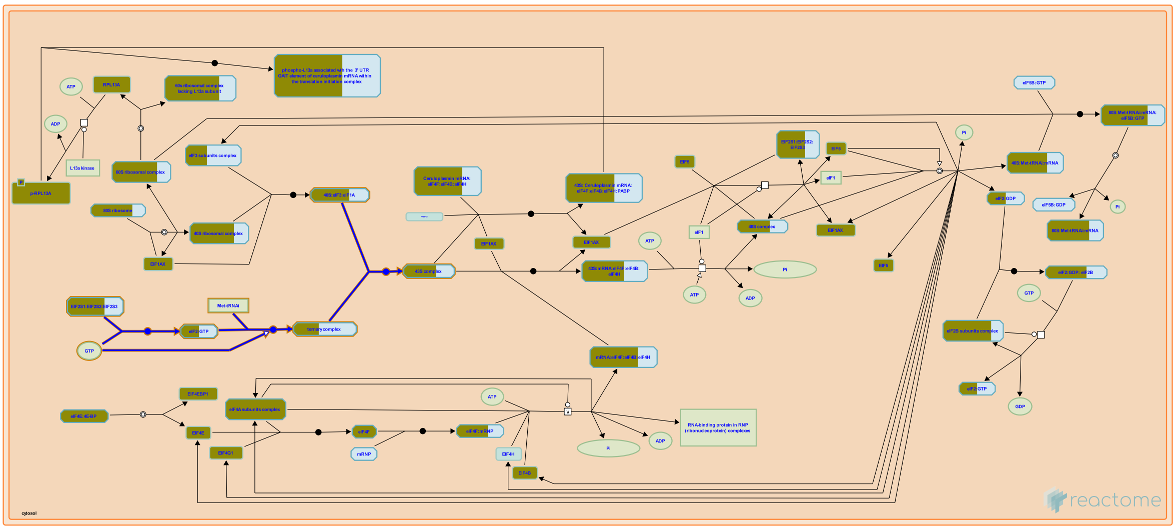
Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ANAPC11	Q9NYG5	ANAPC2	Q9UJX6	ANAPC5	Q9UJX4
ANAPC7	Q9UJX3	APC2	Q9UJX6	AURKB	Q96GD4
BIRC5	O15392	BUB1	O43683, O60566	BUB1B	O60566
BUB3	O43684	CDC20	Q12834	CDC23	Q9UJX2
CDC27	P30260	CDCA5	Q96FF9	CDCA8	Q53HL2
CENPA	P49450	CENPE	Q02224	CENPF	P49454
CENPH	Q9H3R5	CENPI	Q92674	CENPK	Q9BS16
CENPL	Q8N0S6	CENPM	Q9NSP4	CENPN	Q96H22
CENPO	Q9BU64	CENPQ	Q7L2Z9	CENPT	Q96BT3
CENPU	Q71F23	CKAP5	Q14008	CLASP1	Q7Z460
DLC1	P63167	DSN1	Q9H410	DYNC1I2	Q13409
DYNC1LI2	O43237	DYNLL1	P63167	ERCC6L	Q2NKX8
ESPL1	Q14674	INCENP	Q9NQS7	ITGB3BP	Q13352
KIF18A	Q8NI77	KIF2C	Q99661	KNL1	Q8NG31
KNTC1	P50748	MAD2L1	Q13257	NDC80	O14777
NDE1	Q9NXR1	NDEL1	Q9GZM8	NSL1	Q96IY1
NUF2	Q9BZD4	NUP160	Q12769	NUP37	Q8NFH4
NUP43	Q8NFH3	NUP98	P52948-5	PAFAH1B1	P43034
PDS5A	Q29RF7	PLK1	P53350	PPA1	P62714
PPP2CB	P62714	PPP2R5A	Q15172	PPP2R5C	Q13362
PPP2R5D	Q14738	PSMA3	P25788	PSMA4	P25789
PSMA5	P28066	PSMA7	O14818	PSMB2	P49721
PSMB3	P49720	PSMB5	P28074	PSMB6	P28072
PSMB7	Q99436	PSMC1	P62191	PSMC2	P35998
PSMC3	P17980	PSMC4	P43686	PSMC6	P62333
PSMD1	Q99460	PSMD11	O00231	PSMD12	O00232
PSMD14	O00487	PSMD3	O43242	PSMD6	Q15008
PSMD8	P48556	PTTG1	O95997	RANBP2	P49792
RANGAP1	P46060	RCC2	Q9P258	RPN1	Q13200
RPS27	P42677	RPS27A	P62979, P62987	SEM1	P60896
SGO1	Q5FBB7	SGO2	Q562F6	SKA1	Q96BD8
SKA2	Q8WVK7	SMC1A	Q14683	SMC3	Q9UQE7
SPC24	Q8NBT2	SPC25	Q9HBM1	SPDL1	Q96EA4
TAOK1	Q7L7X3	TUBA1B	P68363	TUBA1C	P68363, Q9BQE3

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
TUBB	P04350	TUBB1	Q9H4B7	TUBB2A	Q13885, Q9BVA1
TUBB4B	P04350, P68371	TUBB6	Q9BUF5	UBA52	P62979, P62987
UBB	P0CG47, P62979, P62987	UBE2C	O00762	UBE2S	Q16763
XPO1	O14980	ZW10	O43264	ZWILCH	Q9H900
ZWINT	O95229				

Interactors found in this pathway (14)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
BMP1	P13497	O60216	CCNB1	P14635	Q14674
CDCA5	Q9CPY3, Q96FF9	O60216	CDK1	P11440	Q14674
ESPL1	P60330, Q14674	O95997	FHL3	Q13643	O60216
PDS5A	Q29RF7	O60216, Q7Z5K2, Q96FF9	PLCG2	P16885	O95997
PSMC2	P35998	O60216	PTTG1	O95997	Q14674
SGO1	Q5FBB7	O60216, Q96FF9	SMC1A	Q14683	O60216, Q7Z5K2, Q96FF9
SMC3	Q9CW03, Q9UQE7	O60216, Q7Z5K2, Q96FF9	USP47	Q96K76	O60216

16. Formation of the ternary complex, and subsequently, the 43S complex (R-HSA-72695)



Cellular compartments: cytosol.

Binding of the methionyl-tRNA initiator to the active eIF2:GTP complex results in the formation of the ternary complex. Subsequently, this Met-tRNAi:eIF2:GTP (ternary) complex binds to the complex formed by the 40S subunit, eIF3 and eIF1A, to form the 43S complex.

References

Peterson DT, Merrick WC & Safer B (1979). Binding and release of radiolabeled eukaryotic initiation factors 2 and 3 during 80 S initiation complex formation. J Biol Chem, 254, 2509-16.

Safer B, Adams SL, Anderson WF & Merrick WC (1976). Binding of MET-TRNAf and GTP to homogeneous initiation factor MP. J Biol Chem, 250, 9076-82.

Trachsel H, Erni B, Schreier MH & Staehelin T (1978). Initiation of mammalian protein synthesis. II. The assembly of the initiation complex with purified initiation factors. J Mol Biol, 116, 755-67.

McCarthy JE & Tuite M (1990). *New insights into an old problem: ternary complex (Met-tRNAf.eIF.GTP) formation in animal cells., Post-Transcriptional Control of Gene Expression* , 521-526.

Benne R & Hershey JW (1978). The mechanism of action of protein synthesis initiation factors from rabbit reticulocytes. J Biol Chem, 253, 3078-87.

Edit history

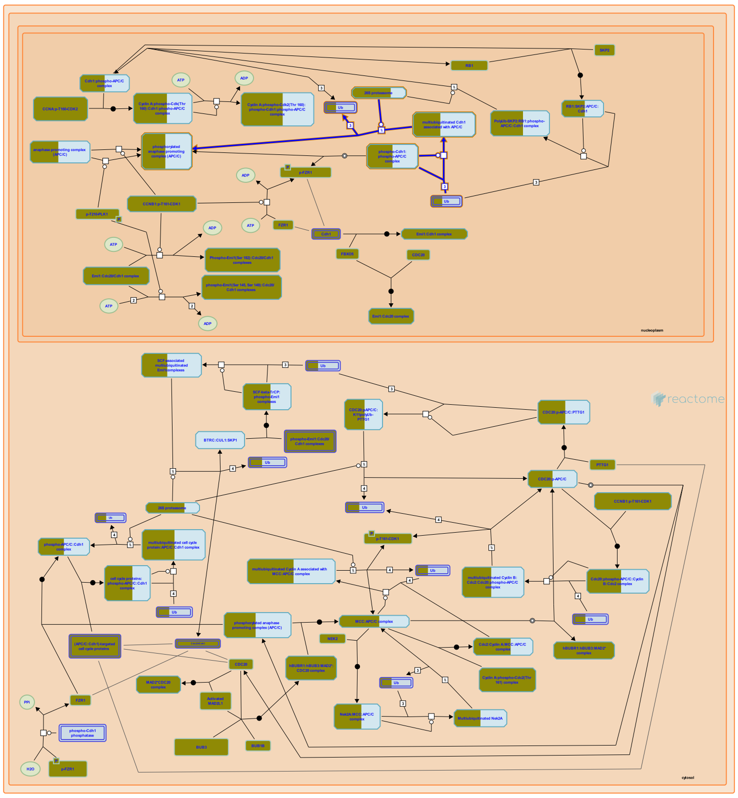
Date	Action	Author
2025-11-15	Modified	Weiser JD

36 submitted entities found in this pathway, mapping to 41 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CRY2	P62263	EIF1AY	P47813	EIF2S1	P05198
EIF2S2	P20042	EIF3B	P55884	EIF3D	O15371
EIF3G	O75821	EIF3I	Q13347	EIF3J	O75822

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
FAU	P62861	RPS10	P46783	RPS11	P62280
RPS13	P62277	RPS14	P62263	RPS15	P62841
RPS16	P62249	RPS17	P08708	RPS18	P62269
RPS23	P62266	RPS24	P62244, P62847	RPS25	P62851
RPS26	P62854	RPS27	P42677, Q71UM5	RPS27A	P62979, Q71UM5
RPS27L	Q71UM5	RPS28	P62857	RPS29	P62273
RPS3	P23396	RPS3A	P61247	RPS4X	P62701, Q8TD47
RPS4Y1	P22090, Q8TD47	RPS6	P62753	RPS7	P62081
RPS9	P46781	UBA52	P62979	UBB	P62979

17. Autodegradation of Cdh1 by Cdh1:APC/C (R-HSA-174084)



Cellular compartments: nucleoplasm.

Cdh1 is degraded by the APC/C during in G1 and G0. This auto-regulation may contribute to reducing the levels of Cdh1 levels during G1 and G0 (Listovsky et al., 2004).

References

Listovsky T, Oren YS, Yudkovsky Y, Mahbubani HM, Weiss AM, Lebendiker M & Brandeis M (2004). Mammalian Cdh1/Fzr mediates its own degradation. EMBO J, 23, 1619-26.

Edit history

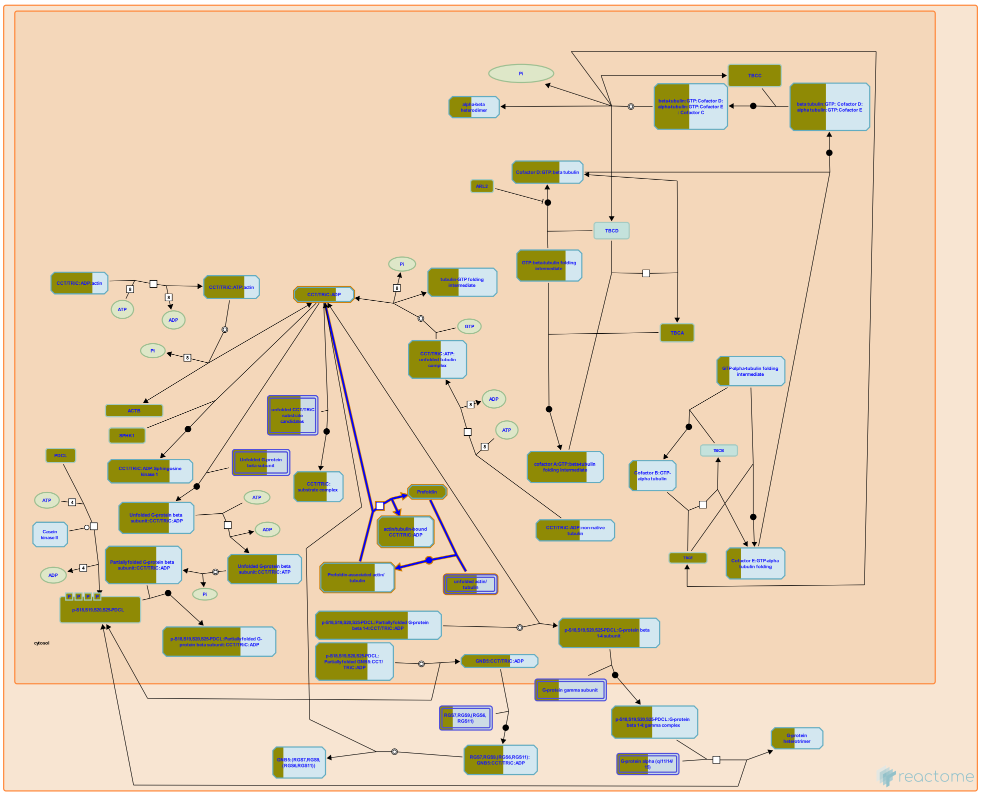
Date	Action	Author
2006-01-26	Authored	Castro A, Lorca T
2006-01-30	Edited	Matthews L
2006-02-17	Created	Matthews L
2006-03-28	Reviewed	Peters JM

36 submitted entities found in this pathway, mapping to 40 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ANAPC11	Q9NYG5	ANAPC2	Q9UJX6	ANAPC5	Q9UJX4
ANAPC7	Q9UJX3	APC2	Q9UJX6	CDC23	Q9UJX2
CDC27	P30260	CDH1	Q9UM11	PSMA3	P25788
PSMA4	P25789	PSMA5	P28066	PSMA7	O14818
PSMB2	P49721	PSMB3	P49720	PSMB5	P28074
PSMB6	P28072	PSMB7	Q99436	PSMC1	P62191

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
PSMC2	P35998	PSMC3	P17980	PSMC4	P43686
PSMC6	P62333	PSMD1	Q99460	PSMD11	O00231
PSMD12	O00232	PSMD14	O00487	PSMD3	O43242
PSMD6	Q15008	PSMD8	P48556	RPN1	Q13200
RPS27A	P62979, P62987	SEM1	P60896	UBA52	P62979, P62987
UBB	P0CG47, P62979, P62987	UBE2C	O00762	UBE2S	Q16763

18. Prefoldin mediated transfer of substrate to CCT/TriC (R-HSA-389957)



Cellular compartments: cytosol.

Unfolded actins and tubulins bound to prefoldin are transferred to CCT via a docking mechanism (McCormack and Willison, 2001).

References

McCormack EA, Llorca O, Carrascosa JL, Valpuesta JM & Willison KR (2001). Point mutations in a hinge linking the small and large domains of beta-actin result in trapped folding intermediates bound to cytosolic chaperonin CCT. J Struct Biol, 135, 198-204.

Edit history

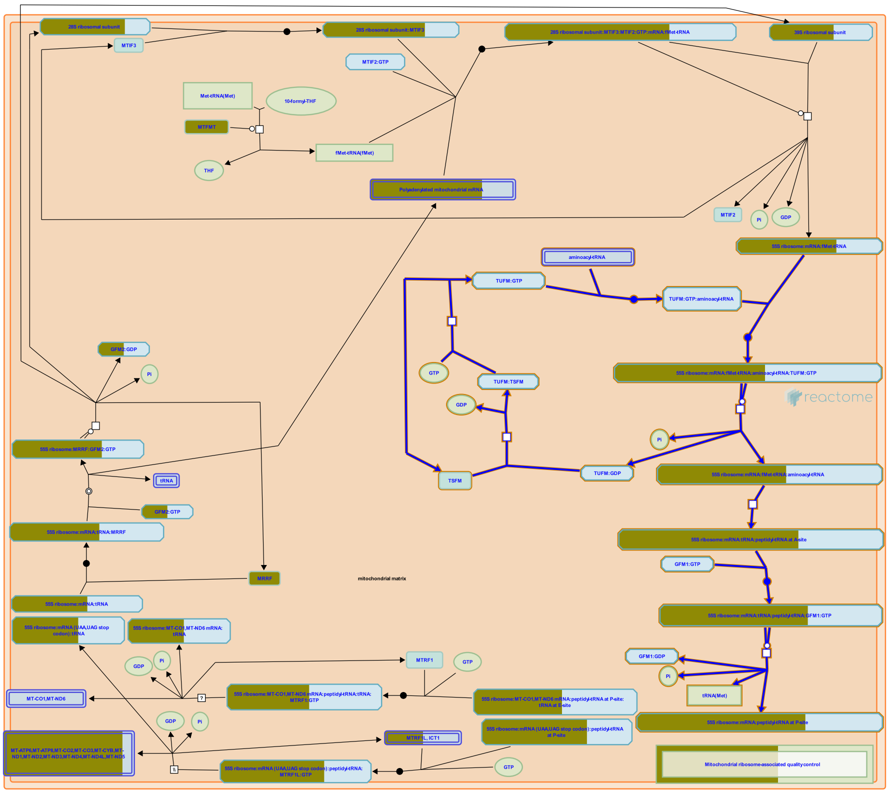
Date	Action	Author
2008-12-01	Authored	Matthews L
2009-01-21	Reviewed	Cowan NJ
2009-01-22	Created	Matthews L
2009-02-21	Edited	Matthews L
2023-10-12	Modified	Weiser JD

21 submitted entities found in this pathway, mapping to 24 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ACTB	P60709	ACTG1	P60709	CCT3	P49368
CCT4	P50991	CCT5	P48643	CCT6A	P40227, Q92526
CCT6B	Q92526	CCT7	Q99832	CCT8	P50990
PFDN1	O60925	PFDN2	Q9UHV9	PFDN4	Q9NQP4

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
PFDN5	Q99471	PFDN6	O15212	TUBA1C	Q9BQE3
TUBB	P04350	TUBB1	Q9H4B7	TUBB2A	Q13885, Q9BVA1
TUBB4B	P04350, P68371	TUBB6	Q9BUF5	VBP1	P61758

19. Mitochondrial translation elongation (R-HSA-5389840)



Cellular compartments: mitochondrial matrix, mitochondrial inner membrane.

Translation elongation proceeds by cycles of aminoacyl-tRNAs binding, peptide bond formation, and displacement of deacylated tRNAs (reviewed in Christian and Spremulli 2012). In each cycle an aminoacyl-tRNA in a complex with TUFM:GTP (EF-Tu:GTP) binds a cognate codon at the A-site of the ribosome, GTP is hydrolyzed, and TUFM:GDP dissociates. The elongating polypeptide bonded to the tRNA at the P-site is transferred to the aminoacyl group at the A-site by peptide bond formation, leaving a deacylated tRNA at the P-site and the elongating polypeptide attached to the tRNA at the A-site. GFM1:GTP (EF-Gmt:GTP) binds, GTP is hydrolyzed, GFM1:GDP dissociates, and the ribosome translocates 3 nucleotides in the 3' direction, relocating the peptidyl-tRNA to the P-site and allowing another cycle to begin. Mitochondrial ribosomes associate with the inner membrane and polypeptides are co-translationally inserted into the membrane (reviewed in Ott and Herrmann 2010, Agrawal and Sharma 2012). TUFM:GDP is regenerated to TUFM:GTP by the guanine nucleotide exchange factor TSFM (EF-Ts, EF-TsMt).

References

Christian BE & Spremulli LL (2012). Mechanism of protein biosynthesis in mammalian mitochondria. Biochim. Biophys. Acta, 1819, 1035-54.

Ott M & Herrmann JM (2010). Co-translational membrane insertion of mitochondrially encoded proteins. Biochim. Biophys. Acta, 1803, 767-75.

Agrawal RK & Sharma MR (2012). Structural aspects of mitochondrial translational apparatus. Curr. Opin. Struct. Biol., 22, 797-803.

Edit history

Date	Action	Author
2014-04-26	Edited	May B
2014-04-26	Authored	May B
2014-04-30	Created	May B
2014-08-29	Reviewed	Chrzanowska-Lightowlers ZM
2014-09-20	Reviewed	Spremulli LL
2025-11-15	Modified	Weiser JD

71 submitted entities found in this pathway, mapping to 77 Reactome entities

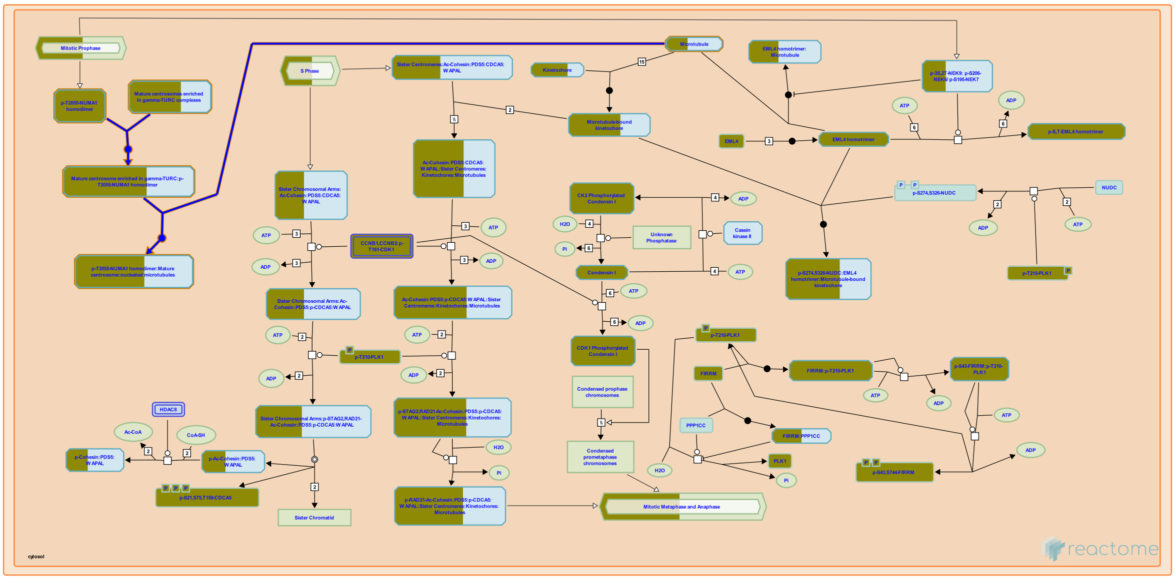
Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
AIP	Q9NWT8	BRIP1	Q9P0J6	CHCHD1	Q96BP2
ERAL1	O75616	GADD45GIP1	Q8TAE8	MRPL10	Q7Z7H8
MRPL11	Q9Y3B7	MRPL12	P52815	MRPL13	Q9BYD1
MRPL14	Q6P1L8	MRPL15	P49406, Q9P015	MRPL16	Q9NX20
MRPL17	Q9NRX2	MRPL18	Q9H0U6	MRPL19	P49406
MRPL22	Q9NWU5	MRPL27	Q8IXM3, Q9P0M9	MRPL28	Q13084, Q8TCC3
MRPL3	P09001	MRPL30	Q8TCC3	MRPL32	Q6P1L8, Q9BYC8
MRPL34	Q9BQ48	MRPL35	Q9NZE8	MRPL36	Q9P0J6
MRPL37	Q9BZE1	MRPL38	Q96DV4	MRPL39	Q9NYK5
MRPL41	Q8IXM3	MRPL44	Q9H9J2	MRPL45	Q9BRJ2
MRPL46	Q9H2W6	MRPL50	Q8N5N7	MRPL51	Q4U2R6
MRPL52	Q86TS9	MRPL55	Q7Z7F7	MRPL57	Q9BQC6
MRPL58	Q14197	MRPS10	P82664	MRPS11	P82912
MRPS12	O15235	MRPS14	O60783	MRPS15	P82914
MRPS16	Q9Y3D3	MRPS18A	Q9NVS2	MRPS18B	Q9Y676
MRPS18C	Q9Y3D5	MRPS22	P82650	MRPS23	Q9Y3D9
MRPS25	P82663	MRPS28	P82673, Q9Y2Q9	MRPS30	Q9NP92
MRPS31	Q92665	MRPS33	Q9Y291	MRPS34	P82930
MRPS35	P82673, Q9Y2Q9	MRPS36	P82909	MRPS5	P82675
MRPS7	Q9Y2R9	MRPS9	P82933	PTCD3	Q96EY7

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
MT-ATP8	ENST00000361851	MT-CO2	ENST00000361739	MT-CO3	ENST00000362079
MT-CYB	ENST00000361789	MT-ND1	ENST00000361390	MT-ND2	ENST00000361453
MT-ND3	ENST00000361227	MT-ND4	ENST00000361381	MT-ND4L	ENST00000361335
MT-ND5	ENST00000361567	MT-RNR1	ENST00000389680		

Interactors found in this pathway (6)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ATXN7L2	Q5T6C5	P43897	GOLGA6L9	A6NEM1	P43897
GTF2E2	P29084	P43897	MRPL12	P52815	P43897
RALBP1	Q15311	P43897	ZG16B	Q96DA0	P43897

20. Recruitment of NuMA to mitotic centrosomes (R-HSA-380320)



The NuMA protein, which functions as a nuclear matrix protein in interphase (Merdes and Cleveland 1998), redistributes to the cytoplasm following nuclear envelope breakdown where it plays an essential role in formation and maintenance of the spindle poles (Gaglio, et al., 1995; Gaglio, et al., 1996; Merdes et al, 1996). The mitotic activation of NuMA involves Ran-GTP-dependent dissociation from importin (Nachury et al, 2001, Wiese et al, 2001). NuMA is transported to the mitotic poles where it forms an insoluble crescent around centrosomes tethering microtubules into the bipolar configuration of the mitotic apparatus (Merdes et al., 2000; Kisurina-Evgenieva et al, 2004). Although NuMA is not a bona fide constituent of the mitotic centrosome but rather a protein associated with microtubules at the spindle pole, specific splice variants of NuMA have been identified that associate with the centrosome during interphase (Tang et al, 1994).

References

Merdes A, Ramyar K, Vechio JD & Cleveland DW (1996). A complex of NuMA and cytoplasmic dynein is essential for mitotic spindle assembly. Cell, 87, 447-58.

Merdes A, Heald R, Samejima K, Earnshaw WC & Cleveland DW (2000). Formation of spindle poles by dynein/dynactin-dependent transport of NuMA. J Cell Biol, 149, 851-62.

Edit history

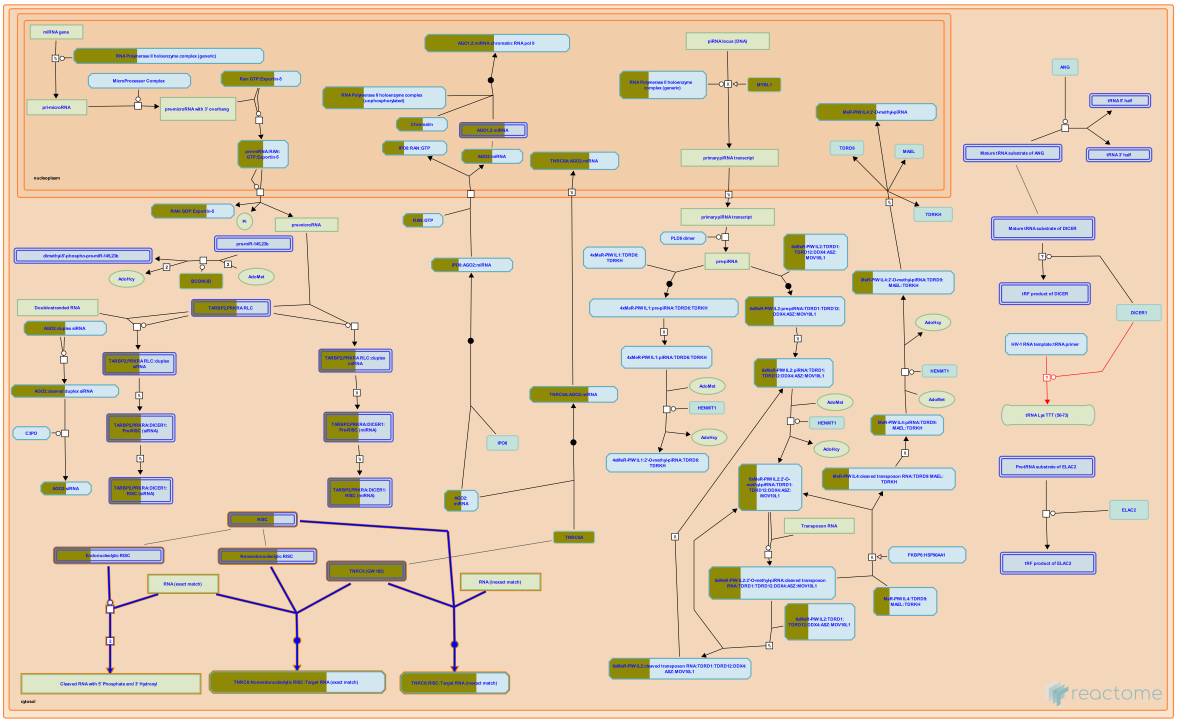
Date	Action	Author
2008-11-09	Edited	Matthews L
2008-11-11	Authored	Matthews L
2008-11-11	Created	Matthews L
2008-11-17	Reviewed	Merdes A
2025-11-15	Modified	Weiser JD

57 submitted entities found in this pathway, mapping to 62 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ACTR1A	P61163	AKAP9	Q99996	CCP110	O43303

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CDK1	P06493	CENPJ	Q9HC77	CEP131	Q9UPN4
CEP152	O94986	CEP164	Q9UPV0	CEP250	Q9BV73
CEP41	Q9BYV8	CEP76	Q8TAP6	CEP78	Q5JTW2
CKAP5	Q14008	CLASP1	Q7Z460	CSNK1D	P48730
CSNK1E	P49674	DBT	P49674	DCTN2	Q13561
DLC1	P63167	DYNC1I2	Q13409	DYNLL1	P63167
FGFR1OP	O95684	FOPNL	O75665	HAUS1	Q96CS2
HAUS2	Q9NVX0	HAUS3	Q68CZ6	HAUS4	Q9H6D7
HAUS7	Q99871	HAUS8	Q9BT25	MZT1	Q08AG7
MZT2A	Q6P582	MZT2B	Q6NZ67	NDE1	Q9NXR1
NEDD1	Q8NHV4	NEK2	P51955	NME7	Q9Y5B8
NUMA1	Q14980	OFD1	O75665	PAFAH1B1	P43034
PLK1	P53350	PLK4	O00444	PRKAR2B	P31323
SDCCAG8	Q86SQ7	SSNA1	O43805	TUBA1B	P68363
TUBA1C	P68363, Q9BQE3	TUBB	P04350, P07437	TUBB1	Q9H4B7
TUBB2A	Q13885, Q9BVA1	TUBB4B	P04350, P68371	TUBB6	Q9BUF5
TUBG1	P23258, Q9NRH3	TUBGCP3	Q96CW5	TUBGCP4	Q9UGJ1
TUBGCP6	Q96RT7	YWHAE	P62258	YWHAG	P61981

21. Post-transcriptional silencing by small RNAs (R-HSA-426496)



Cellular compartments: cytosol.

Small RNAs act with components of the RNA-induced silencing complex (RISC) to post-transcriptionally repress expression of mRNAs (reviewed in Nowotny and Yang 2009, Chua et al. 2009). Two mechanisms exist: 1) cleavage of target RNAs by complexes containing Argonaute2 (AGO2, EIF2C2) and a guide RNA that exactly matches the target mRNA and 2) inhibition of translation of target RNAs by complexes containing AGO2 and an inexactly matching guide RNA or by complexes containing a nonendonucleolytic Argonaute (AGO1 (EIF2C1), AGO3 (EIF2C3), or AGO4 (EIF2C4)) and a guide RNA of exact or inexact match. Small interfering RNAs (siRNAs) and microRNAs (miRNAs) can serve as guide RNAs in both types of mechanism.

RNAi also appears to direct chromatin modifications that cause transcriptional gene silencing (reviewed in Verdel et al. 2009).

References

Hutvagner G & Zamore PD (2002). A microRNA in a multiple-turnover RNAi enzyme complex. Science, 297, 2056-60.

Peters L & Meister G (2007). Argonaute proteins: mediators of RNA silencing. Mol Cell, 26, 611-23.

Liu J, Carmell MA, Rivas FV, Marsden CG, Thomson JM, Song JJ, ... Hannon GJ (2004). Argonaute2 is the catalytic engine of mammalian RNAi. Science, 305, 1437-41.

Su H, Trombly MI, Chen J & Wang X (2009). Essential and overlapping functions for mammalian Argonautes in microRNA silencing. Genes Dev, 23, 304-17.

Maroney PA, Yu Y, Fisher J & Nilsen TW (2006). Evidence that microRNAs are associated with translating messenger RNAs in human cells. Nat Struct Mol Biol, 13, 1102-7.

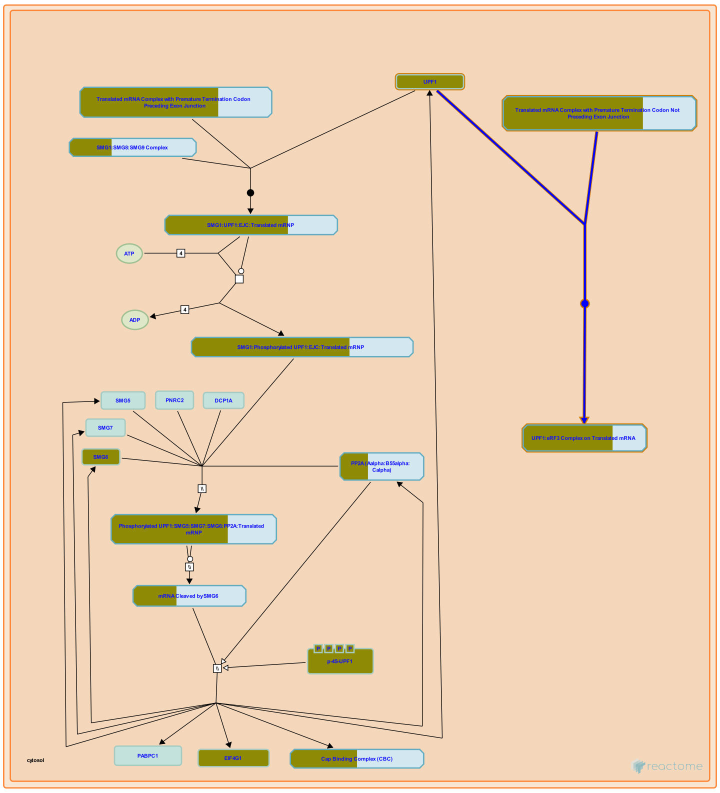
Edit history

Date	Action	Author
2009-06-10	Edited	May B
2009-06-10	Authored	May B
2009-06-17	Created	May B
2012-02-11	Reviewed	Tomari Y
2025-11-15	Modified	Weiser JD

5 submitted entities found in this pathway, mapping to 7 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
AGO1	Q9H9G7, Q9UKV8, Q9UL18	AGO4	Q9HCK5	TNRC6A	Q8NDV7
TNRC6B	Q9UPQ9	TNRC6C	Q9HCJ0		

22. Nonsense Mediated Decay (NMD) independent of the Exon Junction Complex (EJC) (R-HSA-975956)



Cellular compartments: cytosol.

Nonsense-mediated decay has been observed with mRNAs that do not have an exon junction complex (EJC) downstream of the termination codon (reviewed in Isken and Maquat 2007, Chang et al. 2007, Behm-Ansmant et al. 2007, Rebbapragada and Lykke-Andersen 2009, Nicholson et al. 2010). In these cases the trigger is unknown but a correlation with the length of the 3' UTR has sometimes been seen. The current model posits a competition between PABP and UPF1 for access to eRF3 at the terminating ribosome (Ivanov et al. 2008, Singh et al. 2008, reviewed in Bhuvanagiri et al. 2010). Abnormally long 3' UTRs may prevent PABP from efficiently interacting with eRF3 and allow UPF1 to bind eRF3 instead. Long UTRs with hairpin loops may bring PABP closer to eRF3 and help evade NMD (Eberle et al. 2008).

The pathway of degradation taken during EJC-independent NMD has not been elucidated. It is thought that phosphorylation of UPF1 by SMG1 and recruitment of SMG6 or SMG5 and SMG7 are involved, as with EJC-enhanced NMD, but this has not yet been shown.

References

Rebbapragada I & Lykke-Andersen J (2009). Execution of nonsense-mediated mRNA decay: what defines a substrate?. Curr Opin Cell Biol, 21, 394-402.

Nicholson P, Yepiskoposyan H, Metze S, Zamudio Orozco R, Kleinschmidt N & Muhlemann O (2010). Nonsense-mediated mRNA decay in human cells: mechanistic insights, functions beyond quality control and the double-life of NMD factors. Cell Mol Life Sci, 67, 677-700.

Isken O & Maquat LE (2007). Quality control of eukaryotic mRNA: safeguarding cells from abnormal mRNA function. Genes Dev, 21, 1833-56.

Chang YF, Imam JS & Wilkinson MF (2007). The nonsense-mediated decay RNA surveillance pathway. Annu Rev Biochem, 76, 51-74.

Behm-Ansmant I, Kashima I, Rehwinkel J, Saulière J, Wittkopp N & Izaurralde E (2007). mRNA quality control: an ancient machinery recognizes and degrades mRNAs with nonsense codons. FEBS Lett, 581, 2845-53.

Edit history

Date	Action	Author
2010-10-08	Edited	May B
2010-10-08	Authored	May B
2010-10-11	Created	May B
2011-05-19	Reviewed	Neu-Yilik G

72 submitted entities found in this pathway, mapping to 92 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CRY2	P62263	EIF4G1	Q04637	FAU	P62861
GSPT1	P15170, Q8IYD1	GSPT2	Q8IYD1	NCBP1	Q09161
RPL10	P27635, Q96L21	RPL10A	P62906	RPL11	P62913
RPL12	P30050	RPL13A	P40429	RPL15	P61313
RPL17	P18621	RPL18	Q07020	RPL18A	Q02543
RPL19	P84098	RPL21	P46778	RPL22	P35268
RPL23A	P62750	RPL24	P83731	RPL26	P61254, Q9UNX3
RPL26L1	Q9UNX3	RPL27	P61353	RPL27A	P46776, P61353
RPL29	P47914	RPL3	P39023, Q92901	RPL30	P62888
RPL32	P62888, P62910	RPL34	P49207, P62899	RPL35	P42766
RPL35A	P18077, P42766	RPL36A	P83881, Q969Q0	RPL37A	P18077, P61513
RPL38	P63173	RPL39L	Q96EH5	RPL41	P62945
RPL5	P46777	RPL6	Q02878	RPL7	P18124
RPL7A	P18124, P62424	RPL8	P62424, P62917	RPL9	P32969
RPLP2	P05387	RPS10	P46783	RPS11	P62280
RPS13	P62277	RPS14	P62263	RPS15	P62841
RPS16	P62249	RPS17	P08708	RPS18	P62269
RPS23	P62266	RPS24	P62244, P62847	RPS25	P62851
RPS26	P62854	RPS27	P42677, Q71UM5	RPS27A	P62979, P62987, Q71UM5
RPS27L	Q71UM5	RPS28	P62857	RPS29	P62273
RPS3	P23396	RPS3A	P61247	RPS4X	P62701, Q8TD47
RPS4Y1	P22090, Q8TD47	RPS6	P62753	RPS7	P62081
RPS9	P46781	RRP1	P05386	SOS1	P42766
UBA52	P62979, P62987	UBB	P62979, P62987	UPF1	Q92900

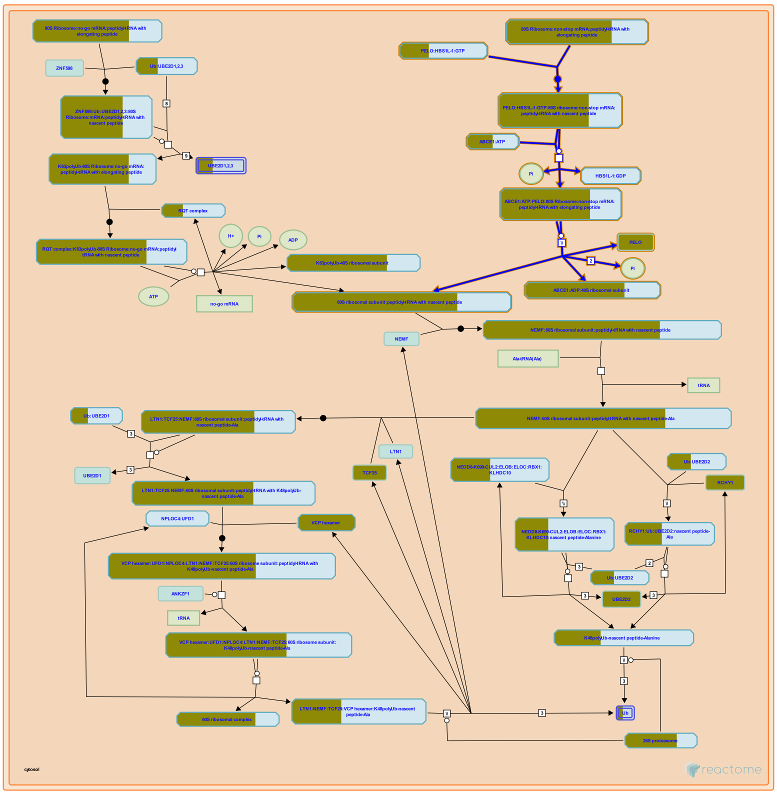
Interactors found in this pathway (13)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
CASC3	O15234	Q92900	DCP2	Q8IU60	Q92900
DDX6	P26196	Q92900	EIF4E	P06730	Q92900
FAM120A	Q9NZB2	Q92900	GSPT1	P15170	Q92900

Input	UniProt Id	Interacts with
GSPT2	Q8IYD1	Q92900
NCBP1	Q09161	Q92900
SLBP	Q14493	Q92900
UPF3B	Q9BZI7, Q9BZI7-2	Q92900

Input	UniProt Id	Interacts with
MAPT	P10636-8	Q92900
RPS6	P62753	Q92900
UPF3A	Q9H1J1	Q92900

23. PELO:HBS1L and ABCE1 dissociate a ribosome on a non-stop mRNA (R-HSA-9954714)



Non-stop mRNAs (mRNAs that lack a stop codon) result in ribosomal stalls proximal to the 3' end of the mRNA, which are resolved by a distinct pathway. In this case, a complex comprising PELO, a paralog of the ribosome release factor eRF1, and HBS1L:GTP, a paralog of the ribosome release factor eRF3:GTP, binds the stalled ribosome near the subunit interface and the mRNA entry site (Shao et al. 2013, and inferred from human PELO:HBS1L and rabbit ribosomes in Pisareva et al. 2011, inferred from yeast homologs DOM34:HBS1 in Shoemaker et al. 2010, Tsuboi et al. 2012, Guydosh and Green 2014, reviewed in Franckenberg et al. 2012). PELO:HBS1L preferentially acts on ribosomes that are bound to mRNAs that have fewer than 12 nucleotides extending 3' of the ribosomal P site (Pisareva et al. 2011).

HBS1L hydrolyzes GTP and dissociates from PELO and the ribosome, exposing a site on PELO to which ABCE1 binds. ABCE1 then hydrolyzes ATP to cause a conformational change that splits the ribosome into 40S and 60S subunits (Shao et al. 2013, Shao and Hegde 2014, and inferred from the yeast homologs DOM34:HBS1 and archaeal homologs in Becker et al. 2012, inferred from the yeast homologs in Saito et al. 2013). ABCE1 and possibly the mRNA remain bound to the 40S ribosomal subunit, while the peptidyl-tRNA remains bound to the 60S ribosomal subunit as in the ASCC-mediated rescue pathway (Becker et al. 2012, reviewed in Franckenberg et al. 2012).

References

Becker T, Franckenberg S, Wickles S, Shoemaker CJ, Anger AM, Armache JP, ... Beckmann R (2012). Structural basis of highly conserved ribosome recycling in eukaryotes and archaea. Nature, 482, 501-6.

Shao S, von der Malsburg K & Hegde RS (2013). Listerin-dependent nascent protein ubiquitination relies on ribosome subunit dissociation. Mol Cell, 50, 637-48.

Pisareva VP, Skabkin MA, Hellen CU, Pestova TV & Pisarev AV (2011). Dissociation by Pelota, Hbs1 and ABCE1 of mammalian vacant 80S ribosomes and stalled elongation complexes. EMBO J, 30, 1804-17.

Shoemaker CJ, Eyler DE & Green R (2010). Dom34:Hbs1 promotes subunit dissociation and peptidyl-tRNA drop-off to initiate no-go decay. Science, 330, 369-72.

Tsuboi T, Kuroha K, Kudo K, Makino S, Inoue E, Kashima I & Inada T (2012). Dom34:hbs1 plays a general role in quality-control systems by dissociation of a stalled ribosome at the 3' end of aberrant mRNA. Mol Cell, 46, 518-29.

Edit history

Date	Action	Author
2025-06-06	Edited	May B
2025-06-06	Authored	May B
2025-06-08	Created	May B
2025-08-09	Reviewed	Bennett EJ, Ford PW
2025-11-07	Reviewed	McGirr T, Jafarnejad SM

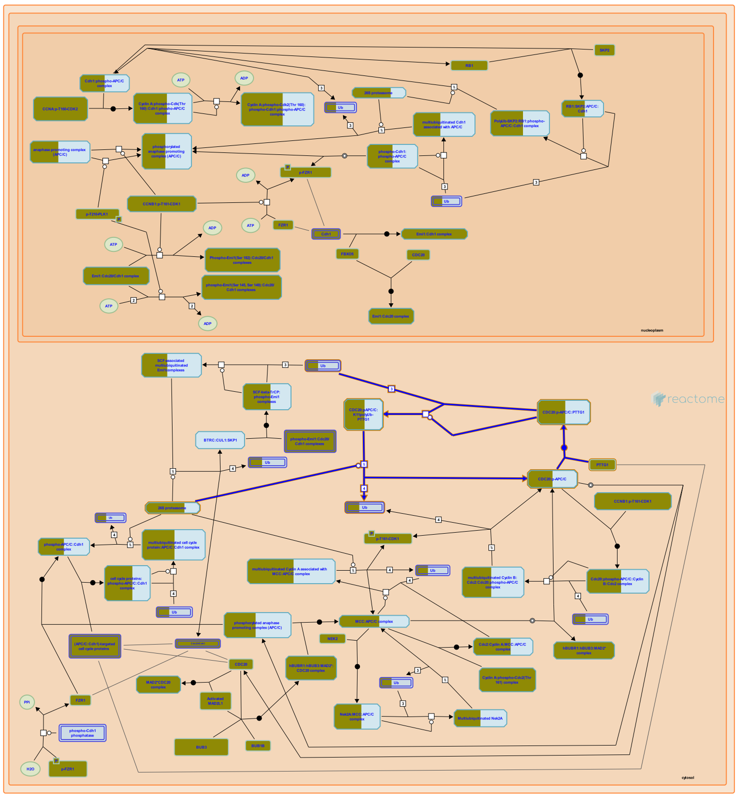
69 submitted entities found in this pathway, mapping to 88 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ABCE1	P61221	CRY2	P62263	FAU	P62861
PELO	Q9BRX2	RPL10	P27635, Q96L21	RPL10A	P62906
RPL11	P62913	RPL12	P30050	RPL13A	P40429
RPL15	P61313	RPL17	P18621	RPL18	Q07020
RPL18A	Q02543	RPL19	P84098	RPL21	P46778
RPL22	P35268	RPL23A	P62750	RPL24	P83731
RPL26	P61254, Q9UNX3	RPL26L1	Q9UNX3	RPL27	P61353
RPL27A	P46776, P61353	RPL29	P47914	RPL3	P39023, Q92901
RPL30	P62888	RPL32	P62888, P62910	RPL34	P49207, P62899
RPL35	P42766	RPL35A	P18077, P42766	RPL36A	P83881, Q969Q0
RPL37A	P18077, P61513	RPL38	P63173	RPL39L	Q96EH5
RPL41	P62945	RPL5	P46777	RPL6	Q02878
RPL7	P18124	RPL7A	P18124, P62424	RPL8	P62424, P62917
RPL9	P32969	RPLP2	P05387	RPS10	P46783
RPS11	P62280	RPS13	P62277	RPS14	P62263
RPS15	P62841	RPS16	P62249	RPS17	P08708
RPS18	P62269	RPS23	P62266	RPS24	P62244, P62847
RPS25	P62851	RPS26	P62854	RPS27	P42677, Q71UM5
RPS27A	P62979, P62987, Q71UM5	RPS27L	Q71UM5	RPS28	P62857
RPS29	P62273	RPS3	P23396	RPS3A	P61247
RPS4X	P62701, Q8TD47	RPS4Y1	P22090, Q8TD47	RPS6	P62753
RPS7	P62081	RPS9	P46781	RRP1	P05386
SOS1	P42766	UBA52	P62979, P62987	UBB	P62979, P62987

Interactors found in this pathway (12)

Input	UniProt Id	Interacts with
C12orf10	Q9HB07	Q9BRX2
EIF3G	O75821	Q9BRX2
FLNA	P21333-2	Q9BRX2
KAT5	Q92993	Q9BRX2
POC1B	Q8TC44	Q9BRX2
WFS1	O76024	Q9BRX2
DYNLT1	P63172	Q9BRX2
FGFR1	P11362-2	Q9BRX2
HAX1	O00165	Q9BRX2
PIK3R3	Q92569	Q9BRX2
PRKCA	P17252	Q9BRX2
YWHAG	P61981	Q9BRX2

24. APC/C:Cdc20 mediated degradation of Securin (R-HSA-174154)



Cellular compartments: cytosol.

The separation of sister chromatids in anaphase requires the destruction of the anaphase inhibitor, securin. Securin associates with and inactivates the protease, separase. Separase cleaves the cohesin subunit, Scc1 that is responsible for the cohesion of sister chromatids (reviewed in Nasmyth et al., 2000). Securin destruction begins at metaphase after the mitotic spindle checkpoint has been satisfied (Hagting et al., 2002).

References

Hagting A, Den Elzen N, Vodermaier HC, Waizenegger IC, Peters JM & Pines J (2002). Human securin proteolysis is controlled by the spindle checkpoint and reveals when the APC/C switches from activation by Cdc20 to Cdh1. J Cell Biol, 157, 1125-37.

Nasmyth K, Peters JM & Uhlmann F (2000). Splitting the chromosome: cutting the ties that bind sister chromatids. Science, 288, 1379-85.

Edit history

Date	Action	Author
2006-01-26	Authored	Castro A, Lorca T
2006-01-30	Edited	Matthews L
2006-02-17	Created	Matthews L
2006-03-28	Reviewed	Peters JM

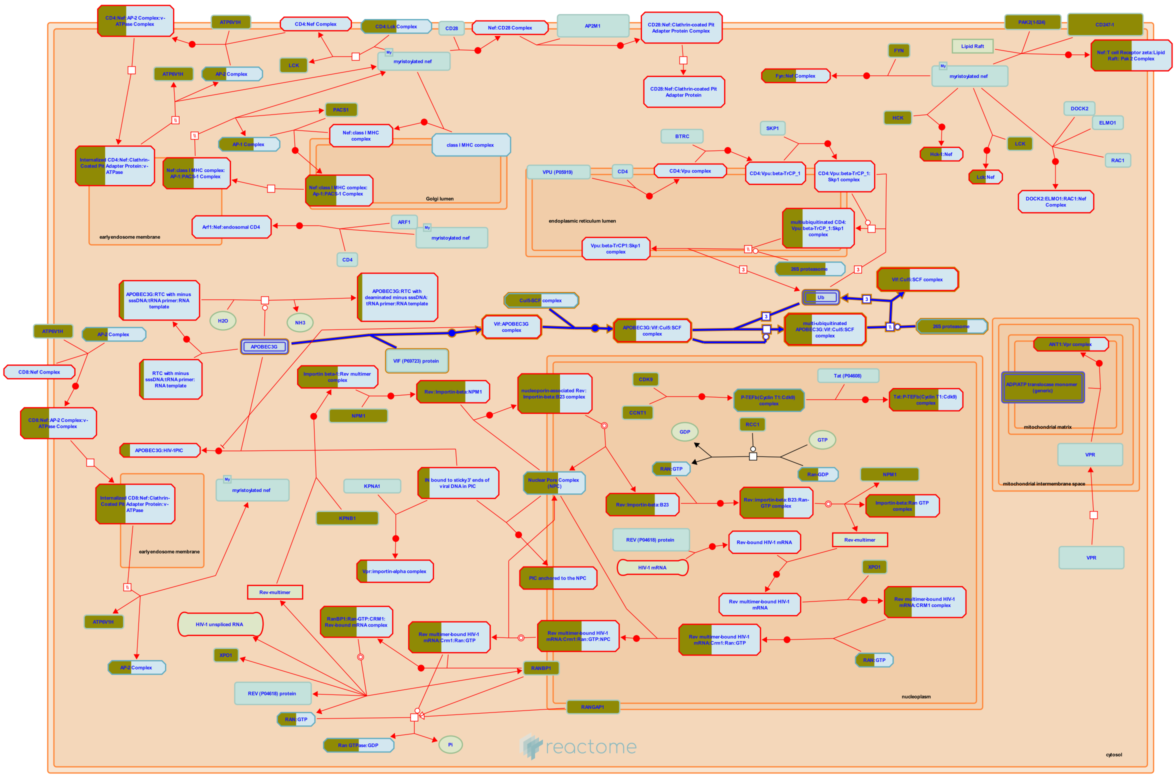
37 submitted entities found in this pathway, mapping to 41 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ANAPC11	Q9NYG5	ANAPC2	Q9UJX6	ANAPC5	Q9UJX4
ANAPC7	Q9UJX3	APC2	Q9UJX6	CDC20	Q12834
CDC23	Q9UJX2	CDC27	P30260	PSMA3	P25788
PSMA4	P25789	PSMA5	P28066	PSMA7	O14818
PSMB2	P49721	PSMB3	P49720	PSMB5	P28074
PSMB6	P28072	PSMB7	Q99436	PSMC1	P62191
PSMC2	P35998	PSMC3	P17980	PSMC4	P43686
PSMC6	P62333	PSMD1	Q99460	PSMD11	O00231
PSMD12	O00232	PSMD14	O00487	PSMD3	O43242
PSMD6	Q15008	PSMD8	P48556	PTTG1	O95997
RPN1	Q13200	RPS27A	P62979, P62987	SEM1	P60896
UBA52	P62979, P62987	UBB	P0CG47, P62979, P62987	UBE2C	O00762
UBE2S	Q16763				

Interactors found in this pathway (2)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ESPL1	P60330, Q14674	O95997	PLCG2	P16885	O95997

25. Vif-mediated degradation of APOBEC3G (R-HSA-180585)



Cellular compartments: cytosol.

Diseases: Human immunodeficiency virus infectious disease.

The HIV-1 accessory protein Vif (Viral infectivity factor) is required for the efficient infection of primary cell populations (e.g., lymphocytes and macrophages) and 'non-permissive' cell lines. Vif neutralises the host DNA editing enzyme, APOBEC3G, in the producer cell. Indeed, in the absence of a functional Vif, APOBEC3G is selectively incorporated into the budding virions and in the next cycle of infection leads to the deamination of deoxycytidines (dC) within the minus-strand cDNA during reverse transcription (Sheehy et al 2003; Li et al., 2005 ; Stopak et al. 2003).

Deamination changes cytidine to uracil and thus results in G to A transitions and stop codons in the provirus. The aberrant cDNAs produced in the infected cell can either be integrated in form of non-functional proviruses or degraded. Vif counteracts the antiviral activity of APOBEC3G by associating directly with it and promoting its polyubiquitination and degradation by the 26S proteasome.

Vif binds APOBEC3G and recruits it into an E3 ubiquitin-enzyme complex composed by the cytoplasmic proteins Cullin5, Rbx, ElonginC and ElonginB (Yu et al., 2003) . Thus, in the presence of Vif, APOBEC3G incorporation into the virion is minimal.

References

Sheehy AM, Gaddis NC & Malim MH (2003). The antiretroviral enzyme APOBEC3G is degraded by the proteasome in response to HIV-1 Vif. Nat Med, 9, 1404-7.

Yu X, Yu Y, Liu B, Luo K, Kong W, Mao P & Yu XF (2003). Induction of APOBEC3G ubiquitination and degradation by an HIV-1 Vif-Cul5-SCF complex. Science, 302, 1056-60.

Kobayashi M, Takaori-Kondo A, Miyauchi Y, Iwai K & Uchiyama T (2005). Ubiquitination of APOBEC3G by an HIV-1 Vif-Cullin5-Elongin B-Elongin C complex is essential for Vif function. J Biol Chem, 280, 18573-8.

Edit history

Date	Action	Author
2006-05-16	Authored	Matthews L
2006-05-23	Created	Matthews L
2007-01-30	Edited	Matthews L
2007-01-31	Reviewed	Mulder L, Simon V
2023-10-12	Modified	Weiser JD

28 submitted entities found in this pathway, mapping to 32 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ELOC	Q15369	PSMA3	P25788	PSMA4	P25789
PSMA5	P28066	PSMA7	O14818	PSMB2	P49721
PSMB3	P49720	PSMB5	P28074	PSMB6	P28072
PSMB7	Q99436	PSMC1	P62191	PSMC2	P35998
PSMC3	P17980	PSMC4	P43686	PSMC6	P62333
PSMD1	Q99460	PSMD11	O00231	PSMD12	O00232
PSMD14	O00487	PSMD3	O43242	PSMD6	Q15008
PSMD8	P48556	RBX1	P62877	RPN1	Q13200
RPS27A	P62979, P62987	SEM1	P60896	UBA52	P62979, P62987
UBB	P0CG47, P62979, P62987				

6. Identifiers found

Below is a list of the input identifiers that have been found or mapped to an equivalent element in Reactome, classified by resource.

4474 of the submitted entities were found, mapping to 5279 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
AACS	Q86V21	AAK1	Q2M2I8	AAMP	Q13685
AARS	P49588	AASDHPPT	Q9NRN7	ABCA10	Q8WWZ4
ABCA13	Q86UQ4	ABCA2	O95477	ABCA5	Q8IUA7, Q8WWZ7
ABCA6	Q8N139	ABCB1	P21439	ABCB10	Q9NRK6
ABCB4	P21439	ABCB6	Q9NP58	ABCB9	Q9NP78
ABCC1	P33527	ABCC10	Q5T3U5	ABCC2	P13569
ABCC3	O15438	ABCD1	P33897	ABCD2	P33897
ABCD3	P28288	ABCD4	O14678	ABCE1	P61221
ABCF2	Q9UG63	ABHD10	Q9NUJ1	ABHD17A	Q96GS6
ABHD17C	Q6PCB6	ABHD3	Q8WU67	ABI1	Q8IZP0
ABL1	P00519	ABL2	P42684	ABLIM1	O14639
ABLIM2	Q6H8Q1	ABRAXAS2	Q15018	ACAA2	P42765
ACACA	Q13085	ACAD8	Q9UKU7	ACAD9	Q9H845
ACADM	P11310	ACADSB	P45954	ACAT1	P24752
ACAT2	O75908	ACBD3	Q9H3P7	ACBD7	Q8N6N7
ACCS	Q93063	ACIN1	Q9UKV3	ACKR2	Q9UNU6
ACLY	P53396	ACOT11	Q8WXI4	ACOT13	Q9NPJ3
ACOT7	O00154	ACOX1	O15254	ACOXL	Q9NUZ1
ACP1	O14561	ACP2	Q9BT43	ACP5	P13686
ACP6	Q9NPH0	ACPP	P15309	ACSL4	O60488
ACSL6	Q9UKU0	ACSM3	Q68CK6	ACTA2	P62736
ACTB	P60709	ACTG1	P60709	ACTL6A	O96019
ACTR10	Q9NZ32	ACTR1A	P61163	ACTR2	P61160
ACTR3	P61158	ACTR5	Q9H9F9	ACVR1B	P36896
ACVR2B	Q13705	ADA	P00813	ADAL	Q6DHV7
ADAM10	O14672	ADAM12	O43184	ADAM19	Q9H013
ADAM22	Q9P0K1	ADAM9	Q99965	ADAMTS10	Q9H324
ADAMTSL4	Q6UY14	ADAMTSL5	Q6ZMM2	ADAT2	Q7Z6V5
ADCY3	O60266	ADCY6	O43306	ADCY7	P51828
ADCY9	O60503	ADD1	P35611	ADD2	P35612
ADD3	Q9UEY8	ADGRE3	Q9BY15	ADGRE5	P48960
ADGRG6	Q86SQ4	ADHFE1	Q8IWW8	ADK	P55263
ADO	Q96SZ5	ADORA2B	P29275	ADPGK	Q9BRR6
ADRB2	P07550	ADSL	P30566	AES	Q08117
AFF4	Q9UHB7	AFMID	Q63HM1	AGBL2	Q5U5Z8
AGBL5	Q8NDL9	AGGF1	Q8N302	AGMAT	Q9BSE5
AGO1	Q9H9G7, Q9UKV8, Q9UL18	AGO4	Q9HCK5	AGPAT1	O15120
AGPAT2	O15120	AGPAT3	Q9NRZ7	AGPAT4	Q9NRZ5
AGPAT5	Q9NUQ2	AGPS	O00116	AGTPBP1	Q9UPW5
AHCY	P23526	AHCYL1	Q96HN2	AHI1	Q8N157

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AHR	P35869	AHRR	A9YTQ3	AIFM2	Q9BRQ8
AIMP2	Q13155	AIP	Q9NWT8	AJUBA	Q96IF1
AK2	P54819	AK3	P27144	AK4	P27144
AK5	Q9Y6K8	AK6	Q9Y3D8	AK8	Q96MA6
AK9	Q5TCS8	AKAP1	Q92667	AKAP10	O43572
AKAP13	Q12802	AKAP8L	Q9ULX6	AKAP9	Q99996
AKR1A1	P14550	AKR1C3	P42330	AKR1E2	Q96JD6
AKR7A2	O43488, O95154	AKR7A3	O43488, O95154	AKT2	P31751
AKT3	Q9Y243	ALAS1	P13196	ALCAM	Q13740
ALDH18A1	P54886-1, P54886-2	ALDH2	P05091	ALDH8A1	Q9H2A2
ALDOA	P04075	ALG1	Q9BT22	ALG10	Q5BKT4
ALG13	Q9NP73	ALG3	Q92685	ALG5	Q9Y673
ALG8	Q9BVK2	ALKAL2	Q6UX46	ALOX12B	O75342
ALOX5	P09917	ALPK1	Q96QP1	AMN	Q9BXJ7
AMOT	Q4VCS5-2	AMPD3	Q01432	AMY2B	P19961
ANAPC11	Q9NYG5	ANAPC2	Q9UJX6	ANAPC5	Q9UJX4
ANAPC7	Q9UJX3	ANKH	Q9HCJ1	ANKRD26	Q9UPS8
ANKRD27	Q96NW4	ANKRD44	Q9H672	ANLN	Q9NQW6
ANO9	A1A5B4	ANP32A	P39687	ANPEP	P15144
ANTXR2	P58335-1, P58335-4	ANXA2	P07355	ANXA3	P12429
ANXA5	P08758	AP1B1	Q10567	AP1G1	O43747
AP1G2	O75843	AP1S1	P56377, P61966	AP2B1	P63010
AP2S1	P53680	AP3M1	Q9Y2T2	AP3S1	Q92572
AP4E1	Q9UPM8	APAF1	O14727	APBA2	Q99767
APBA3	O96018	APBB1	O00213	APBB1IP	Q7Z5R6
APC	P25054	APC2	Q9UJX6	APEH	P13798
APH1B	Q8WW43	APIP	Q96GX9	APLN	Q9ULZ1
APLP2	Q06481	APOBEC3A	P31941	APOBEC3B	Q9UH17
APOBEC3H	Q6NTF7	APOL1	O14791	APOO	Q9BUR5
APOOL	Q6UXV4	APPL2	Q06481	APRT	P07741
AQP9	O14520, O43315	ARAF	P10398	ARAP1	Q96P48
ARAP2	Q8WZ64	ARCN1	P48444	AREG	P01135, P15514
ARF4	P18085	ARFGAP2	Q8N6H7	ARFGEF2	Q9Y6D5
ARFIP2	P53365	ARG2	P78540	ARHGAP11A	Q6P4F7
ARHGAP11B	Q3KRB8, Q6P4F7	ARHGAP12	Q8IWW6	ARHGAP15	Q53QZ3
ARHGAP18	Q8N392	ARHGAP19	Q14CB8	ARHGAP24	Q8N264
ARHGAP25	P42331	ARHGAP26	Q9UNA1	ARHGAP33	A7KAX9
ARHGAP8	P85298	ARHGEF1	Q92888	ARHGEF10	O15013
ARHGEF10L	Q9HCE6	ARHGEF11	O15085	ARHGEF12	Q9NZN5
ARHGEF18	Q6ZSZ5	ARHGEF19	Q8IW93	ARHGEF2	Q92974
ARHGEF25	Q86VW2	ARHGEF28	Q8N1W1	ARHGEF39	Q8N4T4
ARHGEF6	Q15052	ARID1B	Q8NFD5	ARID3A	Q99856
ARID4A	P29374	ARID4B	Q4LE39	ARIH1	Q9Y4X5
ARIH2	O95376	ARL1	P40616	ARL13B	Q3SXY8
ARL2	P36404	ARL3	P36405	ARL4C	P56559
ARL6	Q9H0F7	ARL6IP5	O75915	ARMC10	Q9UH62
ARMC8	Q8IUR7	ARNT	P27540	ARNTL	O00327
ARNTL2	Q8WYA1	ARPC1B	O15143	ARPC3	O15145
ARPC5	O15511	ARPP19	P56211	ARRB1	P49407

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ARSB	P15848	ARSG	Q96EG1	ARSJ	Q5FYB0
ARV1	Q9H2C2	ASAH1	Q13510	ASB2	Q96Q27
ASB6	Q9NWX5	ASB8	Q9H765	ASCC3	Q8N3C0
ASF1A	Q9Y294	ASGR1	P07306	ASGR2	P07307
ASH1L	Q9NR48	ASH2L	Q9UBL3	ASIC1	P78348
ASNA1	O43681	ASPH	Q12797	ASXL2	Q76L83
ATAD2	Q6PL18	ATAD3A	Q5T9A4	ATF1	P18846
ATF4	P18848	ATF6B	Q99941	ATF7IP	Q6VMQ6
ATG101	Q9BSB4	ATG12	O94817	ATG14	Q6ZNE5
ATG3	Q9NT62	ATG4B	Q96DT6, Q9Y4P1	ATG4C	Q96DT6
ATG4D	Q86TL0	ATG9A	Q7Z3C6	ATIC	P31939
ATL2	Q8NHH9	ATM	Q13315	ATOX1	O00244
ATP10A	O60312	ATP10D	Q9P241	ATP11C	Q8NB49
ATP13A4	Q4VNC0, Q4VNC1	ATP1A3	P13637	ATP1B1	P05026
ATP1B2	P14415	ATP2B1	P20020	ATP2B4	P23634
ATP5A1	P25705	ATP5C1	P36542	ATP5E	P56381
ATP5F1	P24539	ATP5G1	P05496	ATP5G3	P48201, Q06055
ATP5H	O75947	ATP5J	P18859	ATP5J2	P56134
ATP5L	O75964	ATP6AP2	O75787	ATP6V0C	P27449
ATP6V0E2	Q8NHE4	ATP6V1A	P38606	ATP6V1B2	P21281
ATP6V1C1	P21283	ATP6V1D	Q9Y5K8	ATP6V1H	Q9UI12
ATP8A1	Q9Y2Q0	ATP8B4	Q8TF62	ATP9A	O75110
ATP9B	O43861	ATR	Q13535	ATRX	P46100
ATXN3	P54252, Q9H3M9	ATXN7	O15265	AUH	Q13825
AUP1	Q9Y679	AURKA	O14965	AURKB	Q96GD4
AUTS2	Q8WXX7	AXIN1	O15169	AXL	P30530
AZIN2	Q96A70	AZU1	P20160	B3GALT2	O43825
B3GAT1	Q9P2W7	B3GNT7	Q8NFL0	B3GNT8	Q67FW5, Q7Z7M8
B3GNT9	Q6UX72	B4GALNT1	Q00973	B4GALT1	P15291
B4GALT2	O60909	B4GALT4	O60513	B4GALT7	Q9UBV7
B4GAT1	O43505	B9D1	Q9UPM9	BACE1	P56817
BACH1	Q9BX63	BAD	Q92934	BAG1	Q99933
BAG5	Q9UL15	BAIAP2	Q9UQB8	BAIAP2L2	Q6UXY1
BAMBI	Q13145	BANP	Q8N9N5	BAP1	Q99496
BARD1	Q99728	BATF	Q16520	BAX	Q07812
BAZ1B	Q9UIG0	BAZ2A	Q9UIF9	BBS10	Q8TAM1
BBS12	Q6ZW61	BBS2	Q9BXC9	BBS7	Q8IWZ6
BBS9	Q3SYG4	BCAT1	P54687	BCAT2	O15382
BCDIN3D	Q7Z5W3	BCKDK	O14874	BCL11A	Q9H165
BCL2	P10415	BCL2A1	Q16548	BCL2L1	Q07817
BCL6	P41182	BCL7C	Q8WUZ0	BCL9L	Q86UU0
BCR	P11274	BDH1	Q02338	BDH2	Q9BUT1
BDP1	A6H8Y1	BECN1	Q14457	BEST1	O76090
BICD1	Q96G01	BICRA	Q9NZM4	BICRAL	Q6AI39
BID	P55957	BIN2	Q9UBW5	BIRC3	Q13489
BIRC5	O15392	BLK	P51451	BLM	P54132
BLMH	Q13867	BLNK	Q8WV28	BLOC1S6	Q9UL45
BLZF1	Q9H2G9	BMF	Q96LC9	BMP1	P13497
BMP2	P12643	BMP2K	Q2M2I8	BMPR2	Q13873

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
BMS1	Q14692	BNIP1	Q12981	BOD1L1	Q8NFC6
BOP1	Q14137	BORA	Q6PGQ7	BPGM	P07738
BPNT1	P49441	BRAF	P15056	BRCA1	P38398
BRCA2	P51587	BRCC3	P46736	BRD1	O95696
BRD2	P25440	BRD4	O60885	BRD7	Q9NPI1
BRD8	Q9H0E9	BRD9	Q9H8M2	BRF1	Q92994
BRF2	Q9HAW0	BRIP1	Q9P0J6	BRK1	Q8WUW1
BRPF3	Q9ULD4	BRWD1	Q9NSI6	BRWD3	Q9NSI6
BSN	Q9UPA5	BST1	Q10588	BST2	Q10589
BTBD1	Q9H0C5	BTD	P43251	BTG1	P62324
BTG2	P78543	BTN2A1	Q7KYR7	BTN3A1	O00481
BTN3A2	P78410	BTN3A3	O00478	BUB1	O43683, O60566
BUB1B	O60566	BUB3	O43684	BUD13	Q9BRD0
BUD23	O43709	BYSL	Q13895	C12orf65	Q9H3J6
C12orf73	Q69YU5	C14orf166	Q9Y224	C16orf62	Q7Z3J2
C19orf70	Q5XKP0	C19orf84	I3L1E1	C1D	Q13901
C1GALT1C1	Q96EU7	C1QA	P02745	C1QBP	Q07021
C1orf112	Q9NSG2	C20orf24	Q9BUV8	C2CD5	Q86YS7
C3	P01024	C3AR1	Q16581	C5AR1	P21730
C5AR2	Q9P296	C8G	P07360	C9orf64	Q5T6V5
CA13	Q8N1Q1	CA3	P07451	CA5B	P35218, Q9Y2D0
CA6	P23280	CAB39	Q9Y376	CACNA1I	Q9P0X4
CACNA2D2	Q9NY47	CACNB3	P54284	CAD	O76075
CADM1	Q9BY67	CALCRL	Q16602	CALHM1	Q8IU99
CALHM3	Q86XJ0	CALM2	P0DP24	CALM3	P0DP23
CALU	O43852	CAMK1	Q14012	CAMK2G	Q13555
CAMK4	Q16566	CAMKK2	Q96RR4	CAND1	Q86VP6
CANT1	Q8WVQ1	CAP1	Q01518	CAPG	Q9BPX3
CAPN10	Q9HC96	CAPN15	O75808	CAPN8	A6NHC0
CAPZA1	P52907	CAPZA2	P47755	CAPZB	P47756
CARD11	Q9BXL7	CARD9	Q9H257	CARNMT1	Q8N4J0
CARNS1	A5YM72	CARS2	Q9HA77	CASC3	O15234
CASP2	P42575	CASP3	P42574	CASP6	Q9BZM1
CASP7	P55210	CASP8	Q14790	CAST	O15446
CASTOR1	Q8WTX7	CATSPERG	Q6ZRH7	CAVIN1	Q6NZI2
CBFA2T2	O43439	CBFB	Q13951	CBL	P22681
CBLB	Q13191	CBLL1	Q75N03	CBR1	P16152
CBR3	P16152	CBX1	P83916	CBX2	Q14781
CBX3	Q13185	CBX4	O00257	CBX5	P45973
CBY1	Q9Y3M2	CC2D2A	Q9P2K1	CCDC115	Q96NT0
CCDC12	Q8WUD4	CCDC59	Q9P031	CCDC88C	Q9P219
CCDC94	Q9BW85	CCL1	P51946	CCL2	P13500
CCL22	O00626	CCL25	O15444	CCL28	Q9NRJ3
CCL3	P10147	CCL3L3	P16619	CCL4	P13236, P13501
CCNA1	P78396	CCNA2	P20248	CCNB1	P14635
CCNB2	O95067	CCND1	P24385	CCND2	P30279
CCNE1	P24864	CCNE2	O96020	CCNF	P41002
CCNG1	P51959	CCNG2	Q16589	CCNH	P51946
CCNK	O75909	CCNL1	P49736	CCNT1	O60563

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CCNT2	O60583	CCP110	O43303	CCR1	P35228
CCR2	P51681	CCR3	P51677	CCR5	P51681
CCR6	O00421, P51684	CCR8	P51685	CCRL2	O00421
CCT3	P49368	CCT4	P50991	CCT5	P48643
CCT6A	P40227, Q92526	CCT6B	Q92526	CCT7	Q99832
CCT8	P50990	CD101	Q93033	CD109	Q6YHK3
CD14	P08571	CD160	O95971	CD163	Q86VB7
CD19	P15391	CD1A	P06126	CD200R1	Q8TD46
CD244	Q9BZW8	CD247	P20963-1	CD27	P26842
CD300E	Q496F6	CD300LD	Q6UXZ3	CD300LF	Q6UXZ3, Q8TDQ1
CD36	P16671	CD38	P28907	CD3D	P04234, P09693
CD3E	P07766	CD3EAP	O15446	CD3G	P09693
CD40	P25942-1	CD40LG	P29965	CD47	Q08722
CD48	Q96DU3	CD52	P31358	CD70	P32970
CD72	P21854	CD80	P33681	CD84	Q9UIB8
CD9	P21926	CD93	Q9NPY3	CD96	P40200
CDA	P32320	CDC14B	O60729	CDC20	Q12834
CDC23	Q9UJX2	CDC25A	P30304	CDC25C	P30307
CDC27	P30260	CDC40	O60508	CDC42BPA	Q5VT25
CDC42BPB	Q9Y5S2	CDC42EP2	O14613	CDC42EP3	Q9UKI2
CDC42SE2	Q9NRR3	CDC45	O75419	CDC6	Q99741
CDC7	O00311	CDC73	Q6P1J9	CDCA5	Q96FF9
CDCA8	Q53HL2	CDH1	Q9UM11	CDH17	Q12864
CDH4	P55283	CDK1	P24941	CDK11A	Q9UQ88
CDK11B	P21127	CDK12	Q14004	CDK13	Q14004
CDK2	P24941	CDK4	P11802	CDK5	Q00535
CDK5RAP1	Q13424	CDK6	Q00534	CDK9	P50750
CDKN1A	P38936	CDKN1C	P49918	CDKN2A	P42771
CDKN2B	P42771	CDKN2C	P42773	CDON	Q4KMG0
CDT1	Q9H211	CEACAM1	P13688, P31997	CEBPA	P49715
CEBPD	P49716	CEBPE	Q15744	CEL	P19835
CENPA	P49450	CENPE	Q02224	CENPF	P49454
CENPH	Q9H3R5	CENPI	Q92674	CENPJ	Q9HC77
CENPK	Q9BS16	CENPL	Q8N0S6	CENPM	Q9NSP4
CENPN	Q96H22	CENPO	Q9BU64	CENPQ	Q7L2Z9
CENPT	Q96BT3	CENPU	Q71F23	CENPW	Q5EE01
CENPX	A8MT69	CEP131	Q9UPN4	CEP152	O94986
CEP164	Q9UPV0	CEP250	Q9BV73	CEP41	Q9BYV8
CEP76	Q8TAP6	CEP78	Q5JTW2	CEP89	Q96ST8
CEP97	Q8IW35	CERS6	Q6ZMG9	CES1	P23141
CFH	P08603, Q92496	CFLAR	O15519-1, O15519-2	CHAC1	Q9BUX1
CHAC2	Q8WUX2	CHCHD1	Q96BP2	CHCHD10	Q8WYQ3
CHCHD2	Q9Y6H1	CHCHD4	Q8N4Q1	CHCHD5	Q9BSY4
CHCHD6	Q9BRQ6	CHCHD7	Q9BUK0	CHD1	O14646
CHD4	Q14839	CHD6	Q8TD26	CHD9	Q3L8U1
CHEK1	O14757	CHEK2	O96017	CHML	P26374
CHMP2A	O43633	CHMP3	Q9Y3E7	CHMP4A	Q9BY43
CHMP5	Q9NZZ3	CHMP7	Q8WUX9	CHN2	P52757
CHRM4	P08173	CHRNA10	Q9GZZ6	CHRNA5	P30532

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CHRNA6	Q15825	CHRNE	Q04844	CHST10	O43529
CHST11	Q9NPF2	CHST14	Q8NCH0	CHST15	Q7LFX5
CHST2	Q9Y4C5	CHST7	Q9NS84	CHTF8	P0CG13
CHTOP	Q9Y3Y2	CIAO1	O76071	CILP	O75339
CIPC	Q9C0C6	CIR1	P38117	CISH	Q9NSE2
CIT	O14578, O14578-3	CKAP4	Q07065	CKAP5	Q14008
CKLF	Q13887	CKS1B	P61024	CLASP1	Q7Z460
CLCF1	Q9UBD9	CLCN2	P51788	CLCN3	P51790
CLCN6	P51797	CLCN7	P51798	CLDN12	P56749
CLDN22	Q8N7P3	CLDN9	O95484	CLEC10A	Q8IUN9
CLEC4A	Q9UMR7	CLEC4E	Q9ULY5	CLIC5	Q9NZA1
CLINT1	Q14677	CLNS1A	Q13153	CLOCK	O15516
CLP1	Q92989	CLPP	Q16740	CLSPN	Q9HAW4
CLTA	P09496	CLTC	Q00610	CLU	P10909
CLUAP1	Q96AJ1	CMAS	Q8NFW8	CMC1	Q7Z7K0
CMC2	Q9NRP2	CMKLR1	Q99788	CMTM6	Q9NX76
CNIH1	O95406	CNKSR2	Q8WXI2	CNN2	Q99439
CNOT11	Q9UKZ1	CNOT2	Q9NZN8	CNOT3	O75175
CNOT4	O95628	CNOT6	Q9ULM6	CNOT6L	Q96LI5
CNTF	P26441	COA1	Q9GZY4	COA3	Q9Y2R0
COA4	Q9NYJ1	COA6	Q5JTJ3	COG1	Q8WTW3
COG3	Q96JB2	COG4	Q9H9E3	COG7	P83436
COL11A2	P13942	COL13A1	Q5TAT6	COL19A1	Q14993, Q9BXS0
COL23A1	Q86Y22	COL25A1	Q9BXS0	COL4A3	Q01955
COL4A4	P53420	COL5A2	P05997	COL5A3	P25940
COL6A2	P12110	COL8A2	P25067	COL9A2	Q14055, Q9UMD9
COLGALT2	Q8IYK4	COMMD4	Q9H0A8	COMMD5	Q9GZQ3
COMMD7	Q86VX2	COMMD8	Q9NX08	COMT	P21964
COPA	P53621	COPB1	P53618	COPB2	P35606
COPG1	Q9Y678	COPRS	Q9NQ92	COPS3	Q9UNS2
COPS4	Q9BT78	COPS8	Q99627	COPZ2	Q9P299
COQ10A	Q96MF6	COQ10B	Q9H8M1	COQ2	Q96H96
COQ3	Q9NZJ6	COQ5	Q5HYK3	COQ8A	Q8NI60
COQ8B	Q96D53	COQ9	O75208	CORO1A	P31146
COTL1	Q14019	COX11	Q9Y6N1	COX15	Q7KZN9
COX17	Q14061	COX19	Q49B96	COX5A	P20674
COX5B	P10606	COX6A1	P12074	COX6B1	P14854
COX6C	P09669	COX7A2	P14406	COX7A2L	O14548
COX7B	P24311	COX7C	P15954	COX8A	P10176
CPD	O75976	CPLX1	O14810	CPNE7	Q9UBL6
CPNE8	Q86YQ8	CPOX	P36551	CPPED1	Q9BRF8
CPSF2	Q9P2I0	CPSF3	Q9UKF6	CPT2	P23786
CR1	P17927	CR2	P20023	CRABP2	P29373
CRBN	Q96SW2	CRCP	O75575	CREB3L2	Q70SY1
CREB3L3	Q68CJ9	CREBBP	Q92793	CREBRF	Q8IUR6
CRHBP	P24387	CRISPLD2	Q9H0B8	CRK	P46108
CRLF2	Q9HC73	CRNKL1	Q9BZJ0	CROT	Q9UKG9
CRTAM	O95727	CRTAP	O75718	CRTC1	Q6UUV9
CRTC3	Q6UUV7	CRY1	Q16526	CRY2	P62263

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CSAD	Q9Y600	CSF1	P09603	CSF1R	P07333
CSF2	P04141	CSF2RA	P15509	CSF2RB	P32927
CSF3R	Q99062	CSGALNACT2	Q8N6G5	CSNK1A1	P48729
CSNK1D	P48730	CSNK1E	P49674	CSNK1G2	P78368
CSRP1	P21291	CST3	P01034	CSTA	P01040
CSTB	P04080	CSTF1	Q05048	CSTF2	P33240
CSTF3	Q12996	CTAGE5	Q96PC5	CTBP1	Q13363
CTBP2	P56545	CTCF	P49711	CTNNA1	P35221
CTNNB1	P35222	CTNNBIP1	Q9NSA3	CTPS1	P17812, Q9NRF8
CTR9	Q6PD62	CTRC	Q99895	CTSB	P07858
CTSC	P53634	CTSF	P56202	CTSO	P43235
CTSS	P25774	CTSW	P56202	CTSZ	Q9UBR2
CTTN	Q14247	CTU1	Q7Z7A3	CTU2	Q2VPK5
CUBN	O60494	CUL3	Q13618	CUX1	Q13948
CWC15	Q9P013	CWC22	Q9HCG8	CWC25	Q9NXE8
CWF19L2	Q2TBE0	CX3CR1	P49238	CXCL16	Q9H2A7
CXCR1	P25024, P25025	CXCR2	P25024, P25025	CXCR3	P49682
CXCR4	P61073	CXCR5	P32302	CXCR6	O00574
CXXC4	Q9H2H0	CXXC5	Q7LFL8	CXorf21	Q9HAI6
CYB5RL	Q6IPT4	CYBA	P13498	CYBB	P04839
CYBRD1	Q53TN4	CYFIP1	Q7L576	CYP1A1	P04798
CYP1B1	Q16678	CYP27A1	Q02318	CYP2E1	P05181
CYP2R1	Q6VVX0	CYP2U1	Q7Z449	CYS1	Q717R9
CYSLTR1	Q9NS75, Q9Y271	CYSTM1	Q9H1C7	CYTH1	Q15438, Q99418
CYTH3	O43739	DAB2	P98082	DAD1	P61803
DAG1	Q14118	DAGLA	Q9Y4D2	DAGLB	Q8NCG7
DAP	P46527	DAPK2	Q9UIK4	DAPP1	Q9UN19
DARS	P14868	DARS2	Q6PI48	DAXX	Q9UER7
DBN1	Q16643	DBT	P49674	DCAF16	Q9NXF7
DCAF17	Q5H9S7	DCAF5	Q96JK2	DCAF6	Q58WW2
DCAF7	P61962	DCAF8	Q5TAQ9	DCK	P27707
DCLRE1A	Q6PJP8	DCLRE1B	Q9H816	DCLRE1C	Q96SD1
DCP1B	Q8IZD4	DCP2	Q8IU60	DCPS	Q96C86
DCTN2	Q13561	DCTN4	Q9UJW0	DCTPP1	Q9H773
DCUN1D1	Q96GG9	DCUN1D5	Q9BTE7	DDB1	Q16531
DDIT3	P35638	DDIT4	Q9NX09	DDR2	Q16832
DDX1	Q92499	DDX11	Q96FC9	DDX39B	Q13838
DDX3X	O00571	DDX4	Q9NQI0	DDX47	Q9H0S4
DDX5	P17844	DDX6	P26196	DECR2	Q9NUI1
DEGS1	O15121	DEK	P51580	DENND1A	Q8TEH3
DENND1B	Q6P3S1	DENND4B	O75064	DENND4C	Q5VZ89
DENND5A	Q6IQ26	DENND6A	Q8IWF6	DENND6B	Q8NEG7
DEPDC1B	Q8WUY9	DEPDC5	O75140	DERL1	Q9GZP9
DERL2	Q9GZP9	DERL3	Q96Q80	DFFA	O00273
DFFB	O76075	DFNA5	O60443	DGAT2	Q96PD6, Q96PD7
DGKA	P23743, P52824	DGKD	Q16760	DGKH	Q86XP1
DGKK	Q5KSL6	DGKQ	P52824	DGKZ	Q13574
DHCR24	Q15392	DHDDS	Q86SQ9	DHFR	P00374

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
DHFR2	Q86XF0	DHODH	Q02127	DHRS7B	Q6IAN0
DHTKD1	Q96HY7	DHX36	Q9H2U1	DHX38	Q92620
DIABLO	Q9NR28	DIAPH3	Q9NSV4	DIEXF	Q68CQ4
DIMT1	Q9UNQ2	DIS3L2	Q8IYB7	DISP2	A7MBM2
DLAT	P10515	DLC1	P63167	DLG2	Q15700
DLG4	P78352	DLG5	Q8TDM6	DLGAP4	Q9Y2H0
DLGAP5	Q15398	DMAC1	Q96GE9	DMAC2	Q7Z7M8
DMAP1	Q9NPF5	DMC1	Q14565	DMD	P11532
DMGDH	Q9UI17	DMKN	Q6E0U4	DMPK	Q09013
DMTN	Q08495	DNA2	P51530	DNAJA4	Q8WW22
DNAJB1	P25685	DNAJB6	O75190	DNAJB9	Q9UBS3
DNAJC2	Q99543	DNAJC7	Q99615	DNASE1L1	P49184
DNM2	P50570	DNMBP	Q6XZF7	DNMT3B	Q9UBC3
DNPH1	O43598	DNTTIP1	Q9H147	DOCK1	Q14185
DOCK10	Q96BY6	DOCK11	Q5JSL3	DOCK3	Q8IZD9
DOCK5	Q9H7D0	DOCK8	Q8NF50	DOHH	Q9BU89
DOK1	Q93070	DOK2	Q7L591	DOLK	Q9UPQ8
DOLPP1	Q86YN1	DPAGT1	Q9H3H5	DPEP2	Q9H4A9
DPF3	Q92784	DPH1	Q9BZG8	DPP3	Q9NY33
DPP4	P27487	DPY30	Q9C005	DPYSL2	Q16555
DRAP1	Q14919	DRG1	Q9Y295	DRP2	Q13474
DSC1	Q08554	DSCAML1	Q8TD84	DSCC1	Q9BVC3
DSG2	Q14126	DSN1	Q9H410	DTL	Q9NZJ0
DTNBP1	Q96EV8	DTX3L	Q8TDB6	DTYMK	P23919
DUSP1	P28562	DUSP10	Q9Y6W6	DUSP16	Q9BY84
DUSP2	Q05923	DUSP4	Q13115	DUSP6	Q16828
DUSP7	Q16829	DUSP8	Q13202	DUT	P33316-2
DVL1	O14640	DXO	O77932	DYNC1I2	Q13409
DYNC1LI2	O43237	DYNLL1	P63167	DYNLRB1	Q9NP97
DYNLT1	P63172	DYRK1A	Q13627	DYRK2	Q92630
DYRK3	Q92630	DZIP3	Q86Y13	E2F1	O00716, Q01094
E2F2	Q14209	E2F3	O00716	E2F5	Q15329
E2F7	Q96AV8	E2F8	A0AVK6	EARS2	Q5JPH6
EBP	Q15125	ECE2	P0DPD6	ECHS1	P30084
ECI1	P42126	ECI2	O75521-2	ECSIT	Q9BQ95
ECT2	Q9H8V3	EDA2R	Q9HAV5	EDC3	Q96F86
EDEM1	Q92611	EDEM3	Q9BZQ6	EDN2	P20800
EDN3	P14138	EED	O75530	EEF1A1	P68104
EEF1AKMT1	Q8WVE0	EEF1AKMT2	Q5JPI9	EEF1B2	P24534
EEF1D	P29692	EEF1E1	O43324	EEF2K	O00418
EEPD1	Q7L9B9	EFEMP2	O95967, Q12805	EFHD2	Q96C19
EFNA1	P20827	EFNA5	P52803	EFNB2	P52799
EGLN3	Q9H6Z9	EHD3	Q6NVY1	EHHADH	Q08426
EID3	Q8N140	EIF1AY	P47813	EIF2AK2	P19525
EIF2B2	P49770	EIF2B3	Q9NR50	EIF2B4	Q9UI10
EIF2B5	Q13144	EIF2S1	P05198	EIF2S2	P20042
EIF3B	P55884	EIF3D	O15371	EIF3G	O75821
EIF3I	Q13347	EIF3J	O75822	EIF4A1	P60842
EIF4A2	Q14240	EIF4B	P23588	EIF4E	P06730

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
EIF4E2	O60573	EIF4EBP1	Q13541	EIF4G1	Q04637
EIF5	P55010	ELAVL1	Q15717	ELF1	O43921
ELF2	Q15723	ELF3	P78545	ELMO2	Q96JJ3
ELOC	Q15369	ELOVL5	Q9NYP7	ELOVL6	Q9H5J4
ELOVL7	A1L3X0	ELP2	Q6IA86	ELP6	Q0PNE2
EMD	P50402	EME1	A4GXA9, Q96AY2	EMILIN1	Q9Y6C2
EML4	Q9HC35	ENGASE	Q8NFI3	ENO1	P06733
ENO2	P09104	ENO3	P13929	ENPEP	Q07075
ENPP2	Q13822	ENPP4	Q9Y6X5	ENTPD1	P49961
ENTPD4	Q9Y227	ENTPD6	O75354	ENTPD7	Q9NQZ7
EOMES	O95936	EP300	Q09472	EPAS1	Q99814
EPB41	O43491, P11171	EPB41L2	O43491	EPB41L3	Q9Y2J2
EPC1	Q9H2F5	EPHA1	P21709	EPHA4	P54764
EPHB4	P54760	EPHX1	P07099	EPHX2	P34913
EPM2A	O95278	EPRS	P07814	EPS15L1	Q9UBC2
ERAL1	O75616	ERBB2	P04626	ERBB3	P21860-1
ERBIN	Q96RT1	ERC1	Q96FL8	ERCC2	P18074
ERCC4	Q92889	ERCC5	P28715	ERCC6L	Q2NKX8
ERCC8	Q13216	EREG	O14944	ERF	P50548
ERI1	Q8IV48	ERLIN1	O75477	ERN1	O75460
ERO1A	Q96HE7	ERO1B	Q86YB8	ERP27	P52566
ERP44	Q9BS26	ESCO2	Q56NI9	ESD	P10768
ESPL1	Q14674	ESRRA	P11474	ESYT1	Q9BSJ8
ESYT2	A0FGR8, A0FGR9	ETFA	P13804	ETFB	P38117
ETHE1	O95571	ETNK1	Q9HBU6	ETNK2	Q9NVF9
ETS1	P14921	ETS2	P15036	ETV4	P43268
ETV6	P41212	EVC	P57679	EVC2	Q86UK5
EXO1	Q9UQ84	EXOC1	Q9NV70	EXOC2	Q96KP1
EXOC5	O00471	EXOSC1	Q9Y3B2	EXOSC3	Q9NQT5
EXOSC4	Q9NPD3	EXOSC5	Q9NQT4	EXT2	Q93063
EXTL3	O43909	EYA3	Q99504	EZH2	Q15910, Q92800
F13A1	P00488	F2RL1	P25116	F2RL3	Q96RI0
F5	P12259	FAAH	Q7L5A8	FAAP100	Q0VG06
FAAP24	Q9BTP7	FABP5	Q01469	FADD	Q13158
FADS2	O95864	FAF2	Q96CS3	FAH	P16930
FAHD1	Q6P587	FAM114A2	Q9NRY5	FAM169A	Q9Y6X4
FAM20B	O75063	FAM213B	Q8TBF2	FAM49B	Q9NUQ9
FAM53B	Q5JPI9	FAM83A	Q86UY5	FAM83D	Q9H4H8
FAM96B	Q9Y3D0	FAM98B	Q52LJ0	FANCA	O15360
FANCB	Q8NB91	FANCD2	Q9BXW9	FANCE	Q9HB96
FANCF	Q9NPI8	FANCG	O15287	FANCI	Q9NVI1
FANCL	Q9NW38	FANCM	Q8IYD8	FAR2	Q96K12
FARP1	Q9Y4F1	FARSA	Q9Y285	FARSB	Q9NSD9
FAS	P25445	FASLG	P48023	FASN	P49327
FASTKD5	Q7L8L6	FAU	P62861	FBN1	P35555
FBN2	P35556, Q75N90	FBP1	P09467	FBXL12	Q9NXK8, Q9UKT6
FBXL15	Q9H469	FBXL17	Q9UF56	FBXL18	Q96ME1
FBXL19	Q6PCT2	FBXL20	Q96IG2	FBXL3	Q9UKT7

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
FBXL5	Q9UKA1	FBXO11	Q86XK2	FBXO2	Q96EF6, Q9H4M3, Q9UK22
FBXO22	Q8NEZ5	FBXO30	Q8TB52	FBXO31	Q5XUX0
FBXO32	Q969P5	FBXO5	Q9UKT4	FBXO6	Q9NRD1
FBXO7	Q9Y3I1	FBXO9	Q9UK97	FBXW11	Q9UKB1
FBXW4	P57775	FBXW5	Q969U6	FBXW7	Q969H0
FBXW9	Q5XUX1	FCAR	P24071	FCER1A	P12319
FCER2	P06734	FCGR3A	P08637	FCGR3B	O75015
FCHO2	Q0JRZ9	FCN1	O00602	FDPS	P14324
FDX1	P10109, Q6P4F2	FDXR	P22570	FEM1B	Q9UK73
FEM1C	Q96JP0	FEN1	P39748	FERMT3	Q86UX7
FGD2	Q7Z6J4	FGD3	Q5JSP0	FGD4	Q96M96
FGF9	P31371	FGFBP2	Q9BYJ0	FGFR1	P11362-1, P11362-19
FGFR1OP	O95684	FGFR1OP2	Q9NVK5	FGL2	Q14314
FH	Q16595	FHL2	Q14192	FIG4	Q92562
FIGNL1	Q6PIW4	FKBP1A	P62942	FKBP1B	P68106
FKBP4	Q02790	FKBP9	O95302	FKBPL	Q9UIM3
FLAD1	Q8NFF5	FLI1	Q01543	FLNA	P21333
FLNB	O75369	FLT3	P36888	FLT3LG	P49771
FMNL1	O95466	FMO5	Q01740	FN1	P02751
FNBP1	Q96RU3	FNIP2	Q9P278	FNTB	P49356
FOPNL	O75665	FOS	P01100	FOSB	P53539
FOXJ2	Q9P0K8	FOXK1	P85037	FOXM1	Q08050
FOXO3	O43524	FOXO3B	O43524	FOXO4	P98177
FOXP1	Q9H334-8	FOXP3	Q9BZS1	FOXP4	Q9BZS1
FOXRED1	Q96CU9	FPGT	O14772	FPR1	P62942
FPR2	P21462, P25090	FRAT1	Q92837	FRAT2	O75474
FRMPD3	Q5JV73	FRS2	Q8WU20	FSCN1	Q16658
FTSJ3	Q8IY81	FUCA1	P04066	FUNDC1	Q8IVP5
FUNDC2	P46736	FUOM	A2VDF0	FUT11	P22083
FUT8	Q9BYC5	FXYD2	P54710	FXYD7	P58549, P59646
FYN	P06241	FZD3	Q9NPG1	FZD4	Q9ULV1
FZD5	Q13467	FZD8	Q9H461	G0S2	P27469
GAA	P10253	GAB1	Q13480	GAB2	Q9UQC2
GAB3	Q8WWW8	GABARAPL1	Q9H0R8	GABARAPL2	P60520
GABBR1	Q9UBS5	GABPA	Q06546	GABRR2	P28476
GADD45GIP1	Q8TAE8	GALC	P54803	GALK1	P51570
GALM	Q96C23	GALNT10	Q86SR1	GALNT11	Q8NCW6
GALNT12	Q8IXK2	GALNT18	Q6P9A2	GALNT3	Q14435
GALNT4	Q8N4A0	GALNT6	Q14435	GALNT8	Q9NY28
GAPDH	P04406	GAPVD1	Q14C86	GAR1	Q9NY12
GARS	P41250	GART	P22102	GAS1	P54826
GATA2	P23769	GATAD2A	Q86YP4	GATAD2B	Q8WXI9
GATM	P50440	GBA	P04062	GBA2	Q9HCG7
GBP1	P32455, Q9H0R5	GBP2	P32456, Q9H0R5	GBP3	Q9H0R5
GCC1	Q96CN9	GCC2	Q8IWJ2	GCHFR	P30047
GCLM	P48507	GCNT1	Q02742	GDI2	P50395
GEM	P55040	GEMIN2	O14893	GEMIN5	Q8TEQ6

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
GEMIN6	Q8WXD5	GEN1	Q17RS7	GFER	P55789
GFI1	Q99684	GFM2	Q969S9	GFOD1	Q9NXC2
GFRA3	O60609	GGCT	O75223	GGCX	P38435
GGH	Q92820	GGPS1	O95749	GGT1	P19440
GID8	Q9NWU2	GINS1	Q14691	GINS2	Q9Y248
GINS3	Q9BRX5	GINS4	Q9BRT9	GJC1	P36383
GLB1	P16278	GLDC	P23378	GLE1	Q53GS7
GLI2	P10070	GLI3	P10071	GLO1	Q04760
GLRX5	Q86SX6	GLTP	Q9NZD2	GLUL	P15104
GLYCTK	Q8IVS8	GM2A	P17900	GMNN	O75496
GMPPB	Q9Y5P6	GMPR	P36959	GMPR2	P36959
GMPS	P49915	GNA13	Q14344	GNA15	P30679
GNAI1	P63096	GNAL	P38405	GNAQ	P50148
GNAS	P63092, Q5JWF2	GNAZ	P19086	GNB2	P62879
GNB3	P16520	GNB4	Q9HAV0	GNG10	P50151
GNG2	P59768	GNG5	P63218	GNG7	O60262
GNLY	P22749	GNPAT	O15228	GNS	P15586
GOLGA1	Q92805	GOLGA4	Q13439	GOLGA7	Q7Z5G4
GOLIM4	O00461	GOLM1	Q8NBJ4	GON7	Q9BXV9
GOPC	Q9HD26	GOSR1	O95249	GOT1	P17174
GOT2	P00505	GPAT2	Q6NUI2	GPAT3	Q53EU6
GPAT4	Q53EU6, Q86UL3	GPBAR1	Q8TDU6	GPC1	P35052
GPCPD1	Q9NPB8	GPD1L	Q8N335	GPHN	Q9NQX3
GPI	P06744	GPKOW	Q92917	GPNMB	Q14956
GPR132	Q9UNW8	GPR15	P49685	GPR18	Q14330
GPR25	O00155	GPR35	Q9HC97	GPR55	Q9Y2T6
GPR68	Q15743	GPR83	Q9NYM4	GPS2	Q13227
GPSM2	P81274	GPX1	P07203	GPX3	O75715, P22352
GPX4	P36969	GRAP	Q13588	GRAP2	O75791
GRB10	Q13322	GREB1	Q4ZG55	GRIK4	Q16099
GRIP1	Q15596	GRK3	P35626	GRK4	P32298-1
GRK5	P34947	GRM2	Q14416	GRPEL2	Q8TAA5
GRSF1	Q12849	GSC	P56915	GSN	P06396
GSPT1	P15170, Q8IYD1	GSPT2	Q8IYD1	GSR	P00390-2
GSS	P48637	GSTM2	P28161, P46439	GSTO1	P78417, Q9H4Y5
GSTP1	P09211	GSTZ1	O43708	GTF2A1	P52655
GTF2A2	P52657	GTF2E1	P29083	GTF2E2	P29084
GTF2F1	P35269	GTF2H1	P32780	GTF2H3	Q13889
GTF3C1	Q12789	GTF3C3	Q9Y5Q9	GTF3C6	Q969F1
GTPBP1	Q9BX10	GTPBP2	Q9BX10	GTSE1	Q9NYZ3
GUCA1B	Q9UMX6	GUK1	Q16774	GYG1	P46976
GYS1	P13807	GZMH	P20718	GZMM	P51124
H2AFJ	Q9BTM1	H2AFX	P16104	H2AFZ	P0C0S5
H3F3A	P84243	H3F3B	P84243	HACD3	Q9P035
HACD4	Q5VWC8	HACE1	Q8IYU2	HADH	P40939, Q16836
HAGH	Q16775-2	HAL	P42357	HAS3	O00219
HAT1	O14929	HAUS1	Q96CS2	HAUS2	Q9NVX0
HAUS3	Q68CZ6	HAUS4	Q9H6D7	HAUS7	Q99871
HAUS8	Q9BT25	HAVCR1	Q96D42	HBP1	Q8IV16

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
HBQ1	P69905	HCAR2	Q8TDS4	HCCS	P53701
HCFC1	P51610	HCFC2	Q9Y5Z7	HCK	P08631
HDAC1	Q13547, Q92769	HDAC11	Q96DB2	HDAC2	Q92769
HDAC4	P56524	HDAC9	Q9UKV0	HDGF	P51858
HDLBP	Q00341	HECW2	Q9P2P5	HELLS	Q9NRZ9
HERC1	Q15751	HERC3	Q15034	HERC4	Q15034, Q5GLZ8
HERPUD1	Q15011	HES1	Q14469	HEXA	P06865
HEY2	Q9UBP5	HFE	Q30201	HGD	Q93099
HGS	O14964	HHEX	Q03014	HIC1	Q14526
HIGD1A	Q9Y241	HIKESHI	Q53FT3	HILPDA	Q9Y5L2
HINT1	P49773	HINT2	Q9BX68	HIP1	O00291
HIPK1	Q86Z02	HIPK2	Q9H2X6	HIRA	P54198
HIST1H1B	P16401	HIST1H1E	P10412	HIST1H2AC	Q93077
HIST1H2AG	P0C0S8	HIST1H2AH	Q96KK5	HIST1H2AI	P20671
HIST1H2AL	P0C0S8	HIST1H2BD	P58876	HIST1H2BG	P62807, Q93079
HIST1H2BH	P58876, Q93079	HIST1H2BJ	P06899	HIST1H2BK	O60814, Q99880
HIST1H2BL	Q99880	HIST1H2BM	Q99879	HIST1H3A	P68431
HIST1H3B	P68431	HIST1H3C	Q71DI3	HIST1H3E	P68431
HIST1H3F	P68431	HIST1H3G	P68431	HIST1H3H	P68431
HIST1H3J	P68431	HIST1H4A	P62805	HIST1H4D	P62805
HIST1H4I	P62805	HIST1H4L	P62805	HIST2H2AB	Q8IUE6
HIST3H2BA	Q8N257	HIST3H2BB	Q8N257	HJURP	Q8NCD3
HK1	P19367	HK2	P52789	HK3	P52790
HLA-B	P01889	HLA-C	P10321	HLA-E	P13747
HLA-F	P30511	HLA-G	P17693	HLTF	Q14527
HMBS	P08397	HMGA1	P17096	HMGB2	P26583
HMGCL	P35914, Q8TB92	HMGCR	P04035-1, P04035-2	HMMR	O75330
HMOX1	P09601	HMOX2	P30519	HNMT	P50135
HNRNPA1	P09651	HNRNPDL	O14979	HNRNPF	P52597
HNRNPH1	P31943, P55795	HNRNPL	P14866	HOMER2	Q9NSB8
HOXA3	O43365	HOXB4	P17483	HOXC4	P09017
HPDL	Q96IR7	HPGD	P15428	HPGDS	O60760
HPRT1	P00492	HPS1	Q92902	HPSE	Q9Y251
HRAS	P01112	HRASLS2	Q9NWW9	HRASLS5	Q96KN8
HS6ST1	O60243	HSBP1	O75506	HSD17B10	Q99714
HSD17B4	P51659	HSD17B6	O14756	HSD3B7	Q9H2F3
HSPA13	P48723	HSPA1A	P0DMV8	HSPA1B	P0DMV8, P0DMV9
HSPA4	P34932	HSPA4L	O95757	HSPA5	P11021
HSPA8	P11142	HSPB11	Q9Y547	HSPD1	P10809
HSPE1	P61604	HSPH1	Q92598	HTATSF1	O43719
HUS1	O60921	HYAL3	P38567	IARS	P41252
IARS2	Q9NSE4	ICAM2	P13598	ICMT	O60725
ICOS	Q9Y6W8-1	ID1	P41134	ID3	Q02535
IDH1	O75874	IDH2	P48735	IDH3B	O43837
IDH3G	P51553	IDI1	Q13907	IDS	P22304
IFI27	P40305	IFI27L2	P09912	IFI30	P13284
IFI6	P09912	IFIH1	Q9BYX4	IFIT1	P09914
IFIT2	P09913	IFNAR1	P17181	IFNGR1	P15260
IFT172	Q9UG01	IFT20	Q8IY31	IFT22	Q9H7X7

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
IFT43	Q96FT9	IFT46	Q9NQC8	IFT81	Q8WYA0
IGF1	P05019	IGF1R	P08069	IGF2BP3	O00425
IGF2R	P11717	IGFBP2	P18065	IGFBP3	P17936
IGFBP5	P24593	IGFBP6	P24592	IGFBP7	Q16270
IGHA1	P01876	IGHA2	P01877	IGHD	P01880
IGHEP2	P19823	IGHG1	P01857, P01861	IGHG2	P01859
IGHM	P01871, P01880	IGHV1-46	P01743	IGHV3-23	P01764
IGHV3-7	P01780	IGHV4-59	P01825	IGKC	P01834
IGKV4-1	P06312	IGLC2	P01834, P0DOY2	IGLC3	P0DOY3
IGLV1-47	P01700	IGLV2-11	P01706, Q5NV84	IGLV2-14	P01704
IGLV3-1	P01715	IGLV3-12	Q5NV85	IK	Q13123
IKBIP	Q70UQ0	IKBKB	O14920	IKBKE	Q14164
IKBKG	Q9Y6K9	IL10RA	Q13651	IL11RA	Q14626
IL12A	P29459	IL13RA1	P78552	IL15	P40933
IL15RA	Q13261	IL16	Q14005	IL17C	Q9P0M4
IL17RA	Q96F46	IL17RC	Q8NAC3	IL17RD	Q8NFM7
IL18RAP	O95256	IL1B	P01583	IL1RL1	Q01638, Q01638-2
IL1RN	P18510	IL21R	Q9HBE5	IL23A	Q9NPF7
IL2RA	P01589	IL2RB	P14784	IL32	P24001
IL4R	P24394	IL6R	P08887, P08887-2	IL7R	P16871
IL9R	Q01113	ILK	Q13418	IMMT	Q16891
IMPA1	P29218	INCENP	Q9NQS7	ING3	Q9NXR8
INO80	Q9ULG1	INO80D	Q53TQ3	INPP4B	O15327
INPP5D	Q92835	INPP5K	Q9BT40	INPPL1	O15357
INTS11	Q5TA45	INTS12	Q96CB8	INTS13	Q9NVM9
INTS14	Q96SY0	INTS2	Q9H0H0	INTS5	Q6P9B9
INTS6	Q9UL03	INTS7	Q9NVH2	INTS8	Q75QN2
IP6K1	Q92551, Q96PC2	IP6K2	Q9UHH9	IPO7	O95373
IQCE	Q6IPM2	IQGAP3	Q86VI3	IRAK2	O43187
IRAK4	Q9NWZ3	IRF1	P10914	IRF2	P14316
IRF3	Q92985	IRF6	O14896	IRF8	Q02556
IRS1	P35568	IRS2	Q9Y4H2	ISCA1	Q9BUE6
ISCA2	Q86U28	ISCU	Q9H1K1-1	ISG15	P05161
ISG20	Q96AZ6	ISM1	Q9NSE4	IST1	P53990
ISY1	Q9ULR0	ITCH	Q96J02	ITFG2	Q969R8
ITGA1	P56199	ITGA2	P17301	ITGA2B	P08514
ITGA3	P26006	ITGA5	P08648	ITGA6	P23229
ITGAM	P11215	ITGAV	P06756	ITGB1	P05556
ITGB2	P05107	ITGB3BP	Q13352	ITGB7	P26010
ITIH4	Q14624	ITPKB	P27987	ITSN2	Q9NZM3
IVD	P26440	IWS1	Q96ST2	JADE1	Q6IE81
JADE2	Q9NQC1	JAG1	P78504	JAG2	Q9Y219
JAK1	P23458	JAK3	P52333	JAML	Q86YT9
JARID2	Q92833	JMJD1C	Q15652	JMJD6	Q6NYC1
JMY	Q8N9B5	JOSD1	Q15040	JOSD2	Q8TAC2
JUN	P05412	JUNB	P17275	KANSL1	Q7Z3B3
KANSL2	Q9H9L4	KANSL3	Q9P2N6	KARS	Q15046
KAT2B	Q92831	KAT5	Q92993	KAT6A	Q92794
KAT6B	Q8WYB5	KAT7	O95251	KBTBD6	Q86V97

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KCNH3	Q9ULD8	KCNH8	Q96L42	KCNK5	O95279
KCNK6	Q9Y257	KCNK7	Q9Y2U2	KCNMB1	Q16558
KCNN4	O15554	KCNQ1	P51787	KCTD10	Q8WZ19
KCTD15	Q96SI1	KCTD7	Q96MP8	KDELR1	P24390
KDELR2	P24390, P33947	KDELR3	O43731	KDM2A	Q9Y2K7
KDM3A	Q9Y4C1	KDM4C	Q9H3R0	KDM4D	Q6B0I6
KDM5A	P29375	KDM5B	Q9UGL1	KDM5D	Q9BY66
KDM6A	O15550	KDM6B	O15054	KDM7A	Q6ZMT4
KEAP1	Q14145	KHDRBS1	Q07666	KHSRP	Q92945
KIAA0141	Q14154	KIAA0355	O15063	KIAA1549	Q9HCM3
KIDINS220	Q9ULH0	KIF11	P52732	KIF13A	Q9H1H9
KIF13B	Q9NQT8	KIF14	Q15058	KIF15	Q9NS87
KIF18A	Q8NI77	KIF18B	Q86Y91	KIF19	Q2TAC6
KIF20A	O95235	KIF20B	Q96Q89	KIF22	Q14807
KIF23	Q02241	KIF24	Q5T7B8	KIF27	Q86VH2
KIF2C	Q99661	KIF3A	Q9Y496	KIF3C	O14782
KIF4A	O95239	KIF4B	O95239, Q2VIQ3	KIFAP3	Q92845
KIFC1	Q9BW19	KIFC3	Q9BVG8-2	KIR2DL1	P43626
KIR2DL3	P43628	KIR2DS4	P43632	KIR3DL1	P43629, Q99706
KIR3DL2	P43630	KIRREL2	Q6UWL6	KIT	P10721
KLC2	Q6P597, Q9H0B6	KLC4	Q9NSK0	KLF16	Q9BXK1
KLF4	O43474	KLF5	Q13887	KLF9	Q13886
KLHL11	Q9NVR0	KLHL12	Q53G59	KLHL2	O95198
KLHL25	Q9H0H3	KLHL3	Q9UH77	KLHL42	Q9P2K6
KLHL5	Q96PQ7	KLHL9	Q9P2J3	KLKB1	P03952
KLRB1	Q12918	KLRD1	Q13241	KLRF1	Q9NZS2
KLRG1	Q96E93	KMT5A	Q9NQR1	KMT5B	Q4FZB7, Q86Y97
KNL1	Q8NG31	KNTC1	P50748	KPNA2	P52292
KPNA5	O60684	KPNA6	O60684	KPNB1	Q14974
KPTN	Q9Y664	KRAS	P01116-1, P01116-2	KRBA1	A5PL33
KRR1	Q13601	KRT18	P05783	KRT2	P35908
KRT7	P08729	KRT72	Q14CN4	KRT73	Q86Y46
KRTAP16-1	A8MUX0	KYAT3	Q6YP21	KYNU	Q16719
L1CAM	P32004	L2HGDH	Q9H9P8	L3MBTL1	Q9Y468
LAG3	P18627	LAGE3	Q14657	LAIR2	Q6ISS4
LAMA2	P24043	LAMP1	P11279	LAMP2	P13473
LAMP3	P34810	LAMTOR2	Q9Y2Q5	LAMTOR3	Q9UHA4
LARGE1	O95461	LARP1	Q6PKG0	LARS2	Q15031
LATS1	O95835	LATS2	Q9NRM7	LBR	Q14739
LCAT	P04180	LCK	P06239	LCMT1	Q9UIC8
LCMT2	O60294	LCN12	Q6JVE5	LCP2	Q13094
LDHAL6A	Q6ZMR3	LDHB	P07195	LDLRAP1	Q5SW96
LEAP2	Q969E1	LEF1	Q9UJU2	LEMD2	Q8NC56
LEMD3	Q9Y2U8	LENG1	Q96BZ8	LEO1	Q8WVC0
LGALS1	P09382	LGALS3	P17931	LGALS3BP	Q08380
LGI4	Q8N135	LGMN	Q99538	LGR4	Q9BXB1
LGR6	Q9HBX8	LHFPL5	Q8TAF8	LIF	P15018

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LIG4	P49917	LILRA1	O75019	LILRA2	Q8N149
LILRA4	P59901	LILRA5	Q8N423	LILRA6	Q6PI73
LILRB2	Q8N423	LILRB3	O75022	LILRB4	Q8NHJ6
LIMA1	Q8TDZ2	LIMD1	Q9UGP4	LIMK2	Q96GF1
LIMS1	P48059	LIMS2	Q7Z4I7	LIN52	Q52LA3
LIN54	Q6MZP7	LIN7A	O14910	LIN9	Q5TKA1
LIPA	P18054	LIPC	P11150	LIPE	Q05469
LIPT1	Q9Y234	LMAN2L	Q9H0V9	LMF1	Q96S06
LMNA	P02545	LMNB1	P20700	LONP1	P36776
LOXL1	P58215, Q08397	LOXL2	Q9Y4K0	LOXL4	Q96JB6
LPAR6	P43657	LPCAT1	Q8NF37	LPCAT3	Q6P1A2
LPCAT4	Q643R3	LPGAT1	Q92604	LRFN3	Q9BTN0
LRFN4	Q6PJG9	LRIG1	Q96JA1	LRMP	Q12912
LRP1	Q07954	LRP5	O75197	LRP8	Q14114
LRPPRC	P42704	LRR1	Q96L50	LRRC41	Q15345
LRRC8A	Q8IWT6	LRRC8B	Q6P9F7	LRRC8D	Q7L1W4
LRRK1	Q96J92	LRTOMT	Q8WZ04	LSM11	P83369
LSM2	Q9Y333	LSM4	Q9Y4Z0	LSM5	Q9Y4Y9
LSM6	P62312	LST1	Q9Y6L6	LTA	P01374
LTB	Q06643	LTBP3	Q14767, Q9NS15	LTBP4	Q8N2S1
LTBR	P36941	LTK	Q9UM73	LUC7L3	O95232
LY6E	Q16553	LY96	Q9Y6Y9	LYL1	P12980
LYN	P07948	LYPD3	O95274	LYPLA1	O75608
LYRM2	Q9NU23	LYRM4	Q9HD34	LYRM7	Q5U5X0
LYZ	P61626	MAD2L1	Q13257	MAD2L2	Q9UI95
MADD	Q8WXG6-3	MAF1	Q9H063	MAFB	Q9Y5Q3
MAFF	Q9ULX9	MAFG	O15525	MAFK	O60675
MAGEF1	Q96MG7	MAGOHB	P61326, Q96A72	MAL	Q969V6
MALSU1	Q96EH3	MALT1	Q9UDY8	MAML1	Q92585
MAML3	Q96JK9	MAMLD1	Q13495	MAN1A1	P33908
MAN1A2	O60476	MAN2A1	Q16706	MAN2B2	Q9Y2E5
MAN2C1	Q9NTJ4	MANBA	O00462	MANEA	Q5SRI9
MAP1LC3B	Q9GZQ8	MAP2K1	Q02750	MAP2K3	P46734
MAP2K4	P45985	MAP2K7	O14733	MAP3K11	Q16584
MAP3K14	Q99558	MAP3K3	Q99759	MAP3K7	O43318
MAP3K8	P41279	MAPK11	Q15759	MAPK6	Q16659
MAPK7	Q13164	MAPKAP1	Q9BPZ7	MAPKAPK2	P49137
MAPKAPK3	Q16644	MAPKAPK5	Q8IW41	MAPRE3	Q9UPY8
MAPT	P10636	MARCO	Q9UEW3	MARK3	P27448
MASTL	Q96GX5	MATK	P42679	MAVS	Q7Z434
MB21D1	Q8N884	MBD3	O95983	MBOAT1	Q6ZNC8
MBOAT2	Q6ZWT7	MBOAT7	Q96N66	MBP	P13727
MBTPS1	Q14703	MBTPS2	O43462	MC1R	Q01726
MCAM	P43121	MCAT	Q8IVS2	MCCC1	Q96RQ3
MCCC2	Q9HCC0	MCEMP1	Q8IX19	MCF2	P10911
MCFD2	Q8NI22	MCM10	Q7L590	MCM2	P33993, P49736
MCM3	P25205	MCM4	P33991	MCM5	P33992
MCM6	Q14566	MCM7	P33993	MCM8	Q9UJA3
MCOLN3	Q8TDD5	MCPH1	Q8NEM0	MCRIP1	C9JLW8

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
MCU	Q8NE86	MCUB	Q9NWR8	MDH1	P40925, P40926
MDH2	P40926	MDK	P21741	MDM2	Q00987
MDM4	O15151	ME2	P23368	ME3	Q16798
MED10	Q9BTT4	MED11	Q9P086	MED13L	Q71F56
MED15	Q96RN5	MED16	Q9Y2X0	MED17	Q9NVC6
MED18	Q9BUE0	MED20	Q9H944	MED21	Q13503
MED26	O95402	MED31	Q9Y3C7	MED7	O43513
MED8	Q96G25	MEF2C	Q06413	MEF2D	Q14814
MEFV	O15553	MEI1	O43521	MELTF	P08582
MERTK	Q12866	METAP2	P50579	METTL21A	Q8WXB1
MEX3C	Q5U5Q3	MFAP1	P55081	MFAP3	P55082
MFSD4B	Q5TF39	MGAM	O43451	MGAT2	Q10469
MGAT3	Q09327	MGAT4A	Q9UM21	MGAT4B	Q9UBM8, Q9UQ53
MGLL	Q99685	MGME1	Q9BQP7	MGRN1	O60291
MIB1	Q86YT6	MICB	Q29980	MID1IP1	Q92748, Q9NPA3
MIF	P14174	MIGA2	Q7L4E1	MINPP1	Q9UNW1
MIS18A	Q9NYP9	MIS18BP1	Q6P0N0	MKKS	Q9NPJ1
MKL1	Q969V6	MKLN1	Q9UL63	MKNK1	Q9BUB5
MKRN1	Q9UHC7	MKS1	Q9NXB0	MLC1	P60660
MLEC	Q14165	MLH1	P40692	MLX	Q9UH92
MMACHC	Q9Y4U1	MMADHC	Q9H3L0	MMP17	Q9ULZ9
MMP25	Q9NPA2	MMRN1	Q13201	MMS19	Q96T76
MNAT1	P51948	MND1	Q9BWT6	MNDA	P41218
MOCS1	Q9NZB8-1, Q9NZB8-5	MOCS3	O95396	MOGS	Q13724
MON1A	Q86VX9	MON1B	Q7L1V2	MORC2	Q9Y6X9
MPC2	O95563	MPDU1	O75352	MPG	P29372
MPHOSPH6	Q99547	MPHOSPH8	Q99549	MPI	P34949
MPIG6B	O95866	MPP1	Q00013	MPRIP	Q6WCQ1
MPST	P25325	MPZL2	Q86YT9	MRC2	Q9UBG0
MRM1	Q6IN84	MRM2	Q9UI43	MRM3	Q9HC36
MRPL10	Q7Z7H8	MRPL11	Q9Y3B7	MRPL12	P52815
MRPL13	Q9BYD1	MRPL14	Q6P1L8	MRPL15	P49406, Q9P015
MRPL16	Q9NX20	MRPL17	Q9NRX2	MRPL18	Q9H0U6
MRPL19	P49406	MRPL22	Q9NWU5	MRPL27	Q8IXM3, Q9P0M9
MRPL28	Q13084, Q8TCC3	MRPL3	P09001	MRPL30	Q8TCC3
MRPL32	Q6P1L8, Q9BYC8	MRPL34	Q9BQ48	MRPL35	Q9NZE8
MRPL36	Q9P0J6	MRPL37	Q9BZE1	MRPL38	Q96DV4
MRPL39	Q9NYK5	MRPL41	Q8IXM3	MRPL44	Q9H9J2
MRPL45	Q9BRJ2	MRPL46	Q9H2W6	MRPL50	Q8N5N7
MRPL51	Q4U2R6	MRPL52	Q86TS9	MRPL55	Q7Z7F7
MRPL57	Q9BQC6	MRPL58	Q14197	MRPS10	P82664
MRPS11	P82912	MRPS12	O15235	MRPS14	O60783
MRPS15	P82914	MRPS16	Q9Y3D3	MRPS18A	Q9NVS2
MRPS18B	Q9Y676	MRPS18C	Q9Y3D5	MRPS22	P82650
MRPS23	Q9Y3D9	MRPS25	P82663	MRPS28	P82673, Q9Y2Q9
MRPS30	Q9NP92	MRPS31	Q92665	MRPS33	Q9Y291
MRPS34	P82930	MRPS35	P82673, Q9Y2Q9	MRPS36	P82909

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
MRPS5	P82675	MRPS7	Q9Y2R9	MRPS9	P82933
MRRF	Q96E11	MRS2	Q9HD23	MSH3	P20585
MSH5	O43196	MSH6	P52701	MSI2	Q96DH6
MSL2	Q9HCI7	MSMO1	Q15800	MSN	P26038
MSR1	P21757	MSRB1	Q9NZV6	MSRB2	Q9Y3D2
MT-ATP8	P03928	MT-CO2	P00403	MT-CO3	P00414
MT-CYB	P00156	MT-ND1	P03886	MT-ND2	P03891
MT-ND3	P03897	MT-ND4	P03905	MT-ND4L	P03901
MT-ND5	P03915	MT1E	P04732	MTAP	Q13126
MTBP	Q96DY7	MTERF3	Q96E29	MTERF4	Q7Z6M4
MTFMT	Q96DP5	MTHFD1	P11586	MTHFD1L	Q6UB35
MTHFD2	P13995	MTHFD2L	Q9H903	MTHFR	P42898
MTHFS	P49914	MTM1	Q13496	MTMR12	Q9C0I1
MTMR2	Q13614	MTMR3	Q96QG7	MTMR6	Q9Y217
MTO1	Q9Y2Z2	MTOR	P42345	MTRR	Q9UBK8
MTX1	Q13505	MUC1	P15941	MUC13	Q9H3R2
MUC4	Q99102	MUS81	Q96NY9	MUT	P22033
MVD	P53602	MX1	P20591	MXD4	Q14582
MYB	P10242	MYBL1	P10243	MYBL2	P10244
MYC	P01106	MYDGF	Q969H8	MYH10	P35580
MYH3	P11055	MYL12A	P19105	MYL12B	O14950
MYL6	P60660	MYL6B	P14649	MYLIP	Q8WY64
MYLPF	Q96A32	MYO18B	P35626	MYO1C	O00159
MYO5B	Q9ULV0	MYO6	Q9UM54	MYSM1	Q5VVJ2
MZT1	Q08AG7	MZT2A	Q6P582	MZT2B	Q6NZ67
N6AMT1	Q9Y5N5	NAA38	Q9BRA0	NABP1	Q96AH0
NABP2	Q9BQ15	NACA	Q13765	NADK	O95544
NADK2	Q4G0N4	NADSYN1	Q6IA69	NAF1	Q15025
NAGS	Q8N159	NAMPT	P43490	NANOS3	P60323
NANS	Q9NR45	NAPA	Q9Y2A7	NAPEPLD	Q6IQ20
NAT10	Q9H0A0	NAT8L	Q8N9F0	NCALD	P61601
NCAPD2	Q15021	NCAPD3	P42695	NCAPG	Q9BPX3
NCAPG2	Q86XI2	NCAPH	Q15003	NCBP1	Q09161
NCEH1	Q6PIU2	NCF2	P19878	NCF4	Q15080
NCKAP1	Q9Y2A7	NCKAP1L	P55160	NCKIPSD	Q9NZQ3
NCOA1	Q15788	NCOA2	Q15596	NCOA6	Q14686
NCOR1	O75376	NCR1	O15118	NCR2	O95944
NCR3	Q06643	NCR3LG1	Q68D85	NCSTN	Q92542
NDC1	Q9BTX1	NDC80	O14777	NDE1	Q9NXR1
NDEL1	Q9GZM8	NDN	Q99608	NDST2	P52849
NDUFA1	O15239	NDUFA11	Q86Y39	NDUFA2	O43678
NDUFA4	O00483	NDUFA5	Q16718	NDUFA8	P51970
NDUFAB1	O14561	NDUFAF1	Q9Y375	NDUFAF3	Q9BU61
NDUFAF5	Q5TEU4	NDUFB1	O75438	NDUFB10	O96000
NDUFB11	Q9NX14	NDUFB3	O43676	NDUFB5	O43674
NDUFB6	O95139	NDUFC1	O43677	NDUFS1	P28331
NDUFS2	O75306	NDUFS4	O43181	NDUFS6	O75380
NDUFS7	O75251	NDUFS8	O00217	NDUFV1	P49821
NDUFV2	P19404	NDUFV3	P56181	NEB	Q15058

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NEDD1	Q8NHV4	NEDD8	Q15843	NEIL1	Q96FI4
NEIL2	Q969S2	NEIL3	Q8TAT5	NEK2	P51955
NEK3	Q9UHY1	NEK4	P45985	NEK7	Q8TDX7
NEO1	O43861	NET1	Q7Z628	NEURL1B	A8MQ27
NF2	P35240	NFAM1	Q8NET5	NFATC2	Q12968, Q13469
NFE2	Q16621	NFE2L2	Q16236	NFIA	Q12857
NFIC	P08651	NFIX	Q14938	NFKB1	P19838
NFKBIA	P25963	NFKBIB	Q15653	NFS1	Q9Y697-2
NFYB	P25208	NGFR	P08138	NHLRC2	Q8NBF2
NHP2	Q9NX24	NIPA1	Q7RTP0	NIPA2	Q8N8Q9
NIPAL2	Q9H841	NIPBL	Q6KC79	NISCH	Q9Y2I1
NKIRAS1	Q9NYS0	NKIRAS2	Q9NYR9	NLE1	Q6Q0C0
NLGN4X	Q8N0W4	NLN	Q9BYT8	NLRC5	Q86WI3
NLRP1	Q9C000	NLRP12	P59046	NME1	P15531
NME2	P22392	NME4	O00746	NME7	Q9Y5B8
NMI	Q13287	NMNAT1	Q9HAN9	NMNAT3	Q96T66
NMRAL1	Q9HBL8	NMT1	P30419	NNMT	P40261
NNT	Q13423	NOC2L	Q9Y3T9	NOCT	Q9UK39
NOL4L	Q86VX2	NOP10	Q9NPE3	NOS3	P29474
NOTCH1	P46531	NOTCH2	Q04721	NPC1	O15118
NPC2	P61916	NPDC1	Q9NQX5	NPHP4	O75161
NPM1	P06748	NPM2	Q86SE8	NPR2	P20594
NPRL2	Q8WTW4	NPRL3	Q12980	NPW	Q8N729
NQO1	P15559	NR1D2	Q14995	NR1H2	P55055
NR1I2	O75469-1, O75469-2, O75469-3, O75469-4, O75469-5, O75469-6, O75469-7	NR1I3	Q14994-1, Q14994-2	NR2F6	P10588
NR3C1	P04150-1, P04150-2, P04150-3, P04150-4, P04150-5, P04150-6, P04150-7, P04150-8, P04150-9	NR3C2	P08235-1, P08235-2, P08235-3, P08235-4	NR4A1	P22736
NR4A2	P43354	NR4A3	Q92570-1, Q92570-2	NRBP1	Q9UHY1
NRF1	Q16656	NRG2	O14511	NRN1	Q9NPD7
NRXN2	P58401, Q9P2S2	NRXN3	Q9HDB5, Q9Y4C0	NSD1	Q96L73
NSD2	O96028	NSDHL	Q15738	NSF	P46459
NSL1	Q96IY1	NSMAF	Q92636	NSMCE1	Q8WV22
NSMCE2	Q96MF7	NSMCE3	Q9UNF1	NSMCE4A	Q8N140, Q9NXX6
NSUN4	Q96CB9	NSUN6	Q8TEA1	NT5E	P21589
NT5M	Q8TCD5, Q9NPB1	NTHL1	P78549	NTNG2	Q96CW9
NTRK1	P04629	NTSR1	P30989	NUAK1	O60285
NUB1	Q9Y5A7-1, Q9Y5A7-2	NUBP1	P53384	NUCB1	Q02818
NUDT1	P36639-1, P36639-2, P36639-3, P36639-4	NUDT10	Q9BW91-1	NUDT11	Q96G61
NUDT13	Q86X67	NUDT2	P50583	NUDT21	O43809
NUDT3	O95989	NUDT5	Q9UKK9	NUDT8	Q8WV74
NUF2	Q9BZD4	NUMA1	Q14980	NUMB	P49757
NUP155	O75694	NUP160	Q12769	NUP188	Q5SRE5
NUP205	Q92621	NUP37	Q8NFH4	NUP43	Q8NFH3
NUP50	Q9UKX7	NUP54	Q7Z3B4	NUP58	Q9BVL2-1, Q9BVL2-2

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
NUP62	P37198	NUP98	P52948-5	NUPL2	O15504
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OAS2	P29728	OAS3	Q9Y6K5	OASL	Q15646
OAT	P04181	OAZ1	P54368	OCA2	Q04671
OFD1	O75665	OGFOD1	Q8N543	OIP5	O43482
OMA1	Q96E52	OPN3	Q9H1Y3	OR2W3	Q7Z3T1
OR7D2	Q96RA2	ORAI2	Q96SN7	ORC1	Q13415
ORC2	Q13416	ORC3	Q9UBD5	ORC4	O43929
ORC5	O43913	ORC6	Q9Y5N6	ORMDL1	Q9P0S3
ORMDL2	Q53FV1	ORMDL3	Q8N138	OS9	Q13438
OSBP	P22059	OSBPL10	Q9BXB5	OSBPL2	Q9H1P3
OSBPL5	Q9H0X9	OSBPL7	Q9BZF2	OSCAR	Q8IYS5
OST4	P0C6T2	OSTC	Q9NRP0	OTOF	Q9HC10
OTUB2	Q96DC9	OTUD1	Q5VV17	OTUD3	Q5T2D3
OTUD5	Q96G74	OVGP1	Q12889	OXCT1	P55809, Q9BYC2
P2RX4	Q99571	P2RX7	Q99572	P2RY1	P47900
P2RY10	O00398	P2RY13	Q9BPV8	P2RY14	Q15391
P2RY2	P41231	P3H1	Q32P28	P4HA2	O15460
PA2G4	Q9UQ80	PAAF1	Q9BRP4	PABPN1	Q86U42
PACS1	Q6VY07	PACSIN1	Q9BY11	PACSIN2	Q9UNF0
PADI2	Q9Y2J8	PADI4	Q9UM07	PAFAH1B1	P43034
PAFAH1B3	Q15102	PAFAH2	Q99487	PAG1	Q9NWQ8
PAGR1	Q9BTK6	PAICS	P22234	PAK1	Q13153
PAK2	Q13177	PAK4	O96013	PAK6	Q9NQU5
PALB2	Q86YC2	PAN3	Q58A45	PANK1	Q8TE04
PANK2	Q9BZ23	PAPD5	Q8NDF8	PAPSS1	O43252
PAPSS2	O95340	PARD6A	Q9NPB6	PARD6G	Q9BYG4
PARG	Q86W56	PARK7	Q99497	PARN	O95453
PARP1	P09874	PARP14	Q460N5	PARP6	Q2NL67
PARP8	Q8N3A8	PARP9	Q8IXQ6	PARS2	Q7L3T8
PARVB	Q9HBI1	PBX2	Q9Y4H4	PCBD1	P61457
PCBP1	Q15365	PCBP2	Q15366	PCBP4	P57723
PCGF5	Q86SE9	PCK2	Q16822	PCLAF	Q15004
PCMT1	P22061	PCNA	P12004	PCP2	Q8IVA1
PCSK6	P29122	PCSK7	Q16549	PCTP	Q9UKL6
PCYOX1L	Q8NBM8	PDAP1	Q13442	PDCD11	Q14690
PDCL	Q13371	PDE2A	O00408	PDE3B	Q13370
PDE4D	Q08499	PDE5A	O76074	PDE7A	Q13946
PDE7B	Q9NP56	PDE8A	O60658	PDGFD	Q9GZP0
PDHB	P11177	PDIA5	Q14554	PDIA6	Q15084
PDK4	Q16654	PDLIM7	Q9NR12	PDP1	Q9P0J1
PDP2	Q9P2J9	PDS5A	Q29RF7	PDSS1	Q5T2R2
PDXK	O00764	PDZD11	Q5EBL8	PEAK1	Q9H792
PEBP1	P30086	PELI1	Q96FA3	PELI2	Q9HAT8
PELO	Q9BRX2	PEMT	Q9UBM1	PER1	O15534
PERP	Q96FX8	PET100	P0DJ07	PEX1	O43933
PEX11B	O96011	PEX12	O00623	PEX2	P28328
PEX5	P50542-1, P50542-2	PFAS	O15067	PFDN1	O60925
PFDN2	Q9UHV9	PFDN4	Q9NQP4	PFDN5	Q99471

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PFKP	Q01813	PFN1	P07737	PFN2	P07737, P35080
PGAM1	P18669	PGAM5	Q96HS1-1	PGAP1	Q75T13
PGD	P52209	PGGT1B	P53609	PGK1	P00558, P07205
PGM2	P36871, Q96G03	PGM2L1	Q6PCE3	PGP	A6NDG6
PGRMC1	O00264	PGRMC2	O15173	PGS1	Q32NB8
PHB	P35232	PHB2	Q99623	PHC1	P78364
PHC2	P78364, Q8IXK0	PHC3	Q8NDX5	PHF1	O43189
PHF19	Q5T6S3	PHF21A	Q96BD5	PHF8	Q9UPP1
PHGDH	O43175	PHKA2	P46019	PHKG2	P15735
PHLDA1	Q8WV24	PHLPP1	O60346	PHLPP2	O43747
PHYH	O14832	PHYKPL	Q8IUZ5	PI4K2A	Q9BTU6
PI4K2B	Q8TCG2	PI4KA	P42356	PI4KB	Q9UBF8
PIAS1	O75925	PIAS2	O75928-1	PIAS4	Q8N2W9
PICK1	Q9NRD5	PIF1	Q9H611	PIGA	P37287
PIGH	Q14442	PIGK	Q92643	PIGL	Q9Y2B2
PIGM	Q9H3S5	PIGN	O95427	PIGP	P57054
PIGT	Q969N2	PIGU	Q9H490	PIGV	Q9NUD9
PIGX	Q8TBF5	PIK3AP1	Q6ZUJ8	PIK3C2A	O00443
PIK3CD	O00329	PIK3CG	P48736	PIK3R1	P27986
PIK3R3	Q92569	PIK3R4	Q99570	PIK3R5	Q8WYR1
PIK3R6	Q5UE93	PILRA	Q9UKJ1	PILRB	Q9UKJ0
PIM1	P11309	PIP4K2A	P48426	PIP4K2C	Q8TBX8
PIP5K1A	Q99755	PIR	O00625	PITPNB	P48739
PITPNM2	O00562, Q9BZ72	PITRM1	Q5JRX3	PIWIL2	Q8TC59
PIWIL4	Q7Z3Z4	PJA1	Q8NG27	PKD1	P98161
PKD2	Q13563	PKM	P14618	PKMYT1	Q99640
PKN1	Q16512	PKN2	Q16513	PKN3	Q6P5Z2
PKP4	Q99569	PLA2G15	Q8NCC3	PLA2G16	P53816
PLA2G4A	P47712	PLA2G6	O60733	PLAC8	Q9NZF1
PLAUR	Q03405	PLBD1	Q6P4A8	PLCB1	Q9NQ66
PLCB2	Q00722	PLCB3	Q01970	PLCD3	Q8N3E9
PLCG2	P16885	PLCH1	Q4KWH8	PLCH2	O75038
PLEK	P08567	PLEKHA1	Q9HB21	PLEKHA2	Q9HB19
PLEKHA3	Q9HB20	PLEKHG2	Q9H7P9	PLEKHG3	A1L390
PLEKHG5	O94827	PLEKHO2	Q8TD55	PLIN1	O60240
PLK1	P53350	PLK2	Q9NYY3	PLK4	O00444
PLOD1	Q02809	PLPP1	O14494	PLPP6	Q8IY26
PLPPR2	Q96GM1	PLRG1	O43660	PLXNA2	O75051
PLXND1	Q9Y4D7	PM20D1	Q6GTS8	PMAIP1	Q13794
PML	P29590	PMM1	Q92871	PMPCA	Q10713
PMPCB	O75439	PMVK	Q15126	PNKP	Q96T60
PNMT	P11086	PNP	P00491	PNPLA2	Q96AD5
PNPLA6	Q8IY17	PNPLA8	Q9NP80	PNPO	Q9NVS9
POC1A	Q16829	POFUT1	Q9H488	POFUT2	Q9Y2G5
POLA1	P09884	POLA2	Q14181	POLB	P06746
POLD1	P28340	POLD2	P49005	POLD3	Q15054
POLE	Q07864	POLE2	P56282	POLG2	Q9UHN1
POLH	Q9Y253	POLK	Q9UBT6	POLM	Q9NP87

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POLQ	O75417	POLR1B	Q9H9Y6	POLR1C	O15160
POLR2D	O15514	POLR2G	P62487	POLR2H	P52434
POLR2L	P62875	POLR3E	Q9NVU0	POLR3F	Q9H1D9
POLR3GL	Q9BT43	POLR3K	Q9Y2Y1	POLRMT	O00411
POM121	Q96HA1	POM121C	A8CG34, Q96HA1	POMP	Q9Y244
POMT1	Q9Y6A1	POP1	Q9UKB1	POP4	O95707
PPA1	P62714	PPARA	Q07869	PPAT	Q06203
PPCDC	Q96CD2	PPFIA1	Q13136	PPFIA3	O75145
PPFIA4	O75335	PPFIBP1	Q86W92	PPFIBP2	Q8ND30
PPID	Q08752	PPIF	P62937	PPIL1	Q9Y3C6
PPIP5K2	O43314	PPM1A	P35813	PPM1L	Q5SGD2
PPME1	Q9Y570	PPOX	O43612	PPP1CA	P62136
PPP1CB	P62140	PPP1R10	Q96QC0	PPP1R12B	O60237
PPP1R13B	Q96KQ4	PPP1R15A	O75807	PPP1R16A	P24298
PPP1R21	Q9Y6Q2	PPP1R8	Q12972	PPP2CB	P62714
PPP2R2A	P63151	PPP2R2D	Q66LE6	PPP2R3B	Q9Y5P8
PPP2R5A	Q15172	PPP2R5C	Q13362	PPP2R5D	Q14738
PPP3CA	Q08209	PPP3CC	P16298	PPP4C	P60510
PPP5C	P53041	PPT1	Q96H96	PQLC2	Q6ZP29
PRC1	O43663	PRCP	P42785	PRDM1	O75626
PRDM16	Q9HAZ2	PRDM4	Q9UKN5	PRDM8	Q9NQV8
PRDX1	Q06830	PRDX3	P30048	PRDX5	P30044-2
PRDX6	P30041	PREB	Q9HCU5	PRELID1	Q9Y255
PRF1	P14222	PRIM1	P49642	PRIM2	P49643
PRKAB1	Q9Y478	PRKAB2	O43741	PRKACB	P22694
PRKAG2	Q9UGJ0	PRKAR1A	P10644	PRKAR1B	P31321
PRKAR2B	P31323	PRKCA	P17252	PRKCE	Q02156
PRKCH	P24723	PRKCZ	Q05513	PRKD3	O94806
PRKDC	P78527	PRKG2	Q13237	PRKN	O60260
PRKRIP1	Q9H875	PRMT3	O60678	PRMT5	O14744
PRMT6	Q96LA8	PRNP	P04156	PROCR	Q9UNN8
PRPF18	Q99633	PRPF19	Q9UMS4	PRPF3	O43395
PRPF38A	Q8NAV1	PRPS2	P11908	PRR29	Q99943
PRR5	P85299	PRSS23	O95084	PSAP	Q8IWL1
PSAT1	Q9Y617	PSEN1	P49768	PSEN2	P49810
PSIP1	O75475	PSMA3	P25788	PSMA4	P25789
PSMA5	P28066	PSMA7	O14818	PSMB2	P49721
PSMB3	P49720	PSMB5	P28074	PSMB6	P28072
PSMB7	Q99436	PSMB8	P28062	PSMC1	P62191
PSMC2	P35998	PSMC3	P17980	PSMC3IP	Q9P2W1
PSMC4	P43686	PSMC6	P62333	PSMD1	Q99460
PSMD10	O75832	PSMD11	O00231	PSMD12	O00232
PSMD14	O00487	PSMD3	O43242	PSMD4	P55036
PSMD5	Q16401	PSMD6	Q15008	PSMD8	P48556
PSME1	Q06323	PSME3	Q06323	PSMG1	O95456
PSMG2	Q969U7	PSPH	P78330	PSPN	O60542
PSTK	Q8IV42	PTCD3	Q96EY7	PTCH1	Q13635
PTCH2	Q9Y6C5	PTDSS1	P48651	PTGDR	Q13258
PTGDR2	Q9Y5Y4	PTGDS	P31025	PTGER2	P35408, P43116

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PTGER3	P43115	PTGER4	P35408	PTGES3	Q15185
PTGIS	Q16647	PTH2R	P49190	PTK2	Q05397
PTK7	P42679	PTPA	P18433	PTPN11	Q06124
PTPN12	Q05209	PTPN14	Q15678	PTPN18	Q99952
PTPN23	Q9H3S7	PTPN3	P26045	PTPN7	P35236
PTPN9	P43378	PTPRA	P18433	PTPRB	P23467
PTPRC	P08575	PTPRK	Q15262	PTPRN2	Q92932
PTRH2	Q9Y3E5	PTS	Q03393	PTTG1	O95997
PUM1	Q14671	PURA	P30520	PUS3	Q9BZE2
PUS7	Q96PZ0	PWP2	Q15269	PXDN	Q92626
PXMP2	Q9NR77	PXMP4	Q9Y6I8	PXN	P11678
PXYLP1	Q8TE99	PYCARD	Q9ULZ3	PYCR1	P32322
PYCR3	Q53H96	PYGL	P06737	PYGM	P11217
PYGO2	Q9BRQ0	QKI	Q96PU8	QPRT	Q15274
QTRT1	Q9BXR0	RAB10	P61026	RAB11A	P62491
RAB11FIP2	Q7L804	RAB11FIP3	O75154	RAB13	P51153
RAB18	Q9NP72	RAB19	A4D1S5	RAB1B	Q9H0U4
RAB20	Q9NX57	RAB23	Q9ULC3	RAB27B	O00194
RAB29	O14966	RAB2A	P61019	RAB31	Q13636
RAB33A	Q14088	RAB33B	Q9H082	RAB34	Q9BZG1
RAB35	Q15286	RAB38	P57729	RAB39B	Q96DA2
RAB3A	P20336	RAB3B	P20337	RAB3GAP1	Q15042
RAB3GAP2	Q9H2M9	RAB3IP	Q96QF0	RAB40B	Q12829
RAB4A	P20338	RAB4B	P61018	RAB9A	P51151
RABGAP1	Q9Y3P9	RABGEF1	Q9UJ41	RABGGTA	Q92696
RABGGTB	P53611	RAC2	P15153	RAC3	P60763
RACGAP1	Q9H0H5	RAD51	Q06609	RAD51AP1	Q96B01
RAD51C	O43502	RAD51D	O75771	RAD54L	P46100
RAF1	P04049	RALA	P11233	RALB	P11234
RALBP1	Q15311	RALGAPA1	Q6GYQ0	RALGAPB	Q86X10
RAN	P62826	RANBP1	P43487	RANBP2	P49792
RANGAP1	P46060	RAP1B	P61224	RAP1GAP2	Q684P5
RAP1GDS1	P52306	RAP2B	P61225	RAP2C	Q9Y3L5
RAPGEF1	Q13905	RAPGEF2	Q9Y4G8	RARS	P54136
RASA3	Q14644	RASGRP1	O95267	RASGRP2	Q7LDG7
RASSF4	Q8WYP3	RB1	P06400	RB1CC1	Q8TDY2
RBBP4	Q09028	RBBP6	Q7Z6E9	RBBP8	Q99708
RBFOX2	O43251	RBFOX3	A6NFN3	RBKS	Q9H477
RBL1	P28749	RBM14	Q96PK6	RBM22	Q9NW64
RBM25	P49756	RBM28	Q9NW13	RBM39	Q14498
RBM4	Q9BWF3	RBM42	Q9BTD8	RBM44	O60894
RBM4B	P52272	RBPJ	Q06330	RBSN	Q9H1K0
RBX1	P62877	RCAN3	Q9UKA8	RCC1	P18754
RCC1L	Q96I51	RCC2	Q9P258	RCCD1	A6NED2
RCE1	Q9Y256	RCHY1	Q96PM5	RCL1	Q9Y2P8
RCVRN	P35243	RDH10	Q8IZV5	RDH11	Q96NR8
RDH16	O75452	RDX	P35241	REG4	Q6UW15
REL	Q00653	RELA	Q04206	REPS2	Q8NFH8
RETREG1	Q9H6L5	REV1	Q9UBZ9	REV3L	O60673

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REXO2	Q9Y3B8	RFC2	P35249, P35250	RFC3	P40937, P40938
RFC4	P35249, P35250	RFC5	P40937, P40938	RFK	Q969G6
RFT1	Q9NWF4	RFWD2	Q8NHY2	RGL1	Q9NZL6
RGL4	Q12967	RGMB	Q6NW40	RGS1	Q9UGC6
RGS10	O43665	RGS14	O43566	RGS17	Q9UGC6
RGS18	Q9NS28	RGS19	P49795	RGS2	Q9UGC6
RGS3	P49796	RGS6	P49758	RHBDF2	Q6PJF5
RHEB	Q15382	RHNO1	Q9BSD3	RHOB	P62745
RHOF	Q9HBH0	RHOQ	P17081	RHOT2	Q8IXI1
RHOU	Q7L0Q8	RHPN1	Q8TCX5	RICTOR	Q6R327
RIDA	P52758	RIF1	Q5UIP0	RILP	Q96MT3
RIN2	Q8WYP3	RIN3	Q8TB24	RINL	Q6ZS11
RIOK3	O14730	RIPK2	O43353	RIPOR2	Q9Y4F9
RIT1	Q92963	RLN2	P04090	RMI1	Q9H9A7
RMI2	Q96E14	RMND5A	Q9H871	RNASEK	Q6P5S7
RNASEL	Q05823	RND1	Q92730	RNF103	O00237
RNF113A	O15541	RNF114	Q9Y508	RNF115	Q9Y4L5
RNF125	Q96EQ8	RNF126	Q9BV68	RNF135	Q8IUD6
RNF139	Q8WU17	RNF144A	P50876	RNF144B	Q7Z419
RNF168	Q8IYW5	RNF185	Q96GF1	RNF19A	Q9NV58
RNF19B	Q6ZMZ0	RNF214	Q86Y13	RNF216	Q9NWF9
RNF219	Q5W0B1	RNF220	Q5VTB9	RNF34	Q969K3
RNF40	O75150	RNF43	Q68DV7	RNF5	Q96GF1, Q99942
RNF6	Q9Y252	RNF7	Q9UBF6	RNGTT	O60942
RNH1	O60930	RNPC3	Q96LT9	RNY1	O00584
ROBO1	Q9Y6N7	ROBO3	Q96MS0-1	ROR1	Q01973
ROR2	Q01974	RORA	P35398	RP2	O75695
RPA1	P27694	RPA3	P35244	RPF1	P46934
RPL10	P27635, Q96L21	RPL10A	P62906	RPL11	P62913
RPL12	P30050	RPL13A	P40429	RPL15	P61313
RPL17	P18621	RPL18	Q07020	RPL18A	Q02543
RPL19	P84098	RPL21	P46778	RPL22	P35268
RPL23A	P62750	RPL24	P83731	RPL26	P61254, Q9UNX3
RPL26L1	Q9UNX3	RPL27	P61353	RPL27A	P46776, P61353
RPL29	P47914	RPL3	P39023, Q92901	RPL30	P62888
RPL32	P62888, P62910	RPL34	P49207, P62899	RPL35	P42766
RPL35A	P18077, P42766	RPL36A	P83881, Q969Q0	RPL37A	P18077, P61513
RPL38	P63173	RPL39L	Q96EH5	RPL41	P62945
RPL5	P46777	RPL6	Q02878	RPL7	P18124
RPL7A	P18124, P62424	RPL8	P62424, P62917	RPL9	P32969
RPLP2	P05387	RPN1	P04843	RPP25	Q9BUL9
RPP38	P78345	RPP40	O75818	RPRD1B	Q9NQG5
RPS10	P46783	RPS11	P62280	RPS13	P62277
RPS14	P62263	RPS15	P62841	RPS16	P62249
RPS17	P08708	RPS18	P62269	RPS23	P62266
RPS24	P62244, P62847	RPS25	P62851	RPS26	P62854
RPS27	P42677, Q71UM5	RPS27A	P62979, P62987, Q71UM5	RPS27L	Q71UM5
RPS28	P62857	RPS29	P62273	RPS3	P23396

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RPS6	P62753	RPS6KA2	Q15349	RPS6KA3	P51812
RPS6KA5	O75582	RPS6KB2	Q9UBS0	RPS7	P62081
RPS9	P46781	RPTOR	Q8N122	RPUSD4	Q96CM3
RRAD	P55042	RRAGA	Q5VZM2, Q7L523	RRAGB	Q5VZM2
RRM1	P23921	RRM2	P31350	RRM2B	Q7LG56
RRP1	P05386	RRP36	Q96EU6	RSAD2	Q8WXG1
RSF1	Q96T23	RSPO4	Q2I0M5	RSU1	Q15404
RTCB	Q9Y3I0	RTN3	O95197	RTN4RL1	Q86UN2
RUNX3	Q13761	RUVBL1	Q9Y265	RUVBL2	Q9Y230
RWDD1	Q9H446	RWDD2B	Q9Y3V2	RXRB	P28702
RYK	P34925	RYR1	P21817	RYR2	Q92736
S100A10	P60903	S100A11	P31949	S100A12	P80511
S100A8	P05109	S100A9	P06702	S100B	P04271
S1PR1	P21453	S1PR4	O95977	S1PR5	Q9H228
SAE1	Q9UBE0	SAFB2	Q15424	SAMD8	Q96LT4
SAMHD1	Q9Y3Z3	SAMM50	Q9Y512	SAP18	O00422
SAP30	O75446, Q9HAJ7	SAP30BP	Q9UHR5	SARM1	Q6SZW1-1
SARNP	P82979	SARS	P49591	SAT1	Q92546
SATB1	Q01826	SATB2	Q9UPW6	SAV1	Q9H4B6
SBF2	Q86WG5	SCAF1	O14548	SCARB1	Q8WTV0-2
SCARB2	Q14108	SCARF1	Q14162	SCCPDH	Q8NBX0
SCD	O00767	SCNN1A	P37088	SCNN1D	P51172
SCO2	O43819	SCOC	Q9UIL1	SCUBE3	Q8IX30
SDC4	P31431	SDCCAG8	Q86SQ7	SDE2	Q6IQ49
SDHAF4	Q5VUM1	SDHC	Q99643	SDHD	O14521
SDSL	Q96GA7	SEC11A	P67812	SEC11C	Q9BY50
SEC22B	O75396	SEC22C	Q9BRL7	SEC23A	Q15436
SEC23IP	Q9Y6Y8	SEC24B	O95487	SEC24D	O94855
SEC31A	O94979	SEC61B	P60468	SEC61G	P60059
SECISBP2	Q96T21	SELENOP	P49908	SELL	P14151
SELP	P14151	SEM1	P60896	SEMA3A	Q14563
SEMA4A	Q9H3S1	SEMA4F	O95754	SEMA7A	O75326
SENP1	Q9P0U3	SENP8	Q96LD8	SEPHS2	Q99611
SERINC1	Q9NRX5	SERINC3	Q13530	SERINC5	Q86VE9
SERP1	Q9Y6X1	SERPINA1	P01009	SERPINB6	P35237
SERPINB8	P50452	SERPINE2	P07093	SERPINH1	P50454
SETD1A	O15047, Q9UPS6	SETD1B	Q9UPS6	SETD2	Q9BYW2
SETD3	Q86TU7	SETD6	Q8TBK2	SETD7	Q8WTS6
SF1	Q13285	SF3A1	Q15459	SF3B1	O75533
SF3B3	Q15393	SF3B6	Q9Y3B4	SFN	P31947
SFXN4	Q6P4A7	SGCB	Q16585	SGCD	Q92629
SGK3	Q96BR1	SGMS1	Q86VZ5	SGO1	Q5FBB7
SGO2	Q562F6	SGPP1	Q9BX95	SGSH	P51688
SGTA	O43765	SH2B1	Q9NRF2-2	SH2B2	O14492
SH2D1B	O14796	SH2D2A	Q9NP31	SH3BP1	Q9Y3L3
SH3BP4	Q9P0V3	SH3GL1	Q99961	SH3KBP1	Q96B97
SHB	Q15464	SHC1	P29353-1, P29353-2, P29353-3	SHC3	Q92529
SHMT1	P34896	SHMT2	P34897	SHPRH	Q149N8

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
SHTN1	A0MZ66	SIAH1	Q8IUQ4	SIGIRR	Q6IA17
SIGLEC1	Q9BZZ2	SIGLEC12	Q96PQ1	SIGLEC7	Q9Y286
SIGLEC9	Q9Y336	SIGMAR1	Q99720	SIN3B	O75182
SIPA1	Q96FS4	SIRPA	P78324	SIRPB1	O00241
SIRPB2	Q9P1W8	SIRPG	Q9P1W8	SIRT1	Q96EB6
SIRT2	Q8IXJ6	SIT1	Q9NP91	SKA1	Q96BD8
SKA2	Q8WVK7	SKIL	P12757	SKIV2L	Q15477
SKIV2L2	P42285	SKP2	Q13309	SLA	Q13239
SLA2	Q9H6Q3	SLAMF7	Q9NQ25	SLBP	Q14493
SLC11A1	P49279	SLC11A2	P49281	SLC12A6	Q9UHW9
SLC14A1	Q13336	SLC15A3	Q8IY34	SLC15A4	Q8N697
SLC16A1	P53985	SLC16A10	Q8TF71	SLC16A3	O15427
SLC19A1	P41440	SLC1A4	P43007	SLC1A5	Q15758
SLC1A7	O00341	SLC20A1	Q8WUM9	SLC22A15	Q8IZD6
SLC22A17	Q8WUG5	SLC22A4	O76082	SLC23A1	Q9UGH3, Q9UHI7
SLC23A2	Q9UGH3	SLC24A3	Q9HC58	SLC24A4	Q8NFF2
SLC24A5	Q71RS6	SLC25A1	P53007	SLC25A10	Q9UBX3
SLC25A11	Q02978	SLC25A12	O75746	SLC25A15	Q9Y619
SLC25A17	O43808	SLC25A19	Q9HC21	SLC25A20	O43772
SLC25A26	Q70HW3	SLC25A28	Q96A46	SLC25A37	Q9NYZ2
SLC25A4	P12235	SLC25A42	Q86VD7	SLC25A5	P05141, P12236
SLC26A11	Q86WA9	SLC26A2	P50443	SLC26A6	Q9BXS9
SLC27A2	O14975	SLC27A4	Q6P1M0	SLC29A1	Q99808
SLC2A10	O95528	SLC2A7	Q6PXP3	SLC2A9	Q9NRM0
SLC30A5	Q8TAD4	SLC31A1	O15431	SLC33A1	O00400
SLC35A2	P78381	SLC35A3	Q9Y2D2	SLC35B2	Q8TB61
SLC35B4	Q969S0	SLC35C1	Q96A29	SLC37A4	O43826
SLC38A1	Q9H2H9	SLC38A5	Q8WUX1	SLC38A9	Q8NBW4
SLC39A1	Q9NP94, Q9NY26	SLC39A8	Q9C0K1	SLC41A1	Q8IVJ1
SLC43A2	Q8N370	SLC44A2	Q8IWA5	SLC44A3	Q8N4M1
SLC44A5	Q8NCS7	SLC4A10	Q6U841	SLC4A2	P04920
SLC4A3	P48751	SLC4A5	Q9BY07	SLC4A7	Q9Y6M7
SLC52A2	Q9HAB3	SLC5A11	Q8WWX8	SLC5A3	P53794
SLC5A6	Q9Y289	SLC6A4	P31645	SLC6A6	P31641
SLC7A1	P30825	SLC7A11	Q9UPY5	SLC7A6	Q92536
SLC7A7	Q9UM01	SLC8B1	Q6J4K2	SLC9A5	Q14940
SLCO4A1	Q96BD0	SLCO4C1	Q6ZQN7	SLIT3	O75094
SLITRK5	O94991	SLK	Q9H2G2	SLU7	O95391
SLX4	Q8IY92	SMAD4	Q13485	SMAD5	Q99717
SMAD6	O43541	SMAD7	O15105	SMARCA2	P51531
SMARCA5	O60264	SMARCB1	Q12824	SMARCC1	Q92922
SMARCD2	Q92925	SMARCD3	Q6STE5	SMC1A	Q14683
SMC2	O95347	SMC3	Q9UQE7	SMC4	Q9NTJ3
SMC5	Q8IY18	SMC6	Q96SB8	SMG6	Q86US8
SMG9	Q9H0W8	SMIM4	Q8WVI0	SMOX	Q9NWM0-3
SMPD2	O60906	SMPD3	Q9NY59	SMS	P52788
SMUG1	Q53HV7	SMURF1	Q9HCE7	SMURF2	Q9HAU4
SNAI1	O95863	SNAPC1	Q16533	SNAPC3	Q92966
SNAPC4	Q5SXM2	SNAPIN	O95295	SNCA	P37840

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
SNIP1	Q8TAD8	SNRNP25	Q9BV90	SNRNP40	Q96DI7
SNRNP70	P08621	SNRPA	P09012	SNRPA1	P09661
SNRPB2	P08579, P09012	SNRPD1	P62314, P62316	SNRPD2	P62316
SNRPD3	P62318	SNRPF	P62306	SNRPG	P62308
SNRPN	P63162	SNTA1	Q13424	SNUPN	O95149
SNX1	O60749	SNX2	O60749	SNX3	O60493
SNX5	Q9Y5X3	SOAT1	P35610	SOAT2	O75908
SOCS1	O15524	SOCS3	O14543	SOCS4	Q8WXH5
SOCS6	O14544	SOD1	P08294	SORBS1	Q9BX66
SORL1	Q92673	SORT1	Q99523	SOS1	P42766
SOS2	Q07890	SOX13	Q9UN79	SP3	Q9Y2H2
SPAG9	O60271	SPARC	Q14515	SPATA13	Q96N96
SPATA2	Q9UM82	SPC24	Q8NBT2	SPC25	Q15005
SPCS1	Q9Y6A9	SPDL1	Q96EA4	SPEN	Q96T58
SPHK1	Q9NYA1	SPHK2	Q9NRA0	SPIDR	Q14159
SPINT1	O43278	SPINT2	O43291	SPN	P16150
SPNS2	Q8IVW8	SPOCK3	Q9BQ16	SPON2	Q9BUD6
SPOPL	Q6IQ16	SPPL2A	Q8TCT8	SPRED2	Q7Z698
SPRY1	O43609	SPRY2	O43597	SPSB2	Q99619
SPSB3	Q6PJ21	SPTB	P11277	SPTBN5	Q9NRC6
SPTLC2	O15270	SPTLC3	Q9NUV7	SPTSSA	Q969W0
SPTSSB	Q8NFR3	SQLE	Q14534	SQOR	Q9Y6N5
SRD5A1	P18405	SRD5A3	Q9H8P0	SRF	P11831
SRGAP2	O75044	SRGN	P10124	SRI	P30626
SRM	P19623	SRP14	P37108	SRP19	P09132
SRP54	P61011	SRP68	Q9UHB9	SRP72	O76094
SRPK1	Q96SB4	SRPK2	P78362	SRPRA	P08240
SRR	Q9GZT4	SRRM1	Q8IYB3	SRRM2	Q9UQ35
SRSF1	Q07955	SRSF10	O75494, Q8WXF0	SRSF11	Q05519
SRSF2	Q01130	SRSF5	Q13243	SRSF6	Q13247
SRSF7	Q16629	SRSF8	Q9BRL6	SS18	Q15532
SS18L1	O75177	SSNA1	O43805	SSPO	A2VEC9
SSR3	Q9UNL2	SSR4	P51571	SSRP1	Q08945
ST14	Q9Y5Y6	ST3GAL2	Q16842	ST3GAL5	Q9UNP4
ST5	P78524	ST8SIA4	Q92187	ST8SIA6	P61647
STAG3	Q9UJ98	STAM2	O75886	STAMBP	O95630
STAMBPL1	Q96FJ0	STAP2	Q9UGK3	STARD3	Q14849
STARD3NL	O95772	STAT1	P42224	STAT3	P40763
STAT4	Q14765	STAT5B	P51692	STAT6	P42226
STBD1	O95210	STEAP3	Q658P3	STIM1	Q13586
STIP1	P31948	STK10	O94804	STK11IP	Q8N1F8
STK24	Q9Y6E0	STK38	Q15208	STK4	O75914
STMN1	P16949	STOML2	Q9UJZ1	STRADA	Q7RTN6
STRN	O43815	STT3A	P46977	STT3B	Q8TCJ2
STUB1	Q9UNE7	STX18	Q9P2W9	STX1A	Q16623
STX1B	P61266	STX3	Q13277	STX4	Q12846
STX6	O43752	STXBP1	P61764	STXBP2	Q15833
SUCLA2	Q9P2R7	SUCLG1	P53597	SUFU	Q9UMX1
SULT1B1	O75897	SUMF1	Q8NBK3	SUMO2	P61956

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
SUMO3	P55854	SUN1	O94901	SUN2	Q9UH99
SUOX	P51687	SUPT16H	Q9Y5B9	SUPT20H	Q8NEM7
SUPT5H	O00267	SUPT7L	O94864	SURF4	O15260
SUV39H1	O43463	SUV39H2	Q9H5I1	SUZ12	Q15022
SV2A	Q7L0J3	SWAP70	Q9UH65	SYCE2	Q6PIF2
SYF2	O95926	SYMPK	Q92797	SYNE1	Q8NF91
SYNE2	Q8WXH0	SYNJ1	O43426	SYS1	Q8N2H4
SYTL1	Q8IYJ3	SYVN1	Q86TM6	SZT2	Q5T011
TACC3	Q9Y6A5	TACO1	Q9BSH4	TADA1	Q96BN2
TADA2A	O75478	TADA2B	Q86TJ2	TADA3	O75528
TAF1	P21675, Q8IZX4	TAF10	Q12962	TAF11	Q15544
TAF12	Q16514	TAF13	Q15543	TAF1C	Q15572
TAF3	Q5VWG9	TAF4	O00268, Q92750	TAF4B	Q92750
TAF5	Q15542	TAF7	Q15545, Q5H9L4	TAF8	Q7Z7C8
TAF9B	Q16594, Q9HBM6	TAGAP	Q8N103	TAGLN2	P37802
TALDO1	P37837	TANK	Q92844	TAOK1	Q7L7X3
TAOK3	Q9H2K8	TAP1	Q03518	TAP2	Q03519
TAPBP	O15533	TARS	P26639	TAS2R5	Q9NYW4
TATDN2	Q93075	TBC1D10B	Q4KMP7	TBC1D2	Q9BYX2
TBC1D25	Q3MII6	TBC1D4	O60343	TBC1D7	Q9P0N9
TBCA	O75347	TBCC	Q15814	TBCE	Q15813
TBL1XR1	Q9BZK7	TBX21	Q9UL17	TCF25	Q9BQ70
TCF4	Q9NQB0	TCF7L2	Q9NQB0	TCTEX1D2	Q8WW35
TDP1	Q9NUW8	TDP2	O95551	TECTA	O75443
TERF1	P54274	TERF2	Q15554	TES	Q9NZU5
TET1	Q8NFU7	TET2	Q6N021	TF	P02788
TFB1M	Q8WVM0	TFDP1	Q14186	TFDP2	Q14188
TFG	Q92734	TFPT	P0C1Z6	TFR2	Q9UP52
TFRC	P02786	TG	P00488	TGFA	P01135
TGFB1	P01137	TGFB3	P10600	TGFBR2	P37173
TGFBR3	Q03167	TGIF1	Q15583	TGM5	O43548
TGOLN2	O43493	TGS1	Q96RS0	THADA	Q6YHU6
THBD	P07204	THBS1	P07996	THG1L	Q9NWX6
THOC1	Q96FV9	THOC2	Q8NI27	THOC3	Q96J01
THOC6	Q86W42	THOC7	Q6I9Y2	THOP1	P52888
THRA	P10827	THRAP3	Q9Y2W1	THTPA	Q9BU02
THUMPD1	Q9NXG2	TIAM1	Q13009	TICRR	Q7Z2Z1
TIFA	P38919	TIMELESS	Q9UNS1	TIMM10	P62072
TIMM10B	Q9Y5J6	TIMM13	Q9Y5L4	TIMM23	O14925
TIMM23B	O14925	TIMM8B	Q9Y5J9	TIMM9	Q9Y5J7
TIMMDC1	Q9NPL8	TINF2	Q9BSI4	TIPIN	Q9BVW5
TIRAP	P58753	TK1	P04183	TKT	P29401
TLE1	Q04724	TLE2	Q04725	TLE3	Q04726
TLE4	Q04727	TLR3	O15455	TLR6	Q9Y2C9
TLR8	Q9NR97	TM7SF2	O76062	TMBIM1	Q969X1
TMED3	Q9Y3Q3	TMEM126A	Q8IUX1, Q9H061	TMEM177	Q53S58
TMEM186	Q96B77	TMEM223	A0PJW6	TMEM258	P61165
TMEM5	Q9Y2B1	TMEM55B	Q86T03	TMEM63A	O94886
TMEM86A	Q8N661	TMEM87B	Q8NBN3	TMLHE	Q9NVH6

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
TMOD1	P28289	TMOD2	Q9NZR1	TMOD3	Q9NYL9
TMPO	P42167	TMPRSS6	Q8IU80	TMSB4X	P62328
TNF	P01374	TNFAIP1	Q13829	TNFAIP3	P21580
TNFAIP8L1	Q8WVP5	TNFAIP8L2	Q6P589	TNFRSF10A	O00220
TNFRSF10B	O14763	TNFRSF10D	Q9UBN6	TNFRSF11A	Q9Y6Q6
TNFRSF12A	Q9NP84	TNFRSF13C	Q96RJ3	TNFRSF14	Q92956
TNFRSF1A	P19438	TNFRSF8	P28908	TNFSF10	P50591
TNFSF14	O43557	TNFSF15	O95150	TNFSF4	P23510
TNFSF8	P32971	TNIK	Q9UKE5	TNKS	O95271, Q9H2K2
TNKS1BP1	Q9C0C2	TNRC6A	Q8NDV7	TNRC6B	Q9UPQ9
TNRC6C	Q9HCJ0	TNS4	Q8IZW8	TNXB	P22105
TOMM20	Q15388	TOMM40	O96008	TOMM40L	O96008
TOMM70	O94826	TOP1	P11387	TOP2A	P11388
TOPORS	Q9NS56	TOR1A	O14656	TOR1AIP1	Q5JTV8
TOR4A	Q9NXH8	TP53	P04637	TP53BP1	Q12888
TP53I3	Q53FA7	TP53RK	Q96S44	TP73	O15350
TPCN1	Q9ULQ1	TPCN2	Q8NHX9	TPD52L1	Q16890
TPGS2	Q68CL5	TPH1	P17752	TPI1	P60174
TPK1	Q9H3S4	TPM1	P09493	TPM2	P07951
TPM4	P67936	TPP2	P29144	TPST1	O60507
TPST2	O60704	TPX2	Q9ULW0	TRA2B	P62995
TRAC	P01848	TRAF2	Q12933	TRAF3	Q13114
TRAF6	Q9Y4K3	TRAF7	Q6Q0C0	TRAIP	Q9BWF2
TRAK1	Q9UPV9	TRAK2	O60296	TRAPPC1	Q9Y5R8
TRAPPC10	P48553	TRAPPC2	P0DI81	TRAPPC3	O43617
TRAPPC5	Q8IUR0	TRAPPC6B	O75865, Q86SZ2	TRAPPC9	Q96Q05
TRAT1	Q6PIZ9	TREM1	Q9NP99	TREML2	Q5T2D2
TRIB3	Q96RU7	TRIM14	Q14142	TRIM17	Q9Y577
TRIM21	P19474	TRIM27	P14373	TRIM3	O75382
TRIM32	Q13049	TRIM33	Q9UPN9	TRIM35	Q9UPQ4
TRIM37	O94972	TRIM39	Q9HCM9	TRIM46	Q7Z4K8
TRIM47	Q8WV44	TRIM56	Q9BRZ2	TRIM58	Q9C037
TRIM62	Q9BVG3	TRIM69	Q86WT6	TRIO	O75962
TRIP10	Q15642	TRIP12	Q14669	TRIT1	Q9H3H1
TRMT10A	Q8TBZ6	TRMT44	Q8IYL2	TRMT5	Q32P41
TRNT1	Q96Q11	TRPC3	Q13507	TRPC4AP	Q8TEL6
TRPM2	O94759	TRPV3	Q8NET8	TRRAP	Q9Y4A5
TRUB2	O95900	TSC1	Q92574	TSC2	P49815
TSC22D3	Q99576	TSEN34	Q9BSV6	TSEN54	Q7Z6J9
TSG101	Q99816	TSPAN15	O95858	TSPAN5	P62079
TSPOAP1	O95153	TSPYL2	Q9H2G4	TST	Q15477
TTBK2	Q6IQ55	TTC21B	Q7Z4L5	TTC26	A0AVF1
TTC30A	Q86WT1, Q8N4P2	TTC30B	Q8N4P2	TTC9	Q9UIM3
TTF1	Q15361	TTLL1	O95922	TTLL11	Q8NHH1
TTLL12	Q14166	TTLL3	A6PVC2, Q9Y4R7	TTLL7	Q6ZT98
TTN	Q8WZ42	TTYH2	Q9BSA4	TTYH3	Q9C0H2
TUBA1B	P68363	TUBA1C	P68363, Q9BQE3	TUBB	P04350
TUBB1	Q9H4B7	TUBB2A	Q13885, Q9BVA1	TUBB4B	P04350, P68371
TUBB6	Q9BUF5	TUBG1	P23258, Q9NRH3	TUBGCP3	Q96CW5

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
TUBGCP4	Q9UGJ1	TUBGCP6	Q96RT7	TWF1	Q12792
TWF2	Q6IBS0	TWIST1	Q15672	TWNK	Q96RR1
TXK	P42681	TXLNA	P40222	TXN	P10599
TXN2	Q99757	TXNDC11	Q6PKC3	TXNIP	Q9H3M7
TXNL1	O43396	TXNRD1	Q16881	TYK2	P29597
TYMP	P19971	TYMS	P04818	TYROBP	O43914
TYW3	Q6IPR3	TYW5	A2RUC4	U2AF1L4	Q8WU68
UACA	Q9BZF9	UAP1	Q16222	UBA2	Q9UBT2
UBA3	Q8TBC4	UBA52	P62979, P62987	UBA6	A0AVT1
UBA7	P22314, P41226	UBAC1	Q9BSL1	UBB	P62979, P62987
UBE2A	P49459	UBE2C	O00762	UBE2D2	P62837
UBE2F	Q969M7	UBE2G2	P60604	UBE2J1	Q9Y385
UBE2J2	Q8N2K1	UBE2K	P61086	UBE2L3	P68036
UBE2M	P61081	UBE2N	P61088	UBE2O	Q9C0C9
UBE2Q2	Q8WVN8	UBE2S	Q16763	UBE2T	Q9NPD8
UBE2V1	Q13404	UBE2V2	Q15819	UBE3B	Q7Z3V4
UBE4A	Q14139	UBL4A	P11441	UBL5	Q9BZL1
UBN1	Q9NPG3	UBOX5	O94941	UBP1	Q70CQ3
UBR2	Q8IWV7, Q8IWV8	UBXN1	Q04323	UBXN7	O94888
UCHL1	P09936	UCK2	Q9BZX2	UCP3	P55916
UFL1	O94874	UGDH	O60701	UGP2	Q16851
UHMK1	Q8TAS1	UHRF1	Q96T88	UHRF2	Q96PU4
ULBP1	Q9BZM5	ULK1	O75385	ULK3	Q6PHR2
UMPS	P11172	UNC119B	A6NIH7	UNC13B	O14795
UPF1	Q92900	UPF3A	Q9H1J1	UPF3B	Q9BZI7
UPP1	Q16831	UQCC1	Q9NVA1	UQCC2	Q9BRT2
UQCR10	Q9UDW1	UQCRH	P07919	UQCRQ	O14949
URM1	Q9BTM9	UROC1	Q96N76	USF1	Q8NFU3
USF2	Q15853	USMG5	Q96IX5	USO1	O60763
USP1	O94782	USP10	Q14694	USP14	P54578
USP16	Q9Y5T5	USP18	Q9UMW8	USP19	O94966
USP24	Q9UPU5	USP28	Q96RU2	USP3	Q9Y6I4
USP30	Q70CQ3	USP33	Q8TEY7	USP34	Q70CQ2
USP39	Q53GS9	USP4	Q13107	USP42	D6RA61, Q9H9J4
USP43	Q70EL4	USP44	Q70CQ1, Q9H0E7	USP47	Q96K76
USP5	P45974	USP7	Q93009	UST	Q9Y2C2
UTP11	Q9Y3A2	UTP14C	Q5TAP6	UTP15	Q8TED0
UTP18	Q9Y5J1	UTP20	O75691	UTP3	Q9NQZ2
UTP6	Q9NYH9	UTRN	P46939	UTY	O14607
UVRAG	Q9P2Y5	UVSSA	Q2YD98	UXS1	Q8NBZ7
VAMP1	P23763	VAMP2	P63027	VAMP3	Q15836
VANGL1	Q8TAA9	VAPB	O95292	VARS	P26640
VAV2	P52735	VAV3	Q9UKW4	VBP1	P61758
VCL	P18206	VCP	P55072	VCPIP1	Q96JH7
VCPKMT	Q9H867	VDAC1	P21796	VDAC3	Q9Y277
VDR	P11473	VENTX	O95231	VGF	O15240
VIM	P08670	VIT	O75030	VKORC1	Q9BQB6
VKORC1L1	Q8N0U8	VPS11	Q9H270	VPS16	Q9H269
VPS25	Q9BRG1	VPS26A	O75436	VPS33A	Q96AX1

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
VPS35	Q96QK1	VPS37B	A5D8V6, Q9H9H4	VPS39	Q96JC1
VPS4A	Q9UN37	VPS52	Q8N1B4	VPS53	Q5VIR6
VPS54	Q9P1Q0	VPS72	Q15906	VRK1	Q99986
VRK3	Q8IV63	VSIG10	Q9NYS7	VTA1	Q9NP79
VTI1A	Q96AJ9	WAC	Q9BTA9	WARS	P23381
WARS2	Q9UGM6	WASF2	Q9Y6W5	WBP1	P39656
WBP4	O75554	WDCP	Q9H6R7	WDR12	Q9GZL7
WDR18	Q9BV38	WDR19	Q8NEZ3	WDR20	Q8TBZ3
WDR26	Q9H7D7	WDR3	Q9UNX4	WDR34	Q96EX3
WDR4	P57081	WDR45	Q9Y484	WDR45B	Q5MNZ6
WDR48	Q8TAF3	WDR5	P61964	WDR59	Q6PJI9
WDR5B	P61964	WDR6	Q9NNW5	WDR60	Q8WVS4
WDR61	Q9GZS3	WDR70	Q9NW82	WDR91	A4D1P6
WDTC1	Q8N5D0	WEE1	P30291, Q99640	WFS1	O76024
WHAMM	Q8TF30	WHRN	Q9P202	WIPF1	O43516
WNK1	Q9H4A3	WNT3	P56703	WRB	O00258
WSB2	Q9NYS7	WTAP	Q15007	WWC1	Q8IX03
XAB2	Q9HCS7	XCL1	P47992, Q9UBD3	XCL2	P47992, Q9UBD3
XPA	P23025	XPNPEP2	O43895	XPO1	O14980
XPO5	Q9HAV4	XPOT	O43592	XRCC1	P18887
XRCC2	O43543	XRCC3	O43542	XRCC4	Q13426
YAF2	Q8IY57	YARS	P54577	YARS2	Q9Y2Z4
YEATS2	Q9ULM3	YEATS4	O95619	YES1	P07947
YIF1A	O95070	YIPF6	Q96EC8	YME1L1	Q96TA2
YOD1	Q5VVQ6	YPEL5	P62699	YTHDC1	Q96MU7
YWHAE	P62258	YWHAG	P61981	YWHAH	Q04917
YY1	P25490	ZAP70	P43403	ZBP1	Q9H171
ZBTB16	Q05516	ZBTB33	Q86T24	ZC3H12A	Q5D1E8
ZC3H15	Q8WU90	ZC3H4	Q9UPT8	ZC3H6	Q8N5P1
ZC3HAV1	Q7Z2W4	ZC3HC1	Q86WB0	ZCCHC11	Q5TAX3
ZCCHC6	Q5VYS8	ZCCHC7	Q8N3Z6	ZCRB1	Q8TBF4
ZDHHC11	Q9H8X9	ZDHHC5	Q9C0B5	ZDHHC7	Q9NXF8
ZDHHC9	Q9Y397	ZEB1	P37275	ZEB2	O60315
ZFHX3	Q15911	ZFP30	Q9Y2G7	ZFP36	P26651
ZFP36L1	Q07352	ZFP36L2	P47974	ZFP37	P17032, Q9Y6Q3
ZFP69	Q49AA0	ZFPM1	Q8IX07	ZFYVE9	O95405-1
ZKSCAN1	P17029	ZKSCAN3	Q9BRR0	ZKSCAN4	Q969J2
ZKSCAN5	Q9Y2L8	ZKSCAN8	Q15776	ZMIZ1	Q92520
ZNF101	Q8IZC7	ZNF124	Q15973	ZNF131	P52739
ZNF135	P52742, Q6ZN57	ZNF136	P52737	ZNF14	P17017
ZNF140	P52738	ZNF141	Q15928	ZNF154	A6NNF4, Q13106
ZNF16	Q7L2R6	ZNF175	Q9Y473	ZNF181	Q9NR11
ZNF184	Q99676	ZNF189	O75820	ZNF197	O14709
ZNF2	Q9BSG1	ZNF200	P98182	ZNF208	O43345
ZNF211	Q13398	ZNF213	O14771	ZNF215	Q8NE65, Q9UL58
ZNF217	O75362	ZNF223	Q9UK11	ZNF225	P18146
ZNF226	Q9NZL3	ZNF23	P17027	ZNF234	Q14588

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
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ZNF256	Q9Y2P7	ZNF26	P17031	ZNF260	Q8TAQ5
ZNF263	O14978	ZNF266	Q14584	ZNF267	Q14586
ZNF268	Q96BR6	ZNF282	Q9UDV7	ZNF286A	Q9HBT8
ZNF300	Q96RE9	ZNF317	Q96PQ6	ZNF324	O75467
ZNF324B	Q6AW86	ZNF331	Q9NQX6	ZNF333	Q96JL9
ZNF335	Q9H4Z2	ZNF33B	Q06730	ZNF34	Q8IZ26
ZNF350	Q9GZX5	ZNF354B	Q96LW1	ZNF37A	P17032
ZNF382	Q96SR6	ZNF383	Q8NA42, Q8NDQ6, Q9BR84	ZNF385A	Q96PM9
ZNF391	Q8NCK3	ZNF394	Q53GI3	ZNF397	Q96MU6
ZNF398	Q9BS31	ZNF41	P51814	ZNF415	Q09FC8
ZNF418	Q8TF45	ZNF419	Q96HQ0	ZNF426	Q9BUY5
ZNF430	Q9H8G1	ZNF432	O94892	ZNF433	Q8N7K0
ZNF442	Q9H7R0	ZNF446	Q9NWS9	ZNF460	Q14592
ZNF467	Q7Z7K2	ZNF470	Q6ECI4	ZNF484	Q5JVG2, Q8N8J6
ZNF486	Q96H40	ZNF490	Q9ULM2	ZNF517	Q6ZMY9
ZNF519	Q8TB69	ZNF521	Q96K83	ZNF543	Q08ER8
ZNF544	Q6NX49	ZNF546	Q86UE3	ZNF547	Q8IVP9
ZNF548	Q8NEK5	ZNF550	Q7Z398	ZNF555	Q96BR6
ZNF558	Q96NG5	ZNF559	Q9BR84	ZNF565	Q8N9K5
ZNF570	Q96NI8	ZNF571	Q7Z3V5	ZNF577	Q9BSK1
ZNF584	Q8IVC4	ZNF595	Q8IYB9	ZNF596	Q8TC21
ZNF600	Q6ZNG1	ZNF613	O94892, Q6PF04	ZNF614	Q8N883
ZNF616	Q08AN1	ZNF628	P51786	ZNF641	Q96N77
ZNF655	Q8N720	ZNF658	Q14586, Q5TYW1	ZNF662	Q6ZS27
ZNF667	Q5HYK9	ZNF669	Q96BR6	ZNF677	Q86XU0
ZNF678	Q5SXM1	ZNF681	Q96N22	ZNF691	Q5VV52
ZNF699	Q32M78	ZNF70	Q9UC06	ZNF703	Q9H7S9
ZNF714	Q96N38	ZNF724	A8MTY0	ZNF726	A6NNF4
ZNF727	A8MUV8	ZNF730	Q6ZMV8	ZNF749	O43361
ZNF75A	Q96N20	ZNF76	Q8TAW3	ZNF761	Q86XN6
ZNF764	Q96H86	ZNF765	Q7L2R6	ZNF771	Q7L3S4
ZNF772	Q68DY9	ZNF774	Q6NX45	ZNF775	Q96BV0
ZNF777	Q9ULD5	ZNF778	Q96MU6	ZNF785	A8K8V0
ZNF786	Q8N393	ZNF790	Q6PG37	ZNF805	O43296
ZNF830	Q96NB3	ZNF84	P51786	ZNF850	Q96PQ6
ZNF853	Q6IV72	ZNF883	P52742	ZNF92	Q03936
ZNF93	P35789	ZNRD1	Q9P1U0	ZNRF1	Q8ND25
ZNRF3	Q9ULT6	ZRANB1	Q9UGI0	ZRSR2	Q15696
ZSCAN22	Q68DY9	ZW10	O43264	ZWILCH	Q9H900
ZWINT	O95229				

Input	ChEBI Id	Input	ChEBI Id
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Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
AAMP	ENSG00000127837	ABCA6	ENSG00000154262	ABCB4	ENSG00000005471
ABCC1	ENSG00000103222	ABCC3	ENSG00000108846	ABCF2	ENSG00000033050

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
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ACOX1	ENSG00000161533	ACTA2	ENST00000224784	ACTB	ENST00000331789
ADD1	ENSG00000087274	AGPAT1	ENSG00000204310	AGPAT2	ENSG00000169692
AIFM2	ENSG00000042286	AIP	ENSG00000110711	ALAS1	ENSG00000023330
ALOX5	ENSG00000012779	AMY2B	ENST00000361355	ANPEP	ENST00000300060
ANXA2	ENST00000451270	ANXA3	ENST00000264908	APAF1	ENSG00000120868
APC2	ENSG00000234906	AREG	ENST00000395748	ARID3A	ENSG00000116017
ARL4C	ENSG00000188042, ENST00000339728	ARNTL	ENSG00000133794	ASAH1	ENSG00000104763
ATF4	ENSG00000128272	ATM	ENSG00000149311, ENST00000278616	ATP6AP2	ENSG00000182220
ATP6V0C	ENSG00000185883	ATP6V0E2	ENSG00000171130	ATP6V1A	ENSG00000114573
ATP6V1B2	ENSG00000147416	ATP6V1C1	ENSG00000155097	ATP6V1D	ENSG00000100554
ATP6V1H	ENSG00000047249	ATR	ENSG00000175054, ENST00000350721	AXIN1	ENSG00000103126
BACH1	ENSG00000156273	BATF	ENSG00000156127	BAX	ENSG00000087088
BCL2	ENSG00000171791	BCL2A1	ENSG00000140379	BCL2L1	ENSG00000171552
BCL6	ENSG00000113916	BDP1	ENSG00000072135	BID	ENSG00000015475
BIRC5	ENST00000301633	BLK	ENSG00000136573	BRCA1	ENSG00000012048, ENST00000357654
BRWD1	ENSG00000185658	BST2	ENSG00000130303	BTG1	ENSG00000133639
BTG2	ENSG00000159388	CAPZA1	ENSG00000116489	CASP6	ENSG00000138794
CBFA2T2	ENSG00000078699	CBX5	ENSG00000094916	CCL2	ENSG00000108691
CCL22	ENSG00000102962, ENST00000219235	CCL3	ENSG00000277632	CCL3L3	ENSG00000277768
CCL4	ENSG00000275302	CCNA1	ENSG00000133101	CCNA2	ENSG00000145386
CCNB1	ENSG00000134057	CCNB2	ENSG00000157456	CCND1	ENSG00000110092
CCNE1	ENSG00000105173	CCNG1	ENSG00000113328	CCNG2	ENSG00000138764
CCNK	ENSG00000090061	CCNL1	ENSG00000073111	CCR1	ENSG00000163823
CCR2	ENSG00000121807	CCR5	ENSG00000160791	CD163	ENSG00000177575
CD36	ENSG00000135218	CD80	ENSG00000121594	CDC25A	ENSG00000164045
CDC25C	ENSG00000158402	CDC45	ENSG00000093009	CDC6	ENSG00000094804
CDC7	ENSG00000097046	CDH1	ENST00000261769	CDK1	ENSG00000170312
CDK2	ENSG00000123374	CDK5	ENSG00000164885	CDKN1A	ENSG00000124762, ENST00000244741
CDKN2A	ENSG00000147889, ENST00000304494, ENST00000579755	CDKN2B	ENSG00000147883	CDT1	ENSG00000167513
CEACAM1	ENSG00000079385	CEBPA	ENSG00000245848	CEBPD	ENSG00000221869
CEBPE	ENSG00000092067	CEL	ENST00000372080	CENPF	ENSG00000117724
CES1	ENSG00000125148	CHAC1	ENSG00000128965	CHD1	ENSG00000153922, ENST00000284049.7
CHEK1	ENSG00000149554, ENST00000438015	CISH	ENSG00000114737	CKLF	ENSG00000102554
CLOCK	ENSG00000134852	CNN2	ENSG00000064666	CNOT6L	ENST00000264903
CPT2	ENSG00000157184	CR1	ENSG00000203710	CRY1	ENSG00000008405
CRY2	ENSG00000121671	CSF1	ENSG00000184371	CSF1R	ENSG00000182578
CSF2	ENSG00000164400	CSF3R	ENSG00000119535	CSRP1	ENSG00000159176
CTRC	ENST00000375949	CXCR3	ENSG00000186810	CXCR4	ENST00000241393
CXXC5	ENSG00000171604	CYP1A1	ENSG00000140465	DDIT3	ENSG00000175197, ENST00000346473
DDIT4	ENSG00000168209	DDX11	ENSG00000013573	DEDD2	ENSG00000160570
DFNA5	ENSG00000105928	DGAT2	ENSG00000062282	DHFR	ENSG00000228716

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
DLGAP5	ENSG00000126787	DNAJB1	ENSG00000132002	DNAJB6	ENSG00000105993
DNAJB9	ENSG00000128590	DRG1	ENSG00000104419	E2F1	ENSG00000101412
E2F7	ENSG00000165891	EDEM1	ENSG00000134109	EED	ENSG00000074266
EEPD1	ENSG00000122547	EIF2AK2	ENSG00000055332	ELOC	ENST00000520242
ELOVL5	ENSG00000012660	ELOVL6	ENSG00000170522	ENPP2	ENST00000427067
EOMES	ENSG00000163508	EPAS1	ENSG00000116016	EPHA1	ENSG00000146904
EPHA4	ENSG00000116106	ERBB2	ENST00000269571	ESRRA	ENSG00000173153
ETS2	ENSG00000157557	EXTL3	ENSG00000012232	EZH2	ENSG00000106462
F13A1	ENSG00000124491	FANCD2	ENSG00000144554, ENST00000419585	FANCI	ENSG00000140525, ENST00000310775
FAS	ENSG00000026103	FASLG	ENSG00000117560	FASN	ENSG00000169710
FBXO32	ENSG00000156804	FBXO5	ENSG00000112029	FCER2	ENSG00000104921
FDPS	ENSG00000160752	FGF9	ENSG00000102678	FHL2	ENSG00000115641
FKBP4	ENSG00000004478	FN1	ENSG00000115414	FOS	ENSG00000170345
FOXO3	ENSG00000118689	FOXP3	ENSG00000049768	FPR1	ENSG00000171051
FSCN1	ENSG00000075618	G0S2	ENSG00000123689	GABPA	ENSG00000154727
GBP1	ENSG00000117228	GBP2	ENSG00000162645	GBP3	ENSG00000117226
GCLM	ENSG00000023909	GEM	ENSG00000164949	GFI1	ENSG00000162676
GGPS1	ENSG00000152904	GLI2	ENSG00000074047	GML	ENSG00000104499
GMPR	ENSG00000137198	GRB10	ENSG00000106070	GREB1	ENSG00000196208, ENST00000381486.6
GSC	ENST00000238558	GSR	ENSG00000104687	GSTO1	ENSG00000148834
HDGF	ENSG00000143321	HERPUD1	ENSG00000051108	HES1	ENSG00000114315
HEY2	ENSG00000135547	HHEX	ENST00000282728	HIGD1A	ENSG00000181061
HLA-B	ENSG00000234745	HLA-C	ENSG00000204525	HLA-E	ENSG00000204592
HLA-F	ENSG00000204642	HLA-G	ENSG00000204632	HMGCR	ENSG00000113161
HMOX1	ENSG00000100292	HNRNPDL	ENSG00000152795	HNRNPF	ENSG00000169813
HOXA3	ENSG00000105997, ENST00000317201	HOXB4	ENSG00000182742, ENST00000332503	HOXC4	ENSG00000198353, ENST00000303406
HSPA1A	ENSG00000204389, ENSG00000215328, ENSG00000234475, ENSG00000235941, ENSG00000237724	HSPA1B	ENSG00000204388, ENSG00000212866, ENSG00000224501, ENSG00000231555, ENSG00000232804	HSPA5	ENSG00000044574
HSPA8	ENSG00000109971	HSPD1	ENSG00000144381	HSPE1	ENSG00000115541
HSPH1	ENSG00000120694	ICOS	ENSG00000163600	ID1	ENSG00000125968
ID3	ENSG00000117318	IDH1	ENSG00000138413	IDI1	ENSG00000067064
IFI27	ENSG00000165949	IFI30	ENSG00000216490	IFI6	ENSG00000126709
IFIT1	ENSG00000185745	IFIT2	ENSG00000119922	IGFBP3	ENSG00000146674
IGFBP7	ENSG00000163453	IGHG1	ENSG00000211896	IKBIP	ENSG00000166130
IL10RA	ENSG00000110324	IL12A	ENSG00000168811	IL1B	ENSG00000125538, ENST00000263341
IL1RN	ENSG00000136689	IL23A	ENSG00000110944	IL2RA	ENSG00000134460
IL4R	ENSG00000077238	IL6R	ENSG00000160712	IP6K2	ENSG00000068745
IRF1	ENSG00000125347	IRF2	ENSG00000168310	IRF3	ENSG00000126456
IRF6	ENSG00000117595	IRF8	ENSG00000140968	ISG15	ENSG00000187608
ISG20	ENSG00000172183	ITCH	ENSG00000078747	ITGA2	ENSG00000164171
ITGA2B	ENSG00000005961	ITGA5	ENSG00000161638	ITGA6	ENST00000442250
ITGAM	ENSG00000169896	ITGB1	ENSG00000150093	ITGB2	ENSG00000160255
JAG1	ENSG00000101384	JUNB	ENSG00000171223, ENST00000302754	KDELR3	ENSG00000100196
KDM4C	ENST00000381306	KDM5A	ENST00000399788	KDM5B	ENST00000235790

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KDM6B	ENSG00000132510	KEAP1	ENSG00000079999	KIT	ENST00000288135
KLF4	ENSG00000136826	KLF5	ENSG00000102554	KPNA2	ENST00000330459.7
KRT18	ENST00000388835	KRT7	ENST00000331817	LDHB	ENST00000350669
LGALS2	ENST00000215886	LGALS3	ENSG00000131981	LIF	ENSG00000128342
LIPE	ENSG00000079435	LMNA	ENSG00000160789	LMNB1	ENSG00000113368
LONP1	ENSG00000196365	LRIG1	ENST00000273261	LYZ	ENST00000261267
MAPK6	ENSG00000069956	MAPKAPK5	ENSG00000089022	MCM2	ENSG00000073111
MCM5	ENSG00000100297	MDM2	ENSG00000135679	MEF2C	ENSG00000081189
MGLL	ENSG00000074416	MIF	ENSG00000104899	MLH1	ENSG00000076242
MRPL18	ENSG00000112110	MSN	ENSG00000147065	MT-ATP8	ENST00000361851
MT-CO2	ENST00000361739	MT-CO3	ENST00000362079	MT-CYB	ENST00000361789
MT-ND1	ENST00000361390	MT-ND2	ENST00000361453	MT-ND3	ENST00000361227
MT-ND4	ENST00000361381	MT-ND4L	ENST00000361335	MT-ND5	ENST00000361567
MT-RNR1	ENST00000389680	MT-TA	ENST00000387392	MT-TP	ENST00000387461
MT-TQ	ENST00000387372	MTAP	ENSG00000099810	MUC1	ENSG00000185499
MVD	ENSG00000167508	MX1	ENSG00000157601	MXD4	ENSG00000123933
MYB	ENSG00000118513	MYBL2	ENSG00000101057	MYC	ENST00000613283
MYDGF	ENSG00000074842	MYLIP	ENSG00000007944	NAMPT	ENSG00000105835
NANOS3	ENSG00000187556	NDN	ENSG00000182636	NFE2	ENSG00000123405
NFE2L2	ENSG00000116044	NOCT	ENSG00000151014	NOS3	ENSG00000187556
NOTCH1	ENSG00000148400	NOTCH2	ENSG00000134250, ENST00000256646	NQO1	ENSG00000181019
NR4A3	ENSG00000119508	NRF1	ENSG00000106459	OAS1	ENSG00000089127
OAS2	ENSG00000111335	OAS3	ENSG00000111331	OASL	ENSG00000135114
OR2W3	ENSG00000238243	OR7D2	ENSG00000188000	ORC1	ENSG00000085840
PAK2	ENSG00000180370	PCBP4	ENSG00000090097	PCNA	ENSG00000132646
PDIA5	ENSG00000065485	PDIA6	ENSG00000143870	PDK4	ENSG00000004799
PER1	ENSG00000179094	PERP	ENSG00000112378	PGD	ENSG00000142657
PIK3R1	ENSG00000145675	PIM1	ENSG00000137193	PLAC8	ENSG00000145287
PLIN1	ENSG00000166819	PLK1	ENSG00000166851	PLK2	ENSG00000145632
PLXND1	ENSG00000004399	PMAIP1	ENSG00000141682	PML	ENSG00000140464
PMVK	ENSG00000163344	PNPLA2	ENSG00000177666	POLA1	ENSG00000101868
POLG2	ENSG00000256525	POLR2D	ENST00000272645	POLRMT	ENSG00000099821
PPARA	ENSG00000186951	PPP1R15A	ENSG00000087074, ENST00000200453	PRDM1	ENSG00000057657
PRDM16	ENSG00000142611	PRDX1	ENSG00000117450	PREB	ENSG00000138073
PRF1	ENSG00000180644	PSAP	ENSG00000122852, ENSG00000185303	PSMB8	ENSG00000204264
PTCH1	ENSG00000185920	PTCH2	ENSG00000117425	PTGDS	ENSG00000107317
PTPN11	ENSG00000179295	PTPN12	ENSG00000127947	PTPN14	ENSG00000152104
PTPN18	ENSG00000072135	PTPN23	ENSG00000076201	PTPN7	ENSG00000143851
PTPN9	ENSG00000169410	PXDN	ENSG00000130508	PXN	ENSG00000089159
RABGGTA	ENSG00000100949	RAD51	ENSG00000051180	RAD51D	ENSG00000185379, ENST00000345365
RALA	ENSG00000006451	RAP1B	ENSG00000127314	RBBP4	ENSG00000162521
RBBP8	ENSG00000101773	RBFOX3	ENSG00000167281	RBL1	ENSG00000080839
RGL1	ENSG00000143344	RHOU	ENSG00000116574	RNASEL	ENSG00000135828
RNU4-1	ENSG00000200795	RNU4ATAC	ENST00000580972	ROBO1	ENSG00000169855
ROBO3	ENSG00000154134	RORA	ENSG00000069667	RRAD	ENSG00000166592
RRM2	ENSG00000171848	RRM2B	ENSG00000048392	RSAD2	ENSG00000134321
S100B	ENSG00000160307	S1PR1	ENSG00000170989	SAMHD1	ENSG00000101347

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
SCD	ENSG00000099194	SCO2	ENSG00000130489	SDC4	ENST00000372733
SEC31A	ENSG00000138674	SERP1	ENSG00000120742	SERPINH1	ENSG00000149257
SFN	ENSG00000175793	SHC1	ENSG00000160691	SIRT1	ENSG00000096717
SLC7A11	ENSG00000151012	SMAD6	ENSG00000137834	SMAD7	ENSG00000101665
SMURF1	ENSG00000198742	SNAI1	ENSG00000124216	SNRPA1	ENSG00000131876
SOCS1	ENSG00000185338	SOCS3	ENSG00000184557	SOCS4	ENSG00000180008
SOD1	ENSG00000142168	SPARC	ENSG00000113140	SQLE	ENSG00000104549
SRF	ENSG00000112658	SRPRA	ENSG00000182934	STAT1	ENSG00000115415
STAT3	ENSG00000168610	STEAP3	ENSG00000115107	STMN1	ENSG00000117632
STT3B	ENSG00000163527	SUZ12	ENSG00000178691	SYVN1	ENSG00000162298
TALDO1	ENSG00000177156	TATDN2	ENSG00000157014	TBX21	ENSG00000073861
TCF4	ENSG00000196628	TET2	ENST00000265149	TFB1M	ENSG00000029639
TFRC	ENST00000360110	TGFA	ENSG00000163235	TGFB1	ENSG00000105329
TGFBR3	ENSG00000069702, ENST00000212355	THBS1	ENSG00000137801	TK1	ENSG00000167900
TKT	ENSG00000163931	TM7SF2	ENSG00000149809	TMEM97	ENST00000226230
TNF	ENSG00000232810	TNFRSF10A	ENSG00000104689	TNFRSF10B	ENSG00000120889
TNFRSF10D	ENSG00000173530	TNFRSF1A	ENSG00000067182	TNIK	ENSG00000154310
TOP2A	ENSG00000131747	TP53	ENSG00000141510	TP53I3	ENSG00000115129
TPH1	ENSG00000129167	TPST2	ENSG00000128294	TRIB3	ENSG00000101255
TRIM14	ENSG00000106785	TRIM17	ENSG00000162931	TRIM21	ENSG00000132109
TRIM3	ENSG00000110171	TRIM35	ENSG00000104228	TRIM46	ENSG00000163462
TRIM62	ENSG00000116525	TRPC3	ENSG00000138741	TSC22D1	ENSG00000102804
TSPYL2	ENSG00000184205	TUBB4B	ENST00000340384	TWIST1	ENSG00000122691
TWNK	ENSG00000107815	TXN	ENSG00000136810	TXNRD1	ENSG00000198431
TYMS	ENSG00000176890	UBB	ENSG00000170315	VGF	ENSG00000128564
VIM	ENST00000544301	WFS1	ENSG00000109501	WWC1	ENSG00000113645
YIF1A	ENSG00000174851	ZAP70	ENSG00000115085	ZCCHC11	ENSG00000134744
ZCCHC6	ENSG00000083223	ZEB1	ENSG00000148516	ZEB2	ENSG00000169554
ZNF225	ENSG00000120738	ZNF385A	ENSG00000161642	ZNF521	ENSG00000198795

Input	NCBI_NUCLEOTIDE Id	Input	NCBI_NUCLEOTIDE Id
MUC1	NM_001371720	SNORD3A	NR_006880

Interactors (5185)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
AAED1	Q7RTV5	Q9NRD5	AAGAB	Q6PD74	O00186, Q12846
AAK1	Q2M2I8	Q9NQ11	AAMP	C9JG97	O14503
AARS	P49588	P13569	AASDHPPT	Q9NRN7	Q9UQK1
ABCB1	P08183	P17931	ABCB10	Q9NRK6	Q71RS6
ABCC1	P33527	P17931	ABCC2	Q92887	P19838
ABCD1	P33897	P46059	ABCD3	P28288	P33897
ABCD4	O14678	P23945	ABCE1	P61221	P13569
ABCF3	A0A0S2Z645	P61289	ABHD10	Q9NUJ1	P10809
ABHD11	Q8NFV4-4	O95231	ABHD14A	Q9BUJ0	Q9UNA3
ABHD15	Q6UXT9	P21673	ABHD16A	O95870	P05177
ABHD17C	Q6PCB6	O14832	ABHD18	Q0P651	Q15116
ABHD3	Q8WU67	P04201	ABI1	Q8IZP0	Q07889
ABL1	P00519	Q13905	ABL2	P42684	P19174

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ABLIM1	O14639	P31947	ABLIM2	Q6H8Q1-8	P14136
ABRAXAS2	Q15018	Q15831	ABT1	Q9ULW3	P06748
ABTB1	Q969K4	P29692	ACAA2	P42765	Q9BQP7
ACACA	Q13085	Q13085	ACAD9	Q9H845	Q8IWL3
ACADM	P11310	P0DTC5	ACAT1	P35610	O75326
ACBD3	Q9H3P7	Q9UBF8	ACBD7	Q8N6N7	Q9NWQ8
ACCS	Q96QU6	P14927	ACIN1	Q9UKV3	P09651
ACKR2	O00590	Q9UKG4	ACLY	P53396	P13569
ACOT11	Q8WXI4-2	Q9NQT4	ACOT13	Q9NPJ3	Q92993
ACOT7	O00154	P13569	ACP1	P24666	P60709
ACP2	P11117	P21453	ACPP	A0ABF7PH85	Q9H1K1
ACSL4	O60488	Q9H2J7	ACTA2	P62736	P15498
ACTB	EBI-5276484	P26358	ACTG1	P63261	P60709
ACTL6A	O96019	P39748	ACTR10	Q9NZ32	Q8N5V2
ACTR1A	P61163	P13569	ACTR2	P61160	P13569
ACTR3	P61157	O00401	ACTR5	Q9H9F9	O96019
ACTR6	Q9GZN1	Q15274	ACVR1	Q04771	Q9NPF0
ACVR1B	P36896	P62942	ACVR2B	Q13705	P27037
ADAL	Q6DHV7-2	P08670	ADAM10	O14672	P12931
ADAM12	O43184	P12931	ADAM22	Q9P0K1	Q14160
ADAM28	Q9UKQ2	P00749	ADAM9	Q13443	Q99962
ADAMTSL4	Q6UY14-3	O75093	ADAMTSL5	Q6ZMM2-2	Q7Z3S9
ADAT2	Q7Z6V5	Q96GM5	ADCK5	Q3MIX3	Q7Z3S9
ADCY3	O60266	P17931	ADCY6	O43306	Q9Y4Z2
ADCY9	O60503	P15514	ADD1	P35611	Q13153
ADD2	P35612	P35611	ADGRA3	Q8IWK6	P34925
ADGRE5	P48960	P08174	ADGRG5	Q8IZF4	Q99808
ADGRL2	O95490	P21926	ADNP	Q9H2P0	Q13257
ADORA2B	P29275	P29274	ADRB2	P07550	Q99685
ADSL	P30566	P13569	AES	Q08117	Q9UH92
AFF1	P51825	P04608	AFF4	Q9UHB7	P55199
AGAP3	Q96P47	P61981	AGFG2	O95081	Q9NQZ5
AGGF1	Q8N302-2	P08670, P06396	AGMAT	Q9BSE5	O43602
AGO1	Q9UL18	P11940	AGO4	Q9HCK5	O15397, Q8NDV7
AGPAT1	Q99943	Q8TDQ0	AGPAT3	Q9NRZ7	Q9H2J7
AGPAT4	Q9NRZ5	Q96BA8	AGPAT5	Q9NUQ2	Q96BA8
AHCY	P9WGV3	P10145	AHCYL1	O43865	P23526
AHDC1	Q5TGY3	Q13185	AHI1	Q8N157	P50570
AHR	P35869	P27540	AHSA1	O95433	P07900
AIF1	P55008	P06396	AIFM1	O95831	P62937
AIMP2	Q13155	P03508	AIP	O00170	Q92985
AJUBA	Q96IF1	O95863	AK2	P54819	P17612
AK3	P27144	P13501	AK4	P27144	P13501
AK6	Q9UIJ7	P02649	AK8	Q96MA6	Q9NPA3
AKAP1	Q92667	P13861	AKAP11	Q9UKA4	P51530
AKAP13	Q12802	P61981	AKAP17A	Q02040	O75152
AKAP8	O43823	P49736	AKAP8L	Q9ULX6	P60484
AKAP9	Q6PJH3	O14641	AKIP1	Q9NQ31	P17612
AKR1A1	P14550	O76024	AKR1C3	P42330	Q04828
AKR1C8P	Q5T2L2	P55212	AKR7A2	O43488	P54252

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
AKR7A3	O95154	P13569	AKT2	P31751	P0CG47
AKT3	Q9Y243	P07900	ALAS1	P13196	Q16822
ALCAM	Q13740	P17931	ALDH2	P05091	P02647
ALDH8A1	Q9H2A2	Q02252	ALDOA	P04075	P12004
ALG1	Q9BT22	P13569	ALG10	P50076	Q96BA8
ALG13	Q9NP73-4	Q01974	ALG3	Q92685	Q9H015
ALG5	Q9Y673	P0DTC3	ALG8	Q9BVK2	Q96BA8
ALKAL2	Q6UX46	Q9UM73	ALOX12B	O75342	Q9BYJ1
ALOX5	P09917	Q9Y6D9	ALS2CR12	Q96Q35-2	Q9UI12
AMIGO1	Q86WK6	Q9H2H9	AMMECR1	Q9Y4X0	Q9Y2T1
AMN	Q9BXJ7	O60494	AMOT	Q4VCS5	Q96PU5
AMPD3	Q01432	P08238	ANAPC11	Q9NYG5-2	Q9H2F3
ANAPC2	Q9UJX6	O60566, Q12834	ANAPC5	Q9UJX4	P51955
ANAPC7	Q9UJX3	O60566, Q12834	ANKHD1	Q8IWZ3	Q92560
ANKMY2	Q8IV38	Q15185	ANKRA2	Q9H9E1	P56524
ANKRD10	Q9NXR5-2	Q15915	ANKRD13D	Q6ZTN6	P43115
ANKRD27	Q96NW4	P25963	ANKRD44	Q8N8A2	P46531
ANKRD49	Q8WVL7	P49674	ANKRD53	Q8N9V6-2	Q8IVS8
ANKS6	Q68DC2	P01031	ANLN	Q9NQW6	P07948
ANO9	A1A5B4	Q12959	ANP32A	P39687	P52294
ANP32B	Q92688	P52294	ANP32E	Q9BTT0	P52294
ANTXR2	P58335	P13423	ANXA2	EBI-8785901, P07355	Q8NBP7
ANXA3	P12429	P13569	ANXA5	P08758	Q96CV9
AP1B1	Q10567	P49757	AP1G1	P22892	Q14677
AP1G2	O75843	Q6UWB1	AP1S1	P61966-2	P16284
AP2B1	P63010	P12236	AP2S1	P53680	P49757
AP3M1	Q9Y2T2	Q9UBH0	AP3S1	Q92572	Q8N307
APAF1	O14727	P99999	APBA2	O35431	P05067-4
APBA3	O96018	Q8N9R8	APBB1	O00213	P05067-4
APBB2	Q92870-2	Q969S8	APC	P25054	P49841
APC2	Q9BV35	Q8N695	APEH	P13798	P13798
APEX2	Q9UBZ4	Q13257	APH1B	Q8WW43	P05067
APIP	Q96GX9	P63208	APLN	APLN	P06396
APLP2	Q06481-5	Q96L34	APMAP	Q9HDC9	P13569
APOBEC3A	P31941	O95630	APOBEC3B	Q9UH17	P11802
APOBEC3F	Q8IUX4	Q86VM9	APOOL	Q6UXV4	P46059
APPL2	Q06481-5	Q96L34	APRT	P07741	Q6DKK2
AQP9	O14520	O60240	ARAF	P10398	Q92985
ARAP1	Q96P48	P62993	ARCN1	P48444	P13569
ARF4	P18085	Q96PM5	ARFGAP2	Q8N6H7	P05155
ARFGEF2	Q9Y6D5	P0DTC5	ARFGEF3	Q5TH69	P15907
ARFIP2	P53365	Q92730	ARGLU1	Q9NWB6	P52298
ARHGAP12	Q8IWW6	P60709	ARHGAP15	Q53QZ3	P63000
ARHGAP19	Q14CB8	P54252	ARHGAP24	Q8N264	P21333
ARHGAP26	Q9UNA1	A1A4S6	ARHGAP33	O14559	P06241
ARHGAP8	P85298-4	Q13526	ARHGEF1	Q92888	P61586
ARHGEF10	O15013	P11142	ARHGEF10L	Q9HCE6	Q8NHY2
ARHGEF11	O15085	O43157	ARHGEF12	Q9NZN5	Q9UM54
ARHGEF2	Q92974	O43524	ARHGEF28	Q8N1W1-4	P27797

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ARHGEF39	Q8N4T4	Q2T9J0	ARHGEF6	Q15052	Q13177
ARID1B	Q8NFD5	P51531	ARID3A	Q99856	P23025
ARID4A	P29374	Q13547	ARID4B	Q4LE39	P14921
ARID5A	Q03989	P61024	ARIH1	Q9Y4X5	Q9BYZ6
ARIH2	O95376	Q93034	ARL1	P40616	P13569
ARL11	Q969Q4	Q96EF0	ARL13B	Q3SXY8	P63252
ARL15	Q9NXU5	O43924	ARL16	Q0P5N6	O43924
ARL2	P36404	Q9BTW9	ARL3	P36405	P36404
ARL4C	P56559	P52306	ARL4D	P49703	P13284
ARL5B	Q96KC2	Q9Y5M8	ARL6	Q9H0F7-1, Q9H0F7	Q96RK4, Q8NFJ9, Q9BXC9
ARL6IP1	Q15041	Q00059	ARL6IP5	O75915	P25103
ARL6IP6	Q8N6S5	O15498	ARMC1	Q9NVT9	Q16143
ARMC10	Q8N2F6-2	Q96GM5	ARMC8	Q8IUR7	Q6VN20, Q96S59
ARMCX5	Q6P1M9	Q15185	ARMT1	Q9H993	P25963
ARNT	P27540	Q16665, Q99814	ARNTL	O00327-8	O14977
ARNTL2	Q8WYA1-3	Q99814	ARPC1B	O15143	O00401
ARPC3	O15145	P07902	ARPC5	O15511	P00533
ARPP19	P56211	P05067	ARRB1	P49407	P12931
ARRDC2	Q8TBH0	Q9UGI0	ARRDC3	Q96B67	P51668
ARSB	P15848	P01185	ARSG	Q96EG1	Q8NBK3
ARSJ	Q8N5N6	Q12933, Q13077	ARV1	Q9H2C2	P01730
ARVCF	O00192	Q96RT1	ASAH1	Q13510, Q13510-2	Q13285
ASB2	Q96Q27-2	Q93034	ASB6	Q9NWX5	O60551
ASB8	Q9H765	Q15369, Q93034	ASCC3	Q8N3C0	P63244
ASF1A	Q9Y294	P49736	ASF1B	Q9NVP2	P49736
ASGR1	P07306	Q96BA8	ASGR2	P07307	P00395
ASH1L	Q9NR48	Q09028	ASH2L	Q9UBL3	Q09472
ASIC1	P78348	Q9NRD5	ASMTL	O95671	Q96CV9
ASNA1	O43681	P10451	ASPH	Q12797-6	Q9HBE4
ASPHD2	Q6ICH7	O75063	ASPM	Q8IZT6	P11802
ASPRV1	Q53RT3	Q9Y324	ASXL2	Q76L83	Q92560
ATAD2	Q6PL18	P03372	ATAD3A	Q9NVI7	Q14197
ATAD5	Q96QE3	P40937, P35249	ATF1	P18846	P18846
ATF4	P18848	P18847	ATF6B	Q99941	Q8TC07
ATF7IP	Q6VMQ6	Q15047	ATG101	Q9BSB4	O75385
ATG12	O94817	Q9H1Y0, Q9H0Y0, Q9NT62, Q676U5	ATG14	Q6ZNE5	P41208
ATG2A	Q2TAZ0	Q9GZQ8	ATG2B	Q96BY7	Q13309
ATG3	P40344, Q9NT62	Q9GZQ8	ATG4B	Q9Y4P1	Q9GZQ8
ATG4C	Q96DT6	O00401	ATG9A	Q7Z3C6	Q96QE2
ATIC	P31939	P00533	ATL2	Q8NHH9	P13569
ATL3	Q6DD88	Q92911	ATM	Q13315	Q9NZQ7
ATMIN	O43313	Q13315	ATOX1	O00244	P35670
ATP1A3	P13637	P21453	ATP1B1	P05026	P21926
ATP23	Q9Y6H3	O14867	ATP2B1	P20020	Q14160
ATP2B4	P23634	P01258	ATP5A1	P25705	P13569
ATP5C1	P36542	Q96H96	ATP5E	P21306	P00747
ATP5F1	P24539	Q14154	ATP5G1	P05496	P21217
ATP5G3	P48201	P06576	ATP5H	O75947	Q9H1C4

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ATP5J	P18859	Q9H2H9	ATP5J2	P56134	P01106
ATP5L	O75964	P13569	ATP6AP2	O75787	Q15904
ATP6V0C	P27449	Q15904	ATP6V1A	P38606	Q9UI12
ATP6V1B2	P21281	Q9UI12	ATP6V1C1	P21283	P21281, P38606
ATP6V1D	Q9Y5K8	Q9UI12	ATP6V1H	Q9UI12	P42772
ATP8A1	Q9Y2Q0	Q9NP91	ATP9A	O75110	P51684
ATPAF1	Q5TC12	P06576	ATPAF2	Q8N5M1	P49247
ATPIF1	Q9UII2	Q96QG7	ATR	Q13535	Q9NZQ7
ATRX	P46100	P84243, Q9UER7	ATXN1L	P0C7T5	Q06330
ATXN3	P54252	Q969S8	ATXN7	O15265	Q8N2S1
ATXN7L1	Q9ULK2-2	P06396	ATXN7L2	Q5T6C5	P43897
AUNIP	Q9H7T9	O14965	AUP1	Q9Y679	P04114
AURKA	O14965	Q9ULW0	AURKB	Q96GD4	P06493
AURKC	Q9UQB9	O15392	AUTS2	Q8WXX7	Q09472
AVPI1	Q5T686	A6NIH7	AXIN1	O15169, Q14DJ8, O35625	P49841
AXL	P30530	P19174	AZI2	Q9QYP6	Q13114
AZU1	P20160	Q99962	B3GALT2	O43825	O94822
B4GALT4	O60513	P11021	B4GALT7	Q9UBV7	P53611
B4GAT1	O43505	P21453	BACE1	P56817	Q9NZ52
BACH1	O14867	O75330	BAD	Q92934	O43521
BAG1	Q99933	P04049	BAG5	Q9UL15	Q9Y6D9
BAIAP2	Q9UQB8	P60709	BANP	Q8N9N5-2	Q96E11
BAP1	Q99496	P51668	BARD1	Q99728	Q05048
BATF	Q16520	P18848	BATF3	Q9NR55	P18847
BAX	Q07812	Q9ULZ3	BAZ1B	Q9UIG0	P17096
BAZ2B	Q9UIF8	Q08379	BBS10	Q8TAM1	Q8IWZ6
BBS12	Q6ZW61	Q8IWZ6, Q9BXC9	BBS2	Q9BXC9	Q16822
BBS7	Q8IWZ6, Q8IWZ6-2, Q8K2G4	Q96RK4, Q8N3I7, Q3SYG4, Q8NFJ9, Q8TAM2, Q9BXC9	BBS9	Q3SYG4	Q8IWZ6, Q96RK4, Q8N3I7, Q8NFJ9, Q8TAM2, Q9BXC9
BBX	Q8WY36	Q13547	BCAS3	Q9H6U6	P08670
BCAT1	P54687	Q8TBF2	BCAT2	O15382	P10809
BCCIP	Q9P287	P06748, Q9HC77	BCDIN3D	Q7Z5W3	Q9NPC8
BCKDK	O14874	P41208	BCL11A	Q9H165	Q13547
BCL2	P10415, P10415-1	Q13794	BCL2A1	Q07440	P55957
BCL2L1	Q07817-1	Q13794	BCL2L12	Q9HB09	Q5JRX3
BCL2L15	Q5TBC7	O00560	BCL6	P41182	P56524
BCL7C	Q8WUZ0	Q9H8M2	BCL9L	Q67FY2	P35222
BCLAF1	Q9NYF8	P10415	BCOR	Q6W2J9, Q6W2J9-1	P42568, Q03111
BCORL1	Q5H9F3	P63208	BCR	P11274	P55735
BDH1	Q02338	P46059	BDP1	Q99952	P00533
BECN1	Q14457	P12931	BEX2	Q9BXY8	P19883
BEX5	Q5H9J7	O14561	BFSP1	Q12934-2	Q13895
BICD1	Q96G01	P49841	BICRA	Q9NZM4	Q96GM5
BICRAL	Q6AI39	Q96GM5	BID	P55957	Q07817
BIRC3	Q13489	Q13546, Q9Y572	BIRC5	O15392	P06493
BIVM	Q86UB2	Q13114	BLK	P51451	P46109
BLM	P54132	P39748	BLMH	Q13867	P05067-4
BLNK	Q8WV28	Q06187	BLOC1S5	Q8TDH9	Q96GM5

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
BLOC1S6	Q9UL45	Q9NUP1, Q9UL45, Q96EV8, P78537, Q6QNY0	BLZF1	Q9H2G9	P36941
BMF	Q96LC9	Q07817	BMP1	P13497	O60216
BMP2K	Q9NSY1	Q8WY64	BMS1	Q14692	P06748
BNIP1	Q12981	Q9NZ43, Q9P2W9, O75396	BOD1L1	Q8NFC6	P52294
BOK	Q9UMX3	P10809	BOLA1	Q9Y3E2	Q86SX6
BOLA3	Q53S33	Q86SX6	BOP1	Q14137	P06748
BORA	Q6PGQ7	P51530	BPGM	P07738	Q9BXK1
BPNT1	O95861	P46199	BPTF	Q12830-4	P62805
BRAF	P15056	P30291	BRCA1	P38398	Q9ULW0, O75330
BRCA2	P51587	Q86YC2	BRCC3	P46736	P38398
BRD1	O95696	Q9UGI0	BRD2	P25440	Q13761
BRD4	O60885	P0DTC4	BRD7	Q9NPI1	Q09472
BRD8	Q9H0E9	O96019	BRD9	Q9H8M2	Q96GM5
BRF1	Q07352	Q16539	BRF2	Q9HAW0	P20226
BRI3BP	Q8WY22	P80370	BRICD5	Q6PL45	P98160
BRIP1	Q9BX63	P54278	BRK1	Q8WUW1	P26641
BRPF3	Q9ULD4-2	P07550	BRWD1	Q9NSI6-4	P07550
BST2	Q10589	Q10589	BTAF1	O14981	P46059
BTBD1	Q9H0C5	P61081	BTBD9	Q96Q07-2	P55212
BTD	P43251	P01008	BTG1	P62324	Q13158
BTG2	P78543	P10275	BTN2A1	Q7KYR7	P22059
BTN3A1	O00481-2, O00481	O60437	BTN3A2	P78410	Q96CV9
BUB1	O43683	P51530	BUB1B	O60566	Q13257
BUB3	O43684	O60566, Q12834	BUD13	Q9BRD0	Q9ULW0
BYSL	Q13895	Q96MT3	BZW2	Q9Y6E2	O14975
C10orf35	Q96D05	O43914	C10orf88	Q9H8K7	Q13163
C11orf49	Q9H6J7-2	P02649	C11orf54	Q9H0W9-3	Q13153
C11orf57	Q6ZUT1	Q9NRZ7	C11orf84	Q9BUA3	O75928-2
C12orf10	Q9HB07	Q9BRX2	C12orf4	Q9NQ89	P55212
C12orf49	Q9H741	P40855	C14orf119	Q9NWQ9	P61587
C14orf166	Q9Y224	P13569	C14orf80	Q86SX3	P35638
C16orf45	Q96MC5	Q8WZA2	C16orf58	Q96GQ5	Q6ZPD8
C16orf59	Q7L2K0	Q7Z3S9	C16orf87	Q6PH81	Q9UM11
C17orf80	Q9BSJ5	Q8N2W9	C18orf54	Q8IYD9	Q53GS7
C19orf38	A8MVS5	O60443	C19orf44	Q9H6X5-2	Q92845
C19orf53	Q9UNZ5	P10636-8	C19orf54	Q5BKX5-3	Q9NQ94
C19orf66	Q9NUL5-4	Q96RU7	C19orf70	Q5XKP0	Q9NR28
C1D	Q13901	Q99547, Q9Y2L1, P42285, Q01780, Q9NQT4, Q9H8H0	C1GALT1C1	Q96EU7	Q6ZMH5
C1QBP	Q07021	Q8N726	C1QTNF3	Q9BXJ4	P42858
C1orf112	Q9NSG2	Q6PIW4	C1orf122	Q6ZSJ8	Q16576
C1orf131	Q8NDD1-6	Q92993	C1orf216	Q8TAB5	P17302
C1orf52	Q8N6N3	Q13257	C1orf74	Q96LT6	Q9GZV5
C20orf24	Q9BUV8	O60704	C21orf2	O43822	O00244
C21orf58	Q0VAL7	Q96RU7	C22orf15	Q8WYQ4-2	P53667
C2CD5	Q86YS7	P43115	C2orf42	Q9NWW7	P13196
C2orf82	Q6UX34	Q15125	C3	P01024	Q16659
C3AR1	Q16581	O43913	C3orf14	Q9HBI5	Q96ST8
C3orf18	Q9UK00	P40855	C3orf52	Q5BVD1	Q15109

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
C5AR1	P21730	P18031	C5AR2	Q9P296	P09601
C5orf22	Q49AR2	P13725	C5orf30	Q96GV9	O00410
C5orf34	Q96MH7	Q14653	C5orf51	A6NDU8	Q6PJG9
C6orf106	Q9H6K1	P11142	C6orf226	Q5I0X4	P09110, O14832
C6orf47	O95873	Q13257	C8G	P07360	P01031
C8orf46	Q8TAG6	Q9NPB3	C8orf76	Q96EF9	P12830
C9orf72	Q96LT7, Q96LT7-2, Q96LT7-1	O75385	CA5B	Q9Y2D0	P11182
CA6	Q8N4G4	Q7Z3S9	CAB39	Q9Y376	Q15831
CABP4	P57796-2	P41182	CACNB3	P54287, P54284	P55040
CACYBP	Q9HB71	P10636-8	CAD	O76075	O00273
CADM1	Q9BY67	P60900	CALCOCO1	Q9P1Z2	Q8TC07
CALCOCO2	Q13137	P26640	CALHM1	Q8IU99	P05067
CALM2	P62161	P52888	CALM3	P62161	P52888
CALU	O43852	Q05513	CAMK1	Q14012	P21709
CAMK1D	Q8IU85	Q14012	CAMK2G	Q13555-5	Q9UBU9
CAMK4	Q16566	Q9Y2X7	CAMKK2	Q96RR4	Q16539
CAMSAP1	Q5T5Y3	P63104	CAMSAP2	Q08AD1	P29692
CAND1	Q86VP6	Q93034, Q9UBF6	CAND2	O75155	Q14192
CANT1	Q8WVQ1	Q96JA1	CAP1	Q01518	P60709
CAPG	Q9BPX3	P13569	CAPN10	Q9HC96	Q16595
CAPN15	O75808	Q9NWF9	CAPRIN1	Q14444	P13569
CAPRIN2	Q6IMN6	O75197	CAPS	Q9ULU8-4	O00560
CAPZA1	P52907	P13569	CAPZA2	P47755	P13569
CAPZB	P47756	P09651	CARD11	Q9BXL7	Q9UDY8
CARD17	Q5XLA6	P29466	CARD9	Q9H257	O95999
CARHSP1	Q9Y2V2	Q13077	CARMIL2	Q6F5E8	P47756
CARNMT1	Q8N4J0	Q13547	CASC3	O15234	Q92900
CASKIN2	Q8WXE0	P16333	CASP2	P42575	O95429
CASP3	P42574	O60551	CASP6	P55212	Q86US8
CASP7	P55210	Q13177	CASP8	Q14790-2, Q14790	O15519-1
CASS4	Q9NQ75-2	Q08379	CAST	O15446	O15160, O95602, P19388, P61218
CASTOR1	Q8WTX7	Q16401	CAVIN1	Q6NZI2	Q9Y6D9
CBFA2T2	O43439-4	Q9BRV8	CBFB	Q13951	Q13761
CBL	P22681	P12931	CBLB	Q13191	P46109
CBLL1	Q9JIY2	P12830	CBR1	P16152	P13569
CBR3	O75828	P16152	CBWD1	Q9BRT8	P62993
CBWD2	Q8IUF1	Q9Y337	CBX1	P83916	Q8N9R8
CBX2	Q14781-2	Q00403	CBX3	Q13185	Q9UI95
CBX4	O00257	Q16665	CBX5	P45973	Q8N9R8
CBX7	Q8VDS3	Q9NZI8	CBY1	Q9Y3M2	Q9UJV9
CCDC102B	Q68D86	P33993	CCDC106	Q9BWC9	P40429
CCDC112	Q8NEF3-2	Q9Y2T1	CCDC114	Q96M63	O14964
CCDC115	Q96NT0	O15342	CCDC117	Q8IWD4	P07900
CCDC12	Q8WUD4	O75152	CCDC121	Q6ZUS5	P36941
CCDC130	P13994	Q92989	CCDC138	Q96M89-2	Q16512
CCDC141	Q6ZP82-1	Q8N2W9	CCDC146	Q8IYE0, Q8IYE0-2	O75496
CCDC150	Q8NCX0-3	Q96JC9	CCDC167	Q9P0B6	P25103
CCDC17	Q96LX7-5	P54252	CCDC174	Q6PII3	P09917

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
CCDC183	G5E9W6	Q969G3	CCDC184	Q52MB2	Q13114
CCDC186	Q7Z3E2	P07947	CCDC28B	Q9BUN5	Q96RK4
CCDC34	Q96HJ3-2	Q92993	CCDC47	Q96A33	Q92985
CCDC50	Q8IVM0	Q13546	CCDC51	Q96ER9	P17900
CCDC59	Q9P031	P98164	CCDC71	Q8IV32	Q13547
CCDC71L	Q8N9Z2	Q9Y4Z2	CCDC74B	Q96LY2-2	Q5T8I9
CCDC77	Q9BR77	P36941	CCDC84	Q86UT8	Q14691
CCDC85B	Q15834	P36873	CCDC88C	CCDC88C	O76024
CCDC90B	Q9GZT6	P04183	CCDC93	Q567U6	P36941
CCDC94	Q9BW85	P52747	CCDC97	Q96F63	P18846
CCHCR1	Q8TD31-3	Q9Y6D9	CCIN	Q13939	P42858
CCL2	P13500	P13501	CCL22	O00626	Q99611
CCL25	O15444	P13501, P48061	CCL28	Q9NRJ3	P13501, P48061
CCL3	P10147	P10147	CCL4	P13236	Q6IN84
CCNA1	P78396	P38936	CCNA2	P20248	Q13415
CCNB1	P14635	Q14674	CCNB1IP1	Q9NPC3	Q93034
CCNB2	O95067	Q99640	CCND1	P24385	Q13761
CCND2	P30279	Q08999	CCNDBP1	O95273	Q96T88
CCNE1	P24864	P61024	CCNE2	O96020	Q08999
CCNF	P41002	O43303	CCNG1	P51959	O95995
CCNH	P51946	P28715	CCNI2	Q6ZMN8	Q00535
CCNK	O75909	Q9NYV4, Q14004	CCNL1	P49736	P33992, P25205
CCNL2	Q96S94	P68400	CCNT1	O60563	P04608
CCNT2	O60583-1, O60583-2, O60583	P50750	CCNY	Q8ND76	P61981
CCP110	O43303	Q8IW35	CCR1	P32246	P13501
CCR5	P51681	P01730	CCR6	P51684	P04035
CCSER2	Q9H7U1	P61981	CCT3	P49368	Q13371
CCT4	P50991	P06493	CCT5	P48643	P06493
CCT6A	P40227	Q8NCC3	CCT6B	Q92526	O96019
CCT7	Q99832	P06493	CCT8	P50990	Q13882
CD109	Q6YHK3	P23946	CD14	EBI-11177705, EBI-11177518	O43474
CD160	O95971	Q92956	CD164	Q04900	P61073
CD19	P15391	Q9Y2A9	CD1A	P06126	P61769
CD200R1	Q8TD46-4	P11686	CD244	Q9BZW8	Q06124
CD247	P20963	P43403, Q9Y2R2	CD248	Q9HCU0	P27361
CD27	P26842	Q9H1U9	CD300E	Q496F6	A8K4G0
CD300LD	Q6UXZ3	A8K4G0	CD300LF	Q8TDQ1	A8K4G0
CD302	Q8IX05	P16871	CD36	P16671	Q9UKG9
CD3D	P04234	P27824	CD3E	P07766	P43403
CD3EAP	O15446	O15160, O95602, P19388, P61218	CD3G	P09693	O43561-2
CD40	P25942	O15397	CD47	Q08722-3	Q9H2K0
CD48	P09326	Q9BZW8	CD5	P06127	P27986
CD52	Q6IBD0	Q9BRI3	CD69	Q07108	P21453
CD70	P32970	O43913	CD72	P21854	Q96BA8
CD80	P33681	Q9NZQ7	CD82	P27701	P01033
CD83	Q01151	P40818	CD84	Q9UIB8	Q9UIB8
CD9	P21926	P05556	CD93	Q9NPY3	Q9BQ95
CD96	P40200	P15151	CDA	P32320	Q96QG7

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
CDC14B	O60729	P43246	CDC20	Q12834, Q9JJ66	Q13257
CDC20B	Q86Y33	Q9NYY3	CDC23	Q9UJX2, Q8BGZ4	P51955
CDC25A	P30304	O96017	CDC25C	P30307	O14757
CDC27	P30260	P51955	CDC40	O60508	P62306, P14678, P62304
CDC42BPA	Q5VT25	P61981	CDC42BPB	Q9Y5S2	Q05513
CDC42BPG	Q6DT37	Q9UHD2	CDC42EP2	O14613	P60953
CDC42EP3	Q9UKI2	Q5S007	CDC42SE1	Q9NRR8	P60953
CDC42SE2	Q9NRR3	P60953	CDC45	O75419	Q9HAW4
CDC6	Q99741	P06493	CDC7	O00311	Q9HBG6
CDC73	Q6P1J9	O00267	CDCA2	Q69YH5	P36873
CDCA3	Q99618	Q9BY43	CDCA4	Q9BXL8	O60341
CDCA5	Q9CPY3, Q96FF9	O60216	CDCA7	Q9BWT1	O76024
CDCA7L	Q96GN5	P46087	CDCA8	Q53HL2	Q96GD4
CDH1	P12830	Q9ULW0	CDHR3	Q6ZTQ4	Q969S2
CDIP1	Q9H305	O43561-2	CDK1	P11440	Q14674
CDK10	Q15131	P15036	CDK11A	Q9UQ88-1	O96017
CDK11B	P21127	Q13153	CDK12	Q9NYV4	O75909
CDK13	Q14004	P11802	CDK16	Q00536	P46527, P24941
CDK17	Q00537	P06493	CDK18	Q07002	Q9Y6D9
CDK2	P24941	O75496, Q9H211	CDK20	Q8IZL9	Q00535
CDK2AP1	O14519	P52294	CDK2AP2	O75956	Q92769
CDK4	P11802	P42772	CDK5	Q03114	Q16620
CDK5RAP1	Q96SZ6	Q8N4J0	CDK5RAP3	Q96JB5	Q9NZQ7
CDK6	Q00534	P42772	CDK9	P50750	Q9HAW4
CDKL1	Q00532	Q96EB6	CDKL4	Q5MAI5	Q13451
CDKL5	O76039	P40692	CDKN1A	P38936	Q99741
CDKN1C	P49918	P24385	CDKN2A	P42771	O75496
CDKN2AIP	Q9NXV6	O94913, Q9H0D6	CDKN2B	P42772	Q9UI12
CDKN2C	P42773	P01185	CDKN3	Q16667, Q00526	P06493
CDON	O35158	Q13635	CDR2	Q01850	P51530
CDT1	Q9H211	Q99741	CDYL	Q9Y232	Q13257
CDYL2	Q8N8U2	Q9UBU9	CEACAM1	P13688	P13688
CEBPA	P49715	P18848	CEBPD	P49716	Q9BXW9
CEBPE	Q15744	P35638	CEBPZ	Q03701	P49841
CELF1	Q92879	P07910	CELSR2	Q9HCU4	Q92838
CENPA	P49450	P49736	CENPE	Q02224	Q9Y6D9
CENPF	P49454	Q96QG7	CENPH	Q9H3R5	P20700
CENPI	Q92674	Q13352	CENPJ	Q9HC77	Q9HC77
CENPL	P38265	P14136	CENPN	Q96H22	Q13352
CENPO	Q9BU64	Q13352	CENPQ	Q7L2Z9	P15336
CENPT	Q96BT3	P56539	CENPU	Q71F23-1, Q71F23	P53350
CEP126	Q9P2H0	Q99728	CEP131	Q9UPN4	P61981
CEP152	O94986	Q13568	CEP164	Q9UPV0	Q6IQ55
CEP170	Q5SW79	Q92845	CEP170B	Q9Y4F5	P61981
CEP19	Q96LK0	P55072	CEP250	Q9BV73	Q96CV9
CEP295	Q9C0D2	Q9NV70	CEP44	Q9C0F1	O43242
CEP55	Q53EZ4	O75419	CEP57L1	Q8IYX8-2, Q6P2R3	P09917
CEP76	Q8TAP6	Q8IW35, O43303	CEP78	Q5JTW2	P61981
CEP85	Q6P2H3-3	O00560	CEP85L	Q5SZL2-5	O00233
CEP89	Q96ST8	P61981	CEP95	Q96GE4	P36941

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CEP97	Q8IW35	O43303	CERS6	Q6ZMG9	Q9H244
CES1	P02795	P09429	CETN3	O15182	Q01831
CFAP20	Q9Y6A4	P53350	CFH	P08603	P01024
CFL2	Q549N0, Q9Y281	Q9NUX5	CFLAR	O15519, O15519-1	Q14790
CGGBP1	Q9UFW8	Q6IN84	CGRRF1	Q99675	Q9H2K0
CHAF1A	Q13111	Q96RU7	CHAF1B	Q13112	Q04637
CHAMP1	Q96JM3	Q9UI95	CHCHD1	Q96BP2	P04618
CHCHD10	Q8WYQ3	Q8N4J0	CHCHD2	Q9Y6H1	P07237
CHCHD4	Q8N4Q1	O43768	CHCHD5	Q9BSY4	P53701
CHCHD6	Q9BRQ6	Q9NX63, Q6UXV4	CHD1	O14646	Q96KQ7
CHD4	Q14839	Q13257	CHD9	Q3L8U1-3	Q07869
CHEK1	O14757	Q9HAW4, P30307	CHEK2	O96017	O96017, P30304
CHIC2	Q9UKJ5	O96015	CHML	P26374	Q92696
CHMP1B	Q7LBR1	P40818	CHMP2A	O43633	Q9Y3E7
CHMP3	Q9Y3E7, Q9Y3E7-1	O95630	CHMP4A	Q9BY43	Q8WUM4
CHMP5	Q9NZZ3	Q9Y3E7	CHMP7	Q8WUX9	Q9H444
CHN2	P52757	P63000	CHORDC1	Q9D1P4, Q9UHD1	P07900
CHRM4	P08173	Q99808	CHRNA5	P30532	P59665
CHST14	Q8NCH0	P27824	CHST15	Q7LFX5	P27824
CHST7	Q9NS84	P40855	CHTF8	P0CG13	P40937, Q9BVC3, P35249
CHTOP	Q9Y3Y2	Q14739	CIAO1	O76071	P51530
CIB1	Q99828	O60422	CIDEB	Q9UHD4	P15954
CIPC	Q9C0C6	Q92905	CIR1	Q86X95	Q07955, Q01130
CIRBP	Q14011	Q92973	CISD2	Q8N5K1	P36941
CISH	Q9NSE2	P00533	CIT	O14578	O14907
CKAP2	Q8WWK9	Q13257	CKAP4	Q07065	O75326
CKAP5	Q14008	Q13627	CKLF	Q13887	P0CG48
CKS1B	P61024	Q99640	CKS2	P33552	Q99640
CLASP1	Q7Z460	P31947	CLCF1	Q9UBD9, Q9UBD9-PRO_0000015616	P26992, O75462
CLCN3	P51790	P17931	CLCN7	P51798	Q53GS7
CLDN22	Q8N7P3	P25025	CLDN9	O95484	Q8IY26
CLDND1	Q9NY35	P25024	CLDND2	Q8NHS1	Q9H2K0
CLEC10A	Q8IUN9	Q9Y5Y5	CLEC11A	Q9Y240	Q14703
CLEC17A	Q6ZS10	Q15436	CLEC4A	Q9UMR7	Q9Y6N7
CLGN	O14967	Q99941, Q14703	CLIC1	O00299	O75832
CLIC3	O95833	Q9NUX5	CLIC4	Q9Y696	O15247
CLINT1	Q14677	O95630, Q96FJ0	CLK1	P49759	P61981
CLK3	P49761	Q7Z6E9	CLK4	Q9HAZ1	P52294
CLN6	Q9NWW5	Q96BA8	CLNK	Q7Z7G1	O14544
CLNS1A	P54105	P18509	CLOCK	O08785	P36873
CLP1	Q92989	Q9UPV0	CLPB	Q9H078	P46199
CLPP	Q16740	P54252	CLSPN	Q9HAW4	O14757
CLSTN1	O94985	Q8IV16	CLTA	P09496	Q13627
CLTC	Q00610	Q9NYZ3	CLU	P10909	P30101
CLUAP1	Q96AJ1	Q9NQC8, Q9BW83, Q9P2H3, Q9NWB7, Q9UG01, Q13099	CMAS	Q8NFW8	P06748
CMKLR1	Q99788	Q9Y6N7	CMTM2	Q8TAZ6	Q9NPL8
CMTM3	Q96MX0	Q6IN84	CMTM6	Q9NX76	Q9NZQ7

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CMYA5	Q8N3K9	O75923	CNIH1	O95406	Q6IN84
CNKSR2	Q8WXI2	P00533	CNN2	Q99439	P13569
CNNM3	Q8NE01	Q12974	CNOT11	Q9UKZ1	Q9BXI6
CNOT2	Q9NZN8	P63244	CNOT3	O75175	P55072
CNOT4	O95628-2	Q9UBH0	CNOT6	Q9ULM6	P47756
CNOT6L	Q96LI5	P47756	CNP	P09543	Q12824
CNPY4	Q8N129	Q9BRQ8	CNTF	P26441	P42702
CNTROB	Q8N137	Q96T60	COA1	Q9GZY4	Q9BQ95
COA3	Q9Y2R0	P00395	COA6	Q5JTJ3	Q9NPB3
COA7	Q96BR5	P29279	COCH	O43405	P08185
COG1	Q8WTW3	Q9NZQ7	COG3	Q96JB2	Q9NZQ7
COG4	Q9H9E3	Q9Y639	COG7	P83436	Q9NZQ7
COL13A1	Q5TAT6	Q6ISS4	COL25A1	Q8NE08	P17612
COL5A2	P05997	Q6ISS4	COL6A2	P12110	P14780
COL8A2	Q4VAQ0	Q00059	COLGALT2	Q8IYK4	Q9BT09
COLQ	Q9Y215	P22303	COMMD4	Q9H0A8	Q14186, O75461
COMMD8	Q9NX08	Q14186, O75461, Q14188	COMT	P21964	P30518
COPA	P53621	Q8WUJ3	COPB1	P53618	P56945
COPB2	P35606	P19838	COPG1	Q9Y678	Q9NTJ5
COPRS	Q9NQ92	P62805	COPS3	Q9UNS2	Q07889
COPS4	Q9BT78	P61081	COPS8	Q99627	Q15831
COQ3	Q9NZJ6	P49638	COQ5	Q5HYK3	Q13304, P35414
COQ8A	Q8NI60	Q9H2K0	COQ8B	Q96D53	P0DTC5
COQ9	O75208	Q14162	CORO1A	P31146	P26640
CORO1C	Q9ULV4	P13569	CORO2A	Q92828	Q8N695
COTL1	Q14019	P09917	COX11	Q9Y6N1	Q9Y337
COX15	Q7KZN9	P27824	COX5A	P20674	P00403
COX5B	P10606	P15954	COX6A1	P12074	P42858
COX6B1	P14854	P00403	COX6C	P09669	P00403
COX7A2	P14406	P00403	COX7A2L	O14548	O14949, P14927
COX7C	P15954	P00403	COX8A	P10176	P06748
CPD	O75976	P0DTC3	CPLX1	O14810	P63027
CPNE7	Q9UBL6-2	Q9UGI0	CPNE8	Q86YQ8-2	P04271
CPSF2	Q9P2I0	Q92797, Q9C0J8, Q6UN15	CPSF3	Q9UKF6	O95639
CPT2	P23786	Q9P0J0	CPVL	Q9H3G5	P47756
CR2	P20023	P01024	CRABP2	P29373	Q9HAW8
CRACR2B	Q8N4Y2-3	Q9UBH0	CRAMP1	Q96RY5	Q9Y4X5
CRBN	Q96SW2	P48729	CRCP	O75575	Q99081
CREB3L2	Q70SY1	Q96BA8	CREB3L3	Q68CJ9	Q02094
CREB5	Q02930-3	Q96PM5	CREBBP	Q92793	Q13568
CREBL2	O60519	P35638	CRELD1	Q96HD1-2	P19404
CRHBP	P24387	O95067	CRIP1	P50238	P04183
CRIP2	P52943	P82979	CRIPT	Q9P021	P78352
CRK	P46108	P16234	CRLF3	Q8IUI8	O14832
CRNKL1	Q9BZJ0	P62306, P14678, P62304	CROCCP2	Q86T23	Q8IY31
CROT	Q9UKG9	P16671	CRTAM	O95727-1, O95727	Q9BY67
CRTAP	O75718	O95076	CRTC1	Q6UUV9	P61981
CRTC3	Q6UUV7	Q9HC77	CRY1	Q16526	P49674
CRY2	Q49AN0	Q9H0D6	CSAD	Q9Y600	Q9NRF2-2
CSE1L	P55060	P04637	CSF1	P09603	P07333

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
CSF1R	P09581, P07333	P09603	CSF2	P04141	P32927
CSF2RB	P32927	O60674	CSF3R	Q99062	P40763
CSGALNACT2	Q8N6G5	Q8NBL1	CSNK1A1	P48729	O75330
CSNK1D	P48730	O75330	CSNK1E	P49674	O75695
CSNK1G2	P78368	Q9Y248, Q9BRT9, Q14691, Q9BRX5	CSPP1	Q1MSJ5-1	Q08379
CSRNP1	Q96S65	P36873	CSRP1	P21291	Q16790
CST3	P01034	P05067	CST7	O76096	P07711
CSTA	P01040	P13569	CSTB	P04080	P13569
CSTF1	Q05048	Q99728	CSTF2	P33240	Q99728
CSTF3	Q12996	Q6P1J9	CTAGE5	Q96PC5	P18846
CTBP1	Q13363	P28749, P06400	CTBP2	P56545	Q8IVP5
CTCF	P49711	P40429	CTDSP1	Q9GZU7	O00560
CTNNA1	P35221	P12830	CTNNAL1	Q9UBT7	O96006
CTNNB1	P35222	Q13761	CTNNBIP1	Q9NSA3	Q9BRT9
CTPS1	P17812	P01106	CTR9	Q6PD62	O00267
CTRC	Q99895	Q8NBK3	CTSB	P07858	P40692
CTSF	Q9UBX1	P59665	CTSO	P43235	P07711
CTSS	P25774	P14316	CTSZ	Q9UBR2	Q7Z3S9, P0DPK4
CTTN	Q14247	P32121	CTTNBP2NL	Q9P2B4	Q9Y6E0, Q9P289
CTU1	Q7Z7A3	P50552	CUBN	O60494	P27352
CUEDC2	Q9H467	Q9UM11	CUL3	Q13618	Q9BYZ6
CUX1	P39880	P06400	CWC15	Q9P013	Q96EB6
CWC22	Q9HCG8	P43115	CWC25	Q9NXE8	P19474
CWF19L2	Q2TBE0	Q9Y6D9	CXCL16	CXCL16	Q96BA8
CXCR1	P25024	P24593	CXCR2	P25025	P10145
CXCR3	P49682	P49682	CXCR4	P61073	Q13315
CXXC4	Q9EQC9	O14640	CXorf38	Q8TB03	Q02548
CYB561	P49447	P09601	CYB561A3	Q8NBI2	Q9NUQ2
CYB561D1	Q8N8Q1	Q05329	CYB561D2	O14569	Q96BA8
CYBRD1	Q53TN4	P09601	CYFIP1	Q7L576	Q9UQB8
CYP1A1	P04798	P62495	CYSLTR1	Q9Y271	Q9NS75
CYSRT1	A8MQ03	Q9Y584	CYSTM1	Q9H1C7	Q6ZMH5
CYTH1	Q15438	P07900	CYTH3	O43739	Q8N0S2
CYTIP	O60759	Q9HAQ2	DAB2	P98078, P98082	Q9UM54
DAD1	P61803	Q14165	DAG1	Q14118	Q14118
DAGLB	Q8NCG7	Q92989	DALRD3	Q5D0E6	P61978
DAPP1	Q9UN19	Q8WY64	DARS	P14868	P05412
DARS2	Q6PI48	Q8NFJ9	DAXX	Q9UER7	P46100
DAZAP2	Q15038	Q9HAU4	DBF4B	Q8NFT6	O00311
DBN1	Q16643	P60709	DBT	P11182	Q96E11
DCAF12	Q5T6F0	P98170	DCAF16	Q9NXF7	P04608
DCAF5	Q96JK2	P26358	DCAF7	P61962	Q13233
DCAF8	Q5TAQ9-2	Q00403	DCLRE1B	Q9H816	Q96EB6
DCLRE1C	Q96SD1	P78527	DCP1B	Q8IZD4	Q9NPJ4, Q9NPI6
DCP2	Q8IU60	Q92900	DCPS	Q96C86	Q96C86
DCTN2	Q13561	P24522	DCTN4	Q9UJW0	P13569
DCTPP1	Q9H773	P25786	DCUN1D1	Q96GG9	P61081
DCUN1D5	Q9BTE7	Q9NRD5	DDAH2	O95865	P48431
DDB1	Q16531	Q8NHY2	DDIT3	P35638	P17676

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
DDIT4	Q9NX09	Q9Y297	DDIT4L	Q96D03	P55957
DDN	O94850	Q96B97	DDR2	Q16832	Q16288
DDX1	Q92499-1	P04618	DDX19A	Q9NUU7	P13569
DDX19B	Q9UMR2	P45973	DDX24	Q9GZR7	Q9NWB7
DDX27	Q96GQ7	Q15646	DDX28	Q9NUL7	Q9NYY8
DDX39B	Q13838	P82979, Q9Y3Y2, O75152	DDX3X	O00571	Q04637, P06730
DDX47	Q9H0S4	Q9Y4X5	DDX5	P17844	Q09472
DDX51	Q8N8A6	P06748	DDX54	Q8TDD1	Q96GD4
DDX56	Q9NY93	P62753	DDX6	P26196	Q92900
DEF8	Q6ZN54-2	Q14957	DEGS1	O15121	Q68D85
DEK	P35659	Q9UER7	DENND1A	Q8TEH3	P61981
DENND4C	Q5VZ89	P61981	DENND6A	Q8IWF6	Q15116
DEPDC1B	Q8WUY9	P61981	DEPDC5	O75140-2	Q6W0C5
DEPTOR	Q8TB45	Q13077	DERL1	Q9BUN8	Q8TB40
DERL2	Q9GZP9	Q9NR28	DERL3	Q96Q80	Q9UKG4
DESI2	Q9BSY9	Q99541	DFFA	O00273	O76075
DFFB	O76075	O00273	DFNA5	O60443	Q86UL3
DGKA	P23743	P12931	DGKD	Q16760	O75923
DGKQ	P52824	Q8TDQ0	DGKZ	Q13574-2	Q08999, P28749, P06400
DHCR24	Q15392	P62736	DHFR	P00374	Q86XF0
DHFR2	Q86XF0	Q03405	DHODH	Q02127	P49638
DHRS1	Q96LJ7	Q9NR28	DHRS7	Q9Y394	Q9NTJ5
DHRS7B	Q6IAN0	P13569	DHX34	Q14147	Q16659
DHX36	Q9H2U1	Q16629, P84103	DHX38	Q92620	Q8WUA2, Q92917, O60231
DHX40	Q8IX18	P52294	DIABLO	EBI-20620429	Q01094
DIAPH3	Q9NSV4	Q6UWR7	DIEXF	Q68CQ4	P32121
DIMT1	Q9UNQ2	P19320	DIP2A	Q14689	P19883
DIRAS1	O95057	P52306	DISC1	Q9NRI5	P13639
DISP1	Q96F81	O76024	DIXDC1	Q155Q3	O14641
DLAT	P10515	P10515	DLC1	P63167	O75330
DLEU1	O43261	Q92993	DLG2	Q15700	O60469
DLG4	P78352	P27037	DLG5	Q8TDM6	P61981
DLGAP4	Q9Y2H0-1	Q8N3R9	DLGAP5	Q15398	P61769
DMAP1	Q9NPF5	Q9BRT9	DMC1	Q14565	Q06609
DMD	P11532	Q13325	DMKN	Q6E0U4	Q05329
DMTN	Q08495	P31947	DMWD	G5E9A7	Q8WUX9
DNA2	P51530	Q15185	DNAAF5	Q86Y56	P43119
DNAJA4	Q8WW22-2	Q8N9N5	DNAJB1	P25685	Q9Y266
DNAJB14	Q8TBM8	P13569	DNAJB5	O75953	P25685
DNAJB6	O75190-2	Q9BRX5	DNAJB9	Q9UBS3	Q13217
DNAJC1	Q96KC8	P13569	DNAJC12	Q9UKB3	Q96LA8
DNAJC18	Q9H819	P43119	DNAJC7	Q99615	Q9BQ52
DNASE1L1	P49184	P12830	DNASE1L2	Q92874	P11021
DNM2	P50570, P39052	Q9P0V3	DNMBP	Q6XZF7, Q6XZF7-1	P50570, O00401
DNMT3B	Q9UBC3	Q9Y6K1	DNPH1	O43598	O43598
DNTTIP1	Q9H147	P20248	DNTTIP2	Q5QJE6	Q6NZI2
DOCK1	Q14185	P46109	DOCK10	Q96BY6	P42357
DOCK11	Q5JSL3	P61981	DOCK3	Q8IZD9	Q92556
DOCK5	Q9H7D0	P46109	DOCK8	Q8NF50	Q99836

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
DOHH	Q9BU89	Q9NQZ5	DOK1	Q99704	P46109
DOK2	O60496	P46109	DOLK	Q9UPQ8	Q9NZQ7
DONSON	Q9NYP3	P33992, P49736	DPAGT1	Q9H3H5	P42858
DPCD	Q9BVM2	Q92696	DPEP3	Q9H4B8	P11021
DPF3	Q92784-1, Q92784, Q92784-2	Q8NFD5, P51531, O96019	DPH1	Q9BZG8	O75386
DPP3	Q9NY33	Q14145	DPP4	P27487	P27487
DPY19L1	Q2PZI1	O75326	DPY19L3	Q6ZPD9	P13671
DPY30	Q9C005	O00487	DPYSL2	Q16555	Q8NFD5
DRAP1	Q14919	Q00403	DRAXIN	Q8NBI3	O95631
DRG1	Q92597-3	Q9Y3D8	DSC1	Q08554	P13569
DSCC1	Q9BVC3	P0CG13, P40938, P40937, P35249	DSCR3	O14972	P23025
DSG2	Q14126	P13569	DSN1	Q9H410	P51587
DSTYK	Q6XUX3	Q9Y639	DTL	Q9NZJ0	Q8N4J0
DTNBP1	Q96EV8	A1L4H1	DTWD1	Q8N5C7	O15431
DTX3	Q8N9I9	P51668	DUS3L	Q96G46	Q96CV9
DUSP1	P28562	Q13309	DUSP10	Q9Y6W6	P60709
DUSP11	O75319	P09651	DUSP12	Q9UNI6	P55010
DUSP16	Q9BY84	Q16539	DUSP2	Q05922	P28482
DUSP4	Q13115	P28482	DUSP6	Q16828	P28482
DUSP7	Q16829	Q16539	DUSP8	Q13202	P45983
DVL1	O14640	Q9Y4D1	DXO	O77932	P08670
DYM	Q7RTS9	P21453	DYNC1I2	Q13409	P43034
DYNC1LI2	O43237	P13569	DYNLL1	P63167	O75330
DYNLRB1	Q9NP97	P10636-8	DYNLT1	P63172	Q9BRX2
DYRK1A	Q13627	P53350	DYRK2	Q92630	P23497
DYRK4	Q9NR20	P07900	DZIP3	Q86Y13	P51668
E2F1	Q01094	P28749, P06400	E2F2	Q14209	P06400
E2F3	O00716	P43246	E2F5	Q15329	Q08999, P06400
E2F7	Q5RIX9	Q16665	E2F8	A0AVK6	P07550
E4F1	Q66K89	Q13547	EAPP	Q56P03	P62306, P14678, P62304
EARS2	Q5JPH6	Q9Y639	EBP	Q15125	Q9H244
ECHDC2	Q86YB7	O75380	ECHS1	P30084	P11182
ECI2	O75521	P13569	ECSIT	Q9BQ95	P46199
ECT2	Q9H8V3	P12830	EDA2R	Q9HAV5	Q13114
EDC3	Q96F86	Q9NPJ4, Q9NPI6	EDEM1	Q92611	Q9UMX1
EDEM3	Q2HXL6	P30101	EDRF1	Q3B7T1	P07900
EED	O75530	Q9Y6K1	EEF1A1	P68104	Q9NYA1
EEF1AKMT1	Q8WVE0	P68104	EEF1B2	P24534	P26641, P68104
EEF1D	P29692-2, P29692	P26641, P29692	EEF1DP3	Q658K8	P07550
EEF1E1	O43324	P04049	EEF2K	O00418	O75344
EEPD1	Q7L9B9	P18827	EFCAB14	O75071	P68400
EFEMP2	O95967	Q9UBX5, P15502	EFHC1	Q5JVL4	P19623
EFHC2	Q5JST6	O94955	EFHD1	Q9BUP0	P03372
EFNA1	P20827	P54764	EFNA5	P52803	O00590
EFNB2	P52799	P48163	EGFL6	Q8IUX8	Q99962
EGLN3	Q9H6Z9	P14618-1	EHBP1L1	Q8N3D4	Q96ES7
EHD3	Q9NZN3	Q9H1K0	EHD4	Q9H223	Q9H1K0
EHHADH	Q08426	P36941	EI24	O14681	Q12982

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
EID3	Q8N140	Q9H4I9	EIF1AD	Q8N9N8	Q9H668, Q2NKJ3
EIF1AY	O14602	P47813	EIF2AK2	P19525	P55265
EIF2B2	P49770	O14975	EIF2B3	Q9NR50	P43115
EIF2B4	Q9UI10	P01019	EIF2B5	Q13144	P43115
EIF2S1	P05198	P19525	EIF2S2	P20042	P01138
EIF3B	P55884	Q04637, P06730	EIF3D	O15371	P21673
EIF3G	O75821	Q9BRX2	EIF3I	Q13347	Q04637
EIF3J	O75822	Q04637	EIF4A1	P60842	Q04637, Q15056, P06730
EIF4A2	Q14240	Q04637, P06730	EIF4B	P23588	P53350
EIF4E	P06730	Q92900	EIF4E2	O60573-1, O60573	Q13541
EIF4EBP1	Q13541	P06730	EIF4G1	Q04637	P06730
EIF5	P55010	P47813	EIF6	P56537	P19525
ELAVL1	Q15717	P09651	ELF1	P32519	P08047
ELF2	Q15723	Q01196	ELF3	P78545	Q16236
ELMO2	Q96JJ3	Q8IZD9	ELOC	Q15369	Q16665
ELOVL5	Q9NYP7	P09601	ELOVL6	Q9H5J4	P09601
ELOVL7	A1L3X0	Q96EV8	EMC10	Q5UCC4	Q5J8M3
EMC8	O43402	P27824	EMC9	Q9Y3B6	Q5J8M3
EMD	P50402	P02545-1	EME1	Q96AY2	Q96NY9
EMILIN1	Q9Y6C2-2, Q9Y6C2	Q96QG7	EML2	A0A0C4DGQ7	P40937
EML3	Q32P44	P63208	EML4	Q9HC35	P63208
ENC1	O14682	Q13501	ENG	P17813	P05556
ENKD1	Q9H0I2	P52747	ENO1	P06733	P03496
ENO2	P09104	P06733	ENO3	P13929	P06733
ENPP4	Q9Y6X5	Q9H8X2	ENTPD1	P49961	Q96S59
ENTPD6	O75354	P27824	EOGT	Q5NDL2	Q03405
EP300	Q09472	Q13761	EPAS1	Q99814	P40818
EPB41	P11171	P61981	EPB41L2	O43491	P42568, Q03111
EPB41L3	Q9Y2J2	P49841	EPC1	Q9H2F5	O96019
EPC2	Q52LR7	P52294	EPHA1	P21709	P21359
EPHA4	P54764	P05177	EPHB4	P54760	P01588
EPHX1	P07099	P13569	EPM2A	O95278, Q9WUA5	Q9UQK1, Q6VVB1
EPRS	P07814	Q8IWL3	EPS15L1	Q9UBC2	P53350
ERAL1	O75616	P08123	ERBB2	P04626	Q9BVV7
ERBB3	P21860	Q14451	ERBIN	Q96RT1	P36873
ERC1	Q8IUD2	P01222	ERCC2	P18074	P15056
ERCC4	Q92889	P23025	ERCC5	P28715	P24522
ERCC6L	Q2NKX8	O14757	ERCC8	Q13216-1	Q03468
EREG	O14944	P00533	ERF	P50548	P28482
ERFE	Q4G0M1	Q07954	ERG28	Q9UKR5	Q6IN84
ERGIC1	Q969X5	P04601	ERGIC2	Q96RQ1	P15907
ERH	P84090	Q9Y5B0	ERI1	Q8IV48	Q14493
ERLIN1	O75477	Q7L5A8	ERMP1	Q7Z2K6	P51684
ERN1	O75460-2	P38936	ERO1A	Q96HE7	P30101
ERO1B	Q86YB8	P0DTC8	ERP27	Q96DN0	P29692
ERP44	Q9BS26	P43251	ERRFI1	Q9UJM3	P31947
ESCO2	Q56NI9	Q08379	ESPL1	P60330, Q14674	O95997
ESRRA	P11474	Q9UBK2	ESYT1	Q9BSJ8	P22309
ESYT2	A0FGR8	P13569	ETFA	P13804	Q8IWL3
ETHE1	O95571	Q99757	ETNK2	Q9NVF9	Q16623

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ETS1	P14921-1	Q8NHY2	ETS2	P15036	Q8NHY2
ETV4	P43268-3	Q13485	ETV6	P41212	Q13257
EVA1B	Q9NVM1	O43826	EVA1C	P58658	O14786
EVI2A	P22794	Q9Y5Z9	EVI2B	P34910-2	P22732
EVI5	O60447	P53350	EXO1	Q9UQ84-1, P39875	P43246
EXOC1	Q9NV70	P18848	EXOC2	Q96KP1	P27930
EXOC3-AS1	Q8N2X6	Q12933	EXOC5	O00471	Q9ULZ3
EXOC6B	Q9Y2D4	O60645, Q8TAG9, Q8IYI6, O00471, Q9UPT5, Q96A65, Q9NV70, Q96KP1	EXOSC1	Q9Y3B2	P52747
EXOSC3	Q9NQT5	P42568	EXOSC4	Q9NPD3	Q99547, Q9Y2L1, Q13901, P42285, Q01780, Q9NQT4
EXOSC5	Q9NQT4	Q8WY64	EXTL3	O43909	P61073
EYA3	Q99504	P10809	EZH2	Q15910	Q9Y6K1
F13A1	P00488	P19388	F2RL1	P55085	Q9BVG9
F5	P12259	P32971	FAAH	Q7L5A8	Q96BA8
FAAP100	Q0VG06	P99999	FAAP24	Q9BTP7	Q8IYD8
FABP5	Q01469	P13569	FADD	Q13158	Q14790
FADS2	Q8NFF5-2	P29373	FADS3	Q9Y5Q0	Q8N5I4
FAF2	Q96CS3	Q9NWR8	FAH	P16930	P00533
FAHD1	Q6P587-2	Q12982	FAHD2B	Q6P2I3	P10809
FAM102A	Q5T9C2-3	P56945	FAM104A	Q969W3	P17612
FAM109A	Q8N4B1-4	P62993	FAM110A	Q9BQ89	P49674
FAM110C	Q1W6H9	P49674	FAM111A	Q96PZ2	P04637
FAM111B	Q6SJ93	Q01105	FAM114A1	Q8IWE2	Q9UBN6
FAM114A2	Q9NRY5	Q969Z0	FAM117A	Q9C073	P61981
FAM118B	Q9BPY3	Q9Y639	FAM120A	Q9NZB2	Q92900
FAM122C	Q6P4D5-2	P27797	FAM126A	Q9BYI3	Q9UHJ6
FAM129A	Q9BZQ8	P06748	FAM129B	Q96TA1	P13569
FAM136A	Q96C01	Q9H668	FAM161A	Q3B820	Q9Y2I6
FAM169A	Q9Y6X4	P20700	FAM171B	Q6P995	P21802
FAM172A	Q8WUF8	P63208	FAM177A1	Q8N128	O14975
FAM184A	Q8NB25	P06396	FAM189B	P81408	Q96PU5
FAM193B	Q96PV7-2	P14373	FAM199X	Q6PEV8	P49674
FAM204A	Q9H8W3	O60341	FAM206A	Q9NX38	Q7L7X3
FAM207A	Q9NSI2	Q13895	FAM20B	O75063	Q9H8X2
FAM210A	Q96ND0	Q9Y366	FAM210B	Q96KR6	P09601
FAM212A	Q96EL1	Q9HAQ2	FAM213A	Q9BRX8	Q9UNA3
FAM213B	Q8TBF2	O43776	FAM219A	Q8IW50	O43924
FAM43A	Q8N2R8	Q9BUR4	FAM46A	Q96IP4	P43405
FAM50B	Q9Y247	P40692	FAM53B	Q14153	P31947
FAM53C	Q9NYF3	P53350	FAM60A	Q9NP50	Q9Y6N5
FAM76B	Q5HYJ3-3	O76024	FAM81A	Q8TBF8	Q9UBB9
FAM83A	Q86UY5	P49674	FAM83D	Q9H4H8	O75330
FAM90A1	Q86YD7	Q15669	FAM92A	A1XBS5-3, A1XBS5	Q9Y3M2
FAM96A	Q9H5X1	O75582	FAM96B	Q9Y3D0	P08670, P06396
FAM98B	Q52LJ0	P10636-8	FAM9B	Q8IZU0	Q96E11
FANCA	O15360	P08173	FANCB	Q8NB91	P99999
FANCD2	Q9BXW9	O00255	FANCE	Q9HB96	Q00597
FANCF	Q9NPI8	P42858	FANCG	O15287	P08684

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
FANCI	Q9NVI1	Q9Y5B0	FANCL	Q9NW38	Q13485
FANCM	Q8IYD8	P49736	FAR2	Q96K12	P40855
FARP1	Q9Y4F1	P18827	FARSA	Q9Y285	P10636-8
FARSB	Q9NSD9	P10636-8	FAS	P25445-1, P25445	Q14790
FASLG	P48023	O75044	FASN	P49327	Q16665
FASTKD1	Q53R41	P07237	FASTKD3	Q14CZ7	Q8N307
FASTKD5	Q7L8L6	P0DTC5	FAT1	Q9QXA3	Q14160
FAU	P62861	P63244	FBN1	P35555	P15502
FBN2	P35556	P04608	FBP1	P09467	Q16665, Q99814
FBRSL1	Q9HCM7	P68400, P67870	FBXL12	Q9NXK8	P63208, Q13616
FBXL15	Q9H469	Q9HAU4	FBXL17	Q9UF56	O14867
FBXL18	Q96ME1	P63208, Q13616	FBXL19	Q6PCT2-2	Q6ZPD8
FBXL20	Q96IG2	P63208	FBXL3	Q9UKT7	Q9UJ68
FBXL5	Q9UKA1	P63208, Q13616	FBXO11	Q86XK2	O95863
FBXO2	Q9UK22	Q99941	FBXO22	Q8NEZ5	O14867
FBXO25	Q8TCJ0	P60709	FBXO28	Q9NVF7	Q13177
FBXO3	Q9UK99	P04608	FBXO30	Q8TB52	Q9UM54
FBXO31	Q5XUX0	Q13315	FBXO33	Q7Z6M2	P63208, Q13616
FBXO34	Q9NWN3	Q9UM54	FBXO38	Q6PIJ6	P63208
FBXO42	Q6P3S6	Q06330	FBXO43	Q4G163	P63208
FBXO45	P0C2W1	P63208	FBXO48	Q5FWF7	P63208
FBXO5	Q9UKT4	P63208	FBXO6	Q9NRD1	P98160
FBXO7	Q9Y3I1	Q9P0P0	FBXO9	Q9UK97-2, Q9UK97	P63208
FBXW11	Q9UKB1	P30304, P30291	FBXW4	P57775	Q13371
FBXW5	Q969U6	P63208, Q13616	FBXW7	Q969H0	P46531
FBXW9	Q5XUX1	P63208, Q13616	FCER2	P06734	P21802
FCGRT	P55899	P22455	FCHO2	Q0JRZ9	P98082
FCHSD2	O94868	P55072	FCN1	O00602	P18433
FCRL2	Q96LA5	O43639	FCRL3	Q96P31-6	P22732
FDPS	P14324	Q14997	FEM1B	Q9UK73	P49593
FEM1C	Q96JP0	Q93034	FEN1	P39748	Q14191
FERMT3	Q86UX7-2	Q99633	FGD4	Q96M96	P17252
FGFR1	P11362-2	Q9BRX2	FGFR1OP	O95684	Q96NL6
FGFR1OP2	Q9NVK5	Q9BRV8	FGL2	Q14314	Q9BYC5
FH	P07954	P11309	FHL2	Q14192	Q9NYA1
FHL3	Q13643	Q96E11	FHOD1	Q9Y613	P19474
FIBP	O43427	P80370	FIGNL1	Q6PIW4	Q9NSG2
FIZ1	Q96SL8	Q04721	FKBP10	Q96AY3	P02452
FKBP1A	Q0VDC6	Q8WUX9	FKBP3	Q00688	Q00987
FKBP4	Q02790	Q9UKV8	FKBP9	O95302	Q8NDV7
FKBPL	Q9UIM3	P49674	FLAD1	Q8NFF5	Q8NFF5
FLI1	Q01543	P84022	FLII	Q13045	P13569
FLNA	P21333-2	Q9BRX2	FLNB	O75369	Q13233
FLT3	P36888	Q06124	FLYWCH2	Q96CP2	P17980
FMNL1	O95466	O75044	FN1	P02751	P05556
FNBP1	Q8R511	P52198	FNBP4	Q8N3X1	O75400
FNDC3A	Q9Y2H6	O43511	FNDC3B	Q53EP0	Q9NPD5
FNTB	P49356	P01185	FOS	P01100	Q8NHY2
FOSB	P53539	Q9BZS1	FOSL2	P15408	Q8NHY2

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
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FOXK1	P85037	P53611	FOXM1	Q08050	Q13416
FOXN2	P32314	Q9Y297	FOXN3	O00409	Q9Y297
FOXO3	O43524	P28749, P06400	FOXO4	P98177	P98177
FOXP1	Q9H334	Q9BZS1	FOXP3	EBI-10097414	P62805
FOXP4	Q8IVH2	Q9BZS1	FOXRED2	Q8IWF2	Q9UBV2
FPR1	P21462	Q13315	FPR2	P25090	Q99808
FRAT1	Q92837	P49841	FRAT2	O75474	P49841
FRG1	Q14331	Q9HCG8	FRMD3	A2A2Y4	Q9Y6X1
FRS2	Q8WU20	Q06124	FSBP	O95073-2	P16885
FSCN1	Q16658	P54278	FSD1	Q9BTV5	Q13177
FTSJ3	Q8IY81	P19525	FUCA1	P04066	Q8NBK3
FUNDC1	Q8IVP5	Q9GZQ8	FUNDC2	Q9BWH2	Q8IVP5
FUT8	Q9BYC5	P26572	FXR2	P51116	Q04637
FYN	P06241	P12931	FZD4	Q9ULV1	P42858
FZD8	Q9H461	P56704	G0S2	P27469	Q07817
GAA	P10253	P98160	GAB1	Q13480	P46109
GAB2	Q9UQC2-1	P62993-1	GABARAPL1	Q9H0R8	O75385
GABARAPL2	P60520	O75385	GABBR1	Q9UBS5	P46459
GABPA	Q06546	P08047	GADD45B	O75293	O14733
GADD45GIP1	Q8TAE8	Q14197	GALK1	P51570	Q9NPJ4
GALNT10	Q86SR1	Q07817	GALNT11	Q8NCW6	Q15116
GALNT12	Q8IXK2	Q9H2A9	GALNT18	Q6P9A2	P01185
GALNT4	Q8N4A0	Q13790	GALNT6	Q8NCL4	O75063
GAPDH	P04406	P35228	GAPVD1	Q14C86	P49674
GAR1	Q9NY12	O14744	GARS	P41250	Q8WXH5
GART	P22102	P13569	GAS2L1	Q99501	Q96B97
GAS2L3	Q86XJ1	Q96GD4	GATA2	P23769	O00712, P08651, Q12857
GATAD1	Q8WUU5	Q13547	GATAD2A	Q86YP4	P52294
GATAD2B	Q8WXI9	P52294	GATB	O75879	Q07954
GATC	O43716	P09917	GATD1	Q8WUU5	Q13547
GATM	P50440	Q9P1W8	GBA	P04062	P51159
GBP1	P32455	P32455	GBP2	P32456	P32455
GBP3	Q9H0R5	P32455	GCAT	O75600	O75380
GCC1	Q96CN9	P08151	GCC2	Q8IWJ2	P18509
GCLM	P48507	P48506	GCNT1	Q02742	Q07954
GDI2	P50395	P02751	GEM	P55040	Q9BUH8
GEMIN2	O14893	P62316, P62306, P14678, P62308, P62314, P62304	GEMIN5	Q8TEQ6	P06730
GEMIN6	Q8WXD5	P62316, P62306, P14678, P62308, P62314, P62304	GFER	P55789	P11802
GFI1	Q07120	Q96KQ7	GFM2	Q969S9	Q8N9N5
GFOD1	Q9NXC2	Q9BX63	GFRA3	O60609	Q5T4W7
GGCX	P38435	P0DTC5	GGH	Q92820	P27824
GGPS1	O95749	O00244	GGT1	P19440-3	Q92993
GID8	Q9NWU2	O60313	GIGYF1	O75420	Q86UK7
GIMAP2	Q9UG22	P42858	GIMAP5	Q96F15	P21579
GINS1	Q14691	P33992	GINS2	Q9Y248	O96017
GINS3	Q9BRX5	P49736	GINS4	Q9BRT9	O75419
GJC1	P36383	P43119	GKAP1	Q5VSY0	Q92993

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GLB1	P16278	Q9UM11	GLCCI1	Q86VQ1	Q13627
GLDC	P23378	Q9BQB4	GLE1	Q53GS7	Q00994
GLI2	P10070, Q0VGT2	Q9UMX1	GLI3	P10071	Q9UMX1
GLO1	Q04760	P13569	GLRX5	Q86SX6	Q8IWL3
GLT8D1	Q68CQ7	P48960	GLTP	Q9NZD2	Q9NQZ5
GLUL	P15104	Q9BZE9	GLYCTK	Q8IVS8	P01189
GM2A	P17900	P62736	GMEB1	Q9Y692	Q13114
GML	Q99445	Q92538	GMNN	O75496	P06493
GMPPB	Q9Y5P6	Q8IVS8	GMPR2	Q9P2T1	P04626
GMPS	P49915	Q93009	GNA13	Q14344	Q92888
GNA15	P30679	P11142	GNAI1	P63096-1	P63211
GNAQ	P50148	P49407	GNAS	P63092	P23786
GNAZ	P19086	P54764	GNB2	P62879	Q13371
GNB4	Q9HAV0	P18827	GNG10	P50151	Q13371
GNG2	P59768	P08173	GNG5	P63218	Q9UBC2
GNL1	P36915	Q9UG63	GNLY	P22749	Q07817
GNPAT	O15228	Q9Y639	GNS	P15586	Q8NFI3
GOLGA1	Q92805	P18848	GOLGA2P5	Q9HBQ8	P04792
GOLGA4	Q13439	Q96ES7	GOLGA6L9	A6NEM1	Q13415
GOLGA7	Q7Z5G4	Q8TDQ0	GOLIM4	O00461	Q9Y2C2
GOLM1	Q8NBJ4	Q07817	GOLPH3L	Q9H4A5	P21453
GON4L	Q3T8J9	P16104	GON7	Q9BXV9	Q96S44
GOPC	Q9HD26	Q6UWP7	GORAB	Q5T7V8	Q96PM5
GOSR1	O95249	Q9NZ43, O75396	GOT1	P17174	P15056
GOT2	P00505	P13569	GPA33	Q99795	Q9NUX5
GPANK1	O95872	Q13190	GPAT3	Q53EU6	Q9H2J7
GPAT4	Q86UL3	O60443	GPATCH11	Q8N954	P62306, P14678, P62304
GPATCH8	Q9UKJ3	P26368	GPBAR1	Q8TDU6	P38646
GPC1	P35052	O15530	GPD1L	Q8N335	P13501
GPHN	Q9NQX3-2	P25786	GPI	P06744	P13569
GPKOW	Q92917	P23193	GPN1	Q9HCN4	O15160, O95602, P19388, P61218, P62875
GPN3	Q9UHW5	P54764	GPNMB	Q14956-1, Q14956	P00533
GPR141	Q7Z602	O00186, Q12846	GPR18	Q14330	P42357
GPR25	O00155	Q9UBD6	GPR35	Q9HC97	Q15800
GPR55	Q9Y2T6	P17931	GPS2	Q13227	P18848
GPX1	P07203	P13569	GRAMD1B	Q3KR37	P68400
GRAMD4	Q6IC98	Q96QG7	GRAP	Q13588	P19474
GRAP2	O75791, O89100	P40818	GRASP	Q7Z6J2	Q9BYF1
GRB10	Q13322-4	P12931	GRIP1	Q9Y3R0	Q14653
GRK3	P35626	P25098	GRK5	P34947	P25963
GRM2	Q14416	Q9H2H9	GRPEL2	Q8TAA5	P11182
GRSF1	Q12849	O75832	GSE1	Q14687	Q13547
GSKIP	Q9P0R6	P49841	GSN	P06396	Q8WUX9
GSPT1	P15170	Q92900	GSPT2	Q8IYD1	Q92900
GSTCD	Q8NEC7	P40763	GSTO1	P78417	P04792
GSTP1	P09211	P13569	GSTZ1	O43708	Q12933
GTF2A1	P52655	P62333	GTF2A2	P52657	P20226
GTF2E1	P29083	Q9Y6K9	GTF2E2	P29084	P43897
GTF2F1	P35269	Q9Y5B0	GTF2H1	P32780	Q9Y266

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GTF2H3	Q13889	P50613	GTF2I	P78347	Q06187
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GTPBP1	O00178	P50552	GTPBP2	Q9BX10	Q9UBU9
GTSE1	Q9NYZ3	P53350	GUCD1	Q96NT3-2	Q13163
GUK1	Q16774	Q6DKK2	GYS1	P13807	Q04446, Q9UQK1
GZF1	Q9H116	P78563	H2AFX	P16104	P49736
H2AFY	O75367	Q01831	H2AFZ	P0C0S5	Q01831
H3F3A	P84243	Q9Y6K1	H3F3B	P84243	Q9Y6K1
HACD3	Q9P035	P0DTC4	HACE1	Q8IYU2	Q96CV9
HADH	P40939	Q16288	HAL	P42357	Q86WX3
HAPLN3	Q96S86	P11021	HASPIN	Q8TF76	Q07954
HAT1	O14929	Q92560	HAUS1	Q96CS2	Q9Y6D9
HAUS2	Q9NVX0	Q13426	HAUS3	Q68CZ6	A1L190
HAUS4	Q8BFT2	Q9BUH8	HAUS7	Q99871	Q6W0C5
HAUS8	Q9BT25	Q6VN20	HAVCR1	Q96D42	P0DTC2
HAX1	O00165	Q9BRX2	HBP1	Q8IV16	Q14703
HBQ1	P09105	P02100, P02042, P68871, P69892	HCCS	P53701	Q96E11
HCFC1	P51610	P08047	HCFC2	Q9Y5Z7	Q92560
HCK	P08631	Q13177	HDAC1	Q13547	Q9UJW3
HDAC11	Q96DB2	P17707	HDAC2	Q92769	P53350
HDAC4	P56524	Q13761	HDAC9	Q9UKV0	Q06413
HDDC2	Q7Z4H3	P01189	HDDC3	Q8N4P3	O00471
HDGF	P51858	P78527	HDGFL2	Q7Z4V5	P32121, P49407
HDLBP	Q00341	Q06609	HEATR6	Q6AI08	P34741
HELB	Q8NG08	P20248	HELLS	Q9NRZ9	P01106
HELZ	P42694	Q8IWL3	HEMK1	Q9Y5R4	P28070
HERC1	Q15751	O95395	HERC3	Q15034	Q9UNU6
HERPUD1	Q15011	Q86TM6	HERPUD2	Q9BSE4	P30825
HES1	Q14469	Q9Y5J3	HES4	Q9HCC6	Q00403
HES6	Q96HZ4	Q14469	HEXIM1	O94992	P50750, O60563
HEXIM2	Q96MH2	P11309	HEY2	Q9UBP5	Q9Y5J3
HFE	Q30201	O14672	HGH1	Q9BTY7	P08238
HGS	O14964	Q9Y5J6	HHEX	Q03014	P06730
HIC1	Q14526	P35222	HIC2	Q96JB3	Q96KQ7
HID1	Q8IV36	O15357	HINT1	P49773	P13569
HINT3	Q9NQE9	P40855	HIP1	O00291	Q86US8
HIPK1	Q86Z02	P04637	HIPK2	Q9H2X6	Q09472
HIRA	P54198	P84243	HIRIP3	Q9BW71	P68400, P67870
HIST1H1B	P16401	P19338	HIST1H1E	P10412	P05455
HIST1H2AG	P0C0S8	O95760	HIST1H2AI	P0C0S8	O95760
HIST1H2AL	P0C0S8	O95760	HIST1H2BD	P58876	Q01831
HIST1H2BG	P62807	Q96CW1	HIST1H2BH	Q93079	P62993
HIST1H2BJ	P06899	P10636-8	HIST1H2BK	O60814	Q12824
HIST1H2BL	Q99880	Q96CW1	HIST1H3A	P68431	P49736
HIST1H3B	P68431	P49736	HIST1H3E	P68431	P49736
HIST1H3F	P68431	P49736	HIST1H3G	P68431	P49736
HIST1H3H	P68431	P49736	HIST1H3J	P68431	P49736
HIST1H4A	P62805	P33992, P25205, P49736	HIST1H4D	P62805	P33992, P25205, P49736
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Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
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HK2	P52789	P19367	HK3	P52790	P19367
HLA-B	P01889	O94822	HLA-C	P10321	A0PJW6
HLA-E	P13747	Q09328	HLA-F	P30511	P61769
HLA-G	P17693	P61769	HLTF	Q14527	Q9NTX7
HMBOX1	Q6NT76	Q02548	HMBS	P08397	Q9NRD5
HMGA1	P17096	P17676	HMGB2	P26583	P48729
HMGB3	O15347	Q02548	HMGCL	P35914	P11182
HMGCR	P04035	P51684	HMGN2	P05204	Q04446
HMGXB4	Q9UGU5	Q09028	HMMR	O75330	Q9ULW0
HMOX1	P09601	Q9NUX5	HMOX2	P30519	P0DTC3
HNRNPA1	P09651	Q8IZH2	HNRNPA1L2	Q32P51	P09651
HNRNPAB	Q99729	Q9NZI8	HNRNPDL	O14979	P13569
HNRNPF	P52597	O43347	HNRNPH1	P31943	P09651
HNRNPL	P14866	P09651	HNRNPLL	Q8WVV9	P61978
HOMER2	Q9NSB8	Q9H4P4	HOMEZ	Q8IX15-3	P52292
HOOK1	Q9UJC3	P53611	HOXA10	P31260	P17947
HOXA5	P20719	Q96RT1	HOXA7	EBI-1801907	Q03164
HOXC4	P09017	Q9Y3P9	HPGDS	O60760	Q13370
HPRT1	P00492	Q9H1K1	HPS1	Q92902	Q9C0C7
HRAS	P01112	P21359	HRASLS5	Q96KN8	Q96JH7
HS6ST1	O60243	Q9Y2A9	HSBP1	O75506	O75385
HSD17B10	Q99714	P13639	HSD17B4	P51659	P13569
HSD3B7	Q9H2F3	Q96FE5	HSF2	Q03933	Q00613
HSPA13	P48723	P01375	HSPA1A	P08107	P27695
HSPA1B	P08107	P27695	HSPA4	P34932	Q53H54
HSPA4L	O95757	P11021	HSPA5	P11021	P01222
HSPA8	P11142	O75496	HSPB11	Q9Y547	Q13114
HSPBAP1	Q96EW2	Q9NUJ1	HSPD1	P10809	Q16822
HSPE1	P61604	P13569	HSPH1	Q92598	P11021
HTATIP2	Q9BUP3-3	Q9H2K0	HTATSF1	O43719	P62306, P14678, P62304
HUS1	O60921	Q03405	HUS1B	Q8NHY5	P61289
HYAL3	O43820	O75553	IARS	P41252	P05412
IARS2	Q9NSE4	Q8IWL3	IBA57	Q5T440	P10809
IBTK	Q9P2D0	Q06187	ICA1L	Q8NDH6-2	O14964
ICAM2	P13598	Q9UI95	ICK	Q9UPZ9	Q9NRG4
ICMT	O60725	O00186	ICOS	Q9Y6W8	O75144
ID1	P41134	A6NI15	ID3	Q02535	P51149
IDH1	O75874	P0DP23	IDH2	P48735	P46059
IDH3B	O43837	P13569	IDH3G	P51553	P10809
IDS	P22304	Q8NBK3, Q8NBJ7	IER2	Q9BTL4	Q53GS7
IFFO1	Q0D2I5, Q0D2I5-5	Q13426	IFFO2	Q5TF58	P21802
IFI27	P40305-2	P22736	IFI30	P13284	O43663
IFIH1	Q9BYX4	P05161	IFIT1	P09914	Q86WV6
IFIT2	P09913	Q86WV6	IFNAR1	P17181	Q13332
IFNGR1	P15260	Q9NPF0	IFT172	Q9UG01	Q9Y266

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IFT20	Q8IY31-3	Q13286	IFT22	Q9H7X7	Q9NQC8, A0AVF1, Q9Y547, Q9BW83, Q8IY31, Q9P2H3, Q9NWB7, Q9Y366, Q9UG01, Q96LB3, Q8WYA0, Q13099
IFT43	Q96FT9, Q96FT9-2	Q7Z4L5, Q8NEZ3, Q96RY7, Q9HBG6, Q9P2L0	IFT46	Q9NQC8	A0AVF1, Q9Y547, Q9BW83, Q8IY31, Q9NWB7, Q9Y366, Q9H7X7, Q8WYA0, Q96AJ1, Q9P2H3, Q9UG01, Q96LB3, Q13099
IFT81	Q8WYA0	P36941	IGF1	P05019	P08833
IGF1R	P08069	P05556	IGF2BP3	O00425	Q9NZI8
IGF2R	P11717	Q9NZ52	IGFBP2	P18065	P59665
IGFBP3	P17936	Q86XT9	IGFBP5	P24593	Q8TB40
IGFBP6	P24592	P53611	IGHA1	P01876	Q16822
IGHA2	P01877	O94844	IGHEP2	P19823	O15520
IGHM	P01871	Q99856	IGSF6	O95976	Q9HD47
IGSF8	Q969P0	Q969Z0	IK	Q13123	Q00403
IKBIP	Q70UQ0	O14920	IKBKB	O14920	Q9ULZ3
IKBKE	Q14164	Q92844	IKBKG	Q9Y6K9	Q9UDY8
IKZF4	Q9H2S9	O00311	IKZF5	Q9H5V7	P61024
IL10RA	Q13651	P22301	IL12A	P29459	Q9Y624
IL13RA1	P78552	Q8TDR0	IL15	P40933	Q13261
IL15RA	Q13261	P40933	IL16	Q14005-2	P52747
IL17RA	Q96F46	Q9NRM6	IL17RD	Q8NFM7	Q9NYV6
IL18RAP	O95256	O76024	IL1RL1	Q01638-2	Q9H2H9
IL1RN	P18510	P14778	IL23A	Q9NPF7	Q9H4P4
IL2RA	P01589	P60568	IL2RB	P14784	P60568
IL32	P24001-2, C6GKH1	Q05655	IL4R	P24394	Q99640
IL6R	P08887-PRO_0000450730, P08887	P05231	IL7R	P16871	O43791
IL9R	Q01113	P63104	ILK	Q13418	P49327, Q13085
ILKAP	Q9H0C8	Q13418	IMMT	Q16891	P54252
IMPA1	P29218-3	P42858	INAVA	Q3KP66-1	Q07157
INCENP	Q9NQS7-1, Q9NQS7	O14965	INF2	Q27J81	P50548
ING3	Q9NXR8	P52294	INO80	Q9ULG1	O96019
INO80D	Q53TQ3	Q9Y265	INPP5D	Q92835, Q92835-2	O75525
INPP5K	Q9BT40	P22102	INPPL1	O15357	P26641
INTS11	Q5TA45	P52294	INTS12	Q96CB8	P61073
INTS13	Q9NVM9	Q13951	INTS14	Q96SY0	P67775
INTS2	Q9H0H0	P67775	INTS5	Q6P9B9	P08173
INTS6	Q9UL03	P33993	INTS7	Q9NVH2	P67775
INTS8	Q75QN2	P67775	IP6K1	Q92551	P21802
IP6K2	Q9UHH9	P07900	IPO11	Q9UI26	Q9NRM0
IPO13	O94829	P47813	IPO4	Q8TEX9	P01106
IPO7	O95373	Q9NZQ7	IQCC	Q4KMZ1	Q14145
IQCD	Q96DY2-2	Q9Y6A5	IQCE	Q6IPM2	P25788
IQGAP3	Q86VI3	P40855	IQSEC1	Q6DN90	Q13309
IRAK2	O43187	Q9Y4K3	IRAK4	Q9NWZ3	Q15109

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IRF2BP2	Q7Z5L9	Q8N6T7	IRF3	Q14653	Q96EN8
IRF8	Q02556	Q96CV9	IRGQ	Q8WZA9	Q9H2D1
IRS1	P35568	P11940	IRS2	Q9Y4H2	P27986
ISCA1	Q9BUE6	Q96RU8	ISCA2	Q86U28	P01160
ISCU	Q9H1K1	Q8IWL3	ISG15	P05161	P41226
IST1	P53990, P53990-3	P13798	ISY1	Q9ULR0	P40692
ITCH	Q96J02	P08151	ITFG1	Q8TB96	P11686
ITFG2	Q969R8	Q5T2D3	ITGA1	P56199	P17931
ITGA2	P17301	P05556	ITGA2B	P08514-1	P21333
ITGA3	P26006	P48509	ITGA5	P08648	P05556
ITGA6	P23229	P05556	ITGAM	P11215	P21246
ITGAV	P06756	P02751	ITGB1	P05556	P08648
ITGB1BP1	O14713	Q15669	ITGB2	P05107	Q9H2K0
ITGB3BP	Q13352	Q00613	ITGB7	P26010, P26010-1	P21333
ITM2C	Q9NQX7	P13569	ITPKB	P27987	Q8N9R8
ITPRIP	Q8IWB1	Q96FT7	ITPRIPL1	Q6GPH6-2	Q12933
ITPRIPL2	Q3MIP1	Q9BRI3	ITSN2	Q9NZM3	Q9UBC2, P00533
IVD	P26440	Q15831	IVNS1ABP	Q9Y6Y0	P53350
IWS1	Q96ST2	O00267	JADE1	Q6IE81-3	Q96CV9
JADE2	Q9NQC1-2	P62993	JAG1	P78504	P46531
JAGN1	Q8N5M9	Q9H2K0	JAK1	P23458	O60674
JAK3	P52333	Q13485	JAML	Q86YT9	Q9NPF0
JARID2	Q92833	Q09028	JAZF1	Q86VZ6	Q9H2F5
JMJD1C	Q15652	P10275	JMJD6	Q6NYC1	O60885
JMY	Q9QXM1	Q09472	JOSD1	Q15040	P14373
JOSD2	Q8TAC2	Q2T9J0	JPT1	Q9UK76	P55212
JRK	O75564-2	Q96MT3	JUN	P05412	P29692
JUNB	P17275	P18847	KANK2	Q63ZY3	P06730
KANSL1	Q7Z3B3	Q9BVI0	KANSL2	Q9H9L4	Q9BVI0
KANSL3	Q9P2N6	Q9BVI0	KARS	Q15046	P13569
KAT2B	Q92831	Q16665	KAT5	Q92993	Q9BRX2
KAT6A	Q92794	P61964	KAT7	O95251	P53350
KATNAL1	Q9BW62	Q14061	KATNAL2	Q8IYT4	P03372
KBTBD4	Q9NVX7-2, Q9NVX7	P17987, P00441	KBTBD6	Q86V97	P61081
KBTBD7	Q8WVZ9	Q86UK7	KCNA2	P16389	Q6UWP7
KCNE3	Q9Y6H6	P40818	KCNH3	Q9ULD8	P04578
KCNK5	O95279	P21964	KCNN4	O15554	P61073
KCTD12	Q96CX2	Q9UJK0	KCTD15	Q96SI1	Q9Y230
KCTD17	Q8N5Z5	P11686	KCTD20	Q7Z5Y7	P36873
KCTD21	Q4G0X4	Q9NWW6	KCTD5	Q9NXV2	Q9UBU9
KCTD7	Q96MP8-2	Q9UBU9	KDELC1	Q6UW63	P0DTC8
KDELC2	Q7Z4H8	P0DTC8	KDELR1	P24390	Q92796
KDM2A	Q9Y2K7	Q13257	KDM3A	Q9Y4C1	O75582
KDM5A	P29375	P06400	KDM5D	Q9BY66-3	P10275
KDM6A	O15550	O43474	KDM6B	Q5NCY0	Q9BVI0
KEAP1	Q14145	Q86YC2	KHDRBS1	Q07666	P12931
KHSRP	Q92945	P05412	KIAA0040	A0A384DVV8	O76024
KIAA0141	Q14154	P15056	KIAA0319L	Q8IZA0	Q8IUQ4

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
KIAA0355	O15063	P0DTC5	KIAA0907	Q7Z7F0	Q9NQZ5
KIAA1024	Q9UPX6	Q9Y5P4-2	KIAA1161	Q6NSJ0	P11021
KIAA1217	Q5T5P2-6	Q13882	KIAA1324	Q6UXG2-3	P14136
KIAA1328	Q86T90	Q96QG7	KIAA1468	Q9P260	P21709
KIAA1524	Q8TCG1	Q9NZQ7	KIAA2013	Q8IYS2	Q15904
KIDINS220	Q9EQG6	P46109	KIF13B	Q9NQT8	P63104
KIF14	Q15058	Q9Y3A6	KIF18A	Q8NI77	P36873
KIF18B	Q86Y91	Q99435	KIF1BP	Q96EK5	P36873
KIF20A	O95235	Q13257	KIF20B	Q96Q89	P52294
KIF22	Q14807	Q8TC07	KIF23	E9Q5G3, Q02241-2, Q02241	Q9H0H5
KIF24	Q5T7B8	P51955	KIF27	Q86VH2	Q9P0N9
KIF2C	Q99661	Q96ES7	KIF3A	Q9Y496	Q5T8I9
KIF3C	O14782	Q9Y496, Q92845	KIFAP3	Q92845	Q13177
KIFC1	Q9BW19	Q14974	KIFC3	Q9BVG8-5	Q99728
KIR2DL1	P43626	Q06124	KIR2DL3	P43628	Q06124
KIR3DL2	P43630	Q5K4L6	KIRREL2	Q6UWL6	P25788
KIT	P10721	Q14451	KIZ	Q2M2Z5	Q96NL6
KLC2	Q9H0B6	O14977	KLC4	Q9NSK0	P31947
KLF10	Q13118	P08047	KLF11	O14901	Q14154
KLF12	Q9Y4X4	Q96KQ7	KLF13	Q9Y2Y9	P43694
KLF16	Q9BXK1	Q12857	KLF2	Q9Y5W3	Q9HCE7
KLF3	P57682	O00712, Q12857	KLF4	O43474	Q13363
KLF5	Q13887	P0CG48	KLF8	O95600	Q12857
KLHDC2	Q9Y2U9	Q9H015	KLHDC4	Q8TBB5	O60341
KLHL11	Q9NVR0	Q9NRD5	KLHL12	Q53G59	O95050
KLHL2	O95198	Q9UF56	KLHL23	Q8NBE8	P29279
KLHL24	Q6TFL4	Q92905	KLHL25	Q9H0H3	Q9NPA3
KLHL28	Q9NXS3	Q9UF56	KLHL3	Q9UH77	P06396
KLHL32	Q96NJ5	P27361	KLHL33	A6NCF5	P10809
KLHL42	Q9P2K6	P61081	KLHL5	Q96PQ7	Q9NYA1
KLHL6	Q8WZ60	Q13485	KLHL8	Q9P2G9-2	P11086
KLHL9	Q9P2J3	Q9UF56	KLKB1	P03952	Q8N0U8
KLRD1	Q13241	P26715	KLRG1	Q96E93	Q6IN84
KMT2E	Q8IZD2	P04637	KMT5A	Q9NQR1	P62805
KMT5B	Q4FZB7	Q9H2G4	KNL1	Q8NG31-2, Q8NG31	O60566
KNOP1	Q1ED39	Q96CW1	KNSTRN	Q9Y448	Q96LB3
KNTC1	P50748	P46059	KPNA2	P52292	O96017
KPNA5	O15131	Q96C86	KPNA6	O60684	P03466
KPNB1	Q14974	Q99558	KPTN	Q9Y664	Q9NYZ3
KRAS	P01116-2	P01116-2	KRBA1	A5PL33	P14373
KRIT1	O00522	Q15185	KRR1	Q13601	Q9H2U1
KRT18	P05783	Q9NUX5	KRT7	P08729	P09429
KRT72	Q14CN4	Q9UL45	KTI12	Q96EK9	Q9HC97
L1CAM	P32004	O00533	L2HGDH	Q9H9P8	O15431
L3MBTL1	Q9Y468	P33992, P49736	L3MBTL4	Q8NA19	Q9Y2I6
LACC1	Q8IV20	O43929	LACTB	P83111	Q9NQ92
LACTB2	Q53H82	Q8IUQ4	LAGE3	Q14657	Q96S44
LAIR2	Q6ISS4	P39060	LAMP1	P11279	Q9Y2Q5

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
LAMP2	P13473-2	Q86US8	LAMP3	Q9UQV4	Q9NWZ5
LAMTOR2	Q9Y2Q5	P61421	LAMTOR3	Q9UHA4	Q9Y2Q5, Q0VGL1, Q6IAA8, O43504, Q8NBW4
LAPTM4A	Q15012	P80370	LAPTM4B	Q86VI4	Q9H2K0
LARGE1	O95461	Q6VN20	LARP1	Q6PKG0	P59595
LARP4	Q6P4E2	Q02548	LARP4B	Q92615	P11940
LARS2	Q15031	O95861	LASP1	Q14847	Q9H2D1
LATS1	O95835	P06493	LATS2	Q9NRM7	O95863
LBR	Q14739	Q9Y4Z2	LCAT	P04180	P02647
LCK	P06239	P43403, Q9Y2R2	LCOR	Q96JN0	Q09028
LCP2	Q13094	O15117	LDHB	P07195	P00338
LDLRAD4	O15165	Q96PU5	LDLRAP1	D3ZAR1	P01130
LDOC1	O95751	Q99728	LEF1	Q9UJU2	Q16236
LEMD2	Q8NC56	Q8WTR4	LEMD3	Q9Y2U8	P16410
LENG1	Q96BZ8	Q9Y6D9	LENG8	Q96PV6	P40429
LEO1	Q8WVC0	P23193	LEPROT	O15243	Q9UBD6
LETMD1	Q6P1Q0	P40855	LGALS1	P09382	P78504
LGALS2	P05162	O00560	LGALS3	P17931	P33527
LGALS3BP	Q08380	P02751	LGALS4	P56470	Q9HAS3
LGALS9B	Q3B8N2	O00308	LGMN	Q99538	O94805
LHFPL5	Q8TAF8	Q9H2K0	LIF	P15018	P40189
LIG4	P49917	P48729	LILRA4	P59901	Q10589
LILRB3	Q8K4V6	P04629	LILRB4	Q8NHJ6	Q08722
LIMA1	Q9UHB6	P13569	LIMD1	Q9UGP4	Q9UKV8
LIMK2	P53671	P53667	LIMS1	P48059	Q9Y3P9
LIMS2	Q7Z4I7-5	Q96PM5	LIN52	Q52LA3	Q08999
LIN54	Q6MZP7	P52747	LIN7A	O14910	Q8N3R9
LIN9	Q5TKA1	Q08999	LINC-PINT	A0A455ZAR2	Q08050
LINGO2	Q7L985	Q9HAU4	LIPT1	Q9Y234	P11182
LITAF	Q99732	Q9NPF0	LLGL2	Q6P1M3	Q05513
LMAN2L	Q9H0V9	P78536	LMBRD2	Q68DH5	P22732
LMNA	P02545-1, P02545-2, P02545	P20700	LMNB1	P20700	P02545-1
LMNB2	Q03252	P20700	LONP1	P36776	Q4G176
LONRF2	Q1L5Z9	P99999	LONRF3	Q496Y0	P35232
LOXL1	Q08397	P06276	LOXL2	Q9Y4K0	Q03405
LOXL4	Q96JB6	P06396	LPAL2	Q16609	P55212
LPAR6	P43657	P23458	LPCAT1	Q8NF37	O14975
LPCAT3	Q6P1A2	P25103	LPCAT4	Q643R3	P08173
LPGAT1	Q92604	P22732	LPP	Q93052	P05120
LPXN	O60711	Q13177	LRCH1	Q9Y2L9	P62258
LRFN3	Q9BTN0	P13385	LRFN4	Q6PJG9	Q9UI95
LRIF1	Q5T3J3	Q96A70	LRIG1	Q96JA1	P22681
LRIG2	O94898	Q96JA1	LRMDA	A0A087WWI0	Q9NQT4
LRMP	Q12912	O94901	LRP1	Q07954	P12236
LRP11	Q86VZ4	Q9Y5M8	LRP5	O75197	P00740
LRP8	Q14114	P35968	LRPPRC	P42704	P06730
LRR1	Q96L50	Q92905	LRRC20	Q8TCA0	P60709
LRRC41	Q15345	Q93034, Q9UBF6	LRRC45	Q96CN5	P13196
LRRC59	Q96AG4	P04049	LRRC6	Q86X45	O14757

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
LRRC61	Q9BV99	O14964	LRRC8A	Q8IWT6	O43914
LRRC8B	Q6P9F7	Q13370	LRRIQ3	A6PVS8	O43639
LRRK1	Q38SD2	P29353	LRRN1	O75427	A1L3X0
LSM11	P83369	P62306, P62304	LSM12	Q3MHD2	P10636-8
LSM14A	Q8ND56	P09651	LSM2	Q9Y333	P62316, P62306, P14678, P62304
LSM4	Q9Y4Z0	P62316, P62304	LSM5	Q9Y4Y9	Q9H4B4
LSM6	P62312	P62316, P62306, P14678, P62314, P62304	LST1	Q9Y6L6	Q9NPD5
LTA	P01374, Q6FG55	Q9BRI3	LTBP3	Q14767	Q9UBX5
LTBP4	Q8N2S1	Q9UBX5	LTBR	P36941	P36941
LTK	P29376	Q9UM73	LUC7L	Q9NQ29-3	Q13247, Q16629
LUC7L3	O95232-2	P40692	LXN	Q9BS40	Q9UQK1
LY96	Q9Y6Y9	O00206	LYAR	Q9NX58	P19525
LYN	P07948	P16671	LYPD3	O95274	O43913
LYPLA1	Q6IAQ1, O75608	Q08379	LYRM2	Q9NU23	O14561
LYRM4	Q9HD34	Q9H1K1	LYRM7	Q5U5X0	Q8IWL3
LYSMD1	Q96S90	Q99081	LYZ	P61626	P61626
LZIC	Q8WZA0	P55072	LZTS1	Q9Y250	P30307
MACF1	Q9UPN3	P13569	MAD2L1	Q13257	O43511
MAD2L1BP	Q15013	O43511	MAD2L2	Q9UI95	Q9UBZ9
MADD	Q8WXG6	Q9NUX5	MAF1	Q9H063	Q9UJK0
MAFB	Q9Y5Q3	O14867	MAFF	Q9ULX9	O14867
MAFG	O15525	O15525	MAFK	O60675	O14867
MAGEE1	Q96JG8	Q15185	MAGI1	Q6P9H4	P42772
MAGI3	Q5TCQ9	Q9NTX7	MAGOHB	Q96A72	Q08117
MAL	P01732	P06493	MALSU1	Q96EH3	Q14197
MALT1	Q9UDY8	Q9UDY8	MAML1	Q92585	P46531
MAML3	Q96JK9	P46531	MAN1A1	P33908	Q9Y2C2
MAN1A2	O60476	O60602	MAN2A1	Q16706	Q03405
MAN2B2	Q9Y2E5	Q96BA8	MANBA	O00462	P08311
MANBAL	Q9NQG1	Q12893	MAP1A	P78559	Q9GZQ8
MAP1LC3B	Q9GZQ8	O75385	MAP1S	Q66K74	O14543
MAP2K1	Q02750	P46937	MAP2K3	P46734	Q16539
MAP2K4	P45985	Q13233	MAP2K7	O14733	O43318
MAP3K11	Q16584	P27986	MAP3K12	Q12852	P00533
MAP3K14	Q99558	P07900	MAP3K2	Q9Y2U5	Q13163
MAP3K20	Q9NYL2	O75582	MAP3K3	Q99759	Q13163
MAP3K4	Q9Y6R4	Q06187	MAP3K7	O43318	P40763
MAP3K8	P41279	P19838, Q8NFZ5	MAP4	P27816	P10636-8, Q12906
MAP4K1	P70218	Q16584	MAP4K3	Q924I2	Q99962
MAP4K5	Q9Y4K4	P62993	MAP7D1	Q3KQU3	P01106
MAP7D2	Q96T17	P10636-8	MAP7D3	Q8IWC1	P61981
MAPK11	Q15759	P49137	MAPK6	Q16659	P53350
MAPK7	Q13164	Q13163	MAPK8IP1	Q9UQF2	O43318
MAPK8IP3	Q9UPT6	Q5SW96	MAPKAP1	Q9BPZ7	P78527
MAPKAPK2	P49137	Q16539	MAPKAPK3	Q16644	P49137
MAPKAPK5	Q8IW41	Q15109	MAPRE2	Q15555	P20618
MAPRE3	Q9UPY8	P05412	MAPT	P10636-8	P13639
MARK3	P27448	Q96L34	MAST3	O60307	P60484

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
MATK	P42679	Q14289	MATR3	P43243	P09651
MAVS	Q7Z434	Q13568	MB21D1	Q8N884	P09874
MBD3	O95983-2	Q9Y5J6	MBNL1	Q86VM6	Q08117
MBNL3	Q9NUK0	O75553	MBOAT7	Q96N66	Q86WV6
MBP	P13727	Q12946	MBTD1	Q05BQ5	P62805
MBTPS1	Q14703	Q8IV16	MCAM	P43121	Q12860
MCAT	Q8IVS2	P13196	MCCC1	Q96RQ3	P47985
MCCC2	Q9HCC0	P13569	MCEMP1	Q8IX19	Q6ZPD8
MCFD2	Q8NI22	P78382	MCM10	Q7L590	Q7L590
MCM2	P33993, P49736	P33992, P25205, P49736	MCM3	P25205	P33992, P49736
MCM3AP	O60318	P60896	MCM4	P33991	P25205, P49736
MCM5	P33992	P25205, P49736	MCM6	Q14566	P25205, P49736, Q9H211
MCM7	P33993	P33992, P25205, P49736	MCM8	Q9UJA3	P43246
MCM9	Q9NXL9	Q9UJA3	MCMBP	Q9BTE3	Q9UJA3
MCOLN3	Q8TDD5	Q86XT9	MCPH1	Q8NEM0	Q01094
MCU	Q8NE86-1, Q8NE86	Q9H4I9, Q9BPX6	MCUB	Q9NWR8	Q15047
MCUR1	Q96AQ8	P23786	MDH1	P40925	P48163
MDH2	P40926	P13569	MDK	P21741	P42568, Q9UHB7
MDM2	Q00987	O96017	MDM4	O15151	P06400
ME2	P23368	P10809	MED10	Q9BTT4	Q15648
MED11	Q9P086	Q15648	MED13L	Q71F56	Q15648
MED15	Q96RN5	Q15648	MED16	Q9Y2X0	Q15648
MED17	Q9NVC6	Q15648	MED18	Q9BUE0	Q15648
MED20	Q9H944	Q15648	MED21	Q13503	Q15648
MED26	O95402	Q15648	MED31	Q9Y3C7	Q15648
MED7	O43513	Q15648	MED8	Q96G25	Q15648
MEF2C	Q06413	P56524	MEF2D	Q14814	Q16539
MEFV	O15553	Q9ULZ3	MEGF8	Q7Z7M0	P22692
MELK	Q14680	Q03405	MELTF	P08582	P13569
MESD	Q14696	Q92989	METAP2	P50579	Q93034
METRNL	Q641Q3	Q01523	METTL13	Q8N6R0	Q12933
METTL15	A6NJ78	P46059	METTL16	Q86W50	P53701
METTL21A	Q8WXB1	Q9UGI0	METTL27	Q8N6F8	Q9NQ94
METTL6	Q8TCB7	Q12948	METTL9	Q9H1A3	Q92504, Q9ULF5, Q6ZMH5
MEX3C	Q5U5Q3	Q93009	MFAP1	P55081	Q9Y6D9
MFAP3	P55082	P43119	MFHAS1	Q9Y4C4	Q9Y496
MFSD10	Q14728	P45973	MFSD14B	Q5SR56	Q9NRZ7
MFSD4B	Q5TF39	Q7L8W6	MFSD6	Q6ZSS7	Q15848
MGAM	O43451	Q96BA8	MGAT2	Q10469	P40855
MGAT4B	Q9UQ53	Q96CV9	MGLL	Q99685	P07550
MGME1	Q9BQP7	Q02790	MGRN1	O60291	P14138
MIB1	Q86YT6	Q9UHD2	MICALCL	O94851	P54252
MICB	Q29980	P17302	MID1IP1	Q9NPA3, Q9CQ20	Q13085, O00763
MID2	Q9UJV3-2	P51668	MIDN	Q504T8	P22736
MIER2	Q8N344	Q13547	MIF	P14174	P13569
MIGA2	Q7L4E1	O14975	MIIP	Q5JXC2	P42568
MILR1	Q7Z6M3	Q92538	MINDY1	Q8N5J2-3	P55212
MIS18A	Q9NYP9	P53350	MIS18BP1	Q6P0N0	P56539
MITD1	Q8WV92	P04618	MKI67	P46013	P53667

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
MKKS	Q9NPJ1	P26641	MKL1	Q969V6	P11831
MKLN1	Q9UL63	P01275	MKNK1	Q9BUB5	Q04637
MKRN1	Q9UHC7	P51668	MKS1	Q9NXB0	O95816
MLC1	Q15049	Q00403	MLEC	Q14165	P13569
MLH1	P40692	P54278	MLLT10	P55197	P29692
MLLT11	Q13015	P04271	MLLT6	P55198	P24534
MLX	Q9UH92	Q08117	MMADHC	Q9H3L0	Q9P0N9
MMD	Q15546	P01350	MMP25	Q9H239	P09958
MMRN1	Q13201	P02649	MMS19	Q96T76	P51530
MMS22L	Q6ZRQ5	Q08945	MNAT1	P51948	P28715
MND1	Q9BWT6	P09917	MNDA	P41218	P08581
MNT	Q99583	O43639	MOB2	Q70IA6	Q9NT62
MOB3C	Q70IA8	Q9NRD5	MOB4	Q9QYW3	Q05193
MOCS1	Q9NZB8	Q9BQ90	MOCS3	O95396	Q9H5K3
MOGS	Q13724	Q14974	MON2	Q7Z3U7	P21453
MORC2	Q9Y6X9	Q14562	MORC3	Q14149	Q9NWF9
MORC4	Q8TE76	P40763	MORN3	Q6PF18	P50458
MOSPD3	O75425	Q3SXY8	MPC2	O95563	Q9NR28
MPDZ	O75970	P46937	MPG	P29372	P04156
MPHOSPH6	Q99547	Q13901, P42285, Q01780, Q9NQT4	MPHOSPH8	Q99549	Q15047
MPI	P34949-2	P55212	MPLKIP	Q8TAP9	Q9ULX6
MPND	Q8N594	P00441	MPP1	Q00013	Q9NRZ7
MPRIP	Q6WCQ1	P53350	MPST	P25325	O96006
MPZL1	O95297	Q06124	MRC2	Q9UBG0	P17931
MRM1	Q6IN84	P46199	MRM2	Q9UI43	P10809
MRM3	Q9HC36	P46199	MRNIP	Q6NTE8	Q9UBU9
MRPL10	Q7Z7H8	Q14197	MRPL11	Q9Y3B7	Q13371
MRPL12	P52815	P43897	MRPL13	Q9BYD1	Q14197
MRPL15	P49406	P40429	MRPL16	Q9NX20	P62241
MRPL17	Q9NRX2	P62241	MRPL18	Q9H0U6	Q14197
MRPL19	P49406	P40429	MRPL22	Q9NWU5	P62241
MRPL27	Q9P0M9, Q8IXM3	Q14197	MRPL28	Q13084	Q14197
MRPL3	P09001	Q14197	MRPL30	Q8TCC3	O00425
MRPL32	Q9BYC8	P62241	MRPL34	Q9BQ48	O14975
MRPL35	Q9NZE8	Q9HD23	MRPL37	Q9BZE1	Q14197
MRPL38	Q96DV4	Q9NPA3	MRPL39	Q9NYK5-2	Q14197
MRPL41	Q8IXM3	Q14197	MRPL44	Q9H9J2	Q15047
MRPL45	Q9BRJ2	Q6NZI2	MRPL46	Q9H2W6	Q9BQP7
MRPL50	Q8N5N7	Q14197	MRPL51	Q4U2R6	Q14197
MRPL52	Q86TS9	Q14197	MRPL57	Q9BQC6	P62241
MRPL58	Q14197	Q00059	MRPS10	P82664	P62753
MRPS11	P82912	Q00839	MRPS12	O15235	P46199
MRPS14	O60783	P40429	MRPS15	P82914	P62753
MRPS16	Q9Y3D3	Q9H9Z2	MRPS18B	Q9Y676	P06400
MRPS18C	Q9Y3D5	Q9UPR3	MRPS22	P82650	P48431
MRPS23	Q9Y3D9	O15160	MRPS25	P82663	Q92845
MRPS28	P82673	P03372	MRPS30	Q9NP92	Q14197
MRPS31	Q92665	P40763	MRPS33	Q9Y291	Q00839
MRPS34	P82930	Q9Y2I6	MRPS35	P82673	P03372

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MRS2	Q9HD23	Q8N5V2	MS4A1	P11836	Q9H2K0
MSANTD4	Q8NCY6	Q96T60	MSH3	P20585	P43246
MSH5	O43196	P00519	MSH6	P52701	P43246
MSI2	Q96DH6	Q9Y5J6	MSL2	Q9HCI7	P61981
MSMO1	Q15800	Q6NT55	MSN	P26038	P13569
MSR1	P21757	Q7L5A8	MSRB1	Q9NZV6	Q13099
MSRB2	Q9Y3D2	P02649	MT-ATP8	P03928	P06576
MT-CO2	P00403	P15954	MT-CO3	P00414	P37840
MT-CYB	P00156	O14949, P14927	MT-ND1	P03886	Q9P0J0
MT-ND2	P03891	O75380	MT-ND4	P03905	Q9P0J0
MT-ND5	P03915	P19526	MTBP	Q96DY7	P49336
MTCH1	Q9NZJ7	Q92611	MTCH2	Q9Y6C9	P02654
MTDH	Q86UE4	Q9UKV8	MTERF2	Q49AM1	P35813
MTERF3	Q96E29	P22732	MTERF4	Q7Z6M4	Q6ZPD8
MTFR2	Q6P444	P13569	MTG1	Q9BT17	P10809
MTHFD1	P11586	P13569	MTHFD1L	Q6UB35	Q96GW9
MTHFD2	P13995	P49137	MTHFD2L	Q9H903	Q5T8I9
MTHFR	P42898	A0PJW6	MTM1	Q13496	Q9C0I1
MTMR12	Q9C0I1	Q13614, Q13496	MTMR2	Q13614	Q9NXD2, Q9C0I1
MTMR3	Q13615	P61981	MTMR6	Q9Y217	Q9Y216, Q96QG7
MTOR	P42345	P06730	MTURN	Q8N3F0	Q9H4P4
MTX1	Q13505	Q05940	MUC1	P15941-PRO_0000317447	P17676
MUS81	Q96NY9	P39748	MUT	P22033	P42858
MX1	P20591	O75928-2	MXD1	Q05195	Q96ST3
MXD3	Q9BW11	Q9UBB9	MYADM	Q96S97	P43115
MYB	P10242	Q92793	MYBL2	P10244	Q09028
MYC	P01106	P26641	MYCL	P12524-2	P54646
MYDGF	Q969H8	Q14554	MYEF2	Q9P2K5-2	Q9UGI0
MYH10	P35580	Q14164	MYH3	P11055	Q16665
MYL12A	P19105	P35579	MYL12B	O14950	P60709
MYL6	P60660	O43809	MYL6B	P14649	P35579
MYLIP	Q8WY64	P54252	MYLK2	Q9H1R3	Q9H4I9
MYLK4	Q86YV6	P07900	MYLPF	Q96A32	P14136
MYO15B	Q96JP2	Q9BRX5	MYO1B	O43795	P13569
MYO1C	Q12965	P50570	MYO1D	O94832	P43251
MYO1E	Q12965	P50570	MYO1G	B0I1T2	P13569
MYO5B	Q9ULV0	P40855	MYO6	Q9UM54	P98082
MYOM1	P52179	O75923	MYOM2	P54296	O75923
MYPOP	Q86VE0	P21673	MZB1	Q8WU39	Q99816
MZT1	Q08AG7	Q6ZMU5	MZT2A	Q6P582	O00571
MZT2B	Q6NZ67	Q9Y6L6	N4BP2L2	Q92802	Q9Y3C6
NAA16	Q6N069-4	P14136	NAA20	P61599	P28908
NABP1	Q96AH0	Q5VWX1	NABP2	Q9BQ15	Q06609
NACA	Q13765	P05067	NACAD	O15069	P42858
NACC1	Q96RE7	P41743	NADK	O95544	P61981
NADSYN1	Q6IA69	O14964	NAF1	Q15025	Q00403
NAMPT	P43490	P43490	NANOS3	P60323-2	P40692

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NAT10	Q9H0A0	P08670	NAT8L	Q8N9F0	P40855
NAV1	Q8NEY1	P31947	NBDY	A0A0U1RRE5	Q9NPI6
NBEAL1	Q6ZS30	Q9Y639	NBL1	P41271-2	P05067
NBPF1	Q3BBV0	O60341	NCALD	P61601	P23352
NCAPD2	Q15021	P13569	NCAPD3	P42695	P61073
NCAPG	Q9BPX3	P13569	NCAPH	Q15003	P03952
NCBP1	Q09161	Q92900	NCDN	Q9UBB6	Q15185
NCEH1	Q6PIU2	P27824	NCF2	P19878	Q92845
NCKAP1	Q9Y2A7	Q9UQB8	NCKIPSD	Q9NZQ3	Q9UQB8
NCOA1	Q15788	Q13569	NCOA2	Q15596	O00482-1
NCOA5	Q9HCD5	P36873	NCOA6	Q14686	P06400
NCOR1	O75376	P56524	NCR1	O76036	P08670
NCR2	O95944	O43914	NCR3	O14931-2	P17931
NCR3LG1	Q68D85	P47813	NCSTN	Q92542	Q15363
NDC1	Q9BTX1	P61073	NDC80	O14777	Q9H211
NDE1	Q9NXR1, Q9CZA6	P43034	NDEL1	Q9GZM8	P23025
NDFIP1	Q9BT67	P43088	NDN	P25233	P08138
NDRG2	Q9UN36	P03372	NDST2	P52849	Q03405
NDUFA1	O15239	O75380	NDUFA11	Q86Y39	O75489
NDUFA2	O43678	Q9P0J0	NDUFA4	O00483	P00533
NDUFA5	Q16718	Q9P0J0	NDUFA8	P51970	Q9P0J0
NDUFAB1	O14561	P26441	NDUFAF1	Q9Y375	Q9P0J0
NDUFAF3	Q9BU61	Q9NPB3	NDUFAF5	Q5TEU4	P01185
NDUFB1	O75438	Q9P0J0	NDUFB10	O96000	P05455
NDUFB11	Q9NX14	Q9P0J0	NDUFB3	O43676	Q9P0J0
NDUFB5	O43674	P19526	NDUFB6	O95139	P19526
NDUFS1	P28331-2	Q14154	NDUFS2	O75306	Q99650
NDUFS4	O43181	P19526	NDUFS6	O75380	Q16822
NDUFS7	O75251	P46199	NDUFS8	O00217	Q8IWL3
NDUFV1	P49821	Q9P0J0	NDUFV2	P19404	Q8IWL3
NDUFV3	P56181	Q9P0J0	NEB	P20929	Q07889
NECAP1	Q8NC96	Q96CW1	NECAP2	Q9NVZ3	P42566
NECTIN3	Q9NQS3	Q92692, P15151	NEDD1	Q8NHV4	Q8IW35, O43303
NEDD8	Q15843	P68104	NEIL2	Q969S2	Q02548
NEIL3	Q8TAT5	Q15185	NEK2	P51955	Q8NG66
NEK3	P51956	P32971	NEK4	P51957	P12235
NEK7	Q8TDX7	Q9ULW0	NEMP1	O14524	P46060
NEPRO	Q6NW34	P06748	NET1	Q7Z628	Q14160
NETO2	Q8NC67-2	P06396	NEURL4	Q96JN8	O43303
NF2	P35240	Q16584	NFAT5	O94916-1	P29350
NFATC2	Q13469-2	P05412	NFE2	Q16621	P06396
NFE2L2	Q16236	O15525	NFE2L3	Q9Y4A8	O15525
NFIA	Q12857	P50458	NFIC	P08651	O00712
NFIX	Q14938-4, Q14938-3	P40763	NFKB1	P19838-PRO_0000030311	O00255
NFKBIA	P25963	Q15813	NFKBIB	Q15653	P19838
NFYB	P25208	P04637	NGDN	Q8NEJ9	P06748
NGFR	P07174, P08138	P01138	NHLRC2	Q8NBF2-2	P04792

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NHLRC4	P0CG21	Q8WY64	NHP2	Q9NX24	O60832, Q9NPE3
NHSL2	Q5HYW2	P53350	NIF3L1	Q9GZT8	P42772
NIFK	Q9BYG3	Q9NS56	NINJ2	Q9NZG7	P17302
NIPBL	Q6KC79	P52294	NIPSNAP3A	Q9UFN0	P10809
NISCH	Q9Y2I1	P03496	NKD2	Q969F2	P01135
NKG7	Q16617	P26715	NKIRAS1	Q9NYS0	P56559
NKTR	P30414	P61981	NLE1	Q9NVX2	Q07020
NLN	Q9BYT8	Q5SW96	NLRC5	Q86WI3	O15111, O14920
NLRP1	Q9C000	Q9ULZ3	NLRP12	P59046	P08631
NME1	P15531	P10911	NME2	P19804	Q05193
NME4	O00746	Q9NPB3	NME7	Q9Y5B8	A0MZ66
NME8	Q8N427	O14867	NMI	Q13287	Q96QG7
NMNAT1	Q9HAN9	P52294	NMRAL1	Q9HBL8	Q96CV9
NMT1	P30419	P04601	NOA1	Q8NC60	P84103
NOC2L	Q9Y3T9	P40429	NOC3L	Q8WTT2	P19525
NOCT	Q9UK39	P11182	NOL4L	Q96MY1	Q13363
NOL7	Q9UMY1	O00410	NOP10	Q9NPE3	Q9NPB3
NOP53	Q9NZM5	P19525	NOS3	P29474	Q12891
NOTCH1	P46531	P78504	NOTCH2	Q04721	Q8N2S1
NPAT	Q14207	Q09028	NPC1	O15118	P61916, 16113
NPC2	P61916	P00751	NPDC1	Q9NQX5	Q9BW85
NPHP4	O75161	O95816	NPM1	P06748	P40429
NPM2	Q86SE8, Q86SE8-2	P06748	NPRL3	Q12980	Q07954
NPTX1	Q15818	P0DTC8	NQO1	P15559	P15559
NR1D2	Q6NSM0	Q8WY64	NR1H2	Q60644	Q96EB6
NR1I2	O75469	Q15788	NR1I3	Q14994	P01100
NR2F6	P10588	P24468	NR3C1	P06536	Q15788
NR3C2	P08235	Q9UBK2	NR4A1	P22736	Q9Y4D7
NR4A2	P43354	P05067	NR4A3	Q92570	Q9H2F3
NRBP1	Q9UHY1	P51530	NRF1	Q14494	O15525
NRM	Q8IXM6	P04000	NRN1	Q9NPD7	Q86XT9
NSA2	P40078	O14641	NSD1	Q96L73	P17096
NSD2	O96028	P17096	NSF	P46459	P59635
NSL1	Q96IY1	Q13185	NSMAF	Q92636	Q9NPI6
NSMCE1	Q8WV22	Q9H668	NSMCE2	Q96MF7	P06396
NSMCE3	Q9CPR8	P08138	NSMCE4A	Q9NXX6	P51587
NSUN3	Q9H649	P10809	NSUN4	Q96CB9	O95861
NT5E	P21589	P36404	NTAN1	Q96AB6	Q13546
NTN5	Q8WTR8	Q8N2S1	NTRK1	P04629	O75385
NTSR1	P20789	P63211	NUAK1	O60285	Q93008
NUAK2	Q9H093	P31947	NUB1	Q9Y5A7-1	Q15843
NUBP1	P53384	Q9Y5Y2	NUCB1	Q02818	P18509
NUCB2	P80303	P24522	NUCKS1	Q9H1E3	Q12824
NUDCD1	Q96RS6	Q92620	NUDCD2	Q8WVJ2	P43034
NUDT1	P36639-2	Q53H54	NUDT2	P50583	Q14566
NUDT21	O43809	O43809, Q16630-1	NUDT6	P53370	P46199
NUF2	Q9BZD4	Q9Y3P9	NUFIP2	Q7Z417	Q9UBX5
NUMA1	Q14980	P03372	NUMB	Q9QZS3-2	P08151
NUMBL	Q9Y6R0	Q9NWB1	NUP155	O75694	P13569
NUP160	Q12769	Q9UBP0	NUP188	Q5SRE5	P13569

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NUP205	Q92621	P13569	NUP37	Q8NFH4	Q8WYP5
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NUP54	Q7Z3B4	Q8IY31	NUP58	Q9BVL2	Q14154
NUP62	P37198	Q6W0C5	NUP98	P52948	Q9UBP0
NUPL2	O15504	Q9NYY8	NUSAP1	Q9BXS6	Q14974
NUTF2	P61970	P12004	NVL	O15381	P40429
NXF1	Q9UBU9	P46060	NXT1	Q9UKK6	Q9Y3Y2
NYAP2	Q9P242	P52594	OAS1	P00973-2	Q96RK4
OAS3	Q9Y6K5	Q7Z434	OASL	Q15646	Q9NW13
OAT	P04181	P13569	OAZ1	P54368	O14977
OCIAD1	Q9NX40	Q13315	ODF2L	Q96GC1	P42858
OFD1	O75665	Q4AC94	OGFOD1	Q8N543	P62266
OGFRL1	Q5TC84	P11802	OIP5	O43482	Q9NQ94
OLFM2	O95897	O14641	OMA1	Q96E52	P46059
ORC1	Q13415	O43929, Q9UBD5, Q13416, O43913	ORC2	Q13416	Q13415
ORC3	Q9UBD5	Q13415	ORC4	O43929	Q13415
ORC5	O43913	Q13415	ORC6	Q9Y5N6	O43929, Q9UBD5, Q13416
ORMDL1	Q9P0S3	Q96BA8	ORMDL2	Q53FV1	Q9Y6L6
ORMDL3	Q8N138	Q9H2K0	OS9	Q13438	Q16665
OSBP	P22059	P05120	OSBPL2	Q9H1P3	P62306
OSBPL5	Q9H0X9	P54802	OSER1	Q9NX31	Q9UJK0
OSGIN1	Q9UJX0	P48507	OST4	P0C6T2	A6NI15
OSTC	Q9NRP0	P80370	OTUB2	Q96DC9	Q9Y4K3
OTUD1	Q5VV17	O43353	OTUD4	Q01804	P42224
OXCT1	P55809	P42566	OXNAD1	Q96HP4	P10809
OXR1	Q8N573-5	Q14145	P2RX4	Q99571	Q10472
P2RX7	Q99572	P14373	P2RY10	O00398	Q13315
P2RY8	Q86VZ1	Q13315	P3H1	Q32P28	Q70UQ0
P3H4	Q92791	Q9H2A9	P4HA2	O15460	P39060
PA2G4	Q9UQ80	P06400	PAAF1	Q9BRP4	P60896
PABPC4	Q13310	P59595	PABPN1	Q86U42	P11940
PACS1	Q6VY07	Q9NZ52	PACS2	Q86VP3	P61981
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PADI4	Q9UM07	Q96RU7	PAFAH1B1	P43034	P07900
PAFAH1B3	Q15102	Q8WUX9	PAG1	Q9NWQ8	P40763
PAGR1	Q9BTK6	Q15788	PAICS	P22234	Q969R5
PAIP2B	Q9ULR5	O15234	PAK1	Q16512	P26641
PAK1IP1	Q9NWT1	P62753	PAK2	Q13177	Q13177
PAK4	O96013	O00401	PAK6	Q9NQU5	P61981
PALB2	Q86YC2	Q06609	PALD1	Q9ULE6	Q92985
PAM	P19021	P08185	PAN3	Q58A45	P11940
PANK2	Q9BZ23-2	Q8WXH5	PAPD5	Q8NDF8	P06748
PAPSS1	O43252	P52294	PAPSS2	O95340	Q96CV9
PAQR6	Q6TCH4	P50876	PAQR8	Q8TEZ7	Q08426
PARD6A	Q9NPB6	Q05513	PARD6G	Q9BYG4	Q05513
PARK7	Q99497	Q13158	PARN	O95453	Q99728
PARP1	P09874	Q9NTX7	PARP14	Q460N5	O43914
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PAXX	Q9BUH6	Q14191	PBK	Q96KB5	O94955
PBX2	P40425	Q9UHB7	PBX4	Q9BYU1	Q9BRT9
PCBD1	P61457	P35680	PCBD2	Q9H0N5	P35680
PCBP1	Q15365	P11940	PCBP2	Q15366	P11940
PCBP4	P57723	P11142	PCDHGB5	Q9Y5G0	Q16850
PCED1A	Q9H1Q7	Q7Z3S9	PCGF1	Q9BSM1	Q03111
PCGF3	Q3KNV8	P68400, P67870	PCGF5	Q86SE9	Q96LA8
PCK2	Q16822	P35558	PCLAF	Q15004	P38936
PCMT1	P22061, P22061-2	P49247	PCMTD2	Q9NV79	Q93034, Q9UBF6
PCNA	P12004	P68104	PCP2	Q92729	Q13685
PCSK6	P29122	P0DTC8	PCSK7	Q16549	P17612
PCYOX1L	Q8NBM8	P27824	PDAP1	Q13442	P10636-8
PDCD10	Q9BUL8	Q9BRV8	PDCD11	Q14690	P19838
PDCD6	P12815	P25445	PDCL	Q13371	O00628
PDCL3	Q9H2J4	Q13371	PDE2A	O00408	O00170
PDE3B	Q13370	O95398	PDE4D	Q08499-2	Q9HBX9
PDE7A	Q13946	P17612	PDF	Q9HBH1	Q9UBF6
PDGFD	Q9GZP0	O95714	PDHB	P11177	P10515
PDIA5	Q14554	Q99538	PDIA6	Q15084	Q16881
PDIK1L	Q8N165	P07900	PDK4	Q16654	Q15119
PDLIM7	Q9NR12	Q96QF0	PDP1	Q9P0J1	P12644
PDS5A	Q29RF7	O60216, Q7Z5K2, Q96FF9	PDZD11	Q5EBL8	Q04656
PDZD2	O15018	P59637	PDZD4	Q17RL8	Q9UJW3
PDZD8	Q8NEN9	P23945	PDZK1	Q5T2W1	Q58EX2
PEAK1	Q9H792	P46109	PEBP1	P30086	P16050
PEF1	Q9UBV8	O14964	PELI1	Q96FA3	P51617, Q9NWZ3
PELI2	Q9HAT8	Q12891	PELO	Q9BRX2	Q92993
PEMT	P15941-PRO_0000317447	P17676	PER1	O15534	P07237
PERP	Q96FX8	P14136	PEX1	O43933	O75925
PEX11B	O96011	P40855	PEX12	O00623	P49638
PEX2	P28328	P40855	PEX5	P50542	O14734
PFDN1	O60925	Q13164	PFDN2	Q9UHV9	Q13164
PFDN4	Q9NQP4	Q13164	PFDN5	Q99471	Q13164
PFDN6	O15212	Q13164	PFKFB3	Q16875	P61981
PFKM	P08237	Q8IUQ4	PFKP	Q01813	Q8NEZ3, Q9HBG6
PFN1	P07737	P60709	PFN2	P35080	P60709
PGAM1	P18669	P12004	PGAM5	Q96HS1	Q9Y572
PGAP2	Q9UHJ9-5	Q9Y624	PGAP3	Q96FM1	Q969Z0
PGBD1	Q96JS3	P22736	PGD	P52209	P13569
PGGT1B	P53609	Q9NQ87	PGK1	P00558	P12004
PGM2	Q96G03	P04792	PGRMC1	O00264	P05177
PGRMC2	O15173	O60906	PHACTR4	Q8IZ21	P36873
PHB	P35232	P40763	PHB2	Q99623	P35232
PHC1	P78364	P68400	PHC2	Q8IXK0	P53350
PHC3	Q8NDX5	Q13363	PHF1	O43189	P04637
PHF12	Q96QT6	Q13547	PHF14	O94880	P16104
PHF19	Q5T6S3	Q14012	PHF21A	Q96BD5, Q96BD5-2	Q13547
PHF23	Q9BUL5	P99999	PHF3	Q92576	P28482

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
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PHKG2	P15735	P0DP23	PHLDA1	A2BDE7	O76024
PHLDB2	Q86SQ0	P21333	PHLPP1	O60346	Q8TAF3, O94782
PHLPP2	Q6ZVD8	Q8TAF3, O94782	PHYH	O14832	O75925
PHYHD1	Q5SRE7	Q9NUX5	PHYKPL	Q8IUZ5	Q9NUX5
PI16	Q6UXB8	P27449	PI4K2A	Q9BTU6	Q96J02
PI4K2B	Q8TCG2	Q03405	PI4KA	P42356	P13569
PI4KB	Q9UBF8	P61981	PIAS1	O75925	Q16143
PIAS2	O75928-2	P15336	PIAS4	Q8N2W9	O60551
PICK1	Q9NRD5	P23025	PID1	Q7Z2X4	Q8N9N5
PIGH	Q14442	O95256	PIGM	Q9H3S5	Q9NQZ5
PIGN	O95427	P43119	PIGP	P57054	Q8N5I4
PIGT	Q969N2	Q8NBL1	PIGV	Q9NUD9	P45973
PIK3AP1	Q6ZUJ8	P62993	PIK3C2A	O00443	O95630, Q96FJ0
PIK3CD	O00329, O00329-2	P27986	PIK3CG	P48736	Q13370
PIK3IP1	Q96FE7	P78382	PIK3R1	P27986	P03496
PIK3R3	Q92569	Q9BRX2	PIK3R4	Q99570	P61981
PIK3R6	Q5UE93	Q13370	PILRA	Q9UKJ1	P29350
PIM1	P11309-2	Q9UNQ0	PIMREG	Q9BSJ6	P04156
PIN4	Q9Y237	P00533	PIP4K2A	P48426	Q96HN2
PIP4K2C	Q8TBX8	Q15831	PIP5K1A	Q99755	P01116
PIR	O00625	P02675	PITPNB	P48739	P55212
PITRM1	Q5JRX3	P0DTC5	PIWIL4	Q7Z3Z4	Q9Y2W6
PJA1	Q8NG27	Q93009	PKD1	Q15139	P12830
PKD2	Q13563	P21453	PKIA	P61925	P17612
PKM	P14618-1	Q16665	PKMYT1	Q99640	P06493
PKN1	Q16512	P26641	PKN2	Q16513	O76075
PKN3	Q6P5Z2	A1A4S6	PKP4	Q99569	Q14160
PLA2G6	O60733	O60733	PLAC8	Q9UHV8	Q12837
PLAGL2	Q9UPG8	O94955	PLAUR	Q03405	Q09328
PLBD1	Q6P4A8	P15498	PLCD3	Q8N3E9	Q96JC9
PLCG2	P16885	O95997	PLCH1	Q4KWH8	P61981
PLCL2	Q9UPR0	P49914	PLEK	P08567	P51659
PLEKHA1	Q9HB21	Q9P0V3	PLEKHA2	Q9HB19	O14908
PLEKHA3	Q9HB20	O43148	PLEKHA7	Q6IQ23-2	Q15669
PLEKHF2	Q9H8W4	Q8IZD9	PLEKHG2	Q9H7P9	Q13526
PLEKHG3	A1L390	P13569	PLEKHG5	O94827-4	P61587
PLEKHH3	Q7Z736	P31152	PLEKHM1	Q9Y4G2	P51149
PLEKHM3	Q6ZWE6	P61981	PLEKHO1	Q53GL0	Q9HCE7
PLEKHO2	Q8TD55	P63104	PLK1	P53350	Q99741
PLK2	Q9NYY3	Q13177	PLK4	O00444	P07237
PLLP	Q9Y342	Q86WV6	PLOD1	Q02809	Q9Y580
PLP2	Q04941	Q96QE2	PLPP1	O14494	P17612
PLPP6	Q8IY26	Q9UBN6	PLPPR2	Q96GM1	Q9BVC3
PLRG1	O43660	O75533	PLS3	P13797	P13569
PLSCR1	O15162	Q9UBR2	PLXDC2	Q6UX71	Q6PIZ9
PLXNA2	O75051	P01375	PLXNB2	O15031	P52198
PLXND1	Q9Y4D7	P22736	PM20D1	Q6GTS8	P01160
PMAIP1	Q13794	P10415	PML	P29590	P06730
PMM1	Q92871	P20340	PMPCA	Q10713	P01160

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PNPLA2	Q96AD5	P09917	PNPLA6	Q8IY17	Q8TDQ0
PNPO	Q9NVS9	Q99551	PNRC1	Q12796	Q9NPI6
POC1A	Q8NBT0	Q13371	POC1B	Q8TC44	Q9BRX2
POC5	Q8NA72, Q8NA72-3	P41208	POFUT1	Q9H488	P0DTC8
POLA1	P09884	Q13257	POLA2	Q14181	Q13257
POLB	P06746	Q01831	POLD1	P28340	P12004
POLD2	P49005	O60673	POLD3	Q15054	O75152
POLDIP2	Q9Y2S7	P13196	POLE	Q07864	Q15185
POLE2	P56282	P52594	POLG2	Q9UHN1	Q02252
POLH	Q9Y253	Q86YC2	POLQ	O75417-1	Q06609
POLR1B	Q9H9Y6	O15160, O95602, P19388, P61218, Q3B726, P62875	POLR1C	O15160	P23025
POLR2D	O15514	Q15648	POLR2G	P62487	Q9Y5B0
POLR2H	P52434	O15160, O95602, P19388, P61218, P62875	POLR2L	P62875	O00267
POLR3E	Q9NVU0	Q9H063	POLR3F	Q9H1D9	P52292
POLR3GL	Q9BT43	O15160, P19388, P61218, P62875	POLR3K	Q9Y2Y1	O15160, P19388, P61218, P62875, P52434
POLRMT	O00411	P36941	POM121	Q96HA1	P36941
POM121C	A8CG34	Q14974	POMP	Q9Y244	Q96FE5
POP1	Q8WXC3	Q9ULZ3	POP4	O95707	P17707
POU2AF1	Q16633	Q06124	PPA1	Q15181	P13569
PPARA	Q07869	P55072	PPAT	Q06203	P61769
PPCDC	Q96CD2	Q13427	PPFIA1	Q13136	P61981
PPFIA3	O75145	Q9UBB9	PPFIBP1	Q86W92	P31947
PPFIBP2	Q8ND30	P61981	PPID	P26882	P07900
PPIF	P30405	P04637	PPIL1	Q9Y3C6	P50458
PPIP5K2	O43314	Q13257	PPM1A	P35813	P32121, P49407
PPM1G	O15355	O96017	PPM1L	Q8BHN0	Q9H3P7
PPME1	Q9Y570	P67775, P63151	PPOX	P50336	O95861
PPP1CA	P62136	P55199	PPP1CB	P62140	Q9UQK1
PPP1R10	Q96QC0	P36873	PPP1R12B	O60237-2	P06396
PPP1R13B	Q96KQ4	Q9BRT9	PPP1R15A	O75807	P62136
PPP1R15B	Q5SWA1	P36873	PPP1R16A	Q96I34	Q92696, P53611
PPP1R18	Q6NYC8	Q9Y6D9	PPP1R21	Q6ZMI0-5	P99999
PPP1R26	Q5T8A7	Q96RU7	PPP1R32	Q7Z5V6-2	P36873
PPP1R3D	O95685	Q9UQK1	PPP1R3F	Q6ZSY5	Q13627
PPP1R8	Q12972, Q12972-2	P36873	PPP1R9A	Q9ULJ8	Q9UM54
PPP2CB	P62714	Q9BRV8	PPP2R2A	P63151	P46531
PPP2R2B	Q00005	O15530	PPP2R2D	Q66LE6	O43768
PPP2R3A	Q06190	Q9NYZ3	PPP2R3B	Q9Y5P8	O14964
PPP2R5A	Q15172	O96017	PPP2R5C	Q13362-3, Q13362-2, Q13362-1	O96017
PPP2R5D	Q14738	P30307	PPP3CA	Q08209	P49841
PPP3CC	P48454	O95644	PPP4C	P60510	O00267
PPP4R3A	Q6IN85	P18509	PPP5C	P53041	P07900
PPT1	P50897	O75925	PPTC7	Q8NI37	P11182
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PRC1	O43663	P36941	PRCP	P42785	O60883

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PRDX1	Q06830	P04156	PRDX3	P30048	O60313
PRDX5	P30044	P00441	PRDX6	P30041	Q7KZN9
PREB	Q9HCU5	P27824	PRELID1	Q9Y255	O43715
PRF1	P14222	P14222	PRICKLE3	O43900	Q9UJK0
PRIM1	P49642	P09884	PRIM2	P49643	Q13257
PRKAB1	Q9Y478	O75385	PRKAB2	O43741	P01185
PRKACB	P22694	P13861	PRKAG2	Q9UGJ0	Q13501, P54646
PRKAR1A	P10644	Q9GZX7	PRKAR1B	P31321	Q13309
PRKAR2B	P31323	P13861	PRKCA	P17252	Q9BRX2
PRKCE	Q02156	P31947	PRKCH	Q9BVQ0	Q8N9N5
PRKCZ	Q05513	O95999	PRKD3	O94806	P63027
PRKDC	P78527	P46531	PRKN	O60260-5	O94844
PRKRIP1	Q9H875	P62304	PRMT3	O60678	P15880
PRMT5	O14744	Q9Y6K1	PRMT6	Q96LA8	Q9Y365
PRNP	P04156	P29372	PROCA1	Q8NCQ7-2	P55212
PROCR	Q9UNN8	P25116	PROSER3	Q2NL68	Q96ES7
PRPF18	Q99633	Q9Y6K9	PRPF19	Q9UMS4	P61981
PRPF3	O43395	Q8WW38	PRPF38A	Q8NAV1	Q6NYC1
PRPF38B	Q5VTL8	P61981	PRPF39	Q86UA1	P0DTC5
PRPS2	P11908	Q9Y3M2	PRR11	Q96HE9	Q15361
PRR14L	Q5THK1	Q15185	PRR5	P85299	P31947
PRR5L	Q6MZQ0	Q9UI95	PRR7	Q8TB68	P46934
PRRC2C	Q9Y520	P13569	PRRG4	Q9BZD6	Q96PU5
PRSS22	Q9GZN4	P35813	PRSS23	O95084	P60709
PRUNE2	Q8WUY3	P07900	PSAP	P07602	P07948
PSAT1	Q9Y617	P12830	PSD4	Q8NDX1-2	Q5SW96
PSEN1	P49768-2	P53350	PSEN2	P49810	P30626
PSIP1	O75475	Q03164	PSMA3	P25788	Q9Y5K5
PSMA4	P25789	Q16665	PSMA5	P28066	Q9Y5K5
PSMA7	O14818	Q16665	PSMB2	P49721	Q9Y5K5
PSMB3	P49720	Q9Y5K5	PSMB5	P28074	Q92989
PSMB6	P28072	Q9Y5K5	PSMB7	Q99436	Q9Y5K5
PSMB8	P28062-2	Q13114	PSMC1	P62191	Q9Y5K5, Q16186
PSMC2	P35998	O60216	PSMC3	P17980	Q9Y5K5, Q16186
PSMC4	P43686	Q9Y5K5, Q16186	PSMC6	P62333	P46109
PSMD1	Q99460	Q9Y5K5, Q16186	PSMD10	O75832	Q12849
PSMD11	O00231	P60896	PSMD12	O00232	Q9Y5K5, Q16186
PSMD14	O35593, O00487	Q9Y5K5, Q16186	PSMD3	O43242	P60896
PSMD4	P55036	Q9Y5K5, Q16186	PSMD5	Q16401	Q9Y5K5, Q16186
PSMD6	Q15008	Q9Y5K5, Q16186	PSMD8	P48556	Q9Y5K5, Q16186
PSME1	Q06323	P60896	PSME3	P61289	Q9HCE7
PSMG1	O95456	Q99436, P28066, O14818, P20618, Q9Y244, P28065, Q969U7, P49721, P49720, P25787, P25786	PSMG2	Q969U7	Q96FE5
PSPC1	Q8WXF1-2	O43809	PSTPIP2	Q9H939	Q05209
PTCD2	Q8WV60	P04004	PTCD3	Q96EY7	P13569
PTCH1	Q13635-1	Q15465	PTGDR	Q13258	Q9NZ52
PTGDS	P41222	P00533	PTGER3	P43115	O15397

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PTGES3	Q15185	Q8TAT5	PTH2R	P49190	P13591
PTK2	Q05397	Q68CZ2	PTK7	Q13308	P35222
PTP4A3	O75365	Q12974	PTPA	P18433	P12931
PTPN11	Q06124, Q06124-2	Q13480	PTPN12	Q05209	P29353
PTPN14	Q15678	P46937	PTPN18	Q99952	P00533
PTPN23	Q9H3S7	P12931	PTPN3	P26045	Q96QG7
PTPN7	P35236	P28482	PTPN9	P43378	Q15116
PTPRA	P18433	P12931	PTPRB	P23467	P27361
PTPRC	P08575	P08575	PTPRE	P23469	P62993
PTPRG	P23470	Q6VY07	PTPRK	Q15262	P04183
PTPRM	P28827	O60469	PTPRR	Q15256	Q16539
PTRH2	Q9Y3E5	Q9H2J7	PTS	Q03393	Q96CW1
PTTG1	O95997	Q14674	PUM1	Q14671	P31947
PURA	Q00577	P06400	PURB	Q96QR8	P13569
PWP2	Q15269	Q8IWZ6	PWWP2B	Q6NUJ5	Q13547
PXDN	Q92626	P01282	PXMP2	Q9NR77	P46059
PXN	P49023	Q13480	PYCARD	Q9ULZ3	Q9C000
PYCR1	P32322	Q99497	PYCR3	Q53H96	P42772
PYGL	P06737	P13569	PYM1	Q9BRP8	Q9NUX5
QKI	Q96PU8	Q9NWB1	QPRT	Q15274	Q8NFD5
QRICH1	Q2TAL8	Q14938	QTRT1	Q9BXR0	P10809
R3HDM2	Q9Y2K5	P61981	RAB10	P61026	Q96CV9
RAB11A	P62491	P61981	RAB11FIP1	Q6WKZ4	P61981
RAB11FIP2	Q7L804	P61981	RAB11FIP3	O75154	Q9H0H5
RAB11FIP4	Q86YS3	Q8NFZ5	RAB11FIP5	Q9BXF6	P63104
RAB13	P51153	O43924	RAB18	Q9NP72	P13569
RAB1B	Q9H0U4	Q92696	RAB20	Q9NX57	P52306
RAB27B	O00194	Q8IYJ3	RAB28	P51157	O43924
RAB29	O14966	Q92696	RAB2A	P61019	Q3MII6
RAB31	Q13636	Q92696, P53611	RAB33A	Q14088	Q92530
RAB33B	Q9H082	Q676U5	RAB35	Q15286	P42858
RAB38	P57729	P14136	RAB39B	Q96DA2	P46059
RAB3A	P20336	Q96QF0	RAB3B	P20337	Q8N695
RAB3GAP1	Q15042	P09917	RAB3GAP2	Q9H2M9	P13569
RAB3IP	Q96QF0	Q96T51	RAB40B	Q12829	P53611
RAB4A	P20338	Q96JA1	RAB4B	P61018	O75419
RAB9A	P51151	Q92696, P53611	RABEP2	Q9H5N1	Q9NWB7
RABGAP1	Q9Y3P9	P40763	RABGAP1L	Q5R372-2	P06400
RABGEF1	Q9UJ41	Q9H0F6	RABGGTA	Q92696	P53611
RABGGTB	P53611	O94955	RABL2B	Q9UNT1-2	P06396
RAC2	P15153	P52306	RAC3	P60763	P52565
RACGAP1	Q9H0H5	O14757	RAD51	Q06609	Q86YC2
RAD51AP1	Q96B01	Q8TAF3	RAD51C	O43502	Q86YC2
RAD51D	O75771	O43543, O43502, O15315	RAD54L	P46100	P84243, Q9UER7
RAD54L2	Q9Y4B4	Q13185	RAF1	P04049	Q13177
RAI14	Q9P0K7	P53350	RALA	P11233	P13569
RALB	P11234	P11182	RALBP1	Q15311	P43897
RALGAPA1	Q6GYQ0	P63104	RALGPS2	Q86X27	P31947
RAN	P62826	P47813	RANBP1	P43487	P18754
RANBP2	P49792, Q9ERU9	P46060, P18754	RANBP6	O60518	O43808, O14975

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RANGAP1	P46060	P61769	RAP1B	P61224	Q12967
RAP1GDS1	P52306	P17480	RAP2B	P61225	P19320
RAP2C	Q9Y3L5	Q13371	RAPGEF1	Q13905	P46109
RAPGEF2	Q9Y4G8	P46937	RAPGEF5	Q92565	Q12955
RAPH1	Q70E73	Q99962	RARS	P54136	P13569
RASD1	Q9Y272	P61978	RASEF	Q8IZ41	P13569
RASL11A	Q6T310	P42858	RASSF1	Q9NS23-4	Q16595
RASSF2	P50749	P27797	RASSF3	Q86WH2	Q9UER7
RASSF4	Q9H2L5	O14964	RASSF5	Q8WWW0	Q8WY64
RASSF8	Q8NHQ8	P36873	RAVER1	Q8IY67	O75791
RB1	P06400	O96017, O14757	RB1CC1	Q8TDY2	O75385
RBAK	Q9NYW8	Q969R5	RBBP4	Q09028	P46531
RBBP6	Q7Z6E9	Q16629	RBBP8	Q99708	Q9UQ84
RBFOX2	O43251	Q9NWB1	RBFOX3	Q8BIF2	Q16665
RBKS	Q9H477	Q9H477	RBL1	P28749	O43524
RBM14	Q96PK6	P09651	RBM15	Q96T37	P52294
RBM19	Q9Y4C8	P06748	RBM22	Q9NW64	P62306, P62304
RBM25	P49756	P13569	RBM28	Q9NW13	P46087
RBM33	Q96EV2	O75152	RBM34	P42696	P40429
RBM38	Q9H0Z9	P31943	RBM39	Q14498-3	Q9UH92
RBM4	Q9BWF3-1	Q04637	RBM41	Q96IZ5	Q9Y2I6
RBM42	Q9BTD8	P09651	RBM45	Q8IUH3	Q9Y316
RBM48	Q5RL73	Q8NFP9	RBM4B	Q9BQ04	P07910
RBMS2	Q15434	O00425	RBP7	Q96R05	Q8WXH5
RBPJ	Q06330-6, Q06330	P46531	RBPMS2	Q6ZRY4	P07910
RBSN	Q9H1K0	Q9NRW7	RBX1	P62877	P61081
RCAN3	Q9UKA8	P11802	RCBTB2	O95199	O60733
RCC1	P18754	O14980	RCC1L	Q96I51	O00425
RCCD1	A6NED2	Q15814	RCE1	Q9Y256	Q96RK4
RCHY1	Q96PM5	Q96PM5	RCL1	Q9Y2P8	Q08379
RCN3	Q96D15	Q9UMX1	RCOR2	Q8IZ40	Q13547
RCOR3	Q9P2K3	Q9Y2I6	RCSD1	Q6JBY9	P52907
RCVRN	P35243	Q53FT3	RDH10	Q8IZV5	Q05329
RDM1	Q8NG50	P26441	RDX	P35241	P52565
RECQL4	O94761	Q9UBX5	RECQL5	O94762	P19388, P61218, P62875
REEP3	Q6NUK4	P31947	REEP5	Q00765	Q8TB40
REEP6	Q96HR9	P0DTC5	REG4	Q9BYZ8	Q9NTG7
REL	Q04864-2	P30622	RELA	Q04206	Q13568
RELL1	Q8IUW5	P26572	RELL2	Q8NC24	P78382
REPS2	Q8NFH8	Q96D71	RETREG1	Q9H6L5-1	Q9GZQ8
REV1	Q9UBZ9	Q9UI95	REV3L	O60673	Q9UI95
REXO5	Q96IC2	Q15185	RFC2	P35250	P40937, P35249
RFC3	P40938	P40937, Q9BVC3, P35249	RFC4	P35249	P35250, P0CG13, P40938, P40937, Q9BVC3, Q8WVB6
RFC5	P40937	P06400	RFT1	Q9NWF4	Q16585
RFWD2	Q8NHY2	Q96RU7	RFWD3	Q6PCD5	P51668
RFX2	P48378	O75928-2	RFX3	P48380	Q12933
RFX5	P48382	Q92769	RFXANK	O14593	P04049
RGS17	Q9UGC6	Q9H2F3	RGS19	P49795	P25786

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RGS6	P49758	P08238	RHBDD2	Q6NTF9-3	P23434
RHBDD3	Q9Y3P4	Q6UWB1	RHBDF2	Q80WQ6	P78536
RHEB	Q15382-PRO_0000082708	O43924	RHNO1	Q9BSD3	Q08379
RHOB	P62745	Q92888	RHOF	Q9HBH0	P52306
RHOT2	Q8IXI1	O00410	RHOU	Q7L0Q8	Q14289
RHPN1	Q8TCX5	Q9NZ53	RIBC2	Q9H4K1	P98164
RIC3	Q7Z5B4-5	Q07817	RIC8A	Q9NPQ8-4	Q96QE2
RICTOR	Q6R327	P22891	RIDA	P52758	P0CG47
RIF1	Q5T3J3	Q96A70	RILP	Q96NA2	P51149
RILPL1	Q5EBL4-3	Q86T03	RIMS3	Q9UJD0	Q92993
RIN3	Q8TB24	P46109	RIOK3	O14730	Q92851
RIPK2	O43353	Q13114	RIPOR2	Q9Y4F9, Q9Y4F9-2	P61586
RIPPLY3	P57055	Q13416	RIT1	Q9C0K0	P46531
RITA1	Q96K30	Q9NPI6	RMDN2	Q96LZ7	Q9Y4Z2
RMDN3	Q96TC7	Q9Y4Z2	RMI1	Q9H9A7	Q14159
RMI2	Q96E14	Q14159	RMND1	Q9NWS8	Q8TDU9
RMND5A	Q9H871	Q6VN20, Q96S59	RNASEH2A	O75792	Q99943
RNASEK	Q6P5S7	Q5QGT7	RNASEL	Q1RLL8	Q99081
RND1	Q92730	O43157	RNF11	Q9Y3C5	Q9HAU4
RNF112	Q9ULX5	O14832	RNF113A	O15541	Q8N4J0
RNF114	Q9Y508	P51668	RNF115	Q9Y4L5	P51668
RNF125	Q96EQ8	P68036	RNF126	Q9BV68	P51668
RNF135	Q8IUD6	Q13426	RNF139	Q8WU17	Q8TCT9
RNF144A	P50876	Q96AA3	RNF144B	Q7Z419	Q07812
RNF145	Q96MT1	P0DTC5	RNF149	Q8NC42	Q30201
RNF166	Q96A37	P36941	RNF168	Q8IYW5	P37840
RNF169	Q8NCN4	Q13627	RNF170	Q96K19	Q13370
RNF185	Q96GF1	P51668	RNF187	Q5TA31	Q93034
RNF19A	Q9NV58	P00441	RNF19B	Q6ZMZ0	Q8TAF3
RNF216	Q9NWF9	P68036	RNF220	Q9BWG1	Q96CV9
RNF24	Q9Y225-2	O95406	RNF26	Q9BY78	P62837
RNF32	Q9H0A6	Q96FJ0	RNF34	Q969K3	Q14790
RNF38	Q9H0F5-2	O76024	RNF40	O75150	P06400
RNF43	Q68DV7	P62837	RNF44	Q7L0R7	Q8TDS5
RNF5	Q99942	P51668	RNF6	Q9Y252	P09917
RNF7	Q9UBF6	Q8WXH5	RNFT1	Q5M7Z0	Q9NRM6
RNGTT	O60942	P52294	RNH1	P13489	Q12834
RNPC3	Q96LT9	P62306, P14678, P62304	ROBO1	Q9Y6N7	O75044
ROBO3	Q96HH0	Q9NRD5	ROGDI	Q9GZN7	P21281
ROR1	Q01973	P12931	ROR2	Q01974	O95231
RORA	P35398-4	Q9BZS1	RP2	O75695	Q16881
RP9	Q8TA86	Q13895	RPA1	P27694	P23025
RPA3	P35244	P23025	RPAIN	Q86UA6	O15315
RPAP3	Q9H6T3	P36873	RPF1	P46934-4	Q9HAU4
RPF2	Q9H7B2	Q15646	RPL10	P27635	P12931
RPL10A	P62906	Q99558	RPL11	P62913	P40429
RPL12	P30050	P13569	RPL13A	P40429	Q9Y3T9
RPL15	P61313	P11388	RPL17	P18621	P63244

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
RPL18	Q07020	P40429	RPL18A	Q02543	Q15361
RPL19	P84098	P40429	RPL21	P46778	P63244
RPL22	P35268	Q96CW1	RPL23A	P62750	Q01105
RPL24	P83731	Q99558	RPL26	P61254	P63244
RPL26L1	Q9UNX3	P42858	RPL27	P61353	Q99558
RPL27A	P46776	P63244	RPL29	P47914	P09429
RPL3	P39023	Q99558	RPL30	P62888	P40429
RPL32	P62910	Q99558	RPL34	P49207	P63244
RPL35	P42766	P05455	RPL35A	P18077	P63244
RPL36A	P83881	P63244	RPL37A	P61513	P17612
RPL38	P63173	P63244	RPL41	P62945	P63244
RPL5	P46777	P40429	RPL6	Q02878	Q99558
RPL7	P18124	Q99558	RPL7A	P62424	P40429
RPL7L1	Q6DKI1	P40429	RPL8	P62917	Q99558
RPL9	P32969	Q99558	RPLP2	P05387	P13569
RPN1	P04843	Q14165	RPP25	Q9BUL9	Q00403
RPP25L	Q8N5L8	Q9UPY3	RPP38	P78345	P06748
RPP40	O75818	P06748	RPRD1B	Q9NQG5	Q9Y5B0
RPS10	P46783	Q9H2U1	RPS11	P62280	Q04637
RPS13	P62277	Q99558	RPS14	P62263	Q15361
RPS15	P62841	P62937	RPS16	P62249	Q8IZH2
RPS17	P08708	P63244	RPS18	P62269	Q99558
RPS23	P62266	Q15843	RPS24	P62847	Q99558
RPS25	P62851	P63244	RPS26	P62854	P63244
RPS27	P42677	Q00987	RPS27A	P62990	P51668
RPS27L	Q71UM5	Q00987	RPS28	P62857	P63244
RPS29	P62273	P63244	RPS3	P23396	P19838
RPS3A	P61247	P40429	RPS4X	P62701	Q00403
RPS6	P62753	Q92900	RPS6KA2	Q15349	Q12849
RPS6KA3	P51812	Q12849	RPS6KA5	O75582	Q16539
RPS6KB2	Q9UBS0	P67870	RPS6KL1	Q9Y6S9	Q96HR3
RPS7	P62081	Q15843	RPS9	P46781	P40763
RPTOR	Q8N122	O75385	RPUSD4	Q96CM3	Q00613
RRAGA	Q7L523	P61916	RRAGB	Q5VZM2-2, Q5VZM2	Q9Y2Q5, Q0VGL1, Q6IAA8, O43504, Q8NBW4, Q9UHA4
RREB1	Q92766	Q13547	RRM1	P23921	P31350
RRM2	P31350	Q9UM11	RRM2B	Q7LG56	O43929
RRP1	P56182	Q13724	RRP36	Q12481	O14641
RRS1	Q15050	P40429	RSAD2	Q8WXG1	P55056, P02654
RSBN1	Q5VWQ0	Q9BYW2	RSF1	Q96T23	P09429
RSL24D1	Q9UHA3	Q9UJW3	RSPO4	Q2I0M5	O14964
RSPRY1	Q96DX4	P00751	RSRC2	Q7L4I2-2	Q08379
RSRP1	Q9BUV0	P61289	RSU1	Q15404	Q9NVD7, Q13418
RTCB	Q9Y3I0	P03496	RTKN2	Q8IZC4	Q96B97
RTL10	Q7L3V2	P21796	RTL8C	A6ZKI3	P42772
RTN1	Q16799-3	O00560	RTN3	O95197-3	Q9NTJ5
RTN4IP1	Q8WWV3	Q9HAU4	RTN4RL1	Q86UN2	Q7Z3S9, P0DPK4
RUNDC3B	Q96NL0	P78368	RUNX3	Q13761	P24385
RUSC1-AS1	Q66K80	P99999	RUSC2	Q8N2Y8	P00533

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RYK	P34925	Q68D85	RYR1	P11716	P62942
RYR2	Q92736	Q00987	S100A10	P60903	P46092
S100A11	P31949	P01375	S100A12	P80511	P01375
S100A5	P33763	P35579	S100A8	P05109	P35228
S100A9	P06702	P35228	S100B	P04271	Q15109
S100Z	Q8WXG8	Q00994	S1PR1	P21453	Q7Z2K8
S1PR4	O95977	Q9NSG2	SACS	Q9NZJ4	Q6UWR7
SAE1	Q9UBE0	Q06710	SAFB2	Q14151	P09651
SAMD1	Q6SPF0	Q13309	SAMD3	Q8N6K7-2	P04156
SAMHD1	Q9Y3Z3	P20248	SAMM50	Q9Y512	Q9BRQ6, Q9NX63, Q6UXV4
SAP18	O00422	Q13547	SAP30	O75446	Q00403
SAP30BP	Q9UHR5	P68400	SAPCD2	Q86UD0	O75496
SARM1	Q6SZW1	Q99836	SARNP	P82979	O75152
SARS	P49591	P38936	SART3	Q15020	P40429
SASH1	O94885	P31947	SASS6	Q6UVJ0	Q9HC77
SAT1	P21673	Q9BQ70	SATB1	Q01826	P63279
SATB2	Q9UPW6	P53350	SAV1	Q9H4B6	P18509
SBDS	Q9Y3A5	Q15149	SBF2	Q86WG5	Q13614
SCAF1	Q9H7N4	Q16629	SCAF8	Q9UPN6	Q08379
SCAMP4	Q969E2	Q8WTR4	SCARB1	Q8WTV0	P11686
SCARB2	Q14108	Q9UBD6	SCARF1	Q14162	Q9HC36
SCCPDH	Q8NBX0	Q92466	SCD	O00767	Q9H2K0
SCIMP	Q6UWF3	P41240	SCML1	Q9UN30-2	O00560
SCNN1A	P37088	P46934	SCNN1D	P51172	P27824
SCOC	Q9UIL1-3	Q9BRV8	SCRN1	Q12765	P10636-8
SCRN2	Q96FV2	P25963	SCRN3	Q0VDG4	Q8NB12
SCUBE3	Q8IX30	P37173	SDC4	P31431	O15530
SDCCAG8	Q86SQ7	P61088	SDF4	Q9BRK5	P50548
SDHAF4	Q5VUM1	Q9NRP4, P31040	SDHC	Q99643	Q92993
SEC11A	P67812	O15118	SEC11C	Q9BY50	P09601
SEC14L1	SEC14L1	Q07955	SEC22B	O75396	O95249, Q13190
SEC23A	Q15436	O94979, P55735	SEC23B	Q15437	Q15436
SEC23IP	Q9Y6Y8	A1L4H1	SEC24B	O95487	Q15436
SEC24D	O94855, O94855-2	Q15436	SEC31A	O94979	Q15436, P55735
SEC61B	P60468	P51572	SEC61G	P60059	P11686
SECISBP2	Q96T21	Q08379	SEL1L3	Q68CR1	Q8IV16
SELENOH	Q8IZQ5	Q9Y4X5	SELENOK	Q9Y6D0	Q9H2K0
SELENOM	Q8WWX9	P61604	SELENON	Q9NZV5	P22891
SELENOO	Q9BVL4	Q6IN84	SELENOP	P49908	P00533
SELL	P14151-2	Q9BRI3	SELP	P49908	P00533
SEM1	P60896	Q00403	SEMA3B	Q13214-2	Q9NRD5
SEMA3C	Q99985	P18827	SEMA4A	Q9H3S1	Q9Y4D7
SEMA4C	Q9C0C4	Q96PM5	SEMA4F	O95754	Q13485
SEMA7A	O75326	O60486	SENP1	Q9P0U3	P46060
SEPHS1	P49903	Q99611	SERGEF	Q9UGK8	O60341
SERINC1	Q9NRX5	Q9H8X2	SERINC3	Q13530	O76024
SERP1	Q9Y6X1	Q9H2K0	SERPINA1	P01009	P08246

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SERPINB8	P50452	Q9Y324	SERPINH1	P50454	P12830
SERTAD1	Q9UHV2	P61024	SERTAD2	Q14140	P61024
SERTAD3	Q9UJW9	Q6W0C5	SESTD1	Q86VW0	O43187
SETD1A	O15047	P51610	SETD1B	Q9UPS6	Q9UER7
SETD2	Q9BYW2	P00740	SETD3	Q86TU7	P12004
SETD7	Q8WTS6	P26358	SETX	Q7Z333	P63279
SF1	Q13285	Q9NPJ4	SF3A1	Q15459	P08047
SF3B1	O75533	O43809	SF3B3	Q15393	P53350
SF3B6	Q9Y3B4	Q9H7X7	SFN	P31947	P30291
SFR1	Q86XK3	Q9UBB9	SFT2D2	O95562	Q99541
SFXN1	Q9H9B4	Q7KZN9	SGCB	Q16585	P01375
SGK3	Q96BR1	Q96J02	SGK494	Q96LW2	O95231
SGO1	Q5FBB7	O60216, Q96FF9	SGO2	Q562F6	Q13257
SGPP1	Q9BX95	Q8WTR4	SGSH	P51688	Q8NBK3
SGSM1	Q2NKQ1-4, Q2NKQ1	P41208	SGTA	O43765	Q13568
SGTB	Q96EQ0	P10451	SH2B1	Q9NRF2	O60674
SH2B2	O14492-2	Q9UBH0	SH2D1B	O14796	P12931
SH2D2A	Q9NP31	O14543	SH2D3C	Q8N5H7-2, Q8N5H7	P56945
SH2D4A	Q9H788	Q9Y2I6	SH3BGRL	O75368	P00533
SH3BGRL3	Q9H299	P00533	SH3BP1	Q9Y3L3	P47756
SH3BP2	P78314	A1A4S6	SH3BP4	Q9P0V3	O14908
SH3BP5	O60239	Q07889	SH3BP5L	Q7L8J4	Q9NWX6
SH3D21	A4FU49	P00533	SH3GL1	Q99961	O14672
SH3GLB1	Q9Y371	P01730	SH3KBP1	Q96B97	P98082
SH3RF3	Q8TEJ3	Q13177	SH3YL1	Q96HL8	Q5HYK7
SHB	Q15464	P00519	SHC1	P29353	P04629
SHC3	Q92529	Q96LC9	SHC4	Q6S5L8	P00533
SHCBP1	Q8NEM2	P05771	SHMT2	P34897	P13569
SHPRH	Q149N8-1	P61088	SHROOM1	Q2M3G4	P61981
SHTN1	A0MZ66	P61981	SIAH1	Q8IUQ4, Q8IUQ4-2	Q96B01
SIGLEC12	Q96PQ1	P11021	SIGLEC9	Q9Y336	P11686
SIGMAR1	Q99720	P14416	SIK2	Q9H0K1	Q9BV73
SIN3B	O75182	Q9UER7	SIPA1	Q96FS4	P38936
SIPA1L3	O60292	P31947	SIRPA	P78324, P97710	Q06124
SIRPB1	O00241	O43914	SIRPB2	Q9P1W8	Q08722
SIRPG	Q9P1W8	Q08722	SIRT1	Q96EB6	Q14469
SIRT2	Q8IXJ6, Q8VDQ8, Q8IXJ6-2	O60566, Q12834	SIRT7	Q9NRC8	Q16656
SIT1	Q9Y3P8	Q06124	SKA1	Q96BD8	A1L4H1
SKA2	Q8WVK7	Q9NWB7	SKIL	P12757	P23025
SKIV2L	Q15477	P06748	SKIV2L2	P42285	P52298, Q09161
SKP2	Q13309	Q13415	SLA	Q13239-3	Q13882
SLA2	Q9H6Q3	P54646	SLAMF1	Q13291	Q06124
SLAMF7	Q8BHK6	P06241	SLBP	Q14493	Q92900
SLC10A7	Q0GE19	P04114	SLC11A2	P49281-3	P00403
SLC12A6	Q9UHW9	P17931	SLC14A1	Q13336-2	Q5QGT7
SLC15A3	F5H1C8	Q14974	SLC15A4	Q8N697	Q9HAI6, Q13568
SLC16A1	P53985	P13569	SLC16A11	Q8NCK7	P35613
SLC16A13	Q7RTY0	Q96BA8	SLC16A3	O15427	Q14160

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
SLC16A6	O15403	O76024	SLC19A1	P41440	Q03135
SLC1A4	P43007	Q15758	SLC1A5	Q15758	P0DTC5
SLC20A1	Q8WUM9	Q9UM47	SLC22A15	Q8IZD6	Q86TM6
SLC22A17	Q8WUG5	P07237	SLC22A4	Q9H015	O76082
SLC23A1	Q9UHI7	Q9UBQ7	SLC24A3	Q9HC58	Q86TM6
SLC25A1	P53007	P13569	SLC25A10	Q9UBX3	P13569
SLC25A11	Q02978	P60709	SLC25A12	O75746	P13569
SLC25A15	Q9Y619	Q15116	SLC25A17	O43808	P40855
SLC25A24	Q6NUK1	P13569	SLC25A28	Q96A46	P46059
SLC25A4	P12235	P12236	SLC25A40	Q8TBP6	P02452
SLC25A43	Q8WUT9	P10809	SLC25A46	Q96AG3	Q8N695
SLC25A5	P05141	P12236	SLC26A2	P50443	P17931
SLC26A6	Q9BXS9-3	P02787	SLC27A2	O14975	O00410
SLC27A4	Q6P1M0	P40855	SLC29A1	Q99808	P43119
SLC2A9	Q9NRM0-1	Q5J8M3	SLC30A4	O14863	P55010
SLC30A5	Q8TAD4	Q9Y2A9	SLC30A6	Q6NXT4	Q8TAD4
SLC30A9	Q6PML9	Q8TC07	SLC31A1	O15431	Q9H9P8
SLC33A1	O00400	P13569	SLC35A2	P78381-1, P78381-2	Q9Y2D2
SLC35A5	Q9BS91	O94805	SLC35B1	P78383	Q96BA8
SLC35B2	Q8TB61	Q96BA8	SLC35B4	Q969S0	O43914
SLC35C1	Q96A29	P05090	SLC35C2	Q9NQQ7-3	Q9NRZ7
SLC35E1	Q96K37	Q9Y366	SLC35F2	Q8IXU6	Q06136
SLC35F6	Q8N357	P38606	SLC35G1	Q2M3R5	Q15125
SLC38A1	Q9H2H9	Q9UKG4	SLC38A9	Q8NBW4	Q9Y2Q5, Q0VGL1, Q6IAA8, O43504, Q9UHA4
SLC39A1	Q9NY26	P43115	SLC39A11	Q8N1S5	Q9NZQ7
SLC39A13	Q96H72	O15173	SLC39A8	Q9C0K1	Q09328
SLC39A9	Q9NUM3	P20585	SLC41A1	Q8IVJ1	P30518
SLC43A2	Q8N370	P0DPK4	SLC43A3	Q8NBI5	P55072
SLC44A2	Q8IWA5	O75695	SLC44A3	Q8N4M1-3	Q8WTS1
SLC44A5	Q8NCS7	Q9P0P0	SLC4A3	P48751	Q12933
SLC4A7	Q9Y6M7	Q14160	SLC5A3	P53794	Q9H244
SLC7A1	P30825	P27352	SLC7A11	Q9UPY5	P16070
SLC7A6	Q92536	Q9UPQ8	SLC9A3R1	O14745	P0DTC4
SLC9A3R2	Q15599	P46937	SLCO4C1	Q6ZQN7	Q14527
SLF1	Q9BQI6	P19838	SLF2	Q8IX21	P25786
SLITRK5	O94991	Q6PJG9	SLMAP	Q14BN4	Q9BRV8
SLU7	O95391	O75152	SLX4	Q8IY92	P53350
SLX4IP	Q5VYV7	P53350	SMAD4	Q13485	Q6VMQ6
SMAD5	Q99717	Q9HAU4	SMAD6	O43541-2, O35182	O95630
SMAD7	O15105	Q9HAU4	SMAGP	Q0VAQ4	Q8IY26
SMARCA2	P51531	Q8NFD5	SMARCA5	O60264	P17096
SMARCAD1	Q9H4L7	P43246	SMARCAL1	Q9NZC9	P27694
SMARCB1	Q12824	Q92925, Q8NFD5, P51531, O96019, O75177, Q15532	SMARCC1	Q92922	Q92925, Q8NFD5, O96019, Q15532
SMARCD2	Q92925	O96019, Q15532	SMARCD3	Q6STE5	Q8NFD5
SMC1A	Q14683	O60216, Q7Z5K2, Q96FF9	SMC2	O95347	P01106
SMC3	Q9CW03, Q9UQE7	O60216, Q7Z5K2, Q96FF9	SMC4	Q9NTJ3	P01106
SMC5	Q8IY18	Q9H668	SMC6	Q96SB8	Q00613
SMCO4	Q9NRQ5	P25025	SMG6	Q86US8	P67870

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
SMG9	Q9H0W8	Q9BRV3	SMIM3	Q9BZL3	Q16585
SMPD2	O60906	P49675	SMPD3	Q9NY59	P04114
SMTN	P53814-5	O76024	SMUG1	Q53HV7-2	Q8IVS8
SMURF1	Q9HCE7-2	O14757	SMURF2	Q9HAU4	Q9HAU4
SNAI1	O95863	P49841	SNAPC1	Q16533	Q8IYI6
SNAPC3	Q92966	P04792	SNAPC4	Q5SXM2	Q9UPP1
SNAPIN	O95295	P20700	SNCA	P37840	Q969S8
SNIP1	Q8TAD8	P59595	SNRNP25	Q9BV90	Q9Y2W6
SNRNP40	Q96DI7	P62316, P62306, P14678, P62304	SNRNP70	P08621	Q09161
SNRPA	P09012	O75152	SNRPA1	P09661	P55199
SNRPB2	P08579	Q14566	SNRPD1	P62316	P09651
SNRPD2	P62316	P09651	SNRPD3	P62318	P62316, P62306, P14678, P62308, P62304
SNRPF	P62306	Q9Y3Y2	SNRPG	P62308	P62306, P14678, P62318, P62304
SNRPN	P63162	O75563	SNTA1	Q13424	P60484
SNUPN	O95149	O14980	SNX1	Q13596	Q8IVP5
SNX10	Q9Y5X0	Q92993	SNX17	Q15036	Q13496
SNX2	O60749	Q9H2K0	SNX20	Q7Z614-3	P50454
SNX24	Q9Y343, Q9Y343-2	P49674	SNX29	Q8TEQ0	Q13077
SNX3	O60493	P56539	SNX30	Q8WV41	Q05193
SNX4	O95219	Q6ZPD8	SNX5	Q9Y5X3	Q9Y639
SNX8	Q9Y5X2	P16435	SOAT1	P35610	O75326
SOBP	A7XYQ1	Q13363	SOCS1	O15524	P46109
SOCS3	O14543	Q15369, Q15370	SOCS4	O14544	P46937
SOCS6	O14544	P46937	SOD1	P00441	P29692
SOGA1	O94964	P61981	SON	P18583	P61981
SORBS1	Q9BX66	Q13177	SORBS2	O94875	Q13177
SORL1	Q92673	Q15669	SORT1	Q99523	P05231
SOS1	Q62245	P62993-1	SOS2	Q07890	P19174
SOX13	Q9UN79	Q12933	SP3	Q02447	P03372
SP4	Q02446	P14859	SPACA9	Q96E40	O00471
SPAG4	Q9NPE6	O00767	SPAG5	Q96R06	Q9Y5N6
SPAG9	O60271	P53350	SPART	Q8N0X7	Q96J02
SPATA13	Q96N96-6	P51570	SPATA2	Q9UM82	Q9Y6D9
SPATA2L	Q8IUW3	P02649	SPATA6	Q9NWH7	Q92466
SPATS2L	Q9NUQ6	Q9H9Z2	SPC24	Q8NBT2	Q96ES7
SPC25	Q9HBM1	P36941	SPDEF	O95238	P25786
SPDL1	Q96EA4	P25963	SPDYA	Q5MJ70	P46527
SPECC1L	Q69YQ0	P25963	SPEN	Q96T58	Q06330
SPG11	Q96JI7	Q05193	SPHK1	Q9NYA1	P68104
SPHK2	Q9NRA0	Q96CV9	SPIDR	Q14159	Q06609
SPIN1	Q9H2V7	Q07817	SPINT1	O43278-2	P02787
SPINT2	O43291	Q99640	SPIRE1	Q08AE8	P60709
SPIRE2	Q8WWL2	P61981	SPN	O75398	P49841
SPNS1	Q9H2V7	Q07817	SPNS3	Q6ZMD2-2	Q8WVV5
SPOPL	Q6IQ16	Q99836	SPPL2A	Q8TCT8	O76024
SPRED2	Q7Z698	P27797	SPRY1	O43609	P54646
SPRY2	O43597	P22681	SPRY3	O43610	Q92504

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
SPRYD4	Q8WW59	P24534	SPSB2	Q99619	Q15369, Q15370, Q93034, Q9UBF6
SPSB3	Q6PJ21	Q93034, Q9UBF6	SPTB	P11277	P42224
SPTLC2	O15270	P13569	SPTY2D1	Q68D10	P78563
SQLE	Q14534	Q96BA8	SQOR	Q9Y6N5	P01106
SREK1	Q9JKL7	Q15424, Q92841	SRF	P11831	Q969V6
SRGAP2	O75044	O75044	SRGN	P10124	P10144
SRI	P30626	P30626	SRM	P19623	P19623
SRP14	P37108	Q96JC9	SRP54	P61011	P11142
SRP68	Q9UHB9	P40429	SRP72	O76094	P48431
SRPK1	Q96SB4	Q9Y3Y2	SRPK2	P78362	P46087
SRPRA	P08240	Q6UWB1	SRR	Q9GZT4	Q9H4P4
SRRM1	Q8IYB3	P40429	SRRM2	Q9UQ35	P04608
SRSF1	Q07955	Q9NYV4, Q14004	SRSF10	O75494	P31947
SRSF11	Q05519	Q92844	SRSF2	Q01130	Q9HD47
SRSF5	Q13243	Q9NYV4, Q14004	SRSF6	Q13247	P05455
SRSF7	Q16629	O95639	SRSF8	Q9BRL6	P61981
SS18	Q15532	Q92925	SS18L1	O75177-5	Q06710
SSBP3	Q9BWW4	P50458	SSBP4	Q9BWG4	P50458
SSH1	Q8WYL5	P60709	SSH2	Q76I76	P63104
SSNA1	O43805	Q9P0N9	SSR3	Q9UNL2	P13569
SSR4	P51571	P13569	SSRP1	Q08945	Q9NTX7
SSSCA1	O60232	Q13459	SSX2IP	EBI-11742485	O14641
ST14	Q9Y5Y6	Q04721	ST3GAL2	Q16842	P11021
ST5	P78524	P31947	ST8SIA4	Q92187	Q92854
STAM2	O75886	O95630, Q92783	STAMBP	O95630	Q9NYZ3
STAMBPL1	Q96FJ0	Q9NYZ3	STAP1	Q9ULZ2	P30281
STAP2	Q9UGK3	Q99836	STARD3	Q14849	P46059
STAT1	P42224	P40763	STAT3	P40763	P35232
STAT4	Q14765	Q99665	STAT5B	P51692	P00519
STAT6	P42226	Q9UHD2	STAU2	Q9NUL3	P03496
STBD1	O95210	O95278, Q04446, Q9UQK1	STIL	Q15468	Q9HC77
STIM1	Q13586	P30101	STIM2	Q9P246	Q8NBK3
STIP1	P31948	P07900	STK17B	O94768	P08173
STK19	B7ZLI8	Q969Z0	STK24	Q9Y6E0	Q9BRV8
STK25	O00506	Q9BRV8	STK38	Q15208	Q9Y5B0
STK4	Q13043	P18509	STK40	Q8N2I9	Q8NHY2
STMN1	P16949	P46527	STOML2	Q9UJZ1	P13569
STOX1	Q6ZVD7	Q01974	STOX2	Q9P2F5	P49674
STRADA	Q7RTN6-1, Q7RTN6	Q15831	STRIP1	Q5VSL9	Q9BRV8
STRN	O55106, O43815	Q9BRV8	STT3A	P46977	P55072
STT3B	Q8TCJ2	Q14165	STUB1	Q9UNE7	P08138
STX18	Q9P2W9	Q13190, Q9NYM9	STX1A	Q16623	P60880
STX1B	P61266	O15554	STX3	Q13277	Q9H2K0
STX4	Q12846	Q13190	STX6	O43752	Q9H2K0
STX8	Q9UNK0	P49675	STXBP1	P61764	P60880
STXBP2	Q15833	Q13277	STYK1	Q6J9G0	P49841
SUCLA2	Q9P2R7	Q9UBN6	SUCLG1	P53597	Q8IWL3
SUFU	Q9UMX1	P30307	SUGP2	Q8IX01	P09651

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
SULF2	Q8IWU5	P17480	SULT1B1	O43704	Q9NUX5
SUMF1	Q8NBK3	Q8NBK3, Q8NBJ7	SUMO2	P61956	Q13569
SUMO3	P55854	P63279	SUN1	O94901	Q8NF91, O94901
SUN2	Q9UH99	Q96PU5	SUOX	P51687	Q15669
SUPT16H	Q9Y5B9	Q9NTX7	SUPT20H	Q8NEM7	Q16539
SUPT5H	O00267	Q9Y6D9	SUPT7L	O94864, O94864-2	Q96ES7
SURF4	O15260	Q86WV6	SUSD3	Q96L08	P17152
SUV39H1	O43463	Q9NPA3	SUV39H2	Q9H5I1	P45973
SUZ12	Q15022	P01106	SV2A	Q02563	P0DPI0
SVIL	O46385	P35579	SWAP70	Q9UH65	Q8WZA2
SYAP1	Q96A49	Q9NUX5	SYCE2	Q6PIF2	Q05329
SYF2	O95926	P40692	SYMPK	Q92797	O95639
SYNE1	Q8NF91-1, Q8NF91	Q9UH99, O94901	SYNE2	Q8WXH0, Q8WXH0-1	Q9UH99, O94901
SYNE3	Q6ZMZ3	Q9UH99	SYNJ1	Q62910, O43426	Q99962
SYNJ2BP	P57105	Q9NR61	SYPL1	Q16563	P63027
SYS1	Q8N2H4	P26951	SYTL1	Q8IYJ3	P51159
SYTL3	Q4VX76	P51159	SYVN1	Q86TM6	P55072
TACC3	Q9Y6A5	Q00994	TACO1	Q9BSH4	P21673
TADA1	Q96BN2	Q9Y4A5	TADA2A	O75478	P52292
TADA2B	Q86TJ2	P52294	TADA3	O75528	P52294
TAF1	P21675	P05067	TAF10	Q12962	Q92831
TAF12	Q16514	Q9Y4A5	TAF1C	Q15572	Q01105
TAF3	Q5VWG9	P84243	TAF4	O00268	O43889
TAF5	Q15542	P20226	TAF7	Q15545	Q13257
TAF8	Q7Z7C8-2	P54652	TAF9B	Q9HBM6	P35249
TAGLN2	P37802	P13569	TALDO1	P37837	P42858
TANC2	Q9HCD6	P36873	TANK	Q92844	P53350
TAOK1	Q7L7X3	P61981	TAOK3	Q9H2K8	Q12933
TAP1	Q03518	P30101	TAP2	Q03519	O15533
TAPBP	O15533	Q30201	TAPBPL	Q9BX59	P0DPK4
TARS	P26639	Q96CV9	TAS2R5	Q9NYW4	Q9BRI3
TATDN2	Q93075	Q03405	TBC1D2	Q9BYX2-5	Q8NC69
TBC1D22B	Q9NU19	P30740	TBC1D25	Q3MII6	P18848
TBC1D2B	Q9UPU7	P09917	TBC1D30	Q9Y2I9-2	Q9UGI0
TBC1D4	O60343	P31947	TBC1D7	Q9P0N9	P23025
TBC1D8	O95759	Q13077	TBC1D9	Q6ZT07	P21709
TBCC	Q15814	P06396	TBCE	Q15813	P25963
TBKBP1	A7MCY6	Q9UKV8	TBL1XR1	Q9BZK7	P36404
TBL2	Q9Y4P3	O15530	TBPL1	P62380	O76024
TBRG1	Q3YBR2	Q8NB12	TBX19	O60806	O95947
TBX21	Q9UL17	P08047	TCEA2	Q15560	Q13158
TCEAL1	Q15170	O14757	TCEAL4	Q96EI5	Q93009
TCEAL8	Q8IYN2	P04792	TCEANC	Q8N8B7-2	P50458
TCF19	Q9Y242	Q15669	TCF25	Q9BQ70	P08311
TCF4	Q9NQB0	Q13761	TCF7L2	Q9NQB0	Q13761
TCL1A	P56279	Q9Y6K1	TCP11L1	Q9NUJ3	P50458
TCP11L2	Q8N4U5	P62736	TCTEX1D2	Q8WW35	P63172
TDP1	Q9NUW8	P06746	TDP2	O95551	O14641
TEFM	Q96QE5	O00411	TEPSIN	Q96N21	O95848

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
TERF1	P54274-2	Q15004	TERF2	Q15554	Q13415
TES	Q9UGI8	P13569	TESC	Q96BS2	Q9H8X2
TESK2	Q96S53	P31947	TESMIN	Q9Y4I5	P08235
TET1	Q8NFU7	O15294	TET2	Q6N021	P19544
TF	P02787	P03372	TFB1M	Q8WVM0	Q00059
TFCP2	Q12800	P03428	TFDP1	Q14186	O75496
TFDP2	Q14188	Q08999, P06400	TFG	Q92734	O94979, P55735
TFPT	P0C1Z6	P55199	TFRC	P02786	Q9P0V3
TGDS	O95455	P42858	TGFA	P01135	P00533
TGFB1	P01137	O60551	TGFB1I1	O43294	Q14289
TGFB3	P10600	P37173	TGFBR2	P37173	Q99741
TGFBR3	Q03167	P22455	TGFBRAP1	Q8WUH2	Q9UBF6
TGIF1	Q15583	P53350	TGM2	P21980	P02751
TGM5	O43548	P0CG47	TGOLN2	O43493	O94822
TGS1	Q96RS0	P52294	THADA	Q6YHU6	P0DTC7
THAP4	Q8WY91	P17707	THAP7	Q9BT49	P12830
THAP8	Q8NA92	Q9UBB9	THBD	P07204	Q9H2K0
THBS1	P07996-PRO_0000035842	P16671	THEM6	Q8WUY1	Q8TC07
THEMIS	Q8N1K5-1	P43403	THEMIS2	Q5TEJ8	Q7Z3S9
THG1L	Q9NWX6	Q9NWX6	THOC1	Q96FV9	P82979, Q9Y3Y2
THOC2	Q8NI27	Q9Y3Y2	THOC3	Q96J01	Q13769
THOC6	Q86W42	Q13838	THOC7	Q6I9Y2	Q86V81
THOP1	P52888	Q96CV9	THRA	P10827	Q15648
THRAP3	Q9Y2W1	Q9UBU9	THUMPD1	Q9NXG2	Q96PZ0
THUMPD2	Q9BTF0	O76024	THUMPD3	Q9BV44	Q9BPX6
THYN1	Q9P016	Q9UHJ6	TIAM1	Q13009	P18827
TIFA	Q96CG3	O14641	TIGD1	Q96MW7	P42858
TIGD4	Q8IY51	Q12933	TIMELESS	Q9UNS1	P09874
TIMM10	P62072	Q9Y5J6	TIMM10B	Q9Y5J6	P18509
TIMM13	Q9Y5L4	Q99558	TIMM23	O14925	Q9BVV7
TIMM8B	Q9Y5J9	Q99558	TIMM9	Q9Y5J7	Q9Y5J6
TIMMDC1	Q9NPL8	O75063	TINF2	Q9BSI4	P52565
TIPIN	Q9BVW5	Q9BRT9, Q14691, Q9BRX5	TIPRL	O75663	P67775
TIRAP	P58753	Q15109	TK1	P04183	Q9UM11
TKT	Q16832	Q16288	TLCD1	Q96CP7	Q9H2J7
TLDC1	Q6P9B6	O60551	TLE1	Q04724	Q13761
TLE2	Q04725	Q5TAB7	TLE3	Q04726	Q08117
TLE4	Q04727	O95231	TLK2	Q86UE8	Q92985
TLN2	Q9Y4G6	P00519	TLNRD1	Q9H1K6	Q9NRD5
TLR3	O15455	O15455	TLR6	Q9Y2C9	O60603
TLR8	Q9NR97	Q9H1C4	TM2D3	Q9BRN9	P0DTC8
TM6SF1	Q9BZW5-2	P06396	TMA16	Q96EY4	Q92973
TMBIM1	Q969X1	Q8NCR0	TMCC1	O94876	Q9BRI3
TMED1	Q13445	Q9H2A9	TMED3	Q9Y3Q3	Q15363
TMED4	Q7Z7H5	P04618	TMED8	Q6PL24	Q9H2K0
TMEM101	Q96IK0	P63027	TMEM102	Q8N9M5	Q96QF0
TMEM106B	Q9NUM4	Q6UX41	TMEM106C	Q9BVX2	P11686
TMEM107	Q6UX40	Q9BRI3	TMEM109	Q9BVC6	Q9H2K0
TMEM14A	Q9Y6G1	P49682	TMEM14B	Q9NUH8	Q9Y584

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
TMEM14C	Q9P0S9	A1L3X0	TMEM150A	Q86TG1	O60563
TMEM154	Q6P9G4	Q15052	TMEM165	Q9HC07	P13569
TMEM167B	Q9NRX6	Q8N130	TMEM183A	Q8IXX5	O14879
TMEM184B	Q9Y519	Q96RK4	TMEM186	Q96B77	Q71RS6
TMEM19	Q96HH6	Q6Q788	TMEM192	Q8IY95	Q9UBV7
TMEM200A	Q86VY9	P16298	TMEM214	Q6NUQ4	O75326
TMEM218	A2RU14	Q9H2K0	TMEM223	A0PJW6	Q9H8X2
TMEM230	Q96A57	P27824	TMEM234	Q8WY98	Q8N130
TMEM243	Q9BU79	Q9H2K0	TMEM25	Q86YD3	P53611
TMEM254	Q8TBM7	Q9H2K0	TMEM255A	Q5JRV8	Q96J02
TMEM255B	Q8WV15	Q6ZPD8	TMEM258	P61165	P0DTC4
TMEM259	Q4ZIN3	Q96A29	TMEM267	Q0VDI3	Q5QGT7
TMEM30B	Q3MIR4	P22455	TMEM33	P57088	P13569
TMEM39B	Q9GZU3	O75326	TMEM41A	Q96HV5	Q7L5A8
TMEM43	Q9BTV4	Q6Q788	TMEM5	Q9Y2B1	O75072
TMEM55B	Q86T03	Q92993	TMEM60	Q9H2L4	Q9UBD6
TMEM63A	O94886	Q99808	TMEM68	Q96MH6	Q16581
TMEM79	Q9BSE2	P11686	TMEM86A	Q8N2M4	O43914
TMEM88	Q6PEY1	P17152	TMEM97	Q5BJF2	Q9H2K0
TMLHE	Q9NVH6	P10809	TMOD1	P28289	P06396
TMOD2	Q9NZR1	Q9UM54	TMOD3	Q9NYL9	P60709
TMPO	P42166	Q93009	TMPRSS3	Q9NRS4	O60238
TMSB10	P63313	Q9NUX5	TMSB4X	P20065	Q8WWQ8
TMTC1	Q8IUR5-4	P06396	TMTC3	Q6ZXV5	Q6PIU2
TMUB2	Q71RG4-4	P06396	TMX1	Q9H3N1	Q9NPF0
TMX2	Q9Y320	P13500	TNF	P01375	Q13546
TNFAIP1	Q13829	P12931	TNFAIP3	P21580	Q14790
TNFAIP8L1	Q8WVP5	Q6NZI2	TNFAIP8L2	Q6P589	P63000
TNFRSF10A	O00220	Q14790	TNFRSF10B	O14763	Q14790
TNFRSF10D	Q9UBN6	Q969Z0	TNFRSF11A	Q9Y6Q6-2	P00533
TNFRSF12A	Q9NP84	Q13490	TNFRSF14	Q92956	Q7Z6A9
TNFRSF1A	P19438	Q13546	TNFRSF8	P28908	Q99741
TNFSF10	P50591	O15519-2, O15519-1	TNFSF14	O43557	Q92956
TNFSF15	O95150-1	O95407	TNFSF4	P23510	P43489
TNFSF8	P32971	Q14118	TNIK	Q9UKE5	O43318
TNKS	O95271	Q9NTX7	TNKS1BP1	Q9C0C2	O14965
TNPO2	O14787	Q9Y639	TNPO3	Q9Y5L0	P13569
TNRC18	O15417	P21741	TNRC6A	Q8NDV7	P11940
TNRC6B	Q9UPQ9	P11940	TNRC6C	Q9HCJ0, Q9HCJ0-1	P11940
TNS1	Q04205	P56945	TNS4	Q8IZW8	P19474
TNXB	P22105	Q8N9N5	TOB1	P50616	P11940
TOM1L2	Q6ZVM7	Q9UM54	TOMM20	Q15388	P19367
TOMM34	Q15785	P07900	TOMM40	O96008	P56705
TOMM70	O94826	P59636	TONSL	Q96HA7	P49736
TOP1	P11387	P00519	TOP2A	P11388	P17096
TOPORS	Q9NS56	P51668	TOR1A	O14656-2	Q14154
TOR1AIP1	Q5JTV8	Q13257	TOX2	Q96NM4-3	O14561
TOX4	O94842	P36873	TP53	P04637	O96017, O14757
TP53BP1	Q12888	Q13315	TP53I3	EBI-1550792	P04637
TP53RK	Q96S44	Q9BTK6	TP73	O15350, O15350-1	P46937

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
TPCN2	Q8NHX9	Q9UJK0	TPD52L1	Q16890	P63104
TPGS2	Q68CL5-3	P27797	TPH1	P17752	O95619
TPI1	P60174	P12004	TPM1	P09493-10, P09493	Q9Y6D9
TPM2	P07951	P13569	TPM4	P67936	P13569
TPP2	P29144	P06746	TPPP	O94811	P37840
TPRG1	Q6ZUI0	P55056, P02654	TPST1	O60507	P61769
TPST2	O60704	Q9BUV8	TPX2	Q9ULW0	O75330
TRA2A	Q13595	O43592	TRA2B	P62995	P61981
TRAC	P01848	Q09328	TRAF2	Q12933	Q13233
TRAF3	Q13114	Q13546	TRAF3IP2	O43734	Q9Y4K3
TRAF6	Q9Y4K3	O75385	TRAF7	Q6Q0C0	Q9NWB7
TRAFD1	O14545	Q96RK4	TRAIP	Q9BWF2	O15160
TRAK1	Q9UPV9	P49841	TRAK2	O60296	P49841
TRAPPC1	Q9Y5R8	Q96QF0	TRAPPC10	P48553	Q96QF0
TRAPPC2	P0DI81	Q96QF0	TRAPPC3	O43617	Q96QF0
TRAPPC5	Q8IUR0	Q9P2M4	TRAPPC6B	Q86SZ2	Q9P2M4
TRAPPC9	Q96Q05	Q96QF0	TRAT1	Q6PIZ9	P49748
TRERF1	Q96PN7	P46937	TRIB3	Q96RU7	Q6W0C5
TRIM14	Q14142	P61024	TRIM17	Q9Y577	O14964
TRIM21	P19474	P19474	TRIM27	P14373	Q96E11
TRIM3	O75382	P51668	TRIM32	Q13049	Q06124
TRIM33	Q9UPN9	Q13257	TRIM35	Q9UPQ4-2	Q9NPB3
TRIM37	O94972	Q7L590	TRIM39	Q9HCM9, Q9HCM9-2	P51668
TRIM44	Q96DX7	Q3SYG4, Q9BXC9	TRIM47	Q96LD4	Q9UPW6, Q01826
TRIM56	Q9BRZ2	P05388	TRIM59	Q8IWR1	P0DTC3
TRIM62	Q9BVG3	P51668	TRIM69	Q86WT6, Q86WT6-2	P08243
TRIO	O75962	Q9NV70	TRIP10	Q15642-2	Q9Y4D1
TRIP12	Q14669	Q8N726	TRIP13	Q15645	Q8NDV7
TRIP6	Q15654	P12236	TRMO	Q9BU70	P12931
TRMT1L	Q7Z2T5	Q00403	TRMT2B	Q96GJ1	P07992
TRMT44	Q8IYL2	P00740	TRNAU1AP	Q9NX07	P27797
TRNT1	Q96Q11-3	O76024	TROAP	Q12815	Q13627
TRPC4AP	Q8TEL6	P01160	TRPS1	Q9UHF7	Q13257
TRRAP	Q9Y4A5	O96019	TRUB2	O95900	P09651
TSC1	Q92574	P49815	TSC2	P49815	Q9P0N9
TSC22D1	Q15714	Q15047	TSC22D3	Q99576-3, Q99576	Q9Y6D9
TSEN54	Q7Z6J9	Q92989	TSG101	Q99816	P21453
TSGA10	Q9BZW7	Q9Y6D9	TSHZ3	Q63HK5	Q9Y6D9
TSNARE1	A0AVG3	Q9NYP7	TSPAN15	O95858	Q9H244
TSPAN2	O60636	P17931	TSPAN31	Q12999	Q9BRI3
TSPAN4	O14817	Q92570	TSPAN5	P62079	P48509
TSPAN9	O75954	P21926	TSPYL1	Q9H0U9	P61289
TSPYL2	Q9H2G4	Q9UGY1	TSPYL4	Q9UJ04	O00273
TSR2	Q969E8	O75928-2	TSSC4	Q9Y5U2	P62306, P62304
TTBK2	Q6IQ55	Q9UPV0	TTC1	Q99614	Q9H4P4
TTC12	Q9H892-2	Q13153	TTC17	Q96AE7	Q8IV16

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
TTC21B	Q7Z4L5	Q8NEZ3, Q96RY7, Q96FT9, Q9HBG6, Q9P2L0	TTC26	A0AVF1	Q9NQC8, Q9Y547, Q9BW83, Q9P2H3, Q9NWB7, Q9Y366, Q9UG01, Q96LB3, Q9H7X7, Q8WYA0, Q13099
TTC28	Q96AY4	P61981	TTC3	P53804	P31751
TTC30A	Q86WT1	Q9NQC8, Q8TDR0, A0AVF1, Q9Y547, Q9BW83, Q8IY31, Q9NWB7, Q9Y366, Q9H7X7, Q8WYA0, Q96AJ1, Q9P2H3, Q9UG01, Q96LB3, Q13099	TTC30B	Q8N4P2	Q86T03
TTC33	Q6PID6	P02649	TTC9C	Q8N5M4	Q06187
TTF1	Q15361	P62263	TTF2	O00358	P19419
TTI2	Q6NXR4	Q9Y639	TTK	P33981	P08173
TTLL12	Q14166	P68104	TTLL7	I6L9D0	Q96CV9
TTN	Q8WZ42	Q9GZQ8	TTYH3	Q9C0H2	P21453
TUBA1B	P68363	Q15398	TUBA1C	Q9BQE3	P36507
TUBB	P07437	P36404	TUBB1	Q9H4B7	P00480
TUBB2A	Q13885	P36404	TUBB4B	P68371	Q9BTW9, P36404
TUBB6	Q9BUF5	P13569	TUBG1	P23258	Q96L34
TUBGCP3	Q96CW5	P0DTC5	TUBGCP4	Q9UGJ1	Q9NTX7
TUFT1	Q9NNX1	Q92993	TUSC5	A5PKU2	Q7Z2H8
TVP23B	Q9NYZ1	Q9NPL8	TWF1	Q12792	P13569
TWF2	Q6IBS0	P60709	TWIST1	Q15672	O60885
TWNK	Q96RR1	Q86VM9	TXK	P42681	O14543
TXLNA	P40222	Q9BRV8	TXLNG	Q8BHN1	Q9UHD2
TXN	P10599	P19883	TXN2	Q99757	P49247
TXNDC11	Q6PKC3	Q99633	TXNDC17	Q9BRA2	Q96T51
TXNDC9	O14530	O00628	TXNIP	Q9H3M7	P10599
TXNL1	O43396	P10636-8	TXNRD1	Q16881	O75695
TYK2	P29597	O75496	TYMP	P19971	Q9BW85
TYMS	P04818	P48059	TYMSOS	Q8TAI1	Q5VWX1
TYROBP	O43914	Q92538	TYW3	Q6IPR3	P01130
TYW5	A2RUC4	Q9NZ43	UACA	Q9BZF9	P13569
UBA2	Q9UBT2	P63279	UBA3	Q8TBC4	Q8TBC4, P61081
UBA52	P62990	P51668	UBA6	A0AVT1	Q8WVK2
UBA7	P41226	P05161	UBAC1	Q9BSL1	Q9BQ52
UBAC2	Q8NBM4	Q8TC07, Q9HA65	UBAP2	Q5T6F2	Q8NFF5
UBASH3A	P57075-2	Q07889	UBB	P0CG47	Q9HAU4
UBE2A	P49459	Q8WZA2	UBE2C	Q5TZN3, O00762	Q13164
UBE2D2	P62837	P22681	UBE2F	Q969M7	P26367
UBE2G2	P60604	P0CG48	UBE2J1	Q9Y385	Q9H2A9
UBE2J2	Q8N2K1	A1L3X0	UBE2K	P61086	Q9H4P4
UBE2L3	P68036	P50876	UBE2M	P61081	O94844
UBE2N	P61088	Q9Y4K3	UBE2O	Q9C0C9	P04608
UBE2Q2	Q8WVN8	P04792	UBE2S	Q16763	P13569
UBE2T	Q9NPD8	Q15078	UBE2V1	Q13404-2, Q13404	P61088
UBE2V2	Q15819	Q14527, P61088	UBE3B	Q7Z3V4	P80370
UBE4A	Q14139	P55072	UBL4A	P11441	O15055
UBL5	Q9I9K6	P24864	UBL7	Q96S82	P55072

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
UBN1	Q9NPG3, Q9NPG3-2	P54198, Q9Y294	UBN2	Q6ZU65	P54198, Q9Y294
UBOX5	O94941	P68036	UBP1	Q9NZI7	O60674
UBR2	Q8IWV8	O14786	UBTD2	Q8WUN7	P07902
UBXN1	Q04323	P55072	UBXN10	Q96LJ8	P55072
UBXN7	O94888	Q16665	UCHL1	P09936	P11802
UCK2	Q9BZX2	Q9NWZ5	UFL1	O94874	P21462
UGDH	O60701	P13569	UGP2	Q16851	O76003
UHMK1	Q8TAS1-2	Q9BU61	UHRF1	Q96T88	Q9Y6K1
UHRF2	Q96PU4	P06400	ULBP1	Q9BZM6	Q6ZMH5
ULK1	O75385	O75385	ULK3	Q6PHR2	O00170
UMPS	P11172	P54764	UNC45A	Q9H3U1	P48729
UNC45B	Q8IWX7	P07900	UNC5CL	Q8IV45	Q96HA1
UPF1	Q92900	P03496	UPF3A	Q9H1J1	Q92900
UPF3B	Q9BZI7, Q9BZI7-2	Q92900	UQCC2	Q9BRT2	P62993
UQCR10	Q9UDW1	P31930, O14949	UQCRH	P07919	P40855
UQCRQ	O14949	P31930, Q9UDW1, P14927, P22695	URB1	O60287	P06748
URI1	O94763, O94763-1	P36873	USF1	P22415	P78527
USF2	Q15853	Q15562	USO1	O60763	Q9Y6L6
USP1	O94782	Q8TAF3	USP10	Q8NFA0	P78423
USP14	P54578	P60896	USP16	Q9Y5T5	O95714
USP18	Q9UMW8	P05161	USP19	O94966	P07900
USP24	Q9UPU5	O75695	USP27X	A6NNY8	Q96B97
USP28	Q96RU2	Q12888	USP30	Q70CQ3	Q6PJG9
USP32	Q8NFA0	P78423	USP33	Q8TEY7	Q9BYE7
USP34	Q70CQ2	P48729	USP35	A2RRA6	O76024
USP36	Q9P275	P06748	USP39	Q53GS9	O75533
USP4	Q13107	Q13546	USP42	Q9H9J4-2	P04637
USP43	Q70EL4	P61981	USP44	Q9H0E7	P41208
USP47	Q96K76	O60216	USP5	P45974-2	Q9NWF9
USP53	Q70EK8	P46109	USP54	Q70EL1-9	O94955
USP7	Q93009	O96017	USPL1	Q6DRC5, Q5W0Q7	P55854, P63165, P61956
UST	Q9Y2C2	Q8WVC6	UTP14C	Q5TAP6	Q9NRX1
UTP15	Q8TED0	Q8TC07	UTP18	Q9Y5J1	P0DTC9
UTP20	O75691	Q9NZQ7	UTP3	Q9NQZ2	P02787
UTP6	Q9NYH9	P32455	UTRN	P46939	Q14118
UTY	O14607	Q14686	UVRAG	Q9P2Y5	P46937
UXS1	Q8NBZ7	P40855	VAMP1	P23763-3	A1L3X0
VAMP2	P63027	P55072	VAMP3	Q15836	P01730
VANGL1	Q8TAA9	O14641	VAPB	O95292	Q8TEY7
VARS	P26640	P26641, P68104	VASN	Q6EMK4	P78504
VAV2	P52735	Q9H3M7	VAV3	Q9UKW4	P62993
VBP1	P61758	Q13164	VCL	P18206	Q9Y490
VCP	P55072	O96017	VCPIP1	Q96JH7	P55072
VCPKMT	Q9H867	P55072	VDAC1	P21796	P0C0U1
VDAC3	Q9Y277	P21796	VDR	P11473	Q15648
VENTX	O95231	Q08117	VEZF1	Q14119	Q96KQ7
VGF	O15240	P01160	VILL	O15195-2	P10636
VIM	P08670	P54819	VIPAS39	Q9H9C1	P12273

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VKORC1L1	Q8N0U8	Q9H2K0	VMA21	Q3ZAQ7	P63027
VMP1	Q96GC9	Q07157	VPS11	Q9H270	Q9UBF6
VPS13B	Q7Z7G8	P49674	VPS13D	Q5THJ4	P61981
VPS16	Q9H269	P0DTC3	VPS25	Q9BRG1	Q96FZ7
VPS26A	O75436	P33993	VPS26B	Q4G0F5	O60493
VPS33A	Q96AX1	Q14790	VPS35	Q96QK1	P33993
VPS37B	Q9H9H4	Q08117	VPS39	Q96JC1	P51149
VPS4A	Q9UN37	P53990, O43633	VPS52	Q8N1B4	P46087
VPS53	Q5VIR6	P51665	VPS54	Q9P1Q0-4	P06396
VPS72	Q15906	O96019	VPS9D1	Q9Y2B5	P19474
VRK1	Q99986	Q9H4B4	VTA1	Q9NP79	Q9UN37, O43633
VTI1A	Q96AJ9	P54920	VWA8	A3KMH1	Q12824
WAC	Q9BTA9	Q16236	WARS	P23381	P13569
WASF2	Q9Y6W5	P00519, Q9UQB8	WASHC2A	Q641Q2	P51530
WASHC3	Q9Y3C0	O75385	WASHC4	Q2M389	O75396
WASHC5	Q12768	P52907	WBP1	P97764, Q96G27	P46937
WBP4	O75554	P62306	WDCP	Q9H6R7	P53350
WDFY1	Q8IWB7	P63027	WDFY2	Q96P53	Q9Y266
WDHD1	O75717	Q9BQP7	WDR12	Q9GZL7	P06748
WDR18	Q9BV38	Q8IZL8	WDR19	Q8NEZ3	Q7Z4L5, Q96RY7, Q96FT9, Q9HBG6, Q9P2L0
WDR20	Q8TBZ3	Q8TAF3	WDR26	Q9H7D7	O15169
WDR27	A2RRH5	P08243	WDR3	Q9UNX4	P06748
WDR34	Q96EX3	P84243	WDR37	Q9Y2I8	Q6VY07
WDR4	P57081	P39748	WDR44	Q5JSH3	P62993
WDR45B	Q5MNZ6	P51813	WDR48	Q8TAF3	Q96B01
WDR5	P61964	O00255	WDR54	Q9H977	Q9Y316
WDR59	Q6PJI9	Q13309	WDR5B	Q86VZ2	P49247
WDR6	Q9NNW5	P03496	WDR60	Q8WVS4	Q9Y266
WDR61	Q9GZS3	Q16143	WDR62	O43379	P45983
WDR70	Q9NW82	Q06330	WDR74	Q6RFH5	Q96PM5
WDR90	Q96KV7	Q13451	WDR91	A4D1P6	P42858
WDR92	Q96MX6	P36873	WDTC1	Q8N5D0	Q13371
WEE1	P30291	P53350	WFS1	O76024	Q9BRX2
WHRN	Q9P202	Q9UMD9	WIPF1	O43516	Q9GZQ8
WNK1	Q9H4A3	Q15149	WNT3	P56703	P56704
WRB	O00258	P21964	WRNIP1	Q96S55	O95786
WSB2	Q9NYS7	Q07817	WTAP	Q15007-2	P54252
WWC1	Q8IX03	Q05513	WWC2	Q6AWC2	P61981
XAB2	Q9HCS7	P23025	XCL1	P47992	P13501, P48061
XCL2	Q9UBD3	P13501, P48061	XKR8	Q9H6D3	P43119
XPA	P23025	Q9P0N9	XPO1	O14980	P04618, P18754
XPO5	Q9HAV4	P0DTC7	XPO6	Q96QU8	Q9NZQ7
XPOT	O43592	P0DTC7	XPR1	Q9UBH6	P21802
XRCC1	P18887	O96017	XRCC2	O43543	O75771, O43502, O15315
XRCC3	O43542	Q86YC2	XRCC4	Q13426	P19838
YAE1D1	Q9NRH1	P08670	YAF2	Q8IY57-5	Q6NZI2
YARS	P54577	O75925	YARS2	Q9Y2Z4	Q12834

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YBEY	P58557	Q9BQ52	YEATS2	Q9ULM3	P17980
YEATS4	O95619	P30291	YES1	P07947	P46109
YIF1A	O95070	Q9NQ11	YIF1B	Q5BJH7	P51684
YIPF4	Q9BSR8	Q8IWU4	YIPF5	Q969M3	O95070, O75460
YIPF6	Q96EC8	Q9H2K0	YME1L1	Q96TA2	P13569
YOD1	Q5VVQ6	Q9Y4K3	YPEL5	P62699	P01275
YTHDC1	Q96MU7	Q16629, Q08170, P84103	YTHDF1	Q9BYJ9	P08311
YTHDF2	Q9Y5A9	Q99558	YTHDF3	Q7Z739	P13569
YWHAE	P62258	Q86UK7	YWHAG	P61981	Q9BRX2
YWHAH	Q04917	P21359	YY1	P25490	Q9BYJ1
ZAP70	P43403	Q9Y2R2	ZBED8	Q8IZ13	Q99856
ZBP1	Q9NZI8	P11940	ZBTB1	Q9Y2K1	P11309
ZBTB10	Q96DT7	O15397	ZBTB16	Q05516	P68104
ZBTB2	Q8N680	P08047	ZBTB21	Q9ULJ3	Q13257
ZBTB22	O15209	Q9Y316	ZBTB3	Q9H5J0	Q8NB12
ZBTB32	A0A0C4DGF1	Q9UH92	ZBTB33	Q86T24	P57764
ZBTB38	Q8NAP3	Q96EB6	ZBTB39	O15060	Q13200
ZBTB4	Q9P1Z0	Q08379	ZBTB40	Q9NUA8	Q99435
ZBTB43	O43298	P49407	ZBTB44	Q8NCP5	P60900
ZBTB45	Q96K62	O95429	ZBTB46	Q86UZ6	Q9UF56
ZBTB48	P10074	Q969R5	ZBTB49	Q6ZSB9	Q13114
ZBTB6	Q15916	P63165	ZC2HC1C	Q53FD0-2	Q9BRV8
ZC3H12A	Q5D1E8	Q14694	ZC3H4	Q9UPT8	P67870
ZC3HAV1	Q7Z2W4, Q7Z2W4-2	O95786	ZC3HAV1L	Q96H79	P53611
ZC3HC1	Q86WB0	P53350	ZCCHC10	Q8TBK6	Q9H8M2
ZCCHC11	Q5TAX3	Q9H9Z2	ZCCHC3	Q9NUD5	P63244
ZCCHC6	Q5VYS8-5	P08670	ZCCHC7	Q8N3Z6	P63279
ZCCHC9	Q8N567	P61247	ZCRB1	Q8TBF4	P62306, P14678
ZDHHC1	Q8WTX9	P42336	ZDHHC11	Q9H8X9	P35813
ZDHHC16	Q969W1	O95070	ZDHHC17	Q8IUH5	P68104
ZDHHC5	Q9C0B5	Q8TDQ0	ZDHHC7	Q9NXF8-2	O43561-2
ZDHHC9	Q9Y397	Q9H8X2	ZEB1	P37275	Q13363
ZEB2	O60315	Q96KQ7	ZER1	Q7Z7L7	Q6IN84
ZFAND2A	Q8N6M9	Q15008	ZFAND2B	Q8WV99	Q9Y5K5
ZFAND5	O76080	P0CG48	ZFAT	Q9P243	Q9UER7
ZFHX2	Q9C0A1	P10636	ZFHX3	Q15911-2	P14373
ZFP36	P26651	Q9NPI6	ZFP36L1	Q07352	Q16539
ZFP36L2	P47974	P31947	ZFP62	Q8NB50	O43159
ZFPL1	O95159	Q9HC36	ZFPM1	Q8IX07	P49841
ZFY	P08048	Q969R5	ZFYVE21	Q9BQ24	Q92993
ZFYVE27	Q5T4F4	Q9NTJ5	ZFYVE9	O95405	P36873
ZG16B	Q96DA0	P43897	ZGPAT	Q8N5A5-3	O14641
ZHX1-C8orf76	Q96EF9	P12830	ZKSCAN1	P17029	Q13164
ZKSCAN3	Q9BRR0	Q6IPR3	ZKSCAN4	Q969J2	Q9UBH0
ZKSCAN5	Q9Y2L8	Q53GS7	ZKSCAN8	Q15776	Q92993
ZMAT4	Q9H898-2	Q15633	ZMIZ2	Q8NF64-2	Q8N9N5
ZMPSTE24	O75844	P02545-1	ZMYM3	Q14202	Q13547
ZMYM5	Q9UJ78-2	Q13158	ZMYM6	O95789	P16035

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ZNF124	Q15973	Q9UGI0	ZNF131	P52739, P52739-2	Q14192
ZNF136	P52737	Q13263	ZNF14	P17017	Q13263
ZNF140	P52738	Q9NPB3	ZNF148	Q9UQR1	P52294
ZNF175	Q9Y473	P13196	ZNF181	Q2M3W8	P05067
ZNF184	Q99676	Q13263	ZNF197	O14709	P05067
ZNF2	Q9BSG1	Q13263	ZNF200	P98182	Q14192
ZNF211	Q13398	Q13263	ZNF213	O14771	Q9UMX1
ZNF217	O75362	Q13547	ZNF223	Q9UK11	Q13263
ZNF225	P18146	Q8N726	ZNF23	P17027	Q13077
ZNF248	Q05CR2	Q92993	ZNF250	P15622-3	O75496
ZNF263	O14978	Q9UMX1	ZNF266	Q14584	Q9HD47
ZNF267	Q14586	Q13263	ZNF268	Q14587	O15431
ZNF276	Q8N554	Q9NRD5	ZNF280D	Q6N043, Q6N043-2	Q13185
ZNF281	Q9Y2X9	P40763	ZNF286A	Q9HBT8	O95714
ZNF316	A6NFI3	Q04721	ZNF317	Q96PQ6	P98175
ZNF32	P17041	Q7Z3S9	ZNF324	O75467	Q04721
ZNF324B	Q6AW86	Q9NS56	ZNF326	Q5BKZ1	P09651
ZNF329	Q86UD4	Q96CV9	ZNF331	Q9NQX6	O15397
ZNF333	Q96JL9-2	P06396	ZNF335	Q9H4Z2	P51610
ZNF34	Q8IZ26	Q13263	ZNF346	Q9UL40	P19525
ZNF350	Q9GZX5	P38398	ZNF382	Q96SR6	O15397
ZNF395	Q9H8N7	P61981	ZNF397	Q8NF99	Q8NHY2
ZNF398	Q8TD17	Q96CV9	ZNF407	Q9C0G0	Q04721
ZNF408	Q9H9D4	Q99611	ZNF41	P51814-6	P55081
ZNF419	Q96HQ0	Q96CV9	ZNF426	Q9BUY5	P57764
ZNF430	Q9H8G1	Q9Y4Z2	ZNF44	M0R160	O14964
ZNF444	Q8N0Y2-2	Q13153	ZNF446	Q9NWS9	Q9BTV6
ZNF451	Q9Y4E5	P11388	ZNF460	Q14592	O15397
ZNF467	Q7Z7K2	Q969R5	ZNF48	Q96MX3	Q13895
ZNF484	Q5JVG2	Q13263	ZNF490	Q9ULM2	Q9UGI0
ZNF497	Q6ZNH5	Q9UGI0	ZNF501	Q96CX3	Q9Y4X5
ZNF502	Q8TBZ5	P0DOE7	ZNF503	Q96F45	O75593
ZNF512	Q96ME7	Q9Y3T9	ZNF516	Q92618	Q13547
ZNF517	Q6ZMY9	Q9H4P4	ZNF518B	Q9C0D4	Q13185
ZNF521	Q96K83	Q13547	ZNF526	Q8TF50	Q04721
ZNF543	Q08ER8	Q13263	ZNF544	Q6NX49	O15228
ZNF547	Q8IVP9	O15160	ZNF550	Q7Z398	Q9UBX5
ZNF558	Q96NG5	P57764	ZNF565	Q8N9K5	P05067
ZNF57	Q68EA5	Q9NPB3	ZNF574	Q6ZN55-2	Q12933
ZNF576	Q9H609	Q7Z6E9	ZNF577	Q9BSK1	O95373
ZNF584	Q8IVC4	P05067	ZNF596	Q8TC21	P60900
ZNF614	Q8N883	Q14938	ZNF622	Q969S3	Q6NZI2
ZNF628	Q5EBL2	Q8IYI6	ZNF629	Q9UEG4	P40429
ZNF639	Q9UID6	Q96T60	ZNF641	Q96N77-2	P04792
ZNF644	Q9H582	Q96KQ7, Q13185	ZNF646	O15015	A8MZ59
ZNF655	Q8N720	P23025	ZNF658	Q5TYW1	Q13263
ZNF662	Q6ZS27-3	Q12933, Q13077	ZNF669	Q96BR6	Q9UBX5
ZNF672	Q499Z4	Q04721	ZNF687	Q8N1G0	Q13547

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
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ZNF70	Q9UC06	Q8IXZ2	ZNF703	Q9H7S9	Q8N695
ZNF724	A8MTY0	P36941	ZNF76	P36508	Q9BYJ1
ZNF761	Q86XN6	Q13263	ZNF763	Q0D2J5	Q9UMX1
ZNF764	Q96H86	Q8N2S1	ZNF765	Q7L2R6	Q13263
ZNF768	Q9H5H4	Q04721	ZNF771	Q7L3S4	P40429
ZNF772	Q68DY9	P50542	ZNF774	Q6NX45	O15397
ZNF775	Q96BV0	Q96KQ7	ZNF777	Q9ULD5	Q9UBB9
ZNF783	Q6ZMS7-2	Q13526	ZNF785	A8K8V0	Q9NPB3
ZNF786	Q8N393	Q9NS56	ZNF800	Q2TB10	Q13547
ZNF808	Q8N4W9	Q13263	ZNF830	Q96NB3	P62304
ZNF835	Q9Y2P0	Q96FJ0	ZNF84	P51523	Q13263
ZNF844	Q08AG5	Q13077	ZNF85	Q49A12	Q14192
ZNF92	Q03936	Q13263	ZNFX1	Q9P2E3	Q9Y3F4
ZNRD1	Q9P1U0	P06396	ZNRF1	Q8ND25	P51668
ZPR1	O75312	P68104	ZRANB1	Q9UGI0	O75419
ZRSR2	Q15696	P42357	ZSCAN12	O43309	Q9BYJ1
ZSCAN18	Q8TBC5	Q9UMX1	ZSCAN22	P10073	Q00403
ZSCAN23	Q3MJ62	P54646	ZSCAN9	O15535	Q92993
ZSWIM3	Q96MP5	Q12933	ZW10	O43264	P08173
ZWINT	O95229	P36941	ZXDA	P98168	P33076
ZXDB	P98169	P38919	ZXDC	Q2QGD7	P61964
ZZEF1	O43149	Q8NBL1			

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ACPP	A0ABF7PH85	29105	BMP1	P13497	29108
BOLA1	Q9Y3E2	33737	BOLA3	Q53S33	33737
CFH	P08603	18132	CIB1	Q99828-1	29108
CRY1	P97784	29105	CSE1L	P55060	15996
GCC2	Q8IWJ2	15996	HADH	P40939	28494
IDH1	O75874	16810	LRP1	Q07954	29108
MMACHC	Q9Y4U1	16856	MMADHC	Q9H3L0, Q99LS1	16856
MUT	P22033	18408	NPC1	O15118	16113
NPC2	P61916	16113	PAK1	Q13153	15996
PLEKHM1	Q9Y4G2	15996	RAB38	P57729	15996
RAB4A	P20338	15996	RAB9A	P51151	15996
RAN	P62826	17552	RCC1	P18754	17552
RILP	Q96NA2	15996	RILPL2	Q969X0	15996
S100A12	P80511	29108	SNCA	P37840	17552
THBS1	P07996	28815	TNF	P01375	29108

7. Identifiers not found

These 2299 identifiers were not found neither mapped to any entity in Reactome.

A1BG-AS1	A2M-AS1	AATBC	ABHD1	ABHD13	ABHD17AP3	ABHD17AP6	ABHD2
ABRACL	ABTB2	AC000123.2	AC000123.3	AC002094.4	AC002310.1	AC002310.2	AC002480.2
AC002558.1	AC002558.3	AC003681.1	AC004076.2	AC004233.3	AC004264.1	AC004453.2	AC004477.1
AC004492.1	AC004585.1	AC004656.1	AC004706.3	AC004801.6	AC004824.1	AC004828.1	AC004846.1
AC004846.2	AC004918.1	AC004918.3	AC004923.4	AC004943.1	AC004943.2	AC004967.1	AC004982.1
AC004982.2	AC005070.3	AC005077.2	AC005077.4	AC005154.1	AC005154.3	AC005224.4	AC005306.1
AC005332.1	AC005332.10	AC005332.5	AC005332.8	AC005534.1	AC005674.2	AC005696.1	AC005726.6
AC005736.1	AC005828.6	AC005833.4	AC005839.1	AC005840.4	AC005912.1	AC005921.3	AC005962.1
AC006001.3	AC006042.3	AC006063.1	AC006064.4	AC006195.1	AC006254.1	AC006330.1	AC006333.2
AC006369.1	AC006441.3	AC006449.3	AC006449.7	AC006504.1	AC006504.5	AC006504.8	AC006548.3
AC006978.1	AC007032.1	AC007038.1	AC007278.1	AC007278.2	AC007342.7	AC007349.2	AC007349.3
AC007382.1	AC007383.2	AC007384.1	AC007485.2	AC007566.1	AC007610.4	AC007611.1	AC007673.1
AC007728.2	AC007780.1	AC007878.1	AC007906.2	AC008033.3	AC008063.1	AC008147.2	AC008434.1
AC008440.2	AC008443.1	AC008467.1	AC008543.1	AC008555.5	AC008560.1	AC008567.3	AC008622.2
AC008669.1	AC008735.1	AC008741.2	AC008750.1	AC008750.2	AC008750.7	AC008763.1	AC008764.10
AC008764.6	AC008764.7	AC008870.2	AC008894.3	AC008906.1	AC008906.2	AC008957.1	AC008966.1
AC008982.2	AC009041.2	AC009061.2	AC009086.3	AC009093.2	AC009108.3	AC009118.3	AC009133.4
AC009145.1	AC009152.1	AC009163.3	AC009237.14	AC009237.15	AC009237.3	AC009275.1	AC009303.4
AC009318.1	AC009403.1	AC009404.1	AC009495.1	AC009495.3	AC009533.1	AC009542.1	AC009686.2
AC009690.1	AC009729.1	AC009779.2	AC009812.1	AC009812.4	AC009831.4	AC009948.1	AC009948.4
AC009951.1	AC010175.1	AC010186.2	AC010186.3	AC010186.4	AC010203.1	AC010247.2	AC010331.1
AC010335.3	AC010422.6	AC010442.2	AC010491.1	AC010536.1	AC010536.2	AC010542.5	AC010615.1
AC010616.1	AC010619.2	AC010642.2	AC010809.1	AC010809.2	AC010883.1	AC010894.2	AC010894.5
AC010973.2	AC011043.1	AC011337.1	AC011379.2	AC011447.3	AC011447.6	AC011447.7	AC011476.3
AC011477.2	AC011497.2	AC011503.2	AC011558.1	AC011603.2	AC011815.1	AC011893.1	AC011912.1
AC011921.1	AC011939.3	AC012020.1	AC012065.4	AC012073.1	AC012074.1	AC012181.2	AC012181.3
AC012213.1	AC012313.10	AC012313.2	AC012313.5	AC012363.1	AC012435.1	AC012467.2	AC012507.1
AC012615.1	AC012615.5	AC012618.3	AC012645.3	AC012651.1	AC013264.1	AC013394.1	AC013468.1
AC013474.1	AC013731.1	AC015468.3	AC015674.1	AC015726.1	AC015799.1	AC015813.1	AC015813.5
AC015849.1	AC015849.5	AC015909.5	AC015911.8	AC015917.2	AC015921.1	AC015959.1	AC015982.1
AC016027.1	AC016065.1	AC016205.1	AC016405.1	AC016588.1	AC016708.1	AC016722.3	AC016747.1
AC016949.1	AC016957.2	AC017083.2	AC017104.1	AC018450.1	AC018529.3	AC018607.1	AC018628.1
AC018629.1	AC018638.2	AC018638.4	AC018647.2	AC018690.1	AC018816.1	AC019069.1	AC019117.1
AC020571.1	AC020604.1	AC020656.1	AC020658.3	AC020663.3	AC020765.2	AC020907.5	AC020910.4
AC020910.5	AC020915.1	AC020917.4	AC020928.1	AC020978.7	AC021016.2	AC021028.1	AC021078.1
AC021127.1	AC021205.3	AC021242.3	AC021755.2	AC021945.1	AC022001.3	AC022075.2	AC022098.1
AC022107.1	AC022126.1	AC022150.4	AC022182.1	AC022211.1	AC022400.7	AC022405.1	AC022532.1
AC022726.2	AC022893.1	AC022916.1	AC023090.2	AC023355.1	AC023794.3	AC023818.2	AC023908.3
AC024060.1	AC024075.2	AC024909.3	AC025164.2	AC025171.1	AC025279.1	AC025884.1	AC026124.2
AC026250.1	AC026271.4	AC026401.3	AC026471.1	AC026471.5	AC026748.1	AC026954.1	AC026979.3
AC027290.1	AC027290.2	AC027307.3	AC027544.1	AC027544.2	AC027601.1	AC027801.1	AC036214.2
AC036222.1	AC037459.2	AC040162.3	AC040169.3	AC044797.1	AC044849.2	AC048341.2	AC048382.5
AC048382.6	AC055822.1	AC055839.1	AC063950.1	AC063965.1	AC063977.1	AC063977.6	AC064801.2
AC067852.2	AC067931.1	AC068205.2	AC068254.1	AC068587.4	AC068722.2	AC068768.1	AC068896.1
AC069234.2	AC069234.5	AC069281.2	AC069499.1	AC069544.2	AC073052.1	AC073130.3	AC073195.1
AC073349.2	AC073548.1	AC073569.3	AC073575.2	AC073655.2	AC074032.1	AC074117.1	AC074135.1

AC074141.1	AC078785.1	AC078795.1	AC078881.1	AC079203.1	AC079331.1	AC079684.1	AC079781.3
AC079950.1	AC080112.2	AC083949.1	AC084033.3	AC084048.1	AC084824.1	AC087164.1	AC087190.2
AC087276.1	AC087276.2	AC087289.5	AC087301.1	AC087500.1	AC087741.1	AC087741.3	AC087752.3
AC087878.1	AC087893.1	AC090114.3	AC090152.1	AC090181.2	AC090181.3	AC090204.1	AC090559.1
AC090589.3	AC090617.5	AC090897.1	AC090922.1	AC090948.2	AC091060.1	AC091516.1	AC091729.3
AC091814.1	AC092045.1	AC092111.1	AC092115.3	AC092117.2	AC092119.2	AC092171.2	AC092535.4
AC092634.3	AC092636.1	AC092653.1	AC092653.2	AC092718.4	AC092747.4	AC092821.1	AC092868.1
AC092910.3	AC092919.2	AC093249.2	AC093297.3	AC093311.1	AC093330.2	AC093503.2	AC093525.7
AC093525.8	AC093567.1	AC093642.2	AC093809.1	AC095055.1	AC096536.3	AC096677.1	AC096734.1
AC096920.2	AC096921.2	AC097372.1	AC097461.1	AC097534.1	AC097534.2	AC097637.2	AC098487.1
AC098613.1	AC098679.1	AC098934.1	AC099489.1	AC099521.3	AC099778.1	AC099804.1	AC099811.1
AC099811.5	AC100793.2	AC103563.7	AC103724.3	AC103724.4	AC103988.2	AC104046.1	AC104109.4
AC104411.1	AC104447.1	AC104564.4	AC104564.6	AC104623.2	AC104695.2	AC104695.3	AC104791.2
AC104794.2	AC104958.1	AC105345.1	AC105429.1	AC105446.1	AC105749.1	AC106782.5	AC106820.5
AC106872.5	AC107871.1	AC107884.1	AC108010.1	AC108062.1	AC108448.2	AC108693.1	AC108704.2
AC108727.1	AC109460.3	AC109779.1	AC109826.1	AC109992.2	AC110285.2	AC110769.2	AC112220.4
AC112512.1	AC112777.1	AC112907.3	AC113146.1	AC113404.3	AC114271.1	AC114490.1	AC114760.2
AC114811.2	AC116050.1	AC116351.1	AC116366.1	AC116366.3	AC116407.4	AC118658.2	AC119044.1
AC119396.1	AC119428.2	AC120053.1	AC121338.1	AC121761.1	AC122718.2	AC123768.3	AC123912.4
AC124045.1	AC124066.1	AC124248.1	AC124312.4	AC124798.1	AC125232.1	AC125257.1	AC125437.1
AC125611.3	AC125612.1	AC126389.1	AC127521.1	AC128688.2	AC130304.1	AC130324.3	AC130371.2
AC130448.2	AC131009.4	AC131212.2	AC132008.2	AC132872.3	AC132872.4	AC132938.6	AC133065.4
AC133435.1	AC133552.6	AC134349.1	AC134407.2	AC134407.3	AC134772.1	AC135048.4	AC135050.5
AC135983.2	AC136475.2	AC136475.9	AC137630.3	AC138028.3	AC138150.1	AC138150.2	AC138207.8
AC138393.1	AC138393.3	AC138517.2	AC138932.1	AC139530.1	AC139530.3	AC139768.1	AC139795.2
AC139887.2	AC140134.1	AC141424.1	AC141557.1	AC144831.1	AC146507.2	AC211476.2	AC233755.1
AC234775.3	AC239868.4	AC240274.1	AC241585.2	AC243829.1	AC243829.4	AC243960.3	AC244093.4
AC244100.2	AC244157.2	AC244197.3	AC244502.1	AC244669.1	AC245297.3	AC245427.5	AC246817.1
ACAP2	ACAP3	ACRBP	ACTBP11	ACTBP2	ACTBP9	ACTG1P20	ACTG1P3
ACTG1P9	ACTRT3	ACYP1	ADAMTS7P1	ADAMTS7P4	ADCK1	ADCY10P1	ADGRB1
ADGRB2	ADGRD1	ADGRF3	ADGRL1	ADIRF-AS1	ADORA2A-AS1	ADPRH	AF001548.1
AF117829.1	AF127936.1	AF165147.1	AFAP1	AFAP1L2	AGAP1	AGPAT4-IT1	AHSA2
AIFM3	AJ003147.3	AJ271736.1	AK4P4	AKAP2	AKNA	AKR1C5P	AKT3-IT1
AKTIP	AL008628.1	AL008721.2	AL021707.1	AL021707.2	AL021878.2	AL022157.1	AL022322.1
AL022322.2	AL022328.3	AL022328.4	AL022341.2	AL023803.2	AL024493.1	AL024498.1	AL031274.1
AL031282.2	AL031283.1	AL031283.2	AL031316.1	AL031587.6	AL031595.3	AL031665.2	AL031667.3
AL031727.1	AL031733.2	AL031846.2	AL031848.2	AL032819.2	AL034397.3	AL034550.2	AL035071.1
AL035563.1	AL035681.1	AL035701.1	AL049796.1	AL049836.1	AL050341.2	AL050343.1	AL078644.1
AL078645.1	AL096816.1	AL096870.2	AL109614.1	AL109811.3	AL109918.1	AL109923.1	AL109955.1
AL109976.1	AL117339.5	AL118516.1	AL121772.1	AL121832.3	AL121845.1	AL121906.2	AL121983.2
AL121985.1	AL121987.2	AL132642.1	AL133215.2	AL133243.2	AL133243.3	AL133330.1	AL133342.1
AL133384.2	AL133415.1	AL133517.1	AL133523.1	AL135818.1	AL135818.2	AL135999.1	AL136038.4
AL136295.7	AL136304.1	AL136368.1	AL136981.2	AL136984.1	AL136985.3	AL137003.1	AL137003.2
AL137060.1	AL137145.1	AL137145.2	AL137186.1	AL138820.1	AL139099.2	AL139241.1	AL139246.5
AL139287.1	AL139288.1	AL139393.2	AL139407.1	AL139415.1	AL139423.1	AL157394.1	AL158163.1
AL158163.2	AL158206.1	AL158211.5	AL159169.2	AL159972.1	AL160313.1	AL160396.2	AL160408.2
AL161421.1	AL161644.1	AL161729.1	AL161756.1	AL161911.1	AL162151.2	AL162457.1	AL163051.2
AL353150.1	AL353593.1	AL353593.2	AL353625.1	AL353743.1	AL353746.1	AL353795.3	AL353804.1
AL353807.3	AL353898.3	AL354707.1	AL354892.3	AL355001.2	AL355075.3	AL355388.2	AL355472.1
AL355488.2	AL355581.1	AL355816.2	AL355987.4	AL356019.2	AL356235.1	AL356273.3	AL356512.1

AL357033.1	AL357033.3	AL357060.2	AL358473.1	AL358781.1	AL359076.1	AL359711.2	AL359878.1
AL359921.2	AL359962.1	AL359962.2	AL360091.1	AL360181.2	AL360270.1	AL365181.3	AL365203.3
AL365273.1	AL365357.1	AL365475.1	AL390066.1	AL390195.2	AL390318.1	AL390728.4	AL390728.6
AL391244.3	AL391684.1	AL391839.2	AL391988.1	AL392172.1	AL441992.1	AL442125.1	AL445228.2
AL445231.1	AL445686.2	AL450326.2	AL450384.2	AL512306.2	AL512306.3	AL513190.1	AL513302.2
AL513534.1	AL590428.1	AL590560.3	AL590666.1	AL590666.2	AL590705.5	AL590764.1	AL590867.2
AL591895.1	AL592148.3	AL592295.1	AL592494.3	AL596202.1	AL596244.1	AL596325.2	AL603839.3
AL604028.2	AL606491.1	AL627171.2	AL645728.1	AL645931.1	AL645939.2	AL662797.2	AL662797.3
AL662907.1	AL669831.5	AL671710.1	AL671855.1	AL671883.3	AL691432.2	AL731563.3	AL731569.1
AL731571.1	AL807752.3	ALKBH1	ALKBH7	AMZ2P1	ANGEL2	ANGPTL6	ANKHD1-EIF4EBP3
ANKMY1	ANKRD10-IT1	ANKRD12	ANKRD20A11P	ANKRD26P4	ANKRD33B	ANKRD35	ANKRD44-IT1
ANP32C	ANTXRL	ANTXRLP1	ANXA2P2	ANXA2R	AOAH-IT1	AP000251.1	AP000769.1
AP000866.1	AP000866.6	AP000873.2	AP000911.1	AP001000.1	AP001020.2	AP001062.1	AP001107.1
AP001157.1	AP001206.1	AP001258.1	AP001273.1	AP001347.1	AP001486.2	AP001767.4	AP001972.5
AP001992.1	AP002004.1	AP002026.1	AP002360.2	AP002364.1	AP002387.1	AP002954.1	AP003086.1
AP003170.1	AP003352.1	AP003392.4	AP003469.2	AP003486.1	AP003717.1	AP003733.4	AP004608.1
AP004609.1	AP005019.1	AP005131.6	AP005482.1	AP006284.1	AP006621.5	AP006623.1	AP3M2
AP5M1	APOBEC3D	APOL4	APOLD1	APTR	ARHGAP26-IT1	ARIH2OS	ARL17A
ARMCX2	ARMCX4	ARMCX6	ASAH2B	ASCL2	ASTL	ATAD3C	ATE1-AS1
ATF4P3	ATOH8	ATP2A1-AS1	ATP2B1-AS1	ATP2C2-AS1	ATP5EP2	ATP5F1P1	ATP6V0E2-AS1
ATRN	ATXN7L3B	AVL9	BAALC	BACE1-AS	BAIAP3	BBOF1	BCAS1
BCLAF3	BCORP1	BEND4	BEX4	BHLHE40-AS1	BICDL1	BICDL2	BIN2P1
BMP8B	BNC2	BOLA3-AS1	BORCS7	BRSK1	BTBD10	BTBD19	BTBD7
BTG1P1	BTG3-AS1	BX284668.5	BX322639.1	BX664727.3	BZW1	C11orf21	C11orf63
C11orf98	C12orf75	C12orf76	C14orf28	C15orf40	C15orf52	C15orf53	C15orf56
C15orf61	C16orf72	C17orf107	C17orf51	C17orf53	C17orf75	C18orf21	C19orf48
C1DP1	C1RL-AS1	C1orf115	C1orf132	C1orf159	C1orf162	C1orf174	C1orf21
C1orf43	C1orf61	C20orf204	C20orf96	C21orf91	C22orf34	C22orf46	C2orf48
C2orf76	C3orf35	C3orf58	C3orf67	C4orf48	C5orf15	C5orf56	C5orf58
C6orf136	C6orf62	C7orf26	C7orf61	C8orf44	C9orf139	C9orf172	C9orf40
C9orf66	CA5BP1	CAAP1	CACNA1C-AS1	CACNA1C-AS2	CALHM2	CALML4	CALML6
CAMK2N1	CARD16	CARD6	CARD8-AS1	CARMIL3	CASC4	CASC8	CASZ1
CBR3-AS1	CBX3P2	CBX3P9	CCDC149	CCDC15	CCDC157	CCDC159	CCDC162P
CCDC173	CCDC18	CCDC18-AS1	CCDC180	CCDC183-AS1	CCDC191	CCDC30	CCDC61
CCDC65	CCDC69	CCDC74A	CCDC74BP1	CCDC91	CCER2	CCND2-AS1	CCNT2-AS1
CCSER1	CCT6P1	CDADC1	CDC123	CDC20P1	CDH26	CDHR1	CDKL2
CDKN2B-AS1	CDRT4	CDV3	CEACAM19	CEACAM4	CEBPB-AS1	CELF2	CELF2-AS1
CELSR1	CENPBD1P1	CEP120	CEP128	CFAP44	CFAP45	CFAP61	CFAP70
CFAP97	CHD2	CHIC1	CHMP4BP1	CHRM3-AS2	CIDECP	CISD1	CKAP2L
CKLF-CMTM1	CLASRP	CLDN24	CLECL1	CLMN	CLN8	CLYBL	CMPK2
CMTM1	CNBD2	CNIH4	CNTD2	CNTLN	COL6A4P2	CORO7	COX6CP1
CPEB3	CPEB4	CPED1	CPQ	CR559946.2	CRAT37	CRIM1	CRIP3
CRISPLD1	CRNDE	CROCC	CROCCP3	CRYBB2P1	CRYBG2	CTBP1-AS	CTBP1-AS2
CTDSPL	CTIF	CU639417.2	CXCR2P1	CXorf65	CYP21A1P	CYP2D7	CYP2D8P
CYP4F29P	CYP4F35P	DARS-AS1	DBNDD2	DDIAS	DDN-AS1	DDX11-AS1	DDX12P
DDX3Y	DDX50P1	DDX59	DEPDC1	DFFBP1	DGKZP1	DGUOK-AS1	DHFRP1
DHX33	DICER1-AS1	DIP2A-IT1	DIP2C	DIRC2	DLEC1	DLEU2	DMTF1

DNAH1	DNAH17	DNAH7	DNAJC19P5	DNAJC25	DNAJC3-AS1	DNAJC9	DNASE1
DNHD1	DOC2GP	DOPEY1	DPY19L2P2	DPY19L4	DSCR9	DSTNP2	DTD2
DTHD1	DUBR	DUS1L	DUSP18	DUSP28	DUTP6	DUX4L50	DYNC1I2P1
DZANK1	E2F3P1	EAF1-AS1	EBF4	EBLN2	EBLN3P	ECHDC1	EEF1A1P3
EEF1A1P30	EEF1DP2	EEF1DP7	EFCC1	EFHB	EFL1	EFR3A	EML2-AS1
EML5	ENDOV	ENOSF1	ENPP5	EPB41L4A	EPB41L4A-AS1	EPG5	ERI2
ERI3	ERICH1	ERICH6-AS1	ERMN	ERV3-1	ERVK13-1	ETV3	EXOSC3P1
EXPH5	FAM107B	FAM120AOS	FAM129C	FAM131A	FAM131C	FAM133B	FAM13A-AS1
FAM159A	FAM160A2	FAM160B2	FAM182B	FAM183BP	FAM193A	FAM198B	FAM19A2
FAM200A	FAM216A	FAM220A	FAM228B	FAM229A	FAM238A	FAM46C	FAM49A
FAM57A	FAM57B	FAM72A	FAM72B	FAM72C	FAM72D	FAR1-IT1	FAT4
FBXW4P1	FCHSD1	FCMR	FCRL1	FCRL6	FCRLB	FGF14-AS2	FHAD1
FILIP1L	FKBP11	FLJ38576	FLJ40194	FNDC10	FNDC7	FOCAD	FOXJ3
FP565260.7	FP565324.1	FRMD4B	FRMD8	FRRS1	FSCN3	FSD1L	FTH1P16
FTH1P22	FUT8-AS1	GANC	GAPDHP1	GAPDHP63	GAPDHP65	GAPT	GARNL3
GATS	GBAP1	GBGT1	GCNT2	GCSAM	GDF11	GDF7	GDPGP1
GFOD1-AS1	GFOD2	GGT4P	GIMAP6	GIMAP7	GIMAP8	GIN1	GK5
GLIPR2	GLOD4	GLT1D1	GNPTG	GOLGA2P10	GOLGA6L10	GPATCH3	GPR137C
GPR146	GPR153	GPR171	GPR174	GPR180	GPR19	GPR34	GPR63
GPR82	GPRASP1	GPRC5D-AS1	GPX1P1	GSAP	GSDMB	GSEC	GTDC1
GTF2IP20	GTPBP6	GTSE1-AS1	GUSBP9	GVINP1	GZMA	H2AFZP3	HARBI1
HCFC1R1	HCG27	HCG4B	HCP5B	HDGFL3	HEATR5B	HECA	HELLPAR
HERC2P10	HIF1A-AS2	HIPK1-AS1	HLA-J	HLA-K	HLA-U	HLCS-IT1	HMGB1P19
HMGB1P5	HMGB2P1	HMGN3	HMMR-AS1	HNRNPA1P21	HOTAIRM1	HOXA-AS2	HOXB6
HPS5	HSDL2	HSPA8P8	IAH1	IFI27L1	IFITM10	IFRD1	IGHJ4
IGHV1-3	IGHV3-15	IGHV3-21	IGLV3-13	IGLV3-9	IGSF9B	IKZF2	IL21R-AS1
IL9RP3	ILF3-AS1	IMMP1L	IMPDH1P5	INE1	INGX	INIP	INVS
IPO5P1	IQCH-AS1	IQCJ-SCHIP1	IRAK1BP1	ISOC2	ITGA9-AS1	ITPKB-IT1	JAZF1-AS1
JCAD	JHDM1D-AS1	JKAMP	JMJD8	JPT2	JRKL	KANSL1L	KBTBD11
KBTBD2	KCNT1	KCTD11	KCTD2	KCTD21-AS1	KHDC1	KIAA0895L	KIAA1109
KIAA1324L	KIAA1586	KIAA1841	KIAA2026	KIR2DP1	KLF3-AS1	KLF7	KLHDC8B
KLHL29	KLRF2	KMT2E-AS1	KRCC1	KRT17P8	KRT73-AS1	KRT8P12	KRT8P33
KRT8P46	KRT8P48	KRTAP5-AS1	KRTCAP2	LBHD1	LCNL1	LCTL	LDAH
LDHAP4	LDHAP7	LDLRAD2	LEF1-AS1	LENG9	LEXM	LILRP2	LIM2
LIMD1-AS1	LINC00094	LINC00116	LINC00271	LINC00294	LINC00298	LINC00299	LINC00324
LINC00337	LINC00469	LINC00484	LINC00539	LINC00565	LINC00599	LINC00629	LINC00641
LINC00659	LINC00665	LINC00672	LINC00847	LINC00852	LINC00885	LINC00892	LINC00893
LINC00920	LINC00921	LINC00944	LINC00959	LINC00987	LINC01001	LINC01006	LINC01023
LINC01024	LINC01091	LINC01126	LINC01128	LINC01132	LINC01176	LINC01187	LINC01221
LINC01222	LINC01238	LINC01260	LINC01278	LINC01355	LINC01358	LINC01366	LINC01422
LINC01451	LINC01480	LINC01481	LINC01504	LINC01505	LINC01547	LINC01572	LINC01578
LINC01588	LINC01619	LINC01694	LINC01759	LINC01771	LINC01775	LINC01786	LINC01801
LINC01814	LINC01819	LINC01825	LINC01871	LINC01881	LINC01934	LINC02001	LINC02012
LINC02033	LINC02035	LINC02055	LINC02084	LINC02088	LINC02132	LINC02256	LINC02320
LINC02321	LINC02328	LINC02341	LINC02384	LINC02422	LINC02446	LINC02458	LINC02470
LINC02486	LIX1-AS1	LIX1L	LMBR1L	LOXL1-AS1	LPP-AS2	LRGUK	LRRC3
LRRC34	LRRC37A15P	LRRC37A2	LRRC37A9P	LRRC39	LRRC42	LRRC43	LRRC63
LRRIQ4	LRRN3	LY6G5B	LY75	LY9	LYRM9	LYSMD4	MACC1
MAFG-AS1	MALAT1	MAN1B1-AS1	MAP10	MAP3K10	MAP3K13	MAP3K14-AS1	MAPKAPK5-AS1

MAST4	MATN1-AS1	MBLAC1	MBNL2	MCPH1-AS1	MCTP2	MCTS1	MDFIC
MEGF9	METRN	METTL2A	METTL2B	METTL4	METTL5	METTL8	MEX3D
MFSD12	MFSD14A	MFSD6L	MFSD7	MGAM2	MGEA5	MGP	MHENCR
MIATNB	MIF-AS1	MIR155HG	MIR17HG	MIR181A2HG	MIR210	MIR210HG	MIR221
MIR222HG	MIR22HG	MIR3142HG	MIR34AHG	MIR3609	MIR3662	MIR3936HG	MIR4258
MIR4420	MIR4458HG	MIR4506	MIR4639	MIR4742	MIR4772	MIR503HG	MIR590
MIR616	MIR646	MIR646HG	MIR6731	MIR6750	MIR6752	MIR6766	MIR6774
MIR6783	MIR6807	MIR6812	MIR6891	MIR7107	MIR7161	MISP3	MMP23B
MMP24-AS1	MN1	MOAP1	MOB3A	MORN1	MOSPD1	MPEG1	MPPE1
MPV17L2	MREG	MRGPRX2	MRPL37P1	MRPS31P5	MS4A6A	MST1P2	MSTO2P
MTATP6P1	MTCO1P11	MTCO2P11	MTCO3P11	MTCYBP18	MTND1P23	MTND5P11	MTND5P14
MTRNR2L12	MXRA7	MYCBP	MYO1F	MYOF	MZF1-AS1	Mar-01	Mar-02
Mar-06	Mar-08	Metazoa_SRP	N4BP2	N4BP2L2-IT2	NAA25	NAALADL1	NACC2
NAGPA	NAIP	NALT1	NANOGP4	NAPA-AS1	NAPSB	NARF-IT1	NBPF11
NBPF12	NBPF14	NBPF15	NBPF17P	NBPF19	NBPF20	NBPF26	NCBP2-AS1
NCBP2-AS2	NCF1B	NCF4-AS1	NCK1-AS1	NDFIP2	NDUFA6-AS1	NDUFB2-AS1	NDUFV2-AS1
NEAT1	NECAB3	NEK10	NEK2P4	NEMP2	NFYC-AS1	NIFK-AS1	NIT1
NKAPP1	NME6	NNT-AS1	NONOP2	NPIPA1	NPIPA5	NPM1P37	NPPA-AS1
NPTXR	NRAV	NREP	NRIR	NSUN5P2	NT5DC2	NUTM2B-AS1	OCIAD2
OIP5-AS1	OLA1P2	OLMALINC	OPA3	ORAI3	OSER1-AS1	OTUD6B-AS1	PA2G4P4
PABPC1P3	PANO1	PAPD4	PAPLN	PAQR4	PARGP1	PARP15	PARPBP
PATL2	PAXIP1-AS1	PCBP2-OT1	PCDH1	PCDHGA10	PCDHGA11	PCDHGA4	PCDHGA6
PCDHGB2	PCDHGB3	PCDHGB6	PCDHGC4	PCED1B-AS1	PCIF1	PCNX3	PDCL3P4
PDLIM2	PDXDC2P-NPIPB14P	PEG13	PEPD	PFN1P1	PGBD4P1	PGGHG	PHC1P1
PHF6	PHKA2-AS1	PHLDA3	PHTF1	PHTF2	PIEZO2	PIK3CD-AS1	PIK3IP1-AS1
PIM3	PINK1-AS	PINLYP	PITHD1	PITPNA-AS1	PKD1L1	PKD1L3	PKD1P6
PKDREJ	PLCL1	PLCXD1	PLCXD2	PLEKHD1	PLEKHF1	PLEKHM1P1	PLPBP
PLXDC1	PMFBP1	PMS2CL	PNISR	PNMA3	PNPLA1	PODN	POLR2J4
POM121B	POU6F1	PPIAP22	PPP1R11P1	PPP1R14B	PPP1R14BP3	PPP3CB-AS1	PRC1-AS1
PREPL	PRICKLE2	PRKAG2-AS1	PRKCZ-AS1	PROB1	PROM2	PRORSD1P	PRPF4B
PRPSAP1	PRR13P2	PRRT3	PRSS30P	PSD2	PSMA3-AS1	PSMD10P1	PSMD6-AS2
PSMG3-AS1	PSRC1	PTCHD3P2	PTGES3P1	PTGES3P3	PTMS	PTOV1-AS2	PTPRVP
PVT1	PWAR5	PWAR6	PXK	PYHIN1	PYROXD2	QRSL1	R3HCC1
R3HCC1L	RAB30-AS1	RABL2A	RAD51-AS1	RAET1K	RALGPS1	RASA3-IT1	RASGEF1B
RBM12B	RBM14-RBM4	RBM18	RBMS3	RC3H1	RC3H2	REM2	REXO1
RFESD	RFLNB	RFX3-AS1	RFX8	RGS17P1	RHBDL3	RHOA-IT1	RHOT1P2
RHOXF1	RHPN1-AS1	RIMS2	RIPOR1	RN7SKP127	RN7SKP173	RN7SKP26	RN7SKP80
RN7SL2	RN7SL280P	RN7SL32P	RN7SL517P	RN7SL650P	RN7SL809P	RN7SL81P	RN7SL834P
RNA5SP155	RNASEH1-AS1	RNF144A-AS1	RNF157	RNF157-AS1	RNF207	RNF215	RNF216P1
RNPEPL1	RNU4-2	RNU6-101P	RNU6-195P	RNU6-790P	RNU6-878P	RNU6ATAC24P	RNY1P16
RPL13AP5	RPL13P5	RPL38P4	RPL3P4	RPL7P49	RPS10P7	RPS27AP12	RPS2P36
RPSAP16	RPSAP36	RRM2P3	RRN3P1	RRN3P2	RSAD1	RSPH4A	RTCA
RTFDC1	RTL5	RTN2	RTP4	SAC3D1	SAMD12	SAMD9L	SAMSN1
SATB1-AS1	SBDSP1	SBF2-AS1	SBK1	SCAMP1-AS1	SCARNA13	SCARNA15	SCARNA21
SCARNA5	SCARNA7	SCART1	SCG5	SCML4	SEC14L1P1	SEC14L2	SEMA6A-AS1
SEMA6C	SENP6	SEPT14P12	SEPT7P2	SERPINI1	SETBP1	SETD4	SETD5
SETP9	SEZ6L	SFSWAP	SGMS1-AS1	SH2D7	SH3BP5-AS1	SH3TC1	SIDT1
SIGLEC18P	SIX4	SKA3	SKOR1	SLC10A3	SLC25A25	SLC25A35	SLC25A36
SLC25A45	SLC25A53	SLC26A4-AS1	SLC35F3	SLC44A3-AS1	SLC45A1	SLC4A11	SLC8A1-AS1

SLC9A7P1	SLC9A9-AS2	SLCO4A1-AS1	SLED1	SLFN13	SLFN5	SLTM	SMCO1
SMCO2	SMG1P1	SMG1P3	SMG1P5	SMIM10L2A	SMIM14	SMIM15	SMIM24
SMIM25	SMIM30	SMIM7	SMPD4P1	SMPDL3B	SNHG1	SNHG14	SNHG15
SNHG17	SNHG22	SNHG25	SNHG26	SNHG3	SNHG5	SNHG6	SNHG8
SNHG9	SNN	SNORA11	SNORA11F	SNORA12	SNORA14	SNORA16B	SNORA28
SNORA2B	SNORA33	SNORA3B	SNORA55	SNORA66	SNORA71A	SNORA71B	SNORA71C
SNORA71D	SNORA73B	SNORA75	SNORA7B	SNORD100	SNORD104	SNORD117	SNORD13E
SNORD14A	SNORD15B	SNORD17	SNORD19	SNORD20	SNORD38A	SNORD46	SNORD4A
SNORD59A	SNORD62B	SNORD63B	SNORD83A	SNORD89	SNORD94	SNRK	SNX18P13
SOS1-IT1	SOWAHD	SPA17	SPAG16	SPAG6	SPATA3-AS1	SPATA6L	SPEF2
SPESP1	SPOCK2	SRGAP2C	SRRD	ST20	ST20-AS1	STAG3L4	STAG3L5P
STAG3L5P-PVRIG2P-PILRB	STAM-AS1	STIP1P3	STK31	STOML1	STPG1	STX16-NPEPL1	STX18-AS1
STYX	STYXL1	SUCLG2-AS1	SUGT1P3	SVOPL	SYCP2L	Sep-01	Sep-04
Sep-06	Sep-07	Sep-08	Sep-11	TAGLN	TAGLN2P1	TATDN3	TBC1D12
TBC1D19	TBC1D23	TCAIM	TCEAL3	TCF3P1	TDRKH-AS1	TESPA1	TEX261
TEX30	TEX48	TFP1	TGFBR3L	TGIF2P1	THAP5	THAP7-AS1	THAP9-AS1
TINCR	TIPARP-AS1	TKTL1	TLE1P1	TM7SF3	TM9SF1	TMC8	TMCC3
TMCO1	TMCO4	TMEM104	TMEM110-MUSTN1	TMEM116	TMEM117	TMEM119	TMEM123
TMEM127	TMEM131L	TMEM132C	TMEM134	TMEM138	TMEM145	TMEM156	TMEM167A
TMEM170A	TMEM170B	TMEM175	TMEM178B	TMEM181	TMEM184C	TMEM191A	TMEM198B
TMEM2	TMEM209	TMEM217	TMEM220-AS1	TMEM241	TMEM245	TMEM250	TMEM41B
TMEM44	TMEM52	TMEM53	TMEM63C	TMEM69	TMEM72-AS1	TMEM8B	TMEM91
TMEM99	TMIGD2	TMPO-AS1	TMPRSS4-AS1	TMSB10P1	TMSB15B	TMSB4XP8	TOGARAM1
TOP1MT	TOR3A	TP53I13	TP53INP2	TP53TG1	TP73-AS1	TPBGL	TPI1P1
TPPP3	TPRA1	TPRG1L	TRABD2A	TRAJ12	TRAJ49	TRAM2	TRAM2-AS1
TRAV10	TRAV22	TRBJ2-2P	TRBV25-1	TRDC	TRDJ2	TRDJ3	TRDJ4
TRDV1	TRG-AS1	TRGV10	TRGV5	TRGV5P	TRGV7	TRGV9	TRGVB
TRIM52	TRIM52-AS1	TRIM60P18	TSC22D2	TSHZ1	TSPAN13	TSPAN32	TSPOAP1-AS1
TTC13	TTC14	TTC16	TTC22	TTC28-AS1	TTC34	TTC38	TTC39B
TTC7A	TTC7B	TTN-AS1	TUBAP2	TUBBP1	TUBD1	TWSG1	TXLNGY
TXNDC15	U1	U73169.1	U91328.2	UAP1L1	UBA52P6	UBA6-AS1	UBALD2
UBAP1L	UBE2F-SCLY	UBE2FP1	UBE2Q2P1	UBE2Q2P2	UBL3	UBR5-AS1	UBR7
UCKL1-AS1	UFC1	UFM1	ULK4	UNQ6494	USF3	USP9Y	VDAC1P8
VDAC2P2	VDAC3P1	VEZT	VIM-AS1	VPS13C	VSIG10L	VSTM2L	VWA3A
VWA5A	VWA7	VWDE	WASH3P	WBP1LP1	WBP1LP2	WDPCP	WDR47
WDR63	WDR76	WDR97	WDSUB1	WHAMMP2	WHAMMP3	WSCD1	XKR3
YJEFN3	YPEL1	Y_RNA	Z69706.1	Z84723.1	Z93930.2	Z94721.1	Z94721.2
Z95115.1	Z97832.2	Z97986.1	Z98882.1	Z99129.4	ZBED3	ZBED6CL	ZBTB20
ZBTB20-AS1	ZBTB37	ZBTB40-IT1	ZC3H12C	ZCCHC14	ZCCHC2	ZCCHC24	ZCWPW1
ZDHHC11B	ZDHHC12	ZEB1-AS1	ZEB2-AS1	ZFAND3	ZFAND4	ZFHX2-AS1	ZFYVE28
ZHX2	ZMAT1	ZNF132	ZNF137P	ZNF142	ZNF146	ZNF174	ZNF185
ZNF204P	ZNF236	ZNF251	ZNF271P	ZNF318	ZNF322	ZNF365	ZNF367
ZNF436-AS1	ZNF487	ZNF491	ZNF528-AS1	ZNF529-AS1	ZNF532	ZNF561-AS1	ZNF578
ZNF587B	ZNF594	ZNF630	ZNF653	ZNF654	ZNF674-AS1	ZNF683	ZNF780A
ZNF780B	ZNF784	ZNF788	ZNF80	ZNF814	ZNF826P	ZNF83	ZNF831
ZNF876P	ZNF878	ZNF887P	ZNHIT1P1	ZNRF2P2	ZSWIM1	ZSWIM4	ZSWIM5
ZSWIM6	gene	uc_338					